

# TFPT — Research Contracts for the Remaining Interfaces:

$$v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$$

( $U_{\text{wall}}$ ): the parabolic flavor wall-selection and ( $G_{\text{metric}}$ ): the full quantum-gravity measure

plus the two continuum contracts CELEST.SEAM.01 and WOIT.OS.TWISTOR.01 (M1–M3 and the  $\delta$ -chains executed and decided;  $\alpha/\beta_1/\beta_2/\beta_3$  executed,  $\gamma$  next)

Stefan Hamann

Alessandro Rizzo

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## The TFPT document set — what is treated where

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and five companions**, best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). (*entry point*)
1. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction, the QBL theorem chain.
2. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor,  $Q_+$  cohomology), the worked closures.
3. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
5. **tfpt\_5\_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions:** **R. tfpt\_research\_contracts** — the open interfaces ( $v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$ ) as numbered contracts; **H. tfpt\_horizon\_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O. origin\_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S. tfpt\_safeguards** — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team); **P. tfpt\_prime\_front** — research documentation of the prime / zeta line: the seven load-bearing modules on the  $E_8$  glue census (Hecke from geometry through the transport ledger), the sandbox localization program that follows them, and the kill list. Documentation, not a result: it makes no claim about the Riemann Hypothesis.

**You are reading companion R (Research Contracts).**

**TFPT status at a glance — read this card the same way in every document**

**Two inputs, one compiler.** From the boundary constant  $c_3 = \frac{1}{8\pi}$  (axiom P1) and the five-slot carrier  $g_{\text{car}} = 5$  (axiom P2) — jointly the elementary symmetric data of one anchor  $a = (1, 1, 2)$  — the discrete compiler  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ( $\alpha^{-1} = 137.0359992$ ) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

**What is closed, and what genuinely remains.** The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

| Interface             | Question                                                                                        | Status        |
|-----------------------|-------------------------------------------------------------------------------------------------|---------------|
| $v_{\text{geo}}$      | the one metrology unit ( $= 1/\sqrt{G} = m/\mu$ ); No-Unit Thm: no compiler scale               | primitive [O] |
| $G_{\text{net}}$      | simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$ | [C]           |
| $F_{\text{transfer}}$ | one functor, four typed interfaces (Koide, $\eta_B$ , axion, $m_p/m_e$ )                        | [C]           |

**Gravity is parameter-free.** The classical metric-sector field equation is no longer only an  $R+R^2$  readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  with *both* coefficients fixed —  $c_3^{-1} = 8\pi$  (no free dimensionless Newton dial; the thermodynamic origin  $2\pi/\eta$  *coincides* with the geometric one  $|\mathbb{Z}_2|2\pi\chi$  via  $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$ , so  $c_3$  is triply over-determined — anchor, geometry, thermodynamics, v358) and  $\Lambda$  from  $\alpha$  ( $\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$ , v60); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (v359). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in v473 without changing the typing) and the one unit  $v_{\text{geo}}$ ; the global measure (C7 = QG.AMB.01) is now a [C] redundancy (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann.

The seam-net structural theorem  $G_{\text{net}}$  (= SEAM.EQUIV.01: the raw seam *is* the holomorphic  $(E_8)_1$  net) now carries two named routes (route split 2026-07-22, no status change in substance): the MMST route SEAM.EQUIV.MMST.01 is *closed modulo cited theorems*; the twistor route SEAM.EQUIV.TWISTOR.01 (Costello–Li, prepared by CELEST.SEAM.01) and the unconditional parent stay [O]. On the MMST route: the explicit lattice model (v367/v368) and the  $S_3$  closure stack pin the target at every computable level — central charge  $c = 8$  (v376), the  $(E_8)_1$  character with 248 currents and one primary (v377), genus-one torus  $\text{GSD} = 1$  (v378) and reflection positivity (v379) — and it is Lean-formalised as FORM.SEAM.MMST.01: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the **#print axioms** check is clean. The *only* residual of that route that stays [O] is the abstract continuum existence of the scaling limit (v336) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package ( $h_s=16/16=1 \in \mathbb{Z}$ , the Longo–Rehren locality criterion; Böckenhauer; KLM  $\mu=4/2^2=1$ ; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route **seamResidualClosed'** (v469); so the MMST route is *closed modulo cited theorems*, not solved. QGEO.SYM.01 (the  $\mu_4$  deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (v335, Lean FORM.QGEO.BW.01). The nonperturbative quantum-gravity measure QG.AMB.01 is no longer carried as an open item: it is a [C] *redundancy* (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin  $1.648 > 0$ , v76/v330). So the honest residual is the one unit  $v_{\text{geo}}$  plus the typed  $F_{\text{transfer}}$  interfaces, above the cited-theorem ceiling on the seam and with QG.AMB.01 a [C] redundancy. Everything carries a claim ID resolving to a row of **verification/status\_ledger.csv** (the single source of truth); **if the text and the ledger ever disagree, the ledger wins**. The historical labels ( $U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$ ) are retained only for ledger continuity — the live residual is  $v_{\text{geo}} \oplus F_{\text{transfer}}$ , with the seam  $G_{\text{net}}$  closed modulo cited theorems on its MMST route (parent [O]) and QG.AMB.01 a [C] redundancy.

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in **verification/status\_ledger.csv**.

**[E]** exact / machine-proven**[C]** conditional**[O]** open / axiom**[X]** kill test

$$\mathbf{P1} \, c_3 = \frac{1}{8\pi} \quad \mathbf{P2} \, g_{\text{car}} = 5 \quad (\text{two inputs, [O] axioms})$$

$$\text{Anchor } a = (1, 1, 2): e_1, e_2, e_3 = 4, 5, 2 \quad ([E] \text{ exact})$$

$$\text{Discrete compiler: } D_5 \oplus A_3 + \mu_4 \Rightarrow E_8, h = 30, \varphi_0 \quad ([E] \text{ exact})$$

$$\text{SM skeleton: three families, hypercharges, flavor matrix } R \quad ([E])$$

$$\text{Physical bridges: } v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}} \quad ([C] \text{ bridges})$$

$$\text{Frontier readouts: } \Lambda, \text{ inflation, } \eta_B, \text{ dark matter, QG} \quad ([C]/[O])$$

### What this document is about

After the compiler closure the entire residual is

$$\text{Rest} = (U_{\text{wall}}) \oplus (G_{\text{metric}}) \oplus (F_{\text{frontier}})$$

This note turns the two genuine research gates into *contracts*: a numbered chain of lemmas, the single theorem that closes each gate, and — for every step — whether it is machine-certifiable today. ( $F_{\text{frontier}}$ ) (Koide,  $\eta_B$ , axion relic scale,  $m_p/m_e$ ) is *not* a gate: those are deliberately typed QFT/cosmology interfaces (see `tfpt_4_frontier`). **Priority:** ( $U_{\text{wall}}$ ) *first* (finite, algebraic, falsifiable), then ( $G_{\text{metric}}$ ) (deep analytic programme).

**Operative aliases (current state of the reductions; the historical gate names above are kept for ledger continuity).** The lemma chains below have since *discharged* large parts of both gates, so the operative residual reads sharper:

$$\text{Rest}_{\text{operativ}} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$$

— ( $U_{\text{wall}}$ )  $\rightarrow v_{\text{geo}}$ : the quark *ratios* are closed combinatorially (Readout Rigidity + Plücker, U2/v49/v71),  $U_{\text{point}}$  collapses to the *one* dimensionful unit  $v_{\text{geo}}$  shared with  $1/G$  (v75/v68) — what remains is a scale anchor, not a flavor gap (the *full* amplitude matrix beyond the ratios is the only place where  $U_{\text{point}}$  still appears); ( $G_{\text{metric}}$ )  $\rightarrow G_{\text{net}}$ : holographically reduced to the single boundary-net statement “the seam-Calderón inclusion has Jones index  $4 = |\mu_4|$ ” — holomorphy,  $(E_8)_1$  and the unique 2D bulk then *follow* (v89/v92/GATE.METRIC.05-07); and the carrier-side QBL input now merges into the *same* statement: the tower carrier  $\rightarrow SO(16)_1 \rightarrow (E_8)_1$  carries one and the same 16-fermion content, whose single quasi-free kernel (rank =  $c$ ; Wick/Pfaffian exact) is the certified Calderón datum — closing  $G_{\text{net}}$  therefore also closes the carrier choice **[E]** (`[verification/v113_quasifree_kernel.py]`); ( $F_{\text{frontier}}$ )  $\rightarrow F_{\text{transfer}}$ : named QFT/cosmology *transfer* interfaces, never compiler powers — and *not* four unrelated interfaces but *one* discrete geodesic on the Sheet Diamond (the  $K \rightarrow C \rightarrow F$  sheet axis  $M(1, t)$ , with the Koide source  $\rightarrow$  pole as its 1-D Möbius projection), along which the external continuous solver merely evaluates (`[verification/v224_diamond_ftransfer_path.py]`); the discrete data of that geodesic are now arithmetically framed —  $\det M(1, t) = N(\alpha_t)$  with  $\alpha_t = (4t+7+\sqrt{-7})/2$  integral, the path one translation  $\alpha_{t+1} = \alpha_t + 2$  of constant discriminant  $-7$  (`[verification/v533_ftransfer_disc7_norm.py]`), and the preregistered next test is whether the four external solvers share *one* fractional-linear action on  $\alpha_t$  (kill: four incompatible arithmetic actions).

**The residual is certification, not construction** (`[verification/v384_residual_certification.py]`). A ledger-CI audit classifies every remaining item into exactly three *external* kinds, with *zero* open *internal* TFPT mechanisms: **[E]** (A) an external math proof — SEAM.EQUIV.01 continuum existence (the cited MMST theorem plus the crossed-product extension package, v336/v469, with AGT/AMT as second witness), ALPHA.QUILLEN.EXACT.01 (v382, its level and normalisation sharpened by v470, the determinant-line/moduli bridge lemma exhibited at the finite level by v472 — the continuum  $\zeta$ -det identification stays the open part) and the constructive QG.AMB.01 measure (a **[C]** redundancy, v369);

[E] (B) theorem-forbidden —  $v_{\text{geo}}$ , the one unit, by the No-Unit theorem (v153/v364); [E] (C) external physics —  $F_{\text{transfer}}$  (v371–v374, firewalled v187). The two load-bearing facts are checked (No-Unit dimensionlessness; the gap-decoupling margin  $1.648 > 0$ ), so the only hard things left are theorems and external physics, not missing dynamics (RESIDUAL.CERTIFICATION.01).

**The Coxeter-clock facet symmetry, resolved as a structural lemma (RES.COXETER.SYMMETRY.01, [verification/v394\_coxeter\_atoms.py]/[verification/v409\_coxeter\_prime2\_lemma.py]).** The order-30 clock  $30 = 2 \cdot 3 \cdot 5 = h(E_8)$  has all three primes facet-typed, and they are exactly the three anchor atoms ( $2=|Z_2|$ ,  $3=N_{\text{fam}}$ ,  $5=g_{\text{car}}$ ) wearing number-field hats ( $\mathbb{Q}(i)/\mathbb{Q}(\sqrt{-3})/\mathbb{Q}(\sqrt{5})$ , v390). The table is *not* symmetric: the seam (prime-2) appears as the *common factor* of both dynamic rates ( $2/3 = |Z_2|/N_{\text{fam}}$  and  $(\varphi+2)/4 = (\varphi+2)/|\mu_4|$ ,  $|\mu_4|=2^2$ ), not as an independent attractor. *The structural lemma (v409) closes the falsifiable corollary:* the sheet involution  $\sigma = \text{diag}(1, -1, -1)$  has spectrum  $\{+1, -1\}$  and projector  $(I+\sigma)/2$  spectrum  $\{0, 1\}$  — none in the open interval  $(0, 1)$ , so prime-2 supplies a *parity/projection*, never a nonzero contraction rate; the genuine rates  $2/3$  and  $(1/3)^6$  lie in  $(0, 1)$  and carry prime-2 only as a *factor*. Hence prime-2 is *necessary* (admissible boundary transport is sheet-paired) but *not autonomous*: **no prime-2-only attractor exists** — dynamics begins only after coupling to family prime-3 or carrier prime-5 ( $|\mu_4|=2^2$  the Galois bridge). This closes the corollary [E] and types the necessity [C], sharpening v383’s “every sector is the same gapped operator” towards “the same *seam-driven* operator”; the full “appears in every gap” universality stays the v383 reading. The corollary is **machine-proved in Lean 4** (TfptCarrier/CoxeterPrime2.lean, FORM.COXETER.PRIME2.01: axiom-pure, lake build clean). [E]/[C].

**QFT layer (this round).** The boundary QFT is now *one relative object*  $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$ , closed modulo the *single keystone* SEAM.EQUIV.01 (whose downstream conformal-deck face is QGEO.SYM.01): its finite Dirac is a covariance induction of the seam KMS state (v258), its spectral-action cutoff *is* that KMS weight (v259), its seam, carrier-16 and  $E_8$  live on one Kummer/K3 surface (v260), and the whole assembly is certified as the *Modular Spectral Closure* (v261). The ambient quantum-gravity measure ( $G_{\text{metric}}$ ) is gap-decoupled and kept separate by design — the QFT layer does not wait on it.

**The two continuum contracts, and how far they are executed.** Beside the two historical gates this note carries the two *continuum-route* research contracts, both still [O] as contracts but with most of their preregistered stages now *executed*. CELEST.SEAM.01 (the celestial and twistor continuum route): WP1–WP4 and both WP5d stages executed, the WP5e stages  $\alpha/\beta/\gamma/\delta_2/\varepsilon_2/\varepsilon_1$  executed, all three back-reaction milestones M1–M3 executed with their preregistered kills evaluated (v515–v517), the  $\delta_1$  chain *decided* (kill under the derived measure, v518), the declared-vs-derived measure question *decided at probe level* for the declared completion reading (single-valuedness from F-independence + the Quillen pairing, v520), and the  $w_m$  normalisation itself *derived constructively* ( $1/\det_j$  the Atiyah–Bott/zeta fixed-point factor from three independent sources,  $\psi = 64$  reproduced, v523) — nothing in the measure chain is declared any more at that level; the residual [O] is the global BCOV integral beyond the fibre zero-mode factor. WOIT.OS.TWISTOR.01 (the Osterwalder–Schrader twistor bridge — the actual bridge from compiler to physics): the  $\alpha$  stage executed ( $\Theta$  exists at the free level and free RP selects the same family, v519),  $\beta_1$  executed (the  $\mu_4$  clock is the euclidean rotation, the gaugeable datum is the GSO  $\mathbb{Z}_2$ , gauge-fixed RP holds, v522),  $\beta_2$  executed (the OS quotient explicit,  $\mathcal{H}_{\text{phys}}$  positive definite at both levels, the clock a positive transfer operator with spectral calculus, v524),  $\beta_3$  executed (the  $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$  duality typed against  $\sigma_{\text{std}}$ : the OS cut and the  $(2, 2)$  signature *forced*, the kill branch empty by algebra, the induced member = the  $\Theta_{\text{phys}}$  carrier, v565) —  $\gamma$  is next, still under the *straddle-law* constraint: the first interacting seam toy fires Kill-Test 2 at toy level (RP breaks on exactly the quartet-straddling cuts, v529) — an honest threat and simultaneously the first hard selection principle for  $\mathcal{A}_{\text{hol}}$ , since *executed* as such: run as a selector on the interacting mark family, reflection positivity keeps exactly *one* member alive —  $\delta = \pi/2$  with positive coupling, the alignment bit itself (v534, toy-fenced; the literal leading-order cone formalization is dead, the protection is nonperturbative). Both contracts stay [O]; no marker moves.

## Research Contract 1 — ( $U_{\text{wall}}$ )

**Goal.** Prove

$$\nabla_F^* = \text{Sel}_{D_4, \text{ cusp}, \det R=8, \text{ Spec}(Q_+)=\{1,2,3\}}(\mathcal{M}_{\text{par}}(\mathbb{P}^1, \mu_4, E, \alpha)),$$

with  $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1) \oplus \mathcal{O}(-1)$  and  $\alpha = \{0, \frac{1}{3}, \frac{2}{3}\}$  at the four points of  $\mu_4$ . The substrate is standard: degree-zero stable parabolic bundles correspond to unitary representations with prescribed

local monodromies (Mehta–Seshadri); tame non-abelian Hodge gives the Higgs / flat-connection side. The TFPT content is to pin the *one*  $D_4$ -symmetric, rank-3, four-point point that realises the selectors.

**Lemma 1** (U0 — normalisation). *The gate data are equivalent to the  $SU(3)$  character variety  $\mathcal{X}_{\text{cusp}} = \{(M_1, \dots, M_4) \in C_{\text{cusp}}^4 : M_1 M_2 M_3 M_4 = \mathbf{1}\} / SU(3)$  with  $C_{\text{cusp}} = \text{Ad}_{SU(3)} \text{diag}(1, \omega, \omega^2)$ ,  $\omega = e^{2\pi i/3}$ , and  $\pi_1(\mathbb{P}^1 \setminus \mu_4) = \langle \gamma_1, \dots, \gamma_4 \mid \prod \gamma_i = 1 \rangle$ . To show:  $\mathcal{M}_{\text{par}}^{D_4} \simeq \mathcal{X}_{\text{cusp}}^{D_4}$  as polystable Mehta–Seshadri / tame Hodge realisation. Machine: the group presentation and  $D_4$ -action are finite and codable (Sage/Wolfram).*

**Finite rigidity — now proven [E].** The finite half of U0 is closed exactly:  $\mu_4$  has cross-ratio 2 and Möbius stabiliser  $\langle z \mapsto iz, z \mapsto 1/z \rangle$  of order  $8 = |D_4|$  (a 4-cycle of the marks + a reflection, faithful);  $H^1(\mathbb{P}^1 \setminus \mu_4)$  has rank  $4 - 1 = N_{\text{fam}} = 3$ ; and the forms  $\omega_k = z^{k-1} dz / (z^4 - 1)$ ,  $k = 1, 2, 3$ , are  $\mu_4$ -eigenforms ( $z \mapsto iz : \omega_k \mapsto i^k \omega_k$ ) carrying the three nontrivial characters  $\chi_1, \chi_2, \chi_3$  with weights  $(1, 2, 3) = \text{Spec}(Q_+)$ . Hence a genus-zero boundary with faithful  $D_4$ , four parabolic marks and this  $\mu_4$ -decomposition is Möbius-equivalent to  $\mathbb{P}^1 \setminus \mu_4$  with the  $(1, 2, 3)$  grading (QGEO.RIGID.01, [E], [verification/v168\_qgeo\_rigidity.py]). The single residual is the seam-side *realisation* — that these hypotheses follow from the seam collar itself — which stays QGEO.REALIZE.01 ([C]).

**Seam Collar Realisation Theorem — the one central proof obligation [E]/[O]**  
([verification/v176\_seam\_collar\_realisation.py])

*Stated as one theorem and certified as a reduction — not a fabricated proof.*

**Theorem (target).** *Given (1) an oriented one-sided RP seam collar  $\Sigma$ ; (2) a sheet-odd Calderón involution; (3) quasi-free one-particle seam data with gap  $(2/3)^6$ ; (4) a faithful  $D_4$  action on the boundary marks; (5) four parabolic marked points with  $\mu_4$ -character decomposition; (6) an admissible  $c=8$  boundary net — then the conformal boundary of  $\Sigma$  is Möbius-equivalent to  $\mathbb{P}^1 \setminus \mu_4$ , its  $H^1$  character basis is canonically the generation basis with grading  $(1, 2, 3)$ , the seam Calderón contraction is the rank-8  $K_\Sigma$  polarisation, and the index-4 simple-current extension is  $(E_8)_1$ .*

**The proof decomposes into six steps, five already [E]:**

- 1–2 Geometry / uniformisation [E] (v168):**  $\mu_4$  has cross-ratio 2 and a faithful Möbius  $D_4$  stabiliser of order 8; four genus-0 marks with faithful  $D_4$  force the  $\mu_4$  square up to Möbius;  $H^1(\mathbb{P}^1 \setminus \mu_4)$  has rank  $4 - 1 = N_{\text{fam}} = 3$ .
- 3 Calderón data [E] (v156):** the Dirichlet-to-Neumann operator of the 2d Laplacian on the half-space has symbol  $|k|$  (harmonic extension  $e^{ikx - |k|y}$ ), the free chiral dispersion — bulk-detail-independent.
- 4  $H^1$  character = generation grading [E] (v137/v168):**  $\omega_k = z^{k-1} dz / (z^4 - 1)$  carry  $\mu_4$  characters  $i^k$  with weights  $(1, 2, 3)$  = the  $A_3$  exponents =  $\text{Spec}(Q_+)$  — natural, not merely isomorphic.
- 5 One-particle contraction [E] (v162):** a  $\mu_4$ -equivariant operator is diagonal in the cusp basis  $(\frac{1}{|\mu_4|} \sum_j U^j X U^{-j} = \text{diag } X$  for  $U = \text{diag}(1, i, -1, -i)$ ); its spectrum is the cusp weights  $\{0, \frac{1}{3}, \frac{2}{3}\}$ , so the transfer eigenvalues  $(1 - \alpha)^{p_2}$  ( $p_2=6$ ) are  $\{1, (2/3)^6, (1/3)^6\}$ , gap  $(2/3)^6$  — forced, no free knob.
- 6 AQFT closure [E] (v154/v175):**  $(D_5)_1 \otimes (A_3)_1$  with the isotropic  $\mu_4$  glue has index  $4 = |\mu_4|$ ,  $c = 5+3 = 8$ ,  $\mu$ -index  $16/16 = 1 \Rightarrow$  holomorphic  $\Rightarrow (E_8)_1$  ( $248 = 120+128$ ); full-cone RP holds for all  $m$  via the CAR second-quantisation functor.

**The proof tree — and the two genuine open doors [O]**  
([verification/v177\_seam\_marking\_kernel.py]). Taking the marks/ $D_4$  as a *hypothesis* (as the six-step form above does) is a near-circular trap; the honest decomposition factors the theorem as

$$\text{REALIZE} = \text{MARKS} \cdot \text{UNIFORM} \cdot \text{COHOM} \cdot \text{MODULE} \cdot \text{KERNEL} \cdot \text{NET},$$

of which *four nodes are now closed constructively*: UNIFORM [E] (an order-4 Möbius map conjugates to  $z \mapsto iz$ , a non-fixed orbit scales to  $\mu_4$ , and  $\sigma\rho\sigma = \rho^{-1}$  for  $\sigma : z \mapsto 1/z$  gives a faithful  $D_4$  — so a genus-0 four-marked faithful- $D_4$  boundary *is*  $(\mathbb{P}^1, \mu_4)$ , strengthening v168's cross-ratio to the full uniformisation — and now **Lean-formalised** as FORM.QGEO.02:  $\rho^4 = \text{id}$ ,  $\rho^2 \neq \text{id}$ ,  $\sigma^2 = \text{id}$ ,  $\sigma\rho\sigma = \rho^{-1}$ , the orbit scaling to  $\mu_4$ , the  $\mu_4$  permutation, and the multiplier classification “order-4  $\Leftrightarrow \zeta = \pm i$ ” are machine-proved in TftptCarrier/MobiusUniformisation.lean, AUDIT: PASS); COHOM [E] ( $\rho^* \omega_k = i^k \omega_k$ , grading  $(1, 2, 3)$  — also **Lean-formalised** as FORM.QGEO.03: the pullback characters  $i^1, i^2, i^3$  and the reflection parity

$\sigma^*\omega_1=\omega_3$ ,  $\sigma^*\omega_2=\omega_2$ ,  $\sigma^*\omega_3=\omega_1$  are machine-proved in `TfptCarrier/CohomologyGrading.lean`, AUDIT: PASS); MODULE [E] (a *unique*  $\mu_4$ -equivariant iso — multiplicity-free + residue normalisation, reflection  $\omega_1 \leftrightarrow \omega_3$ ,  $\omega_2$  fixed; the parity is the Lean FORM.QGEO.03 above, the *uniqueness* stays symbolic); and NET [E] (v154/v175). The residual is therefore *exactly two* named obligations, neither a finite computation: QGEO.MARKS.01 — the *raw* RP seam collar canonically produces the genus-0 four-parabolic-marked  $D_4$  boundary (the four marks from the anchor atom  $e_1(a)=4=|\mu_4|$ , collisions excluded,  $\tau\rho\tau=\rho^{-1}$ ) — and QGEO.KERNEL.01 — the *raw* seam Calderón operator equals the  $\mu_4$ -equivariant free  $c=8$  gapped one-particle contraction *as an operator* ( $C_\Sigma = U^{-1}C_{\mu_4}U$ ), not merely the same spectral list. **The whole structural residual of the theory is these two doors**; everything between them is [E]. They need a human constructive-geometry/AQFT argument (or a proof assistant), not more arithmetic.

**How far each door opens by computation** ([verification/v178\_marks\_kernel\_attempt.py]). Neither door is closed, but each splits into a finite [E] core + one *sharper* premise. For MARKS: *given* a genus-0 double with an order-4 clock  $\rho$  (fixed points  $\{0, \infty\}$ ) and reflection  $\tau$ , the marks are forced to be a generic  $\rho$ -orbit  $\{1, i, -1, -i\} = \mu_4$  (not the fixed points), distinct iff  $b_1 = 3 = N_{\text{fam}}$  (any collision drops  $b_1$ ), with  $\tau\rho\tau = \rho^{-1}$  giving faithful  $D_4$  — so MARKS reduces to the *single* topological premise “the seam double is genus-0 with an order-4 clock whose marks form one orbit”. For KERNEL: on the 3-dim multiplicity-free  $H^1$  the  $\mu_4$  characters  $(1, 2, 3)$  are distinct, so by Schur a  $\mu_4$ -equivariant operator is forced *diagonal* and is *completely determined by its eigenvalue per character* — two such with the forced transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  and the unique residue-normalised basis are *equal as operators*, not merely isospectral; the “spectrum vs operator” distinction *vanishes* on the finite block. So KERNEL reduces to the *single* premise “the full DtN on  $L^2$  reduces to this 3-dim gapped transfer block + the fixed rank-8 polarisation” (principal symbol  $|k|$ , Lee–Uhlmann v156; no drift v158). The two doors are thereby *milder*, but still open.

**Marks without a geometric posit: a Lefschetz/character argument** ([verification/v195\_marks\_lefschetz\_character.py]). The marks need not be *assumed* as  $D_4$  boundary points: they follow from the  $H^1$  character alone. The order-4 clock  $\rho: z \mapsto iz$  fixes exactly  $\{0, \infty\}$ , so a puncture at a fixed point would contribute a *trivial* character to  $H^1$ ; but the carrier datum is  $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$  (the  $A_3$  exponents  $(1, 2, 3)$ , with  $\text{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$ ), which has *no* trivial component — so no puncture sits at a fixed point. With  $\text{rank } H^1 = n - 1 = 3 \Rightarrow n = 4$  punctures, all off the fixed locus and each in a free order-4 orbit, the only possibility is a *single* free  $\mu_4$  orbit (cross-ratio 2), and the closed-surface Lefschetz number  $L(\rho) = 1 + 1 = 2 = |\{0, \infty\}|$  is consistent. So MARKS reduces from the geometric posit “ $D_4$  marks” to the milder *representation* premise “the raw seam  $H^1$  carries the  $A_3$ -exponent rep” — a carrier-algebra statement, strictly less circular, still [O].

**And the two doors are in fact ONE** ([verification/v179\_conformal\_realisation.py]). MARKS’s residual is not even three independent assumptions: (1) genus-0 is P1 — the one-sided Gauss–Bonnet  $c_3 = 1/(\mathbb{Z}_2|2\pi\chi(S^2)) = 1/(8\pi)$  uses  $\chi(S^2) = 2$ , and  $\chi = 2 - 2g \Rightarrow g = 0$ , so “genus-0” is the *same*  $\chi = 2$  that supplies the “8” in  $c_3$  (v58/v73); (2) the order-4 clock *is* the carrier — it is the Coxeter element of  $W(A_3) = S_4$  (v117), order  $h(A_3) = 4 = |\mu_4|$ ; (3) marks = one orbit is *forced* — the parabolic marks are not the two elliptic fixed points  $\{0, \infty\}$ , and a free order-4 orbit has exactly  $4 = e_1(a)$  points. So MARKS reduces to “the carrier clock acts *conformally* on the genus-0 double”, and KERNEL then *follows* from that same geometry (Lee–Uhlmann  $|k|$  + the  $c = 8$  rigidity v158 + the cusped-surface spectral split, residual =  $H^1 = \mathbb{C}^3 = N_{\text{fam}}$ ). **Hence the entire structural residual of the theory is a single geometric premise** QGEO.CONF.01: *the raw RP seam normal double is conformally* ( $\mathbb{P}^1, \mu_4$ ) *with the carrier clock as the order-4 Möbius automorphism*  $z \mapsto iz$ . Given it, everything — genus-0, the four  $D_4$  marks, the  $(1, 2, 3)$  grading, the DtN block, the rank-8 polarisation and the  $(E_8)_1$  net — follows. This single premise is a constructive-geometry statement (the conformal class of the raw seam *is* the cusped sphere), *not* a finite computation, and is left honestly [O].

**And even that conformal premise reduces to a milder isometry premise** ([verification/v180\_clock\_is\_mobius.py]). QGEO.CONF.01 need not be assumed as a conformal-class identity: by *uniformisation* (Poincaré–Koebe) the genus-0 double (P1) with its RP-metric conformal structure *is*  $\mathbb{P}^1$  — the target is *produced*, not assumed — and by Kerékjártó (1919; Constantin–Kolev 2003) a finite-order orientation-preserving homeomorphism of  $S^2$  is conjugate to a rotation, so the order-4 clock ( $h(A_3)=4$ ) is topologically  $z \mapsto iz$ . An orientation-preserving *isometry* of a Riemannian surface is holomorphic (angle-preserving), hence a Möbius map of  $\mathbb{P}^1$ , and an order-4 Möbius map is  $z \mapsto iz$  (elliptic, multiplier  $i$ , free orbit  $\mu_4$ ). So QGEO.CONF.01 is *implied* by the strictly milder, non-circular premise QGEO.ISO.01: “the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric” — which does *not* name  $\mathbb{P}^1/\mu_4$  and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). **That isometry premise is the**

single, now milder, structural residual of the whole theory; its surface realisation from the raw seam stays honestly [O].

**And one step further reaches bedrock** ([verification/v181\_clock\_is\_conformal\_symmetry.py]). By *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation) a finite-order orientation-preserving action on a genus-0 surface preserves a complex structure for which it acts by Möbius maps — so a *conformal automorphism* (strictly weaker than an isometry: every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form  $z \mapsto iz$ . Since the clock is the Coxeter element of  $W(A_3) = S_4$  (v117), i.e. the  $\mu_4$  deck, and a deck transformation preserves the pulled-back conformal structure *by definition*, the residual collapses to the *definitional* identification QGEO.SYM.01: *the seam's conformal structure is the  $\mu_4$ -deck structure of the carrier clock* (the carrier  $\mu_4$  glue is the conformal monodromy of the seam). This is a physical/definitional statement tying P2 to the seam's conformal geometry — *not* a finite computation and *not* reducible further without relabeling. **This is bedrock:** the chain  $\text{REALIZE} \rightarrow \text{theorem} \rightarrow \text{MARKS} + \text{KERNEL} \rightarrow \text{CONF} \rightarrow \text{ISO} \rightarrow \text{SYM}$  (v175–v181) ends at a single definitional premise, left honestly [O]; everything above it is a theorem or an established citation.

**A concrete geometric model for the bedrock premise: the  $\mu_3$ -cover program** ([verification/v597\_mu4\_cover\_model.py]–[verification/v601\_equivariant\_dual.py]). The bedrock premise now has its first explicit *candidate realisation*: the cyclic cover  $y^3 = x^4 - 1$  of  $\mathbb{P}^1$  branched at  $\mu_4$  (order-3 cusp monodromies, exactly the contract's cusp structure). Machine-exact, in five rounds: the cover carries the rank-3 Eisenstein  $H_1$ , the projective  $D_4$  via the deck, the cusp-class torsion  $(\mathbb{Z}/3)^4$  of order 81, and the unique invariant hermitian form of Lorentz signature  $(1, 2)$  with  $\det J = 8$  (v597/v599); the anti-holomorphic reflection (the real structure  $R$ ,  $R^2=1$ , unique up to scale) exists and closes the operator dictionary (v599); the compiler pair  $(U, V)$  itself is *realised* —  $\mathbb{Z}[\omega]$ -integral,  $\text{GL}_3$ -conjugate to the compiler pair, integer on the saturated  $\text{Fix}(R)$  lattice, with saturation index  $81 = 3^4$  in the exact v566 divisor pattern  $(1, 1, 1, 3, 3, 3, 3)$  (v600, Lean-certified as FORM.QGEO.04 in `TfptCarrier/CoverEmbedding.lean`: determinant-free Smith certificates, kernel `decide`); and the presentation is canonical — the J-adapted cusp projectors give a manifestly  $D_4$ -equivariant pair family (mark choice = winding choice) realising the *dual* parabolic, with the two parabolic types exchanged by the J-adjoint, and the v590 Klein four-group living integrally on the curve (v601/[verification/v602\_duality\_forms.py]). Honest status: this is a *model*, not yet the seam — the identification of the raw RP seam with the cover (the Weil-twisted positivity comparison against the v572 Gram data, and the geometric identity of the seam line) stays [O]; QGEO.SYM.01 does not move.

**A sharper, operator-checkable restatement:** QGEO.ENERGY.01 ([verification/v192\_qgeo\_energy\_clock.py]). The bedrock can be re-expressed as an energy identity rather than a conformal-class statement: *the carrier clock  $\rho$  preserves the Calderón / Dirichlet-to-Neumann energy form of the RP seam double*,  $\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$ . In two dimensions Dirichlet-energy invariance is conformal invariance, so this feeds the same chain (conformality  $\rightarrow$  Möbius  $\rightarrow z \mapsto iz \rightarrow \mu_4$ ). Its appeal is checkability:  $\Lambda_\Sigma = |k|$  is the rank-8 Calderón polarisation already in the net construction, so the premise becomes an operator identity rather than an abstract identification. **Honest scope, however:** “ $\rho$  preserves  $\Lambda_\Sigma$ ” is plausibly *equivalent* to “ $\rho$  is the conformal deck” (QGEO.SYM.01), so QGEO.ENERGY.01 *relocates* the bedrock to a more falsifiable surface but does *not* eliminate it — it becomes a theorem only if  $\rho$  is defined *independently* (from the carrier algebra) and the invariance is then *proven*, which is open. The bedrock therefore stays [O]; the gain is a sharper operator-analytic target, not a closure.

**The non-circular sharpening:** QGEO.ENERGY.02 ([verification/v193\_qgeo\_energy\_commutator.py]). The way to make “ $\rho$  defined independently” precise is the *energy commutator*  $[\rho, \Lambda_\Sigma] = 0$  on the rank-8 Calderón polarisation, with three sub-claims: (1)  $\rho$  is the Coxeter element of  $W(A_3) = S_4$  (order  $4 = |\mu_4|$ , v117), defined from the *carrier* algebra, not the seam; (2) *the non-circularity hinge* —  $\Lambda_\Sigma$  must be the DtN of the *raw* RP seam (definable from the RP data alone), not the  $\mu_4$  normal-form operator; (3)  $[\rho, \Lambda_\Sigma] = 0$  forces the  $\mu_4$  character grading  $(1, 2, 3)$  on  $H^1$ , already established (QGEO.COHO.01). An exact consistency rider pins the direction: the induced EH coefficient  $k = (\ln m)/(12\pi)$  (v150) reproduces  $c_3/2$  iff  $\ln m = q(A_3) = \frac{3}{4}$  (the *family* glue norm), while  $q(D_5) = \frac{5}{4}$  (the carrier norm) misses by  $g_{\text{car}}/N_{\text{fam}} = \frac{5}{3}$  — so the seam energy form must reproduce the *family* norm (gravity is family-geometry-induced). If sub-claim (2) holds,  $[\rho, \Lambda_\Sigma] = 0$  turns QGEO.SYM.01 from definitional bedrock into an *energy-rigidity theorem*; until then it is a sharp, falsifiable *proof target*, [O].

**Subclaim (2) tackled — one notch closer, not closed** ([verification/v194\_raw\_seam\_dtn\_rp.py]). The non-circularity hinge is essentially *met* at the canonical level: Osterwalder–Schrader reconstruction

yields a canonical transfer operator  $T = e^{-H}$  from the RP state and reflection alone — no coordinates (v54; §(4) of `tfpt_2`) — on a *quasi-free* seam state (v155: the boundary marginal of a Gaussian bulk is the compression  $PTP$ , a contraction). So  $\Lambda_\Sigma$  is RP-definable without ever naming  $\mathbb{P}^1 \setminus \mu_4$ . The commutator  $[\rho, \Lambda_\Sigma] = 0$  then *closes on the finite cohomology*  $H^1$  [E]:  $\rho: z \mapsto iz$  acts on  $\langle w_1, w_2, w_3 \rangle$  ( $w_k = z^{k-1} dz / (z^4 - 1)$ ) as  $\rho^* w_k = i^k w_k$ , distinct characters  $i, -1, -i$ , so any  $\rho$ -equivariant operator is diagonal there (v177). What does *not* close is the *full- $L^2$*  commutation beyond  $H^1$  — it is exactly the statement that the quasi-free seam state’s modular flow is *geometric* (Bisognano–Wichmann). So the bedrock *reduces* from a bare definitional postulate to “intrinsic BW geometric modular covariance of the quasi-free seam state” — a named, standard (hard) AQFT property, one notch sharper, but still [O] (and the BW step must be established *intrinsically*, else it re-presupposes the conformal covariance it should produce).

**A concrete variational handle on the residual** ([`verification/v196_seam_energy_functional.py`]). The open BW statement can be made a discretisable extremum problem via the failure functional  $E_{\text{fail}}(\Sigma, \rho) = \|[\rho, \Lambda_\Sigma]\|_{HS}^2 + \|\rho^4 - 1\|^2 + \|\Theta \rho \Theta - \rho^{-1}\|^2$ , whose zero locus is exactly the QGEO.SYM.01 conditions (the energy commutator, the order-4 clock, and the RP-reflection twist). On the finite  $H^1$  block  $E_{\text{fail}} = 0$  at the  $\mu_4$  configuration ( $\rho = \text{diag}(i, -1, -i)$ ,  $\Lambda$   $\mu_4$ -equivariant,  $\Theta = \text{conjugation}$ ) and  $> 0$  for any  $\mu_4$ -breaking perturbation, so  $\mu_4$  is a genuine minimiser [E]; minimising  $E_{\text{fail}}$  over the *full* raw-seam DtN is precisely the BW residual above. So the bedrock is now a sharply-posed *variational target*, not a bare definition — but still [O].

**The decisive crack: principal symbol exact + Tomita–Takesaki** ([`verification/v198_modular_commutator_reduction.py`]). Two observations collapse the residual to a single non-circular state-invariance. *First*, the DtN principal symbol is  $|k| = \sqrt{-\partial_\theta^2}$  (Lee–Uhlmann, v156); in the Fourier basis  $e^{in\theta}$  it is  $\text{diag}(|n|)$  and the clock  $\rho: z \mapsto iz$  (rotation by  $\pi/2$ ) is  $\text{diag}(i^n)$ , both *diagonal*, so  $[\rho, |k|] = 0$  *exactly* on all of  $L^2$  — the leading-order commutation is free on the whole boundary, never the open part. *Second*, by Tomita–Takesaki the modular operator  $\Delta_\omega$  is intrinsic to  $(M, \Omega)$ , so a state-preserving symmetry ( $\rho\Omega = \Omega$ ) commutes with  $\Delta$  and hence with  $\log \Delta \sim \Lambda_\Sigma$ : thus  $[\rho, \Lambda_\Sigma] = 0$  *follows from*  $\omega \circ \rho = \omega$  — using only the state and reflection (which RP supplies) and *no* conformal covariance, which **removes the BW circularity worry above**. Since the finite  $H^1$  block is already  $\mu_4$ -invariant (v177), the entire residual is now the single *state-invariance* “the raw quasi-free seam state is  $\mu_4$ -invariant” ( $\omega \circ \rho = \omega$ , equivalently the Gaussian bulk measure is deck-invariant) — the maximally-operational, checkable form of QGEO.SYM.01. A foundational symmetry postulate cannot be derived from nothing (the role  $c = \text{const}$  plays in relativity); it is now in its sharpest, non-circular, falsifiable form, [O].

**The residual, fully localised and numerically probed** ([`verification/v199_seam_state_invariance.py`], [`verification/v200_seam_variational_scan.py`]). Attacking  $\omega \circ \rho = \omega$  on the *raw* seam (not the  $\mu_4$  normal form) confirms it is non-circular —  $\rho$  is the Coxeter element of  $W(A_3) = S_4$  ( $\rho^4 = 1$ ), an automorphism of the raw CAR algebra, defined entirely carrier-side — and reduces it, via the quasi-free covariance  $C_\Sigma = \frac{1}{2}(1 + \text{sgn } H_1)$ , to  $[\rho, H_1] = 0$ , i.e.  $H_1$  is *block-diagonal in the four  $\mu_4$ -character classes*  $\{n \equiv r \pmod{4}\}$ . The principal symbol  $|k| = \text{diag}(|n|)$  passes automatically, so the entire residual is the single bounded statement “the sub-principal symbol of the raw DtN has no off-character matrix elements”. The companion variational scan minimises  $E_{\text{fail}}(\theta) = \|\rho(\theta)^4 - 1\|^2 + (\text{faithfulness})$  over a free clock angle and finds its *unique* minimum at  $\theta = \pi/2$  — the  $\mu_4$  clock  $z \mapsto iz$  (the only faithful order-4 grading  $i, -1, -i$ ) — a numerical sign that the variational principle selects  $\mu_4$ . This is the “next executable theorem” a working mathematician can take up; it does not close the postulate, [O].

**The bounded sub-principal residual is the image of the  $\mu_4$  marks, not a free postulate** ([`verification/v201_seam_subprincipal_marks.py`]). One genuine step further: the seam DtN is  $\Lambda = |k| + M_f$  with  $M_f$  the bounded sub-principal piece = multiplication by a curvature function  $f(\theta)$  on the seam circle (the standard Steklov/DtN structure), so  $\langle n | M_f | n' \rangle = f_{n-n'}$ . Then  $M_f$  is  $\mu_4$ -character-block-diagonal *iff*  $f$  has Fourier support only on modes  $\equiv 0 \pmod{4}$ , i.e. *iff*  $f$  is  $\mathbb{Z}_4$ -invariant (the principal  $|k|$  already commutes, v198). But a curvature sourced by the four marks,  $f(\theta) = \sum_{j=0}^3 g(\theta - 2\pi j/4)$  for *any* profile  $g$ , has  $f_m = 4 g_m$  [ $m \equiv 0 \pmod{4}$ ] — automatically  $\mathbb{Z}_4$ -invariant. So the  $\mu_4$ -mark orbit (forced by v195) *forces* the sub-principal symbol block-diagonal; 3 equally-spaced marks ( $\mathbb{Z}_3$ ) or 4 generic marks both break it (negative controls). The residual of  $\omega \circ \rho = \omega$  thus collapses to “the DtN sub-principal symbol is *mark-local* (the seam is flat away from the  $\mu_4$  marks = the conformal-deck structure, v177)” — a locality/flatness property, structurally definitional, the narrowest form yet of QGEO.SYM.01, [O].

**Certified on realistic Steklov profiles, and at the state level**

(`[verification/v210_mark_local_dtn.py]`). The mode-level argument above is carried to concrete, smooth boundary curvature: a real von-Mises bump  $g(\theta)$  localised at a puncture, summed over the four marks  $f(\theta) = \sum_{j=0}^3 g(\theta - j\pi/2)$ , has *every* off-(mod 4) Fourier coefficient vanishing to  $< 10^{-16}$  (exact 4-fold cancellation), the assembled DtN  $\Lambda = |D_\theta| + M_f$  has  $\|[\rho, \Lambda]\| < 10^{-15}$ , and — the genuinely new step — the *quasi-free state* it defines, with covariance  $C = \frac{1}{2}(1 + \text{sgn } H_1)$ , is clock-invariant:  $\rho C \rho^\dagger = C$  to  $< 10^{-13}$ , i.e. the actual  $\omega \circ \rho = \omega$ , not merely  $[\rho, \Lambda] = 0$ . A single off-centre bump, a  $\mathbb{Z}_3$  (three-mark) source and four unequally-spaced marks each break both by  $O(1)$ , and the mark-local commutator stays  $< 10^{-10}$  as the truncation grows ( $N \in \{16, 32, 64\}$ ) — not an artefact. This *numerically certifies the implication* “mark-local DtN  $\Rightarrow \omega \circ \rho = \omega$ ” for real profiles; the premise that the physical seam *is* mark-local stays **[O]** (it does not close QGEO.SYM.01).

**The same residual, read dynamically: modular flow + a recovery Lindblad generator** (`[verification/v238_modular_lindblad_dynamics.py]`). The whole QGEO.SYM.01 discussion above is *static* — a commutator condition  $[\rho, \Lambda_\Sigma] = 0$ . Read as a *dynamics* it is the statement that the modular flow  $\sigma_t = \Delta^{it}$  (the Tomita–Takesaki one-parameter group generated by the modular Hamiltonian  $\Lambda_\Sigma = \log \Delta_\omega$ ) *conserves* the  $\mu_4$  clock:  $\rho \sigma_t \rho^{-1} = \sigma_t \forall t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$  — so the open premise is exactly “modular time conserves the carrier charge”. The dissipative half is the recovery channel  $T$  of v221 read as a generator:  $L = \log T$  is a doubly-stochastic (CPTP-classical) Markov generator ( $e^L = T$ ,  $e^{L\tau} \rightarrow$  the democratic projector = the arrow of time), and its dissipative gap is  $-\log((2/3)^6) = 6 \log(3/2) = \Delta$ , the OS mass gap of v64 — the static recovery bound and the dynamical mass gap are *one* number, one exponentiated. Together  $G = i\Lambda_\Sigma + L$  is the canonical Hamiltonian + Lindbladian (GKSL) split, the reversible (imaginary) part conserving and the dissipative (real  $\leq 0$ ) part setting the arrow. This is a finite-block exhibit of where the static compiler becomes a dynamics; the residual it leaves — that  $\sigma_t$  is the *geometric* modular flow (Connes–Rovelli “thermal time = physical time”) on the full seam algebra — is the same QGEO.SYM.01 bedrock, **[O]**.

**From the dynamics to a QFT skeleton on the seam** (`[verification/v239_kms_thermal_time.py]`, `[verification/v240_gns_os_reconstruction.py]`, `[verification/v241_dhr_sectors.py]`, `[verification/v242_spin_statistics_modular.py]`, `[verification/v243_haag_ruelle_braiding.py]`, `[verification/v244_spectral_action_contract.py]`). With the static  $\rightarrow$  dynamic flip in hand, the standard AQFT machinery assembles a finite-block QFT skeleton, each step typed honestly. *Thermal time* (v239): the modular flow is KMS at  $\beta = 1$ , and the seam unit  $2\pi = 1/(4c_3)$  fixes the modular temperature as the horizon temperature  $T_H = c_3/M$ . *Reconstruction* (v240): GNS gives the Hilbert space and a cyclic vector from  $(\mathcal{M}, \omega)$ , and the OS Hamiltonian  $H_{\text{OS}} = -\log T = -L$  is positive with gap  $\Delta$  — the v64 “ $H \geq 0 + \text{mass gap}$ ” is the recovery generator read as Euclidean time. *Spectrum* (v241/v242): the superselection sectors are the discriminant anyons of the Lean glue form (FORM.GLUE.01); fusion is recovered from the modular  $S$  by Verlinde, the Gauss–Milgram sum returns  $c = 8$ , and the condensation tower  $16 \rightarrow 4 \rightarrow 1$  (v235/v237) selects the holomorphic  $(E_8)_1$ . *Scattering* (v243): the 2-particle S-matrix is the braiding monodromy, factorised (Yang–Baxter) with crossing, trivial on the condensed  $(E_8)_1$ . The one genuinely 4d step is typed as a contract, CONTRACT.QFT4D.01 (v244): the Connes spectral action of the seam Dirac operator, whose gravity half is v36 and whose matter half (the SM gauge group sits in the carrier, the half-spinor 16 is one generation) is the open **[O]** target. Its representation-theory core is now discharged (`[verification/v245_spectral_matter_content.py]`): the **16** is exactly one *anomaly-free* SM generation ( $\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$ ) with the forced SM hypercharges and the spectral-action value  $\sin^2 \theta_W = 3/8$  ( $g_3 = g_2 = \sqrt{5/3} g_1$ ), so only the seam  $\rightarrow$  Dirac realisation and the run-down couplings stay open. So the residual to a full 4d QFT is *one premise* (QGEO.SYM.01) plus *one contract* (CONTRACT.QFT4D.01), not a diffuse list.

**The perturbative 4d S-matrix is constructible** (`[verification/v269_spert_paqft_skeleton.py]`). The 4d scattering layer is typed precisely as three *distinct* objects:  $S_{\text{top}}$  (the 2d DHR braiding monodromy,  $|M|=1$  statistics, v243),  $S_{\text{pert}}$  (the 4d Epstein–Glaser / BV S-matrix of the spectral action), and  $S_{\text{phys}}$  (LSZ on the OS-reconstructed Wightman functions, v240) — the 2d braiding is *not* the 4d cross-section S-matrix. **[E]** the spectral-action  $a_4$  interaction is power-counting renormalizable (every operator mass dimension  $\leq 4$ : 7 marginal + 3 relevant, a *finite* counterterm basis), so **[C]** by the Epstein–Glaser theorem (causal perturbation theory; Brunetti–Fredenhagen pAQFT) the perturbative S-matrix  $S(g) = T \exp(i \int g L_{\text{int}})$  is built order by order by causal factorisation + finite local extension — no path integral, *no* UV divergences. **[C]** the gap  $\Delta = 6 \log(3/2)$  gives an IR-safe adiabatic limit. **[C]** this is *perturbative* only: the nonperturbative ambient measure QG.AMB.01 is now a **[C]** redundancy (v369), a certification object rather than missing dynamics, and the canonical 4d reading remains boundary-only (v265);  $S_{\text{pert}}$  is the perturbative cross-section layer on top (QFT4D.SPERT.01).

**A concrete one-loop EG number** ([[verification/v271\\_eg\\_oneloop\\_quartic.py](#)]). The skeleton is taken from “exists” to an actual Epstein–Glaser renormalization: the order-1 product  $T_1 = :L_{\text{int}}:$  needs no extension, and the first one is the order-2  $T_2$ , whose UV behaviour is the Steinmann *scaling degree*. [E] the massless propagator has  $\text{sd} = 2$  in  $d = 4$ , the one-loop bubble (two propagators) has  $\text{sd} = 4 = d$ , so the singular order  $\omega = \text{sd} - d = 0 \Rightarrow$  *exactly one* logarithmic local counterterm ( $C(\omega+d, d) = 1$  — finite, renormalizable, no infinite tower); the one-loop factor is  $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$  exactly, the *same*  $1/(16\pi^2)$  that normalises the spectral-action  $a_4$  geometric quartic. [C] the  $\phi^4$  one-loop running is then  $\beta_\lambda = 3\lambda^2/(16\pi^2)$  (symmetry factor 3 = the  $s, t, u$  channels), with the EG initial condition fixed by the spectral action,  $\lambda_\Phi(\text{UV}) = 1/(16\pi^2)$ . [O] this is order-2/one-loop, one diagram class; the all-order perturbative construction stays open (QFT4D.SPERT.02), while the nonperturbative measure is discharged as a [C] redundancy (below).

*Red-team caveat, now discharged* ([[verification/redteam/rt\\_F\\_qft4d.py](#)]). Power-counting renormalizability is *not* unitarity for the gravity sub-sector: the local  $R^2/\text{Weyl}^2$  *truncation* gives a four-derivative propagator  $1/(p^2(p^2 + M^2))$  whose massive spin-2 mode has a negative residue (the Stelle ghost). But that ghost is a Seeley–DeWitt *truncation artefact*: the entire seam KMS form factor  $a(p^2) = e^{p^2/M^2}$  keeps its only pole at  $p^2=0$  (v304), the spin-2 sector is ghost-free via the Barnes–Rivers decomposition (v370), and the nearest truncation-zero modulus runs to infinity (v380), so *perturbative* graviton unitarity is established. The matter+gauge  $S_{\text{pert}}$  is unitary outright; the *nonperturbative* QG.AMB.01 (asymptotic safety v207 / the boundary  $(E_8)_1$  net) is then a [C] *redundancy* (v369; gap-decoupled from every TFPT readout, not a TFPT structural item, v335), a certification object rather than a new gap.

**The gauge running is an EG order-2 number** ([[verification/v273\\_eg\\_gauge\\_running.py](#)]). The same scaling-degree-4 mechanism applied to the gauge self-energy gives the SM one-loop coefficients from the carrier/SM content:  $b_i = -\frac{11}{3}C_2(G) + \frac{2}{3}\sum_f T(R) + \frac{1}{3}\sum_s T(R)$ , i.e. [E]  $b_3 = -7$ ,  $b_2 = -\frac{19}{6}$ ,  $b_1 = \frac{41}{10}$  (GUT-normalised; the same 41 as the carrier algebra  $10b_1 = g_{\text{car}} 2^{g_{\text{car}}-2} + 1$ , v159). [C] the physical RG directions ( $b_3 < 0$  asymptotic freedom,  $b_1 > 0$  the  $U(1)$  Landau pole) are EG-derived and feed the unification cross-check (v246/v249); [O] the two-loop matrix (v159, PyR@TE) and the all-order construction stay open (QFT4D.SPERT.03).

**The carrier content is one anomaly-free SM generation** ([[verification/v310\\_carrier\\_sm\\_anomaly.py](#)]). Those one-loop coefficients sit on top of an exact spectrum/anomaly closure: the carrier half-spinor  $16 = \Lambda^{\text{even}}(\mathbb{C}^5) = 1+10+5$  decomposes under  $SU(5) \subset SO(10)$  as  $10+\bar{5}+1 =$  exactly one Standard-Model generation ( $Q, u^c, d^c, L, e^c, \nu^c$ ), and that content is [E] gauge-anomaly-free ( $\sum Y = 0$ ,  $\sum Y^3 = 0$ ,  $[SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$ , exact) while reproducing  $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ ; a fourth family would shift  $b_1$  off  $41/10$ , so  $N_{\text{fam}}=3$  *pins* the RG. [E] for the spectrum, anomalies and one-loop  $\beta$ ; the full Epstein–Glaser interacting  $S$ -matrix and the Yukawa sector stay [C]/[O].

**The bridge to the physical S-matrix and one-loop unitarity** ([[verification/v278\\_lsz\\_bridge\\_unitarity.py](#)]). The perturbative layer connects to the physical asymptotic S-matrix: [E] the EG time-ordered products of  $S_{\text{pert}}$  are the OS Wick-rotated correlators (v240), so amputating external legs and taking the on-shell residue gives  $S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$  (the three-layer S-matrix  $S_{\text{top}}|S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$ ), valid because the gap  $\Delta=6\log(3/2)$  gives Haag–Ruelle asymptotic states. [E] *one-loop unitarity* is the optical theorem: the  $s$ -channel bubble discontinuity  $\text{Im} B(s)/\pi = x_+ - x_- = \sqrt{1-4m^2/s}$  is *exactly* the two-body phase space (turning on at the physical threshold  $s=4m^2$ ), so  $2\text{Im} M = \sum \int d\Pi |M|^2$  holds and the matter+gauge sector is unitary order by order. [C] the gravity sector’s Stelle ghost is a truncation artefact (perturbative spin-2 unitarity established, v304/v370/v380); the nonperturbative QG.AMB.01 is a [C] redundancy (v369) (QFT4D.SPERT.04).

**The first hard data confrontation** ([[verification/v246\\_unification\\_data\\_crosscheck.py](#)]). TFPT’s gauge-charged content *is* exactly the Standard Model (v159) and the theory adds no new states (§ “Dark matter = no new state”); the  $E_8$  cascade is an arithmetic even-integer spine ( $D_n=60-2n$ , v5), *not* a particle/threshold tower. So a “TFPT-adapted” RGE run *is* the SM run — and the SM does not unify. Tested against the *measured* couplings (v159 betas, plus the standard two-loop gauge matrix): at one loop the pairwise crossings spread over  $\sim 10^{13}$ – $10^{17}$  GeV with residual  $\Delta\alpha^{-1} \approx 9$  at  $2 \times 10^{16}$  GeV; at two loops (RK4) the minimal spread is  $\Delta\alpha^{-1} \approx 3.2$  at  $\mu \approx 1.7 \times 10^{14}$  GeV — the three couplings never meet (Fig. 2). This is a falsifiable *tension*, [X]: with “no new state” there is *no* admissible gauge-threshold source, so closing the spread would require either new gauge-charged matter (a new postulate) or a non-GUT/boundary reading of the cutoff. The spectral-action 4d-GUT route is thus in genuine tension — the same status as the Connes–Chamseddine NCG Standard Model — and is likely

falsified unless the theory is extended; the AQFT skeleton, DHR structure and SM rep-core survive regardless as an emergent boundary QFT. Recorded as a kill-test, never as a confirmation.

**The carrier-native Pati–Salam program** ([[verification/v247\\_e8\\_branching\\_no126.py](#)], [[verification/v248\\_ps\\_algebra.py](#)], [[verification/v249\\_ps\\_unification.py](#)], [[verification/v250\\_dirac\\_triple\\_contract.py](#)]). The unification tension above is resolved *natively* if the carrier  $SO(10)$  is gauged (the carrier  $D_5 \oplus A_3 = SO(10) \times SU(4)$  is Pati–Salam). *First*, the  $E_8$  audit hull *forbids* the **126**: the **248** branches as  $(45, 1) + (1, 15) + (10, 6) + (16, 4) + (\overline{16}, \overline{4})$ , so the available  $SO(10)$  reps are exactly  $\{1, 10, 16, 45\}$  — the renormalisable  $126_H$  is algebraically absent, making the “ $16_H$  over  $126_H$ ” selection a theorem of the hull, not a fit (v247). *Second*, the Pati–Salam NCG algebra  $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})$  gives the unimodular group  $SU(2)_L \times SU(2)_R \times SU(4)_c$ , the exact SM hypercharges  $Y = T_{3R} + (B-L)/2$  and the matching  $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$ ,  $\sin^2 \theta_W = 3/8$  (v248). *Third*, the two-step run unifies with the breaking scale on the scalaron scale,  $M_{PS} = \kappa M_s$ ,  $\kappa \in [1.0, 1.7]$  at one loop ( $\rightarrow [1.0, 1.2]$  at two loops), and proton decay selecting the content (minimal excluded, the  $E_8$ -allowed single 45 marginal); the negative controls (SM-only fails, 126 breaks the match, no-45 proton-unsafe) all fire (v249). *Fourth*, the residual “carrier is gauged” reduces to ONE constructive object, the spectral triple  $(A, H, D_F, J, \gamma)$  with  $A_F$  Pati–Salam and  $H_F = 3 \times 16$ , whose NCG axioms (the first-order condition above all) and the  $\sigma$  = scalaron identification are the open contract **CONTRACT.QFT4D.DIRAC.01** (v250). *Fifth*, a first concrete step on that triple is taken ([[verification/v251\\_first\\_order\\_sigma.py](#)]): on the lepton block the first-order condition is verified to *hold* for the Yukawa operator and to be *violated exactly* by the Majorana/ $\sigma$  term — the Chamseddine–Connes–van Suijlekom “beyond first order” mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate  $\sigma$  (honest correction:  $\sigma$  does *not* rescue the first-order condition; its controlled violation *is* the mechanism). *Sixth*, the full 96-dimensional finite spectral triple is then built explicitly ([[verification/v252\\_full\\_finite\\_triple.py](#)]) —  $H_F$  the particle+antiparticle doubling of three  $SO(10)$  **16**-plets — and the NCG axioms are verified by direct computation: reality  $J$ , grading  $\gamma$ , KO-dimension 6, order-zero, the first-order condition (*holding* for the SM algebra including the Majorana, *violated* for the Pati–Salam algebra exactly by the Majorana on  $\{\nu_R, e_R\}$ ), and the spectrum of  $D_F$  reproducing the fermion masses and the type-I seesaw  $m_{\text{light}} = m_D^2/M_R$  — thereby discharging five of the seven obligations of **CONTRACT.QFT4D.DIRAC.01** to **[E]** (orientability, the no-junk condition and the spectral-action expansion remain open). *Seventh*, the scalaron–Pati–Salam ratio  $\kappa$  in  $M_{PS} = \kappa M_s$  is pinned to a tight  $O(1)$  interval ( $\approx 1.3$ ) by the RG and shown to be a scale-free ratio of heat-kernel coefficients,  $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}}$ , over the same 96-dim  $H_F$  ([[verification/v253\\_kappa\\_scalaron\\_ps.py](#)]). *Eighth*, the remaining NCG obligations are then discharged or honestly typed: the inner fluctuations  $\Omega_D^1$  give exactly the one SM Higgs doublet  $(1, 2)_{1/2}$  with no junk and no **126**, while  $\sigma$  (the  $\nu_R$  Majorana scalar,  $B-L=2$ ) appears only as the beyond-first-order Pati–Salam field ([[verification/v254\\_inner\\_fluctuations.py](#)]); the spectral-action Seeley–DeWitt expansion  $a_0, a_2, a_4$  yields the cosmological, Einstein–Hilbert and Yang–Mills+ $R^2$ +Higgs terms with  $b/a^2 = 1/3$ ,  $\sin^2 \theta_W = 3/8$  and the  $R^2$  scalaron, making  $\kappa$  an explicit moment ratio ([[verification/v255\\_spectral\\_action\\_expansion.py](#)]); only strict orientability and strict Poincaré duality genuinely *fail* for this triple — a published feature of the NCG-Standard-Model class (KO-dimension 6 with three  $\nu_R$  forces a skew intersection form, corank 1; arXiv:0902.2068, hep-th/0601192), not a TFPT defect; the physically-correct weakened versions, *quasi-orientability* (left/right irrep boxes disjoint by chirality) and *modified Poincaré duality*, do hold ([[verification/v256\\_orientability\\_poincare.py](#)], [[verification/v257\\_quasiorient\\_modpd.py](#)]). *Ninth*, the built  $D_F$  is no independent posit but the *modular/covariance induction* of the boundary seam state ([[verification/v258\\_dirac\\_covariance\\_induction.py](#)]): a quasi-free (KMS,  $\beta=1$ ) state is fixed by its covariance  $C$  (a positive contraction), with one-particle modular Hamiltonian  $H = \log((1-C)C^{-1})$  (Araki; Casini–Huerta) inverse to the KMS occupation  $C = (1+e^H)^{-1}$ ; inducing the carrier covariance  $C_F = P_F C_\Sigma P_F$  returns the v252 operator *exactly* — same spectrum, chirality-odd, the Majorana an off-diagonal covariance entry with the seesaw  $m_D^2/M_R$  recovered — so  $D_F$  is the Kasparov shadow  $[D_\Sigma] \otimes_{(E_8)_1} [K_{\text{car}}]$  of the boundary geometry and the v18 Yukawas are the *readout* of  $C_\Sigma$ , **[E]** modulo the one **[O]** seam realisation. *Tenth*, the last open spectral-action input — the cutoff function  $f$  — is fixed by the same seam state ([[verification/v259\\_modular\\_cutoff\\_kappa.py](#)]): by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at  $\beta=1$  (v239) and the seam unit  $2\pi=1/(4c_3)$  (v58) fixes that  $\beta$ , so the heat-kernel cutoff  $f(u) = e^{-u}$  is *derived*, not chosen, with moments  $f_2/f_0 = f_4/f_2 = 1$  *exactly* (a generic Gaussian cutoff gives  $\sqrt{\pi}/2 \neq 1$ , so the choice is non-vacuous); hence  $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}} = \sqrt{c_{PS}/c_{\text{grav}}}$  loses its scheme factor and the v253/v255 open **[O]** cutoff freedom becomes a **[C]** finite trace ratio over the 96-dim  $H_F$  (the RG  $\kappa \approx 1.3$  pinning  $c_{PS}/c_{\text{grav}} \approx 1.7$ ). So

unification is now *one* theory decision (gauge  $SO(10)?$ ) plus *one* constructive object, both falsifiable — not a fog. The colour  $SU(4)_c$  here sits inside  $SO(10)=D_5$  and is distinct from the carrier family factor  $A_3$ .

**The implication is now Lean-formalised (FORM.QGEO.01).** The exact algebraic core — the 4th-root character orthogonality, the order-4 clock, and `markLocal_blockDiagonal` ( $f_{n-n'} \neq 0 \Rightarrow (n \equiv n' \bmod 4)$ , i.e.  $[\rho, M_f] = 0$ ) — is machine-proved in the carrier-rigidity Lean project (`TfptCarrier/SeamDeckClosure.lean`; AUDIT: PASS, axioms `propext`, `Classical.choice`, `Quot.sound` only). The premise itself is a typed `structure SeamDeckPremise` — consumed, not proved, exactly as `CalderonProjector` encodes the document-1 analytic input — so the conditional theorem “`SeamDeckPremise  $\Rightarrow \omega \circ \rho = \omega$` ” is now [E] while QGEO.SYM.01 stays the one open [O] postulate.

**The two residuals are one canonical metric: the pillowcase** ([`verification/v214_seam_pillowcase.py`]). The isometry residual QGEO.ISO.01 (v180) and the mark-locality residual QGEO.SUBPRIN.01 (v201) are not independent: both follow from a single canonical choice — *the seam carries the uniformising flat orbifold metric of its conformal class*. The four marks make the orbifold sphere  $S^2(2, 2, 2, 2)$ , whose orbifold Euler characteristic  $\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$  is *Euclidean*: by Troyanov’s cone-metric uniformisation it carries a flat metric, unique up to scale, with curvature concentrated at the four cone points (Gauss–Bonnet: four deficits  $\pi$  sum to  $4\pi = 2\pi\chi(S^2)$ ) — this is the v201 “flat away from the marks”. And the order-4 clock is no longer an extra assumption but a *consequence* of the cross-ratio 2 that v168 already proved:  $j(\lambda) = 256(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda - 1)^2)$  gives  $j(2) = 1728$ , so the modulus is the *square* torus  $\tau = i$ , whose complex multiplication by  $\mathbb{Z}[i]$  supplies the order-4 isometry  $z \mapsto iz$  (explicitly  $(x, y) \mapsto (-x, iy)$  on  $y^2 = x^3 - x$ , order 4); a generic cross-ratio gives  $j \notin \{0, 1728\}$  and only  $\mathbb{Z}/2$ , so the clock is specific to  $\mu_4$ . Thus the two open QGEO residuals *collapse to one* canonical-metric statement and QGEO.ISO.01 ceases to be an imposed symmetry. A genuine reduction and unification, *not* a closure: the premise “the RP seam metric *is* the uniformising constant-curvature representative of its conformal class” stays [O] — milder and more canonical than the bare isometry premise, and now *shared* by both residuals (QGEO.PILLOW.01).

**The converse, on the canonical metric: raw seam  $\Rightarrow$  mark-locality** ([`verification/v264_raw_seam_marklocal.py`]). Composing the chain explicitly closes the gap between v216, v201 and v198 on the uniformising metric: the flat pillowcase has its curvature *only* at the four cone points (v216), so the DtN sub-principal symbol  $M_f$  is a  $\mu_4$ -orbit sum, hence  $\mathbb{Z}_4$ -invariant (Fourier support on  $4\mathbb{Z}$ , verified by FFT), hence block-diagonal in the clock-character basis, hence  $[\rho, M_f] = 0$ ; with the principal symbol  $|k|$  already commuting (v198) this gives  $[\rho, \Lambda] = 0$  and  $\omega \circ \rho = \omega$  *on this metric*. A negative control (curvature *off* the  $\mu_4$  orbit) breaks  $\mathbb{Z}_4$ -invariance and the chain fails. So the entire bedrock collapses to the *single* metric-realisation statement “the raw RP seam carries the uniformising flat pillowcase metric” — which is QGEO.SYM.01. A genuine reduction/sharpening (QGEO.ENERGY.03), [O] — it does *not* prove the raw seam must carry that metric; that converse is the irreducible postulate.

**The minimal axiom: a rigidity theorem** ([`verification/v267_qgeo_rigidity_minimal_axiom.py`]). One can sharpen the bedrock to its irreducible kernel. [E] the *single* bare statement “the seam admits the carrier’s order-4 clock as a conformal symmetry cyclically permuting its four marks” forces everything: an order-4 Möbius orbit  $\{a, ia, -a, -ia\}$  has cross-ratio 2 *independent of  $a$* ; cross-ratio 2  $\Leftrightarrow j=1728$  (only  $\lambda \in \{-1, \frac{1}{2}, 2\}$ )  $\Leftrightarrow$  the order-4 CM automorphism  $(x, y) \mapsto (-x, iy)$  on  $y^2 = x^3 - x$ ; and from the square configuration the unique flat pillowcase metric (Troyanov), mark-locality,  $[\rho, \Lambda]=0$  and  $\omega \circ \rho = \omega$  all follow (v214/v201/v264/v198). [E] neg controls: a generic four-mark set has  $j \neq 1728$  and only a  $\mathbb{Z}_2$  symmetry (no clock), the hexagonal point  $j=0$  has a  $\mathbb{Z}_6$  symmetry (order 6  $\neq 4$ ) — only the square (order-4) point realises the  $\mu_4$  clock. [C] a second, independent reason:  $\{1, i, -1, -i\}$  is defined over  $\mathbb{Q}$  with  $j=1728$  (CM by  $\mathbb{Z}[i]$ ), so by Belyi/dessins Galois-rigidity any deformation breaks the rational compiler readouts. [O] So QGEO.SYM.01 is now in its minimal sharpest form — the bare existence of that order-4 conformal symmetry, which (like  $c=\text{const}$  in relativity) cannot be derived from nothing. The *reduction* is closed (axiom  $\Rightarrow$  everything); the axiom itself stays the one foundational postulate (QGEO.SYM.02).

**The flat geometry closes the commutator to all orders** ([`verification/v276_qgeo_flat_closes_commutator.py`]). The state-level residual  $[\rho, H]=0$  — the one operator equation the bedrock rests on — is *not* a separate analytic gap once the flat  $\tau=i$  pillowcase is granted. [E] if  $H = f(\Delta_{\text{flat}})$  is a function of the intrinsic flat Laplacian (the DtN of a flat metric, no curvature corrections) and the order-4 deck  $\rho$  ( $z \mapsto iz$ ) is a flat- $\tau=i$  isometry, then  $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0$  *exactly*, to all orders (an isometry commutes with every spectral function of its Laplacian; verified on the discretised square torus, with a generic non-isometry as a failing control).

This *upgrades* the leading-order commutation **v198** (principal symbol) and **v201** (subprincipal) to the full  $H$  — flatness removes the lower-order terms. **[O]** so **QGEO.SYM.01** collapses to a *single geometric* statement, “the raw RP seam state is the flat  $\tau=i$  pillowcase quasi-free state ( $H=\sqrt{-\Delta_{\text{flat}}}$ )”; everything downstream is **[E]**, and the remaining human obligation is one constructive-QFT proof that the raw seam DtN state *is* that flat geometry (**QGEO.SYM.03**).

**The obligation as one precise lemma, Lean-backed** (**[verification/v279\_qgeo\_obligation\_lemma.py]**). The bedrock is now written as a single constructive-QFT *lemma*: *given* the premise “the raw seam carries the flat  $\tau=i$  pillowcase metric  $g_{\text{flat}}$  (Trojanov-unique, cone angles  $\pi$ ), so  $\Lambda_{\Sigma} = \sqrt{-\Delta_{g_{\text{flat}}}}$ ”, the conclusion  $\omega_{\Sigma} \circ \rho = \omega_{\Sigma}$  (and hence mark-locality, the metric inclusion,  $E_8$ ) follows. **[E]** the reduction DAG from “free RP seam” to this premise has *exactly one* un-discharged leaf — “the raw seam state *is* the flat state” — every other node being closed; and the geometric side is fixed (the flat DtN is Fourier-diagonal), so the obligation is a *state-identification*, not a geometric ambiguity, with two candidate routes (**[C]**): RP-state uniqueness on the flat  $\tau=i$  orbifold, or Trojanov + OS reconstruction. **[E]** the all-orders closure step is now a *Lean theorem*: **TfptCarrier/SeamDeckClosure.lean** (**flat\_all\_orders\_clock**) proves that any spectral function  $g(\Delta_{\text{flat}})$  of the Fourier-diagonal flat Laplacian commutes with the carrier clock  $\rho = \text{diag}(i^n)$  — upgrading **v198/v201** to the full operator with only the standard axioms, no **sorry** (**FORM.QGEO.03**). **[O]** the one premise stays the honest axiom (**QGEO.OBLIG.01**).

**Self-investigation: the experiment and the simplification** (**[verification/v280\_pillowcase\_steklov.py]**, **[verification/v282\_e8\_tau\_i\_unification.py]**). Two things can be done *ourselves*. (i) A direct numerical experiment on the actual flat  $\tau=i$  pillowcase orbifold realises the whole chain: the order-4 deck permutes the four cone points (2 fixed + 1 swapped) and  $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0 \Rightarrow [\rho, C]=0 \Rightarrow \omega \circ \rho = \omega$  is numerically exact, with the Steklov DtN positive and metric-determined — route-(i) evidence (RP-state uniqueness), the premise still **[O]**. (ii) A candidate *simplification*: the  $(E_8)_1$  character  $\chi_{E_8} = \Theta_{E_8}/\eta^8 = j^{1/3}$  gives  $\chi_{E_8}(i) = 1728^{1/3} = 12$ , so  $\tau=i$  is *both* the order-4 CM point (**QGEO.SYM.01**, flat geometry) *and* the  $(E_8)_1$  partition-function modulus (**QG.AMB.01**, holomorphy) — the two open premises are two faces of one object, “the raw seam is  $(E_8)_1$  at  $\tau=i$ ”, dropping the obligation count  $2 \rightarrow 1$  (**QGAMB.UNIFY.01**).

**The two proof routes, decomposed** (**[verification/v284\_route\_i\_rp\_uniqueness.py]**, **[verification/v285\_route\_ii\_seam\_condensation.py]**). The keystone — now *closed modulo cited theorems on its MMST route* **SEAM.EQUIV.MMST.01** via the lattice/ $S_3$  stack (below; the parent stays **[O]**) — can be attacked from two sides, and both are now lemma chains with the dischargeable steps discharged. *Route (i)* (RP-state uniqueness) is a six-lemma chain — **[E]** five are already **[E]/[O]** (**RP**→**DtN v194**; **marks**→**pillowcase v195/v216**; the flat Steklov operator  $= \sqrt{-\Delta_{\text{flat}}}$ , principal symbol  $|\xi|$  with no curvature term; covariance from the DtN **v258**;  $\mu_4$ -invariance **v276**), and **[C]** Trojanov ( $\chi_{\text{orb}}=0$ ) reduces the residual to “the raw seam is the constant-curvature geometric state”. *Route (ii)* (seam-net condensation) is a four-lemma chain — **[E]** three are discharged (no abelian sector  $\Leftrightarrow \det K=1 \Leftrightarrow$  holomorphic  $\Leftrightarrow E_8$  **v234/v277/v281**; the tower **16**→**4**→**1 v92**), leaving “the free RP seam condenses the  $\mu_4$ -Lagrangian glue” = **QGEO.SYM.01**. **[E]** **the two open lemmas coincide** — both say “the raw seam is the  $(E_8)_1$  at  $\tau=i$ ” (**v282**), so the canonical equivalence between the raw seam KMS/DtN state and the holomorphic  $(E_8)_1$  net at  $\tau=i$  is the *single* **[O]** statement that closes both (**QGEO.ROUTEI.01/QGAMB.ROUTEII.01**).

**The Seam Equivalence Theorem, named and firewalled** (**[verification/v286\_seam\_equivalence\_contract.py]**). That single statement is now the named target **SEAM.EQUIV.01**: *the raw RP seam state reconstructs canonically the holomorphic  $(E_8)_1$  boundary net at the  $\tau=i$  pillowcase* (respecting the DtN, the KMS flow, the  $\mu_4$  deck, the Q-system extension,  $\det K=1$  and the  $(E_8)_1$  character). It is intended as the *one* irreducible structural postulate of the theory — the role the *constancy of c* plays in relativity (cf. **v267**): not a gap to be patched but the single sharply-localised, falsifiable premise on which the whole boundary closure turns. **[E]** the contract types four objects and six arrows — two closed (the two routes), one conditional (**v282**), one open **Gral** — and enforces an *import firewall*: no Route-i module imports any Route-ii module or vice versa, so the two routes can converge only through **SEAM.EQUIV.01** (this kills the “ $E_8$  smuggled into the geometry and pulled back out” attack). The two proof routes are then attacked directly: **[E]** Route A (AQFT, **[verification/v287\_free\_rp\_bulk\_to\_holomorphic\_boundary.py]**) reduces the **Gral** to *one* standard theorem, “invertible (SRE) Gaussian bulk  $\Rightarrow$  single-sector boundary” (4/5 lemmas discharged); and **[E]** Route B (DtN, **[verification/v288\_full\_l2\_subprincipal\_z4.py]**) *proves* the full- $L^2$  lift of the sub-principal  $\mathbb{Z}_4$  block-diagonality (a mark-sourced curvature has Fourier support only

on  $m \equiv 0 \pmod{4}$ , so  $[\rho, \Lambda] = 0$  on all of  $L^2$ ), shrinking the residual to the single sharper question “why is the raw seam sub-principal term mark-local?” (SEAM.EQUIV.A01/SEAM.EQUIV.B01).

**The route split (2026-07-22; ID restructuring, no status change in substance).** The target statement now carries *two named routes* as ledger rows of their own: SEAM.EQUIV.MMST.01, the conditional scaling-limit route (the lattice/ $S_3$ /Lean stack documented in this section — *closed modulo cited theorems*, its only [O] residual the abstract continuum existence v336), and SEAM.EQUIV.TWISTOR.01, the open Costello–Li construction (prepared by CELEST.SEAM.01, bridged to physics only by WOIT.OS.TWISTOR.01; [O]). The parent claim SEAM.EQUIV.01 remains, with exactly this logic as its claim text: *closed if MMST.01 or TWISTOR.01 closes; currently: MMST conditional-closed modulo cited theorems, TWISTOR open  $\Rightarrow$  the parent stays [O] as an unconditional claim*. Every “closed modulo a cited theorem” statement in the papers and on the website is henceforth attached to the MMST route ID; the assertions themselves are unchanged.

**Route B sharpened to one analytic lemma; both routes precisely scoped.** That last question is decomposed ([verification/v289\_marklocal\_raw.py], MARKLOCAL.RAW.01) into a five-lemma chain whose only open link is [O] *Flat-Away* — RP + gap + the four branch marks force the seam curvature to vanish away from the marks — while the conic parametrix ([C], b-calculus/Melrose), the  $\mu_4$ -orbit source, the  $4\mathbb{Z}$  Fourier support (v264/v288) and the full- $L^2$  commutation (v288) are discharged: 4/5 done, so Route B closes the moment Flat-Away does. The two claims are firewalled against over-reading: Route B does *not* assert “full- $L^2$  QGEO” — its exact-label split is Fourier-lemma (Formal/Exact), operator test (Numerical Exact), transfer to the raw conic DtN (Conditional, v264), mark-locality from the raw seam (Open Selection, v289); and Route A’s “one standard theorem” is itself a composition of citable results (Kitaev/Freed–Hopkins, Muger/Kawahigashi–Longo–Muger, Kawahigashi–Longo, [verification/v287\_free\_rp\_bulk\_to\_holomorphic\_boundary.py]) whose only open hypothesis — seam-bulk invertibility — coincides with Route B’s Flat-Away. Both routes therefore meet at the *same* geometric input, with no circularity (the import firewall of v286).

**Red-team:**  $\mathbb{Z}_4$ -invariance is not mark-locality. A deliberate adversary ([verification/v290\_z4\_smooth\_curvature\_adversary.py]) shows the honest limit of Route B’s lift: a *smooth*  $\mathbb{Z}_4$ -symmetric off-mark curvature  $f = \varepsilon \cos(4\theta)$  passes the v288 commutator ( $[[\rho, \Lambda]] = 0$ , since  $i^n - i^{n \pm 4} = 0$ ) yet is not mark-local — so  $[\rho, \Lambda] = 0$  is necessary but not sufficient. The modulus is instead excluded by the spectral data: it shifts the DtN spectrum off the flat ladder and the heat trace ( $\max |\Delta\lambda| = 0.155$ ,  $\Delta\text{Tr} = 0.019$  at  $\varepsilon = 0.3$ ), i.e. the KMS weight and  $(E_8)_1$  character that SEAM.EQUIV.01 must match. Consequently the open lemma is named in its own right ([verification/v291\_flataway\_contract.py], FLATAWAY.RP.01): *the smooth curvature density in the DtN sub-principal symbol vanishes away from the four branch marks*, with three independent attack routes — spectral-rigidity and heat-kernel (the flat seam is the *unique* spectral/heat match of the smooth  $\mathbb{Z}_4$  family) and RP-energy/Troyanov (the constant-curvature minimiser) — closing any one of which closes Route B.

**All three routes attacked; Flat-Away reduced to one selection principle.** Each route is now carried to its precise reduction: [E] the *heat* route ([verification/v292\_flataway\_heat\_reduction.py]) shows the heat-trace deviation  $\Delta\text{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$  is a positive-definite quadratic form in the off-mark curvature (leading invariant  $\|f\|_{L^2}$ ), so Flat-Away  $\Leftarrow$  “the seam fixes the  $a_2$  heat coefficient”; [E] the *spectral* route ([verification/v293\_flataway\_spectral\_hessian.py]) lifts this to the full smooth- $\mathbb{Z}_4$  space — mode  $4k$  splits the degenerate Steklov level  $|2k|$ , the mismatch Hessian over  $\{4, 8, 12\}$  is positive-definite ( $\{0.497, 0.500, 0.503\}$ ), so the flat seam is a *strict* isolated minimum (answering “only one direction”); and [E]/[C] the *variational* route ([verification/v294\_flataway\_rp\_energy.py]) gives, with the rigid mark divisor and Troyanov uniqueness, the flat cone metric as the unique strictly-convex curvature-energy minimiser. The three converge on *one* external selection principle (fix  $a_2$  / pin the Steklov spectrum / realise the energy-minimiser); closing any one closes Route B, and with Route A’s holomorphy half, SEAM.EQUIV.01.

**The heat route made exact and Lean-formalised.** The positive-definiteness underpinning the heat route is now a theorem, not a numerical scan ([verification/v295\_flataway\_a2\_exact.py]): the first variation of  $\text{Tr } e^{-t\Lambda}$  vanishes (the smooth off-mark curvature has zero mean, Gauss–Bonnet), and since  $\Lambda \mapsto \text{Tr } e^{-t\Lambda}$  is convex (Klein/Peierls),  $\varepsilon=0$  is the *global* minimum — so  $\Delta\text{Tr} \geq 0$  for every deformation,  $= 0$  iff  $f = 0$ ; the exact second-order coefficient is the divided-difference kernel of  $e^{-tx}$  (validated to  $10^{-6}$ ). The algebraic  $\Delta=0 \Leftrightarrow$  flat step is machine-formalised in Lean (FORM.FLATAWAY.01, SeamDeckClosure.lean: a positive-weighted sum of squares is positive-definite; axiom-clean). So Flat-Away’s heat route is reduced to the single external invariant “the seam fixes  $a_2$ ”, with its positivity both proved and for-

malised. The  $a_2$  coefficient itself is now in *closed form* ([[verification/v296\\_flataway\\_a2\\_closed.py](#)]): the operator tails telescope geometrically to  $W_k^{\text{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$ , leaving a finite  $(4k-1)$ -term middle, e.g.  $W_1(t) = t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8$  (manifestly positive; small- $t$   $W_k = t/2 + O(t^3)$ ). Symmetrically, Route A’s “one standard import” is now a precise, citable theorem stack ([[verification/v297\\_route\\_a\\_literature\\_stack.py](#)): LIT-A (Kitaev; Freed–Hopkins)  $\rightarrow$  LIT-B (Müger; Kawahigashi–Longo–Müger)  $\rightarrow$  LIT-C (Conway–Sloane; Dong–Xu) compose into “invertible bulk  $\Rightarrow (E_8)_1$ ”, all three legs established, the one shared hypothesis (seam-bulk invertibility) coinciding with Route B’s Flat-Away — so both routes are reduced to citable/established machinery plus the *one* shared geometric input, which the lattice/ $S_3$  stack (below) now pins at every computable level, making the keystone’s MMST route (SEAM.EQUIV.MMST.01) *closed modulo cited theorems*.

**Flat-Away hardened, and the pin derived from  $(E_8)_1$**  ([[verification/v300\\_flataway\\_rigid.py](#)], FLATAWAY.RIGID.01). The spectral route was a *soft* minimum (“the mismatch grows with  $\varepsilon$ ”); it is now a *discrete* obstruction. [E] the flat fingerprint is the integer Steklov ladder with every level  $n \geq 1$  of multiplicity exactly 2; [E] the resonant mode  $4k$  splits level  $2k$  by gap  $= g_k$  (the exact  $2 \times 2$  block  $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$ , validated against the full Toeplitz operator), so [E] any  $g_k \neq 0$  lifts the degeneracy ( $2 \rightarrow 1+1$ ) and changes the spectral *multiset* — preserving the flat ladder *with its multiplicities* forces every  $g_k = 0$ , a degeneracy obstruction robust to rescaling and strictly stronger than the v293 “deviation grows” scan; with the v295 convexity, flat is isolated analytically *and* discretely. Moreover the external pin is now *derived*: [C] the  $(E_8)_1$  vacuum character  $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$  has integer  $q$ -coefficients ( $E_8$  even unimodular  $\Rightarrow$  integer conformal weights), so by 2d Steklov conformal rigidity the integer ladder  $\Leftrightarrow$  flat seam — i.e. “the seam fixes its spectrum” *is* the  $(E_8)_1$  rationality of Route A. [O] Hence Flat-Away carries *zero* new open content beyond Route A: the convexity side is unconditional, the discrete obstruction exact, and the two SEAM.EQUIV.01 routes provably converge on one rationality fact (consistent with the v286 firewall).

**Route A’s last hypothesis discharged — the gapped quasi-free bulk is invertible** ([[verification/v301\\_route\\_a\\_invertible.py](#)], SEAM.EQUIV.A.INV.01). Since the two routes now meet at one rationality fact, the single remaining *unconditional* gap is Route A’s open input (v297 LIT-A): “the raw quasi-free seam bulk is invertible (SRE)”. A *gapped* free-fermion (Gaussian) 2+1D bulk cannot carry intrinsic topological order — free fermions realise only the integer (IQHE/Chern) phases, all *invertible*; intrinsic topological order requires interactions (Kitaev’s periodic table, class D, the only invariant being the chiral central charge). [E] the carrier is 16 Majoranas, so  $c = g_{\text{car}} + N_{\text{fam}} = 8 = 16/2$  (v148); [E]  $SO(16)_1 \supset (E_8)_1$  ( $\dim \mathfrak{so}(16) = 120$  currents + 128 spinor = 248, v154); [C] hence a gapped quasi-free bulk is invertible automatically, and [E] the invariant  $|\det K| = \#\text{anyons}$  is 1 for  $E_8$  (invertible) against 4 for the gauged  $D_8=SO(16)$  (anyons), which free fermions cannot produce. [O] So LIT-A’s ‘invertible bulk’ *follows* from ‘gapped 16-Majorana quasi-free bulk’ (v148/v160 + the mass gap): the residual changes from a topological-order statement to a spectral one. *Bottom line*: together with v300, the entire open content of SEAM.EQUIV.01 reduces to the single spectral statement “the quasi-free seam bulk is gapped” — the topological-order obstruction (hidden anyons) is removed, and the gap itself is not manufactured.

**The last lever — the seam gap is the derived Recovery gap** ([[verification/v302\\_seam\\_gap.py](#)], SEAM.EQUIV.GAP.01). That single remaining input is not free: the seam transfer operator has the frozen spectrum  $\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$  (v160/v162), a simple Perron eigenvalue 1 separated from the rest, and the separation is the *Recovery gap*  $\Delta = -\ln((2/3)^6) = 6 \ln(3/2) = 2N_{\text{fam}} \ln(3/2) \approx 2.4328 > 0$  — a number forced by the carrier  $((2/3)^6$  the  $\mu_4$ -deck transfer eigenvalue), with the decoupling margin  $\Delta - 31/(4\pi^2) \approx 1.648 > 0$  (v76) making it robust. [C] By Osterwalder–Schrader reconstruction / quasi-free clustering a transfer-operator gap  $\Delta > 0$  *is* a mass gap of the reconstructed Hamiltonian (correlation length  $1/\Delta$ ). *Bottom line*: with v300/v301, the whole residual of SEAM.EQUIV.01 is now a composition of standard cited theorems (OS/quasi-free clustering + Kitaev free-fermion invertibility + the v297 AQFT stack) over established TFPT facts (16 quasi-free Majoranas; transfer gap  $6 \ln(3/2)$ ): *no undischarged TFPT-internal assumption remains*. It stays [O] (not machine-proved end-to-end), but the open front is reduced to citable machinery over a spectral gap that is provably positive.

**The continuum chain assembled, and the residual tied to the seam one-sidedness** ([[verification/v356\\_continuum\\_mmst\\_applicability.py](#)], SEAM.EQUIV.CONTINUUM.03). Writing the chain out link-by-link: **S1** the gapped (v302) quasi-free 16-Majorana collar  $\rightarrow$  **S2** invertible (Kitaev free fermions, v301)  $\rightarrow$  **S3** the *nontrivial* invertible phase ( $c_- = 8 \neq 0$ )  $\rightarrow$  **S4** a non-gappable anomalous chiral edge (anomaly inflow / bulk-edge)  $\rightarrow$  **S5** its chiral-CFT scaling limit (MMST, v336; range rank  $\leq c \leq D$ , i.e.  $8 \leq 8 \leq 16$ )  $\rightarrow$  **S6** the order-4  $\mu_4$  clock + the AQFT stack pin  $(E_8)_1$  (v351/v297). Every link is a

theorem or an established TFPT fact *except* S3, and S3 is precisely the seam being *one-sided* (chiral,  $c_- \neq 0$ ) — the *same* one-sidedness that makes  $c_3 = 1/(8\pi)$  a one-sided Gauss–Bonnet ( $|\mathbb{Z}_2| \nmid K$ , v58). So the lone realization input is tied to the constant that *defines* the theory, not a free dial; the residual is now “the abstract collar is realized as a genuine lattice chiral free-fermion invertible phase”, [O] but theorem-bounded on both sides.

**MMST in-class verified for the collar, so the scaling limit is  $(E_8)_1$**  ([verification/v366\_mmst\_seam\_collar.py], SEAM.MMST.INCLASS.01). The continuum leg is now checked *hypothesis-by-hypothesis* for the explicit seam collar: it is a  $D = \dim S^+ = 16$  Majorana quasi-free (CAR) state (v155/v160), gapped with  $\Delta = 6 \log \frac{3}{2} > 0$  *derived* (v302), and chiral with  $c_- = D/2 = 8 = g_{\text{car}} + N_{\text{fam}}$ , so it sits squarely in the MMST free-lattice-fermion class with rank  $\leq c \leq D$  reading  $8 \leq c \leq 16$  (*in* range, saturating rank  $= c = 8$ ). The MMST massless-scaling-limit theorem then applies as a cited result, and the order-4  $\mu_4$  clock pins the target to  $(E_8)_1$  ( $\det K = 1$ , one primary) over the same- $c$  rival  $SO(16)_1$  ( $\det K = 4$ , v351). [E] the class arithmetic, the derived gap, the chirality and the  $(E_8)_1$  pinning; [C] the CAR-class typing and the cited MMST theorem; [O] the lone residual is exactly S3 above (the lattice realization). The continuum side of SEAM.EQUIV.01 is thereby reduced to a single, named, theorem-bounded input. The lattice→conformal-net step itself is now a *rigorous AQFT construction*: Adamo–Giorgetti–Tanimoto (arXiv:2506.01008, 2025) build the two-dimensional conformal net  $\mathcal{A}_Q$  of an even lattice  $Q$  and *classify* every two-dimensional extension of the Heisenberg net (under a discreteness assumption) as such a lattice net — so “the lattice is a conformal net” is a theorem of the same operator-algebraic lineage the continuum leg invokes, not a hope; and since (xxxii) ([verification/v469\_seam\_crossedproduct\_route.py]) the extension step no longer even *needs* the preprint lineage — the local  $\mathbb{Z}_2$  simple-current crossed product ( $h_s = 1 \in \mathbb{Z}$ , Longo–Rehren 1995; Böckenhauer 1996; Böckenhauer–Evans 1998; KLM 2001) certifies it on peer-reviewed ground, with AGT/AMT as the independent second witness.

**S3 made concrete: an explicit gapped chiral-Majorana lattice model** ([verification/v367\_seam\_s3\_lattice.py], [verification/v368\_seam\_s3\_inflow.py]). The S3 input is no longer abstract. The  $p+ip$  Bloch model  $h(k) = \sin k_x \sigma_x + \sin k_y \sigma_y + (M - \cos k_x - \cos k_y) \sigma_z$  is, at  $M=1$ , a gapped free-fermion phase with numerically-computed Chern number  $|C|=1$  ( $C=0$  trivially at  $M=3$ ), so  $N_{\text{Maj}} = \dim S^+ = 16$  copies give a chiral edge of  $c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}}$ , with the edge K-matrix  $SO(16)_1$  ( $\det 4$ ) condensed to  $(E_8)_1$  ( $\det 1$ ) by the order-4  $\mu_4$  clock (v351/v369). A finite STRIP of the same model carries an edge-localised in-gap chiral branch (absent in the trivial phase) – so by anomaly inflow the chiral edge *exists* as a consequence of the invertible bulk (the S4 leg), discharging the “does the edge exist?” question on the lattice. [E] the gapped Chern model, the  $c_- = 8$  counting and the edge existence; [C] the  $\mu_4$  condensation to  $(E_8)_1$ ; [O] only the genuine *continuum scaling limit* (MMST, v336) stays open — S3 is a runnable lattice model that pins the target at every computable level, so the keystone’s MMST route (SEAM.EQUIV.MMST.01) is *closed modulo cited theorems*.

**The S3 closure stack:  $c = 8$ , the  $(E_8)_1$  character, the torus count and reflection positivity — all computed on the lattice** ([verification/v376\_seam\_s3\_centralcharge.py], [verification/v377\_seam\_s3\_e8character.py], [verification/v378\_seam\_s3\_modular.py], [verification/v379\_seam\_s3\_rp.py]). On that explicit model the target of the scaling limit is now pinned at every level a computation can reach. *Central charge* (v376): the Calabrese–Cardy finite-size entanglement scaling gives  $c = 1.0000$  per critical complex mode, so the 16-Majorana ( $= 8$  complex) collar has  $c = 8 = g_{\text{car}} + N_{\text{fam}}$ , and it is chiral ( $c_- = 8$ ). *Convergence* ([verification/v392\_seam\_s3\_scalinglimit.py]): across five sizes  $L = 66 \dots 514$  the slope  $c_1(L)$  is a monotone Cauchy-like sequence approaching 1 and  $1/L$ -Richardson-extrapolates to  $c_1(\infty) \approx 1.000$  ( $c \approx 8.00$  in the limit) — the existence-relevant *convergence* the two-size value did not test (evidence, not a proof of existence; the residual stays [O]). *The net* (v377): the exact affine character  $\chi = E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$  gives  $c = 8$  and 248 level-1 currents (one primary), and the  $\mu_4$ /GSO promotion  $248 = 240 + 8 = 120 + 128$  ( $240 = 16 \cdot 5 \cdot 3$ ) pins  $(E_8)_1$  over the same- $c$  rival  $SO(16)_1$  (120 currents, four primaries). *Genus-1* (v378): the KLM condensation  $\mu(\text{carrier}) = 16 \rightarrow \mu(E_8) = 16/4^2 = 1$  and the single character’s modular  $T$ -eigenvalue  $e^{-2\pi i/3}$  give torus ground-state degeneracy 1 — the holomorphy /  $\det K = 1$  statement that the genus-0 arguments could not reach (v344). *Reflection positivity* (v379): the gapped collar’s Euclidean two-point function has a positive spectral measure, so the OS reflection Gram matrix is a *positive-mixture Gram*  $M = V^T \text{diag}(1/K)V$ ,  $V_{ki} = e^{-\varepsilon_k \tau_i}$  — hence  $x^T M x = \sum_k w_k (v_k \cdot x)^2$  is a sum of squares and positive semidefiniteness holds *exactly*, as an identity rather than to a tolerance (hardened 2026-07-28 after T133 certified that the *assembled double-precision* matrix has a negative direction at  $-7.75 \times 10^{-18}$ , inside the old  $10^{-10}$  tolerance; the min-eigenvalue check is retained as a diagnostic measurement, and a ghost mode still breaks the RP test). [E] all four

(numerical  $c$ , exact character, genus-1 count, RP); [O] the one thing a computation cannot supply is the abstract *continuum existence* of the limit (the cited MMST theorem, v336). With RP in hand the ambient measure C7 is then discharged by the redundancy theorem (v369), so S3 is the master key: the target is pinned exactly, only the existence proof is external.

**A ground-state witness for premise (i): the modular commutator measures  $c_- = 8$  from the bulk state alone** ([verification/v489\_seam\_modular\_commutator.py], SEAM.CCC.01). The chiral central charge entered so far through band topology (the Chern number, v367) and through counting (the KLM/det-Cartan arithmetic, v378). The Kim–Shehab–Kim modular commutator  $J(A, B, C) = i \text{Tr}(\rho_{ABC}[K_{AB}, K_{BC}]) = (\pi/3) c_-$  (PRL 128 (2022)) reads  $c_-$  off the *entanglement data of the exact BdG ground state* of the same collar — no band bundle, no representation theory enters — via the Peschel covariance ( $W_X = -2 \text{artanh } \Gamma_X$ ) on a disk cut into three 120° wedges: per layer  $c_- = 0.499998$  at  $L=32$  (converged 0.500000 at  $L=48$ , deviation  $-1.8 \times 10^{-7}$ ), the trivial  $M=3$  control gives 0.000000, the orientation flip  $M=-1$  gives  $-0.499986$  (sign flip), layer additivity is exact ( $10^{-14}$ ), and the 16-layer carrier lands on  $c_- = 8 = g_{\text{car}} + N_{\text{fam}}$  (direct run 7.9967 at  $L=16$ , additivity route 8.0000). [E] the third, mutually independent lattice witness of premise (i) of SEAM.EQUIV.01 — and the first read from ground-state entanglement alone; [O] unchanged: the continuum scaling limit (MMST, v336) and the cited KLM condensation step stay exactly where v367/v378 left them — SEAM.EQUIV.01 stays [O].

**The spin-structure parity census:**  $\mu_{\text{pre}}=4$  **witnessed on the lattice** ([verification/v490\_seam\_parity\_census.py], SEAM.MU.01). Step 1 of the KLM chain (v378) — “the carrier sits in the  $SO(16)_1$  class,  $\mu_{\text{pre}}=4$ , before the  $\mu_4$  condensation” — was arithmetic; it is now a lattice ground-state measurement. The  $T^2$  fermion parity per spin structure ( $AA, AP, PA, PP$ ) is determined *twice* (Read–Green TRIM product and real-space Majorana Pfaffian, spectra matching to  $9 \times 10^{-15}$ ): per layer  $(+, +, +, -)$  — the  $\nu=1$  Ising signature — and for the 16-layer carrier parity<sup>16</sup> = + in *all four* sectors, so the gauged torus GSD is  $4 = \mu(SO(16)_1) = |\det \text{Cartan}(D_8)|$ . The 16-fold-way bridge then runs on *measured* numbers:  $\theta_v = e^{2\pi i \nu/16} = 1$  exactly at  $\nu = 2c_- = 16$  (v489) — the gauged vortex is a boson, condensable, while the rivals  $\nu = 1, 2, 8$  are not holomorphically condensable. Honest fences: the fermionic Kitaev–Preskill TEE is identically zero for *both*  $M=1$  and  $M=3$  ( $\gamma_f = 0.000001/0.000000$ ) — an invertible fermionic phase has no TEE, so  $\gamma_f$  does *not* discriminate  $(E_8)_1$  vs  $SO(16)_1$ ; and the KLM step  $\mu: 4 \rightarrow 4/2^2 = 1$  stays the cited theorem ([C], the condensed model is interacting). The numerics pin exactly the premise package  $\{c_-=8, \mu_{\text{pre}}=4, \theta_v=1\}$  of the condensation; SEAM.EQUIV.01 stays [O].

**And that external step is now Lean-formalised as “closed modulo a cited theorem”** (FORM.SEAM.MMST.01, TftptCarrier/SeamScalingLimit.lean). The continuum leg is machine-formalised in the audit-contract style of SeamEquivChain.lean: the collar’s MMST applicability hypotheses are *provable* arithmetic facts by Lean kernel *decide* (no axioms) —  $D=16$  Majoranas, rank  $=c=8$ , the range  $8 \leq 8 \leq 16$ , the chirality  $c_-=8 \neq 0$ , and the holomorphy discriminator  $\det K=1$  vs  $SO(16)$ ’s 4; the MMST scaling-limit theorem and the Adamo–Moriwaki–Tanimoto OS reconstruction are declared as *named cited axioms*; and the conclusion  $(E_8)_1$  follows by a well-typed composition whose `#print axioms` pins the residual to exactly those two published theorems (plus the carrier gap), with no hidden assumption and no *sorry*. Every TFPT-internal hypothesis is thus kernel-checked; only the two cited theorems are assumed — the honest closure ceiling for SEAM.EQUIV.01. *Two of the seven convergent routes below are now kernel-hardened the same way*: FORM.SEAM.BW.HSMI.01 (TftptCarrier/SeamStandardPair.lean) proves the four Borchers/Wiesbrock *standard-pair relations* exactly over  $\mathbb{Z}$  on the centred ladder ( $[K, P]=P$ ,  $J^2=1$ ,  $JKJ=-K$ ,  $JPJ=P_+$ , all by kernel *decide*, no domain axiom), with the cited Borchers step and the one continuum input as named axioms; and FORM.SEAM.APPLICABILITY.01 (TftptCarrier/SeamApplicabilityLedger.lean) kernel-checks the audit *count* itself — of the 12 cited-theorem hypotheses exactly 11 are internal and 1 external (and  $8 \leq 8 \leq 16$ ,  $\det K=1 \neq 4$ ), all axiom-free — so the very bookkeeping the keystone’s honesty rests on is machine-verified, leaving only the one named continuum fact open.

**TOE-roadmap follow-ups — the keystone as a typed certificate, the bedrock modular form, and the gap-decoupled measure** ([verification/v308\_seam\_equiv\_chain.py], [verification/v309\_modular\_bedrock.py], [verification/v311\_gap\_decoupled\_measure.py]). Three machine-checked steps push the keystone arc further. (i) *Composition certificate* ([verification/v308\_seam\_equiv\_chain.py]): the closing implication is one well-typed logical chain  $\text{gap} > 0 \rightarrow \text{invertible} \rightarrow \text{holomorphic } c=8 \rightarrow (E_8)_1$  whose arithmetic discriminators are exact —  $|\det \text{Cartan}(E_8)|=1$  against  $|\det \text{Cartan}(D_8)|=4$ , the carrier tower  $4 \cdot 4=16 \rightarrow 4 \rightarrow 1$ ,  $c=g_{\text{car}}+N_{\text{fam}}=8$  — so the only undischarged premise is the cited OS step; [E] the discriminators, [O] the keystone.

(ii) *Modular bedrock* ([verification/v309\_modular\_bedrock.py]): with the four marks derived from Gauss–Bonnet (v216) and the order-4 clock from cross-ratio 2 (v214), the quasi-free modular Hamiltonian  $K = \log((1-C)/C)$  is built explicitly and  $[\rho, K]=0 \Leftrightarrow \omega \circ \rho = \omega$  (a  $\mathbb{Z}_2$  profile breaks it), so the bedrock reduces *non-circularly* to the Bisognano–Wichmann intrinsicity of the raw collar state; [E] the model, [O] the intrinsicity. (iii) *Gap-decoupled measure* ([verification/v311\_gap\_decoupled\_measure.py]): the admissible sector clusters exponentially at rate  $(2/3)^6$  (correlation length  $1/\Delta$ , finite susceptibility 729/665) and the v76 margin  $\Delta - 31/(4\pi^2) > 0$  decouples it from the ambient, so the physical sector is a convergent object while QG.AMB.01 is a [C] redundancy (v369). Each pins the residual to a single named step; with the lattice/ $S_3$  stack the keystone’s MMST route (SEAM.EQUIV.MMST.01) is now *closed modulo cited theorems*. (iv) *The intrinsic-BW residual as a cited reduction* ([verification/v424\_seam\_lto\_rp\_reduction.py]): the open bedrock lemma of (ii) — the raw collar’s intrinsic Bisognano–Wichmann property — is reduced to the recent commuting-projector theorem of Naaijken–Penneys–Wallick (arXiv:2605.10693), whose reflection-positivity axiom (LTO-RP) is exactly  $u_\Theta = J$  (the reflection *is* the modular conjugation), i.e. the BW condition  $\theta K \theta = -K$  — *not*  $\omega \circ \rho = \omega$ . [E] the  $\mu_4$  clock is a modular *symmetry* ( $[\rho, K]=0$ ), not the flow ( $\sigma^\psi = \rho$  is type-wrong, a one-parameter group versus a finite order-4 element); a positive character-block-diagonal  $K$  has  $[\rho, K]=0$  yet, being positive definite, never  $\theta K \theta = -K$ , so clock-invariance does *not* imply (LTO-RP). [O] make-or-break: NPW26 prove (LTO-RP) for *topologically ordered* commuting-projector models (Toric Code via a trace, Levin–Wen anyonic), whereas the seam is the opposite corner — invertible ( $|\det \text{Cartan}(E_8)|=1$ , no anyons) and a  $\beta=1$  KMS state (not a trace) — so two sub-steps stay open: realising the raw seam as a  $\mathbb{Z}_2$ -reflection commuting-projector LTO, and extending (LTO-RP) to the invertible KMS case. An MMST-style reduction (cf. v308/v336), *not* a closure. *Sub-step (ii) is now constructively exhibited* ([verification/v426\_seam\_lto\_rp\_kms.py]): a gapped reflection-symmetric  $\mu_4$ -even collar at  $\beta=1$  KMS satisfies  $\Theta K \Theta^{-1} = -K$  ( $u_\Theta = J$ ) and  $[\rho, K]=0$ , so (LTO-RP) holds there by direct Tomita–Takesaki — the corner NPW26 leave open — with  $|\det \text{Cartan}(E_8)|=1$  (invertible, no anyons) and the state a genuine KMS, not a trace; only the continuum sub-step (i) remains [O]. On the covariance side, the recent theorem of Adamo–Giorgetti–Tanimoto (arXiv:2508.07109, 2025) — every DHR representation of a conformal net extending a loop-group/Virasoro net is *automatically* positive-energy and diffeomorphism-covariant — defuses the old “Bisognano–Wichmann presupposes the very covariance it should produce” circularity (v215): once the seam net is in hand, geometric covariance is a *consequence*, not an input.

**Twenty-six convergent attacks on the one continuum residual — intrinsic BW, a bulk continuum motor and its chiral-edge counterpart, the  $\beta$ -homotopy, the applicability audit, uniform rigidity with its forcing converse and the derivation of clock-invariance, three independent reads of the edge central charge, the four-point and uniform-in- $N$  pinning of the external fact, the Ising operator tower, the  $(E_8)_1$  torus modular data, the  $\mu_4$ -from-marks derivation, the kill tests, the lattice Sugawara current algebra, a uniform-in- $N$  Tomita–Takesaki tower, the chirality-from- $P_1$  forcing, an  $(E_8)_1$  character kill test, the exact MMST citation audit, the complementary lattice-VOA route, the explicit flat-band commuting-projector LTO realisation, the strict-locality (Kapustin–Fidkowski) obstruction, the character-level 128-spinor extension, the  $c=8$  holomorphic-uniqueness pin and the one-particle realisation rigidity** ([verification/v438\_seam\_hsmi\_borchers.py], [verification/v439\_seam\_lieb\_robinson.py], [verification/v440\_seam\_lto\_rp\_beta.py], [verification/v441\_seam\_applicability\_ledger.py], [verification/v442\_seam\_rigidity\_uniform.py], [verification/v443\_seam\_e8\_discriminator.py], [verification/v444\_seam\_edge\_scaling.py], [verification/v445\_seam\_rigidity\_forcing.py], [verification/v446\_seam\_clock\_invariance.py], [verification/v447\_seam\_edge\_chern.py], [verification/v448\_seam\_edge\_fourpoint.py], [verification/v449\_seam\_mmst\_uniform.py], [verification/v450\_seam\_edge\_entanglement.py], [verification/v451\_seam\_edge\_cardy\_tower.py], [verification/v452\_seam\_e8\_modular.py], [verification/v453\_seam\_mu4\_from\_marks.py], [verification/v454\_seam\_edge\_virasoro.py], [verification/v455\_seam\_bw\_uniform.py], [verification/v456\_seam\_chirality\_from\_c3.py], [verification/v457\_seam\_character\_killtest.py], [verification/v458\_seam\_mmst\_citation\_audit.py], [verification/v459\_seam\_lattice\_voa\_route.py], [verification/v460\_seam\_s3\_lto.py], [verification/v461\_seam\_strict\_locality.py], [verification/v462\_seam\_spinor\_continuum.py], [verification/v463\_seam\_c8\_holomorphic\_uniqueness.py], [verification/v464\_seam\_oneparticle\_rigidity.py], [verification/v469\_seam\_crossedproduct\_route.py]). None closes SEAM.EQUIV.01; together they ground the one continuum input on both its halves, reduce it to a single analytic fact and make the claim falsifiable. (v) *The intrinsic Bisognano–Wichmann route* ([verification/v438\_seam\_hsmi\_borchers.py]): the seam supplies a Longo *standard pair* /

+half-sided modular inclusion — a positive lightlike translation  $P \geq 0$ , the boost  $K$  with  $[K, P]=P$  (the Borchers scaling  $\Delta^{it} P \Delta^{-it} = e^{-t} P$ ), and a reflection  $J$  with  $JKJ=-K$ ,  $JPJ=P_-$  — so by Borchers’ theorem the modular flow is geometric *intrinsically, without* presupposing the rotation-covariance of the vacuum; [E] the ladder algebra and the  $sl(2, \mathbb{R})$  positivity (min eig  $> 0$ , with the indefinite boost as control), [C] the synthesis, [O] assembling the finite algebra and the positive-energy rep into one continuum standard pair (the MMST/NPW26 wall). (vi) *The continuum-existence motor* ([verification/v439\_seam\_lieb\_robinson.py]): the explicit  $p+ip$  collar (v367) has a uniform-in- $N$  gap, exponential clustering with  $N$ -independent  $\xi$ , and a finite Lieb–Robinson light cone, so by Nachtergaele–Sims its *bulk* thermodynamic limit exists — grounding the bulk half of the cited MMST existence in our model and narrowing the open step to the chiral massless *edge* scaling limit; [E] gap/clustering/light-cone, [C] the lift, [O] the edge CFT. (vii) *The  $\beta$ -homotopy* ([verification/v440\_seam\_lto\_rp\_beta.py]): the (LTO-RP)/BW data ( $\Theta K \Theta = -K$ ,  $[\rho, K]=0$ ) are  $\beta$ -independent, and along the whole path  $C_\beta = 1/(1+e^{\beta K})$  from the NPW26-proven *trace* state ( $\beta=0$ ) to the seam KMS state ( $\beta=1$ ) every covariance is reflection-related,  $\mu_4$ -invariant and valid — a norm-continuous, obstruction-free homotopy, upgrading the single point of v426 to a path argument for sub-step (ii). (viii) *The applicability audit* ([verification/v441\_seam\_applicability\_ledger.py]): a hypothesis-by-hypothesis ledger of MMST/NPW26/Adamo finds 11 of 12 hypotheses are TFPT-internal computed facts ([E]) and exactly one is external — the continuum existence of the chiral scaling limit — covariance being an Adamo consequence, not an input; the keystone’s external dependence is pinned to that one analytic fact. (ix) *Uniform rigidity* ([verification/v442\_seam\_rigidity\_uniform.py]): the band-limited  $\mu_4$ -character rigidity of v398 is uniform in  $N$  — exact 4-sector commutation to  $N=256$ , a uniform gap, and an  $N$ -independent clock-breaking lower bound  $\|[\rho, V]\|=2$  — so it is not a small- $N$  artifact (the full- $L^2$  forcing stays [O]). (x) *The kill tests* ([verification/v443\_seam\_e8\_discriminator.py]): mode D of the closure-mode classification (v347) as concrete *falsifiable* discriminators against the nearest  $c=8$  rival  $SO(16)_1$  — torus GSD 1 vs 4, level-1 currents 248 vs 120, lattice theta 240 vs 112 norm-2 roots — with  $c=8$  alone unable to decide (v277) and only  $\det K=1$  (the genus-1 statement, v90) selecting  $E_8$ ; a seam with GSD $>1$  would falsify  $(E_8)_1$ . [X] the tests, [O] the realisation. (xi) *The chiral-edge scaling motor* ([verification/v444\_seam\_edge\_scaling.py]): the counterpart of (vi) on the *edge*, attacking exactly the chiral massless scaling limit that the bulk motor left open. On the same  $p+ip$  collar (v367/v368) the lattice edge already carries the conformal data of a chiral  $c=\frac{1}{2}$  Majorana CFT — a linear, chiral in-gap dispersion (velocity  $v \approx 0.98$ ,  $R^2=0.9999$ , with *opposite* per-edge velocities), an *algebraic* edge two-point  $\sim r^{-\eta}$  with  $\eta \approx 1$  (versus the *exponential* bulk row and the exponential trivial-phase control), and a Cauchy-convergent scaling-collapse under refinement ( $\eta=1.001 \pm 0.007$ , successive spread  $0.028 \rightarrow 0.013$ ) — so the 16-copy edge is a chiral  $c_- = 8$   $(E_8)_1$  CFT; [E] the kinematics, correlator and collapse, [C] the central-charge reading, [O] the rigorous uniform-convergence theorem (cited MMST). With (vi) for the bulk and (xi) for the edge, *both halves* of the one continuum residual are numerically grounded and pinned to a single cited theorem. (xii) *The rigidity forcing converse* ([verification/v445\_seam\_rigidity\_forcing.py]): the converse of (ix), upgrading rigidity from “block-diagonal *permits* commuting” to “commuting with the order-4 clock *forces* block-diagonal”. For  $\rho = \text{diag}(i^n)$ ,  $[\rho, K]=0$  iff  $K_{ij}=0$  whenever  $i \neq j \bmod 4$  (verified entrywise, swept  $N=4..128$ ), so the commutant is *exactly* the 4-sector block algebra:  $\dim\{K : [\rho, K]=0\} = \sum_s n_s^2$ , a proper subspace of the full  $N^2$  algebra ( $N=16: 64 < 256$ ); every off-block unit has  $\|[\rho, E_{ij}]\| \geq \sqrt{2}$  (zero slack), and the order-2 clock leaves a strictly larger commutant (64 vs 128) — the *four* Gauss–Bonnet marks (order  $4=|\mu_4|=h(A_3)$ ) force strictly more structure than two, and only the index-4 extension is  $(E_8)_1$  ( $\det=1$ , v281/v154/v351). [E] the entrywise iff, exact commutant dimension, off-block sharpness and order discriminator; [C] the synthesis (RP-definability + clock-invariance *force* the block-diagonal  $\Lambda_\Sigma$ , v442’s “permit” upgraded to “force”); [O] the single residual is now precisely that the physical transfer lies in the commutant (clock-invariance = QGEO.SYM.01, v323/v335). (xiii) *Clock-invariance derived* ([verification/v446\_seam\_clock\_invariance.py]): closing (xii)’s residual one notch further — the operator condition  $[\rho, \Lambda_\Sigma]=0$  is itself a *consequence* of a structural symmetry, not an extra assumption. On explicit finite OS data, the canonical transfer  $\Lambda = C(1)C(0)^{-1} = e^{-h}$  (the transfer fixed by RP+reflection, v54) of a  $\mu_4$ -invariant covariance satisfies  $[\rho, \Lambda]=0$  (swept  $N=4..32$ ; iff-sharp  $\mu_4$ -breaking neg control),  $\Lambda$  is a positive contraction, and the free modular Hamiltonian  $K = \log((1-C)C^{-1})$  (v258) of the invariant state has  $[\rho, K]=0$  (Tomita–Takesaki). [E] the inheritance, contraction and modular commutation; [C] so with (xii), RP-definability *plus* the four marks force the block-diagonal  $\Lambda_\Sigma$ ; [O] the residual reduces from an operator commutation to the structural “the seam RP data is  $\mu_4$ -symmetric” (QGEO.SYM.01, v216/v323). (xiv) *The independent topological edge reading* ([verification/v447\_seam\_edge\_chern.py]): a second, fit-free route to the edge central charge of (xi). On the same  $p+ip$  collar (v367/v368) the

bulk Chern number is the integer  $C=+1$  (Fukui–Hatsugai–Suzuki link variables;  $C=0$  in the trivial phase,  $C=-1$  at the opposite mass, matching (xi)’s opposite per-edge velocities), the open strip carries exactly  $|C|=1$  chiral edge branch per boundary, and the Li–Haldane entanglement spectrum has in-gap modes at  $\frac{1}{2}$  only in the topological phase; [E] the Chern integer, channel count and entanglement signature, [C] the bulk–edge reading  $c_- = N_{\text{Maj}} \cdot \frac{1}{2} \cdot |C| = 16 \cdot \frac{1}{2} = 8 = g_{\text{car}} + N_{\text{fam}}$  — the *same*  $c_- = 8$  as (xi) from a logically independent (topological) observable. (xv) *The four-point pinning of the external fact* ([verification/v448\_seam\_edge\_fourpoint.py]): extending (xi)’s two-point scaling limit to the next order to support exactly the one external fact (viii) isolates. The edge state is quasi-free (the ground-state correlation matrix is an exact projector  $G^2 = G \sim 10^{-13}$ , the MMST CAR hypothesis), and the edge four-point cross-ratio  $Q = [G_{01}G_{23}]/[G_{02}G_{13}]$  at commensurate positions is Cauchy-convergent under refinement to the free-fermion conformal value  $\sin(5\pi/12)/\sin(\pi/12) = 2 + \sqrt{3} \approx 3.732$  (the bulk control off-scale, exponential); [E] the quasi-free property, four-point convergence and conformal form, [C] so the one external MMST fact is empirically supported at *both* two- and four-point order. (xvi) *The uniform-in- $N$  pinning* ([verification/v449\_seam\_mmst\_uniform.py]): strengthening (xv) from convergence at a single cross-ratio to the *uniformity* the cited theorem asserts — two *independent* edge four-point cross-ratios converge to *distinct* conformal values ( $Q(1, 2, 4, 7)/12 \rightarrow 2 + \sqrt{3}$  and  $Q(1, 3, 5, 9)/12 \rightarrow 2$ ) at one  $N$ -independent  $1/N$  rate ( $|Q(N) - Q_\infty| \cdot N \leq 4$  for both,  $N_x \in \{36, 48, 60\}$ ), the bulk control off-scale; [E] the two limits and the uniform rate, [C] so the MMST uniform-convergence *hypothesis* is empirically met. (xvii) *The entanglement reading of  $c_-$*  ([verification/v450\_seam\_edge\_entanglement.py]): a *third*, correlator-independent route to the edge central charge — the Calabrese–Cardy block-entropy law gives  $c_{\text{Dirac}} = 1.000$  for the critical free-fermion chain (vs a saturating,  $\ell$ -independent gapped control), so  $c$  per Majorana  $= \frac{1}{2}$  and  $c_- = 16 \cdot \frac{1}{2} = 8 = g_{\text{car}} + N_{\text{fam}}$ ; [E] the law and the control, [C] the assembly. (xviii) *The Ising operator tower* ([verification/v451\_seam\_edge\_cardy\_tower.py]): naming the edge CFT by its *operator content* — the Cardy finite-size spectrum reproduces the exact Ising chiral primaries  $h_\sigma = \frac{1}{16}$  (the periodic–antiperiodic ground split) and  $h_\epsilon = \frac{1}{2}$  (the lowest NS pair gap), both  $L$ -independent, so  $\{c, h_\sigma, h_\epsilon\} = \{\frac{1}{2}, \frac{1}{16}, \frac{1}{2}\}$  is the free-Majorana minimal model  $M(3, 4)$ ; [E] the dimensions and conformal scaling, [C] the identification. (xix) *The  $(E_8)_1$  torus modular data* ([verification/v452\_seam\_e8\_modular.py]): pinning the  $(E_8)_1$  identity on the *torus*, complementing the edge reads (xi)/(xiv)/(xvii)/(xviii) — the character  $\chi = E_4/\eta^8$  is  $S$ -invariant ( $\chi(-1/\tau) = \chi(\tau)$ , one primary),  $T$ -multiplies by  $e^{-2\pi i/3} = e^{-2\pi i c/24}$  ( $c=8$ ) and is holomorphic ( $c \equiv 0 \pmod{8}$ ,  $T$ -phase order 3); [E] the full modular signature. (xx) *The  $\mu_4$ -from-marks derivation* ([verification/v453\_seam\_mu4\_from\_marks.py]): folding the last structural premise into the existing one — the four Gauss–Bonnet marks *are*  $\mu_4$  (roots of  $z^4 - 1$ ), the form basis  $\omega_k$  is  $\mu_4$ -graded ( $\rightarrow i^k \omega_k$ , weights  $(1, 2, 3) = A_3$  exponents) and the order-4 clock  $z \rightarrow iz$  fixes their cross-ratio  $\lambda = 2$  (the  $D_4$  stabiliser, v168), so the residual QGEO.SYM.01 of (xiii) follows from marks  $= \mu_4$  plus the existing QGEO.REALIZE.01 — [E] the three mark facts, [C] no new premise: the chain (xx)  $\rightarrow$  (xiii)  $\rightarrow$  (xii) closes modulo realisation. The integer backbone of (xiv)–(xix) ( $c_- = 8$ , the Chern integers, the order-3  $T$ -phase) is kernel-hardened in Lean axiom-free as FORM.SEAM.EDGE.CHERN.01. (xxi) *The lattice Sugawara current algebra* ([verification/v454\_seam\_edge\_virasoro.py]): testing the cited MMST “Koo–Saleur generators  $\rightarrow$  continuum Virasoro with the correct  $c$ ” fact on the edge *algebra*, not just its central charge. The critical free-fermion ring has the Affleck–Cardy Casimir energy  $E_0(L) = e_\infty L - \pi v c / (6L)$  fitting  $c_{\text{Dirac}} = 1.000$  ( $\frac{1}{2}$  per Majorana,  $L = 64.1024$ ), the lattice  $U(1)$  current closes its affine *level* exactly ( $\langle 0 | J_n J_{-n} | 0 \rangle = n$ ,  $n = 1..8$ ; vanishing for a gapped control), and the level-1 Sugawara central charge  $c = \dim G / (1 + h^\vee) = 8$  for *both*  $SO(16)_1$  (120/15) and  $(E_8)_1$  (248/31); [E] Casimir, Kac–Moody level and Sugawara, [C]  $16 \cdot \frac{1}{2} = 8$  is the level-1 current-algebra  $c_-$ . (xxii) *The uniform-in- $N$  Tomita–Takesaki tower* ([verification/v455\_seam\_bw\_uniform.py]): the inductive-limit companion of v426 for sub-step (i) of (iv) — a family of reflection-odd,  $\mu_4$ -even, gapped collars ( $N = 8..128$ ) on which the intrinsic Bisognano–Wichmann condition  $u_\Theta = J$  ( $\|\Theta K \Theta + K\| = 0$ ) holds *exactly at every*  $N$ , with an  $N$ -independent gap ( $6 \log \frac{3}{2}$ ) and an  $N$ -independent neg-control margin ( $\|\Theta K \Theta + K\| = 2\|K\|$  for a side-even reflection); [E] the uniform BW, gap, clock symmetry and teeth, [C][O] so the inductive limit inherits  $u_\Theta = J$  (the abstract continuum theorem stays the cited backbone). (xxiii) *The chirality-from- $P_1$  forcing* ([verification/v456\_seam\_chirality\_from\_c3.py]): the S3 input of the continuum chain (v356) is removed as a free choice —  $c_3$ ’s “8” *is* the one-sided count  $8 = |\mathbb{Z}_2| \cdot (\int K/\pi) = 2 \cdot 4$  ( $8\pi = |\mathbb{Z}_2| \int K = 2 \cdot 2\pi \chi(S^2)$ , v73), and a reflection sends the edge Chern integer  $C \rightarrow -C$  (verified  $C = +1 \rightarrow -1$  on the explicit  $p+ip$  model), so any *two-sided* (reflection-symmetric) gapped boundary forces  $C = -C = 0$  while the *one-sided* seam — the same  $|\mathbb{Z}_2|$  quotient that gives  $c_3$  its  $8\pi$  — has no such reflection and forces  $C \neq 0$ ; [E] the one-sided count and the  $C \rightarrow -C$  flip, [C] chirality (S3) is a *consequence* of  $P_1$ , magnitude  $c_- = 8$ . (xxiv) *The  $(E_8)_1$  character kill*

*test* ([[verification/v457\\_seam\\_character\\_killtest.py](#)]): a *falsifiable* sharpening of the (x)/(x”) discriminators — the  $(E_8)_1$  tower  $\chi=E_4/\eta^8=q^{-1/3}(1+248q+4124q^2+34752q^3+\dots)$ , the weight-1 count  $248=120+128$  (vs the free 120) and the sector count  $|\det \text{Cartan } E_8|=1$  (vs  $|\det \text{Cartan } D_8|=4$ ); a measured 120, four sectors, or a wrong tower coefficient would *falsify*  $(E_8)_1$ , and the data give  $248/1/\{1, 248, 4124\}$  (passed). [X] the kill criterion, [E] the exact discriminators. (xxv) *The exact MMST citation audit* ([[verification/v458\\_seam\\_mmst\\_citation\\_audit.py](#)]): a hypothesis-by-hypothesis reading of the cited MMST theorem (arXiv:2107.13834) against the collar — H1 CAR class, H2 massless limit, H3 the per-copy  $c=\frac{1}{2}$  Thm 4.11, H4 decoupled additivity to  $c=8$ , and H5 the WZW range  $8 \leq 8 \leq 16$  with its 120 bilinear currents all *match*, isolating the single residual H6 to the 128-spinor extension  $SO(16)_1 \rightarrow (E_8)_1$  (the  $\det K 4 \rightarrow 1$  holomorphy step), *outside* MMST’s bilinear method; [E] the per-copy theorem, [C] the additivity and range, [O] the one open spinor extension — since (xxxii) certified at net level by the peer-reviewed crossed product. (xxvi) *The complementary lattice-VOA route* ([[verification/v459\\_seam\\_lattice\\_voa\\_route.py](#)]): the second, independent construction that supplies exactly H6 — for the even unimodular  $E_8$  the lattice net  $A_{E_8}$  (Adamo–Giorgetti–Tanimoto, arXiv:2506.01008; OS arXiv:2407.18222) has weight-1 space  $248=8+240$  with the 240 roots splitting 112 integer + 128 *half-integer* (the spinor,  $248=120+128$ ), so the lattice route builds the 128-spinor currents MMST (xxv) leaves open; [E] the even unimodularity and root split, [C] AGT constructs the holomorphic extension — since (xxxii) an *independent second witness*, the primary net-level certification being the peer-reviewed crossed product — [O] the routes leave one shared realisation input (the collar’s scaling limit  $= A_{E_8}$ ), tied to  $P_1$  by (xxiii). The G-block arithmetic (the one-sided  $8=|\mathbb{Z}_2| \int K/\pi$ , the reflection  $C \rightarrow -C$ ,  $c_-=8$ , the range  $8 \leq 8 \leq 16$ ,  $248=8+240=120+128$ ,  $240=112+128$ ,  $\det K 1$  vs 4) is kernel-hardened in Lean (FORM.SEAM.RESIDUAL.01, TfptCarrier/SeamResidualAxiom.lean), reducing the residual to ONE named TFPT-internal realisation axiom (CollarRealizedAsLatticePhase) plus the two cited existence theorems. (xxvii) *The explicit flat-band commuting-projector (LTO) realisation* ([[verification/v460\\_seam\\_s3\\_lto.py](#)]): NPW26 prove the intrinsic Bisognano–Wichmann lemma of (iv) for *commuting-projector* (local-topological-order) models, so sub-step (i) asks for the seam realised as such a  $\mathbb{Z}_2$ -reflection LTO. On the same v367  $p+ip$  collar the flattened Bloch  $Q=h/|h|$  is a frustration-free flat-band parent —  $Q^2=I$  and the occupied  $P=(I-Q)/2$  is an exact projector EQUAL to the ground-state projector — whose real-space kernel is gap-protected quasi-local ( $\xi \approx 1.14$  at  $M=1$ ,  $|P(20,0)| < 10^{-6}$ ; a clean power law  $|R|^{-2.1}$  at the gapless point, so locality needs the gap), and whose flattened modular Hamiltonian  $K_{\text{flat}} = \text{sign}(K)$  inherits the  $\mathbb{Z}_2$  reflection  $\Theta K_{\text{flat}} \Theta = -K_{\text{flat}}$  and the  $\mu_4$  clock  $[\rho, K_{\text{flat}}]=0$  EXACTLY; [E] the flat-band/projector and reflection algebra, [C] the quasi-locality, [O] the chiral  $c_-=8 \neq 0$  obstructs *strictly* finite-range commuting-projector locality (Kapustin–Fidkowski, arXiv:1810.07756), so the realisation is exactly the quasi-local LTO net NPW26 use — advancing sub-step (i) of (iv) from abstract existence to an explicit gap-protected net, strict locality the cited residual. (xxviii) *The strict-locality obstruction made explicit* ([[verification/v461\\_seam\\_strict\\_locality.py](#)]): upgrading (xxvii)’s “strict locality is the cited residual” from a citation to an EXHIBITED topological obstruction. On the same v367 collar the FHS Chern is  $C=+1$  (chiral) / 0 (trivial) /  $-1$  (opposite mass), and the hybrid Wannier-centre (Wilson-loop) phase  $\theta(k_y)$  winds by  $|C|=1$  (chiral) and 0 (trivial) over the BZ — exponentially-localised Wannier functions exist *iff* that winding is 0 (Brouder–Panati et al.; Thouless), so the chiral collar has none; both phases are gapped, yet only the trivial one (winding 0) admits a strict finite-range commuting projector, so it is the *chirality*  $c_-=8 \neq 0$ , not the gap, that obstructs. [E] the Chern integer, the winding  $= |C|$  identity and the gapped neg-control, [C][O] Kapustin–Fidkowski (arXiv:1810.07756): a strictly finite-range commuting projector would force winding 0 hence  $C=0$ , so for  $C=1 \neq 0$  none exists — sub-step (i)’s realisation is *provably* the quasi-local NPW26 LTO net, strict locality topologically forbidden rather than a missing premise. (xxix) *The character-level 128-spinor extension* ([[verification/v462\\_seam\\_spinor\\_continuum.py](#)]): the constructive companion of (xxv)/(xxvi), exhibiting the 128-spinor extension three ways — the exact Jacobi/ $E_8$  identity  $\theta_2^8 + \theta_3^8 + \theta_4^8 = 2E_4$  makes the holomorphic character the sum of the  $SO(16)_1$  vacuum and spinor sectors,  $\chi_{(E_8)_1} = E_4/\eta^8 = \chi_o^{SO16} + \chi_s^{SO16}$  (weight-1 counts  $120+128=248$ ); the finite 16-Majorana Fock space already carries them ( $C(16,2)=120$  currents and  $2^{16/2}=256=128+128$  Ramond states); and a critical free-fermion ring converges to the chiral CFT (Casimir  $c_- \rightarrow 8$ , lowest scaled gap  $x_1 \rightarrow 1$  with  $|x_1-1| \sim 1/N^2$ ). [E] the  $\theta$  identity, the  $120+128=248$  split and the finite- $L$  convergence, [C][O] the rigorous holomorphic extension is certified at net level by the crossed product (xxxii), with AGT/AMT (xxvi) as the independent second witness. (xxx) *The  $c=8$  holomorphic-uniqueness pin* ([[verification/v463\\_seam\\_c8\\_holomorphic\\_uniqueness.py](#)]): the positive lever that makes the *identification* half classification-forced rather than assumed. At level 1 the central charge  $c = \dim \mathfrak{g}/(1+h^\vee)=8$  has *three* simple solutions —  $A_8$  (dim 80),  $D_8=\mathfrak{so}(16)$  (dim 120) and

$E_8$  (dim 248) — so  $c=8$  is necessary but *not* sufficient, and the  $\det K=4$   $SO(16)$  rival of (xxiii) is a genuine same- $c$  competitor, not a strawman. But a holomorphic (single-sector)  $c=8$  theory has an  $SL(2, \mathbb{Z})$ -covariant character, and the only weight-0 form with leading  $q^{-1/3}$  and integer coefficients is  $j^{1/3}=E_4/\eta^8$ , whose  $q^1$  coefficient (the one-dimensional candidate space, vacuum  $a_0=1$ ) is 248 — so  $\dim V_1=248$  is *forced*, selecting  $E_8$  uniquely and excluding  $A_8/D_8$ . [E] the three candidates, the forced 248, the tower  $\{1, 248, 4124, 34752, 213126\}=E_4/\eta^8=j^{1/3}$ ; [C] Dong–Mason/ Schellekens give the holomorphic  $c=8$  VOA unique  $= V_{E_8}$  (the even unimodular rank-8 lattice unique  $= E_8$ , Mordell–Witt), so “ $c=8$  holomorphic limit  $=(E_8)_1$ ” is classification-forced. (xxxi) *The one-particle realisation rigidity* ([verification/v464\_seam\_oneparticle\_rigidity.py]): attacking the *realisation* input  $R_1$  itself. Since the seam is quasi-free (v113/v160/v161), by Araki’s self-dual CAR the whole net is the second quantisation of its one-particle symbol  $P$ , so  $R_1$  collapses to a finite-per-mode statement about  $P$ : the gapped ground state is the *unique* quasi-free state with  $P=$  the spectral projection onto  $K<0$  ( $P^2=P$  to  $<10^{-12}$ ,  $K$  fixed by gap + chirality + reflection v426); its real-space kernel is Cauchy on every fixed window ( $\max|P_{2N}-P_N|\sim 1/N\rightarrow 0$ , so  $P_\infty$  exists), and the limit carries  $c=1$  per Dirac (entanglement  $S(L)=(c/3)\ln L$ , 16 copies  $\Rightarrow c_-=8$ , tying v450). [E] idempotency, Cauchy convergence and the  $c$ -fit; [C] Araki/Shale–Stinespring give the state $\leftrightarrow$ symbol bijection and implementable Bogoliubov map, so  $R_1$  is upgraded from a bald axiom to “the unique quasi-free realisation, its one-particle limit exhibited, modulo the named CAR functor”. (xxxii) *The net-level crossed-product route + the invariant-level realisation* ([verification/v469\_seam\_crossedproduct\_route.py]): both halves of the residual re-founded on harder-to-reject ground. The 128-spinor extension needs no preprint: the  $SO(16)_1$  spinor has  $h_s=N/16=16/16=1\in\mathbb{Z}$ , so its statistics phase is  $+1$  — *exactly* the Longo–Rehren locality criterion — and the  $\mathbb{Z}_2$  simple-current crossed product  $SO(16)_1\rtimes\{1, s\}$  is a *local* index-2 extension by 1995–2001 *peer-reviewed* subfactor theory (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1  $so(N)$  net with its DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions of nets: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM (Comm. Math. Phys. 219 (2001)) gives  $\mu(B)=4/2^2=1\Rightarrow$  holomorphic  $\Rightarrow (E_8)_1$  by the (xxx) pin — so the AGT/AMT route (xxvi) is *demoted to an independent second witness*. The index-4  $\mu_4$  glue runs on the same integer,  $h(J^k)=\{1, 1, 1\}$  for  $k=1, 2, 3$  (the v125 isotropy  $q(k(1, 1))=k^2$  at net level,  $\mu=16/4^2=1$ ). And the realisation input  $R_1$  is reduced from model fiat to *invariants* ( $R_1'$ ): quasi-free [C] (v155/v160) + gap [E] (v302) + class D +  $c_-=8$  [E] ((xxiii), from  $P_1$ ) — computed on the collar as FHS Chern  $|C|=1$  (control 0),  $\nu=16\times|C|=16$ ,  $c_-=\nu/2=8$ ; and  $\nu\equiv 0 \pmod{16}$  is exactly Kitaev’s class-D phase whose edge admits a purely bosonic description = the  $(E_8)_1$  level-1 state (the sixteen-fold way;  $c_-$  a *phase* invariant, Kapustin–Spodyneiko; Kim et al.), so any realisation with the  $R_1'$  invariants is phase-equivalent to the v367 stack. [E] the locality integers, the KLM  $\mu$ -arithmetic, the computed invariants and the composed chain; [C] the re-founded citation package; [O] the cited theorems stay cited. The full G-block arithmetic — now including the (xxviii) winding obstruction, the (xxix) finite-Fock spinor counts, the (xxx) holomorphy selector and the (xxxii) locality/ $\mu$ /sixteen-fold joints — is kernel-hardened in Lean (FORM.SEAM.RESIDUAL.01), the two formerly separate cited theorems are *merged* into one combined external axiom `seam_realisation_theorem`, so the residual reduces to ONE realisation axiom plus ONE cited theorem (#print axioms drops from six to four); and the (xxxii) route is Lean-carried as the *parallel* derivation `seamResidualClosed'` (the invariant-level axiom `collar_invariants` plus the single combined package `crossedproduct_route_theorem`, its joints — `lr_locality_integer`, `glue_locality_integers`, `klm_holomorphic`, `sixteenfold_pin` — kernel facts with no axioms). Across (v)–(xxxii), SEAM.EQUIV.01 stays [O]: with (xxx) pinning the identification, (xxxi) reducing the realisation to its one-particle data and (xxxii) re-founding the extension leg on peer-reviewed ground, the residual is now *entirely certification* — a named, hypothesis-audited package of published theorems (MMST, Longo–Rehren/Böckenhauer/Böckenhauer–Evans/KLM, Dong–Mason, Araki, Shale–Stinespring; AGT/AMT as second witness) with no open internal mechanism — yet still rests on cited continuum-existence theorems we do not re-prove.

The TOE/QFT closure contracts as three citable certificates ([verification/v398\_seam\_state\_rigidity.py], [verification/v399\_gravity\_complete.py], [verification/v401\_metrology\_closure.py]). Three companion contracts package the closure status as named targets without fabricating any closure. (i) *Seam State Rigidity* (SEAM.RIGIDITY.01): the scattered state-invariance reductions (v177/v199/v308/v309/v335) as ONE theorem target — [E] the RP-definability hinge (OS canonical transfer from RP + reflection alone, v54, quasi-free v155), [E] the band-limited  $\mu_4$ -character-block-diagonal core swept  $N=4..64$  (beyond the 3-dim  $H^1$ , made uniform in  $N$  to  $N=256$  by v442), [E] holomorphy  $\Rightarrow (E_8)_1$ ; and [E] the forcing converse (v445) that commuting with the order-4 clock is *exactly*  $\mu_4$ -block-diagonality, so the residual is no longer

“does block-diagonal permit” but only  $[\rho, \Lambda_\Sigma] = 0$  on the full  $L^2$  — the physical transfer lying in that commutant (intrinsic Bisognano–Wichmann). [E] Moreover (v446) that operator commutation is itself *derived*: the OS canonical transfer of  $\mu_4$ -symmetric RP data inherits the clock (with the modular Hamiltonian commuting, v258), so the residual reduces further to the *structural* statement that the seam RP data is  $\mu_4$ -symmetric (the four marks, QGEO.SYM.01), leaving [O] only that structural symmetry on the raw continuum collar — for which the four marks now have an *established* continuum frame: the four-interval multilocal modular geometry (v480, keybox below), where the invariance is manifest at the state level in the exactly solvable class. (ii) *Gravity-complete* = *Boundary-complete* + *Ambient-redundancy* (GRAVITY.COMPLETE.01): the local equation parameter-free (v358/v359) plus the redundancy discriminators (v369), with  $\mathcal{O}_{\text{phys}} \subset \mathcal{A}_\Sigma$  checked over five gravitational readout classes. (iii) *Metrology closure* (METROLOGY.CLOSURE.01): the final input set  $\{a, \pi, v_{\text{geo}}\}$  — 0 dimensionless dials + 1 unit, the dimensional bookkeeping that every dimensionful observable is a number  $\times v_{\text{geo}}^d$  (No-Unit, v153), and  $v_{\text{geo}}$  theorem-forbidden ( $\text{TOE}_{\text{strict}} = \text{TOE}_{\text{dimensionless}} + [v_{\text{geo}}]$ , v384).

**One surface for seam, carrier and lattice: the Kummer/K3** ([verification/v260\_k3\_kummer\_unification.py]). The pillowcase is the elliptic-involution quotient  $T_{\tau=i}^2/(z \mapsto -z)$ , its four marks the four 2-torsion fixed points; squaring the construction to complex dimension two gives the *Kummer surface*  $\text{Km}(T^2 \times T^2)$ , a K3, whose three facets reproduce three TFPT objects at once. [E] the 16 nodes (the 2-torsion  $|A[2]| = 2^4 = 16$  of the abelian surface) resolve to 16 exceptional  $\mathbb{P}^1$ 's =  $\dim S^+ = \Lambda^{\text{even}}(\mathbb{C}^5) = 16$  (v197), the carrier-16 as the *square* of the seam-4; [E] the cup-product form on  $H^2(\text{K3}, \mathbb{Z})$  is the even unimodular lattice  $U^3 \oplus E_8(-1)^2$  of signature (3, 19), so the same  $E_8$  the carrier glues to (v1) is the surface's intersection form. (Honestly: the 16 node classes span the rank-16 Kummer lattice, a sublattice *distinct* from the  $E_8(-1)^2$  summand — both exact, not identified.) Thus the v1 “glue  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ ” is the lattice shadow of a single smooth object that already carries the seam and the carrier;  $E_8$  appears both *locally* as the du Val  $\mathbb{C}^2/2I$  singularity (the Poincaré-sphere link, v232/v236; by Kronheimer that local model *is* the hyper-Kähler quotient of the marks quiver, v479) and *globally* as this  $H^2$  summand. A unification (ARCH.K3.01), *not* a closure: the lattice arithmetic is [E], the carrier  $\leftrightarrow$  node identification [C].

**Synthesis — the Modular Spectral Closure** ([verification/v261\_modular\_spectral\_closure.py]). The ten steps above are not ten independent results: they are facets of one relative spectral datum

$$\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$$

(seam algebra, KMS state, modular generator,  $\mu_4$  clock; the carrier finite triple; the relative spectral action), governed by a single *Modular Spectral Universality* statement: *the unique reflection-positive, holomorphic, index-4 KMS seam state  $\omega_\Sigma$  minimises the modular clock defect, induces via the Kasparov/covariance map (v258) the 96-dim KO-6 finite triple  $D_F$  (v252), whose inner fluctuations (v254) generate the gauged Pati–Salam spectral action with the seam KMS cutoff (v259) — so Dirac, cutoff, gauging, glue and orientability are not separate choices but readouts of one seam state. The certificate v261 pins the cross-consistency: one number  $4 = [B:A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$  (index = marks = fixed points = glue order), one carrier-16 ( $\dim S^+ = \Lambda^{\text{even}}(\mathbb{C}^5) = \text{Kummer nodes}$ ,  $H_F = 96$ ), and one gap  $\Delta = 6 \log(3/2)$  (OS mass gap = recovery rate = Lindblad gap, static and dynamic). The honest account of completeness is therefore:*

- [E]/[C] **QFT-complete (modulo one premise)**. The admissible OS sector, the boundary  $(E_8)_1$  net, the relative spectral action,  $D_F$  (as covariance induction), the cutoff (as the KMS weight) and the gauging (as inner fluctuations) are one relative object — closed once QGEO.SYM.01 (the seam realisation: *the raw quasi-free seam state is  $\mu_4$ -invariant*) is granted. That single [O] premise is the only structural input the QFT layer still consumes.
- [C] **Ambient QG — a redundancy, separate by design**. The full covariant metric-sector / ambient quantum-gravity measure (QG.AMB.01) is gap-decoupled ( $\Delta_{\text{eff}} = 1.648 > 0$ ) and is now a [C] *redundancy* (v369), *not* a blocker for the QFT layer: a certification object, not a missing piece of the boundary QFT.

So “solving the QFT” here means exactly this: the boundary QFT is a single relative object reduced to one foundational symmetry postulate, with ambient QG held apart — not a diffuse list of open interfaces. The MSC closes the boundary relative QFT object modulo QGEO.SYM.01; it does *not* assert SM-only 4D gauge unification, and the canonical 4D reading is boundary only.

#### 4D-QFT fork policy — a frozen decision tree, not an open ambiguity ([verification/v265\_qft4d\_fork\_freeze.py])

The 4D reading is a *branch policy*, machine-frozen in `qft4d_fork_freeze.json`:

- **A (canonical) — boundary only.** [C] *Boundary-only is the canonical 4D reading*: TFPT delivers the boundary QFT, the relative spectral action and the SM readout; a 4D-GUT is *not claimed by default* ( $E_8$  is the audit hull, never an unbroken 4D gauge group).
- **SM-only 4D-GUT unification is killed.** [X] the plain SM couplings do not meet (at the  $\alpha_1=\alpha_2$  scale  $\alpha_3^{-1}$  is off by  $\sim 5.6$ ; spread  $\sim 8.8$  at  $2 \times 10^{16}$  GeV, v246). This is the canonical *expectation*, not a failure.
- **B (conditional) — carrier-native Pati–Salam UV branch.** [C]/[O] permitted *only* with the  $E_8$ -allowed content  $\{1, 10, 16, 45\}$  (no **126**, v247),  $\sin^2 \theta_W = 3/8$  (v248),  $\kappa = M_{PS}/M_s$  in the frozen  $O(1)$  band (v249/v253), and a threshold model with a proton-decay kill test; carrier gauging itself stays the contract `CONTRACT.QFT4D.DIRAC.01`.
- **Proton-decay rule, now applied** ([verification/v266\_ps\_threshold\_proton.py]). [X] the explicit two-step thresholds give  $d=6$   $\tau_p(p \rightarrow e^+ \pi^0)$ : the *minimal* 16-Higgs content ( $M_{GUT} \sim 2.4 \times 10^{15}$ ) gives  $\tau_p \sim 4.4 \times 10^{33}$  yr < Super-K ( $2.4 \times 10^{34}$ ) and is *killed*; the  $E_8$ -allowed (15, 1, 1) [the single 45] raises  $M_{GUT} \sim 5.9 \times 10^{15} \Rightarrow \tau_p \sim 1.6 \times 10^{35}$  yr, proton-safe *now* but sitting at the Hyper-K reach ( $\sim 1.4 \times 10^{35}$  yr) — so branch B is *content-selected* (must be  $+(15, 1, 1)$ ) and *Hyper-K-decisive within the decade*, not an open-ended hope (only one 45 in the hull, so it stays structurally marginal;  $\tau_p$  carries an  $O(3)$  hadronic band).

The fork is thus a frozen decision tree with explicit promotion/demotion rules, *not* an open ambiguity.

**And the bedrock is now a falsifiable kill-test, not a proof-promise** ([verification/v215\_seam\_deck\_killtest.py]). A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the very covariance it should produce); the honest move is to recast `QGEO.SYM.01` as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (v198/v199/v201) gives  $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$  for the quasi-free covariance, with the principal symbol  $|k|$  commuting exactly, so the whole residual sits in the bounded sub-principal  $M_f$  (equivalently: the raw Gaussian bulk form  $A$  is  $\mu_4$ -deck-invariant). The four levers are **K1** exactly four marks ( $b_1 = \# \text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \# \text{marks} = 4 = |\mu_4|$ ); **K2** the *square* configuration (cross-ratio  $2 \Rightarrow j=1728$ ,  $\text{Aut} = \mathbb{Z}/4$ ; the clock order  $h(A_3)=4$  selects  $\mathbb{Z}/4$ , not the  $j=0$   $\mathbb{Z}/6$ , and a generic mark set has only  $\mathbb{Z}/2$ ); **K3** the mod-4 sub-principal support ( $[\rho, M_f] = 0 \Leftrightarrow f$  is  $\mathbb{Z}_4$ -invariant, automatic for a  $\mu_4$ -mark sum); and **K4** the transfer gap  $(2/3)^6 = 64/729$ . These are frozen in `freeze_file.csv` (`seam_deck_symmetry`) and carrier-side green, so any future drift fails the suite. The *decisive* kill is deferred to `QGEO.REALIZE.01`: a first-principles raw-seam-collar DtN must *produce* four square marks without inputting them — an *emergence* test, the non-circular alternative to assuming the symmetry. The *light* version of that deferred test is now *executed* ([verification/v503\_qgeo\_emergence\_light.py], `QGEO.EMERGE.LIGHT.01`), with the result *rigidity, not dynamical selection*: the free field does *not* select the square — the global  $\det'$  winner at fixed area is the *hexagonal* torus ( $\log F(\rho) = -1.03352 > \log F(i) = -1.05469$ , Osgood–Phillips–Sarnak),  $\tau = i$  is a saddle of the full moduli space, the only critical points are the CM points ( $E_2^*(i) = E_2^*(\rho) = 0$ ), and criticality at the square is Palais symmetric criticality — while the square *is* pinned sharply by the clock: the raw free-field DtN is mod-4 block-diagonal *iff*  $\tau = i$  (an isolated zero of the K3 indicator as a modulus function,  $8 \times 10^{-15}$  at  $\tau = i$ , strictly monotone away), and  $\text{Aut}(E_\rho) = \mathbb{Z}_6$  carries no order-4 element. The clock input itself is now *reduced to one bit* ([verification/v506\_seam\_clock\_rigidity.py], `SEAM.CLOCK.RIGIDITY.01`): the deck has exactly two Möbius square roots, both automatically of order 4, and a marks-preserving root exists *iff* the deck is the central involution of the mark-stabiliser  $D_4$  — given that alignment bit, order, uniqueness, the free 4-cycle and  $\tau = i$  are all theorems (harmonicity alone is insufficient on bare mark data, exact counterexample — on free-deck circles it becomes sufficient, v510); on the Fock side the NS deck implementation is *forced* to order 4 ( $U^2 = (-1)^F$  exactly, nonsplit  $\mathbb{Z}_4$ ; the R sector splits), so the fermionic  $\mathbb{Z}_4$  explains the *order* and only the spatial alignment remains the carrier input. That bit has now survived its *tautology attack* ([verification/v507\_seam\_bit\_origin.py], `SEAM.BIT.ORIGIN.01`): the marks are the four half-periods of the seam double and *every* mark-free collar deck is a half-period translation — the deck's *class* is derived from the common torus origin — yet the origin does not force the alignment (the marks-preserving root exists *iff* the deck pairs separate harmonically,  $\lambda \in \{-1, 2, \frac{1}{2}\}$ , and the silver counterexample sits in-family as “ $\tau = i$  with

a non-CM-fixed half-period”); the bit is an equivalence tetrad (clock  $\iff \tau=i$  with the CM-fixed half-period  $\iff$  deck central in the mark- $D_4 \iff$  harmonic deck pairs  $\iff$  nonsplit NS Fock lift) whose physical face is the Fidkowski–Kitaev-type extension class (central  $U^2 = (-1)^F$  vs edge  $U_e^2 = +1$ , split) — distinguished, but not derived: only the *position* (which half-period) remains the carrier input. That position half is now *topology* ([[verification/v510\\_seam\\_bit\\_freedom.py](#)], SEAM.BIT.FREEDOM.01): the collar deck is a *covering* deck, hence free on the seam circle; the unique free involution is the antipode ( $\text{Aut}(C_{16}) = D_{16}$ , complete census), nonsplit  $\iff$  free over all 17 involutions in exact  $\text{Cl}(16)$ , and the edge/silver class — whose mark circle passes through the deck poles, the restriction of the *branched* pillowcase involution — is excluded; harmonic + free solves exactly to  $b = \pm i$ , so the bit reduces to the square-modulus datum  $\tau = i$  alone. That datum now carries an equivalent *discrete* form ([[verification/v512\\_seam\\_tau\\_flag.py](#)], SEAM.TAU.FLAG.01): on the free seam circle bare mark-transitivity is automatic for every deck-invariant configuration  $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$  (the pair-exchanging  $V_4$ ; no local jet sees the side bit), and what selects  $\delta = \pi/2 \iff \tau = i$  is exactly *flag* transitivity ( $V_4 \rightarrow D_4$  symmetry lift), welded into a 13-fold exact equivalence web (since 15-fold by [v528](#) facets #14/#15) — the remaining input is *one discrete symmetry-lift bit*, its negation concretely measurable (odd-doublet split  $(2/\pi) \ln \cot(\delta/2) \neq 0$ ). The candidate [E]-closure route through free OS positivity is since *decided negative* ([[verification/v521\\_seam\\_bit\\_rp\\_blind.py](#)], SEAM.BIT.RPBLIND.01): free reflection positivity and anti-linear  $\Theta$  existence (the [v519](#) battery) are *side-blind* along the whole  $M(\delta)$  family — bond-cut Gram spectra identical to 40 digits for every tested  $\delta$ , the  $N = 16$  odd- $m$  failure a lattice-placement artifact resolved at  $N = 32$ , the continuum kernel exactly  $1/\sin((s+t)/2)$  with the axis position dropping out — so the free RP/ $\Theta$  battery is the *eighth* side-blind test on the [v512](#) scoreboard ( $7 \rightarrow 8$ ); the only  $\delta$ -sensitive clauses presuppose the clock (the group generated by the admissible OS reflections closes to  $D_4$  exactly at  $\delta = \pi/2$ : face #13 of the web, a typed [C] reformulation), and the honest residual [O] gap is the mark-decorated *state* class (defect/twist insertions, the interacting  $\mathcal{A}_{\text{hol}}$ ) — the mark-decorated half of that gap is since decided side-blind too ([[verification/v525\\_seam\\_bit\\_twist\\_blind.py](#)], SEAM.BIT.TWISTBLIND.01: every well-defined mark-decorated state functional in the free-plus-twist class keeps RP positive along the whole  $M(\delta)$  ladder — the *ninth* side-blind test,  $8 \rightarrow 9$ ; the free-plus-twist class is exhausted) and the interacting half is since decided too ([[verification/v529\\_seam\\_interacting\\_toy\\_fk.py](#)], SEAM.INT.FKTOY.01: the *tenth* side-blind test,  $9 \rightarrow 10$ , on genuinely non-Wick dynamics —  $\Theta_{15}$  mirrors  $H(m) \rightarrow H(8-m)$  exactly despite  $\delta$ -seeing spectra) — while the bit itself is now *physically defined* as the twist-class choice ([[verification/v528\\_seam\\_bit\\_twist\\_class\\_definition.py](#)], SEAM.BIT.TWISTCLASS.01: facets #14/#15 both directions, order parameter  $O$ , unification with flag-transitivity, web  $13 \rightarrow 15$ ; [C] interferometric measurement sketch; the bit remains formal input) — and since carries the first *positive* selection datum: the straddle filter, run as a selector on the interacting mark family, keeps exactly *one* member alive —  $\delta = \pi/2$  with positive coupling ([[verification/v534\\_seam\\_straddle\\_cone.py](#)], SEAM.STRADDLE.CONE.01; toy-level evidence for a dynamical origin, not a derivation). The full QGEO.REALIZE.01 stays [O]; no marker moves. [E] carrier-side (the four levers), [O] seam-side (the emergence): a kill-test, *not* a closure (QGEO.KILL.01).

**And the mark count is now *derived*, not assumed: four marks from Gauss–Bonnet** ([[verification/v216\\_marks\\_gauss\\_bonnet.py](#)], [[verification/v217\\_marks\\_emergence\\_scan.py](#)]). The K1 lever (“exactly four marks”) is the one that can be *executed* today. If the marks are  $\mathbb{Z}_2$  branch points (cone angle  $\pi$ , deficit  $\pi$ ) on the flat seam sphere ( $\chi=2$  — the same  $\chi$  that supplies the 8 in  $c_3$ ), Gauss–Bonnet *forces* their number:  $n\pi = 2\pi\chi = 4\pi \Rightarrow n = 2\chi = 4 = |\mu_4| = N_{\text{fam}}+1$ . The closed Euclidean sphere 2-orbifolds are exactly  $(2,3,6)$ ,  $(2,4,4)$ ,  $(3,3,3)$ ,  $(2,2,2,2)$ , and *three independent* criteria — all cone orders = 2 (the  $|\mathbb{Z}_2|$  sheet branch), rank  $H^1 = \#\text{marks} - 1 = 3 = N_{\text{fam}}$  (the three generations pick the 4-mark square over the 3-mark hexagonal  $(3,3,3)$ ), and a free order-4 deck — converge on the pillowcase  $(2,2,2,2)$ . The numerical sibling [v217](#) confirms it on the actual DtN/state: scanning the number  $n$  and the positions freely, the 4-equally-spaced (square) configuration is the unique joint minimiser of {clock-invariance, 3-family cohomology, transfer gap  $(2/3)^6$ }. So the mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence QGEO.REALIZE.01 stays [O]. The light emergence test ([[verification/v503\\_qgeo\\_emergence\\_light.py](#)]) sharpens this residual to a *convergence*: given the clock, the modulus freezes automatically ( $P(iZ) = P(Z)$  forces  $a_3 = a_2 = a_1 = 0$  with  $j = 1728$  identically, [v493](#)), and the order-4 element on  $H^1$  is the Coxeter element of  $W(A_3)$  — exactly the P2 cusp-class carrier datum of [v499](#): the QGEO modulus rest and the P2 typing rest R1 are *one* common residual (“why does the raw seam carry the order-4 clock”), not two. And the clock-rigidity theorems ([v506](#)) reduce that common residual to *one alignment bit* — “the collar deck is the central involution of

the mark- $D_4$ ” — with the clock’s *order* fermionically forced ( $U^2 = (-1)^F$  exact); the bit-origin theorems (v507) then derive the deck’s *class* (every mark-free collar deck is a half-period translation of the seam double) and characterise the bit as an equivalence tetrad with a physical extension-class face, so only its *position* — which half-period — stays the input; the freedom theorems (v510) finally type that position half as *topology* (the collar deck is a covering deck, hence free; the edge class is excluded), so the residual is the square-modulus datum  $\tau = i$  alone — restated by the flag-transitivity web (v512) as one discrete symmetry-lift bit ( $V_4 \rightarrow D_4$ : flag transitivity  $\iff \tau = i$ ), and since v528 equivalently as the *twist-class choice*, in principle interferometrically readable (SEAM.BIT.TWISTCLASS.01; the bit remains formal input). [E] (count + equality), [O] (the square modulus and the raw-seam realisation) — QGEO.MARKS.03 / QGEO.EMERGE.01.

And the four marks are a *four-interval modular geometry* ([verification/v480\_multilocal\_four\_interval.py]). The seam circle minus its four marks is the  $n=4$  disjoint-interval region of a chiral free fermion — the one continuum class whose *multi-interval* modular theory is exactly known (Casini–Huerta mixing term; Longo–Xu multi-interval subfactors; Rehren–Tedesco multilocal fermions; and Kawahigashi–Longo–Müger: *holomorphic*  $\iff$  trivial multi-interval inclusion,  $\mu=1$  — the same one bit as  $\det K=1$ ). Until now the suite used only the single-ball CHM boost (v358); v480 exhibits the multilocal mechanism on the exactly solvable antiperiodic critical ring: [E] the quarter-turn clock is a signed permutation with  $\rho^4=-1$  *exactly* (the fermionic  $\mu_4$  clock is the order-8 double-cover lift — the same  $\mathbb{Z}_2$ -lift pattern as  $2I$  over  $I$ ); [E] its primitive-8th-root character projectors decouple the 4-interval covariance at  $10^{-15}$  (the diagonal 4-fold-cover isomorphism), each sector a *single-interval* quarter-ring kernel with boundary twist ( $10^{-14}$  spectra); [E]  $\rho C \rho^{-1} = C$  at  $10^{-14}$ , so  $\omega \circ \rho = \omega$  holds *manifestly* at the 4-interval state level — and a one-site displacement of a single interval re-couples the sectors at  $10^{-2}$  (the invariance is the  $\mu_4$ -symmetric configuration, not an artifact); and the cross-interval modular coupling peaks on the conjugate-point diagonal, the lattice shadow of the Casini–Huerta mixing term (numerical). [C] the continuum multi-interval theory is cited, not re-derived; [O] the raw-seam premise (QGEO.SYM.01) is untouched — this puts the bedrock and the AP2 continuum leg (v476/v478) in one established, exactly solvable frame; it does not close them.

#### QGEO.SYM.01 is *not* a theorem — it is the final seam-identification premise

The single load-bearing bolt of the whole construction, on one page.

1. **What is identified.** That the conformal structure of the raw RP seam double *is* the  $\mu_4$ -deck structure of the carrier clock — i.e. the carrier (P2) and the seam’s conformal geometry are the same object. Equivalently, in Riemann–Roch *index* language: the seam’s degree-4 mark divisor on the genus-0 double has  $h^0 = \deg + 1 = 5 = g_{\text{car}}$  and  $\text{rank } H_1 = \deg - 1 = 3 = N_{\text{fam}}$  (with  $\Lambda^{\text{even}}(\mathbb{C}^5) = 1+10+5 = 16 = \dim S^+$ ), so P2 is the genus-0 mode count of the four-mark seam divisor ([verification/v228\_rr\_index\_gate.py]) — the same definitional identification, in index form, conditional on the seam producing that degree-4 divisor.
2. **Why it is weaker than QGEO.CONF.01/ISO.01.** It does *not* name  $\mathbb{P}^1$  or  $\mu_4$  (uniformisation *produces* them) and asks only for a conformal *symmetry*, not a metric isometry or a pre-named conformal class — strictly the mildest of the three.
3. **What follows automatically (the conditional theorem).** *Given* QGEO.SYM.01, equivariant uniformisation puts the order-4 clock in the form  $z \mapsto iz$  on  $\mathbb{P}^1$ , the four parabolic marks are its free orbit  $\mu_4$ ,  $H^1$  carries the  $(1, 2, 3)$  grading, the seam Calderón contraction is the rank-8  $K_\Sigma$  polarisation, and the index-4 simple-current extension is  $(E_8)_1$  — the *entire* AQFT closure of v175–v181.
4. **What would falsify or unseat it.** A seam double of genus  $> 0$  (uniformisation to  $\mathbb{P}^1$  fails); a carrier clock that is *not* a conformal symmetry of the seam geometry (a non-deck transport); or a fourth chiral generation ( $N_{\text{fam}} \neq 3$  collapses  $b_1 = N_{\text{fam}}$  and the four-mark  $\mu_4$  structure). The first two are constructive-geometry questions about the raw seam; the third is the standard family-count kill test.

**Honest framing.** The correct external statement is *conditional*: “*Given* QGEO.SYM.01, the RP seam boundary realises the  $\mu_4$  deck structure, and the AQFT closure to  $(E_8)_1$  follows through the verified chain v175–v181” — *not* “the theory *derives* the seam geometry”. The premise stays [O]; everything downstream of it is [E]. Per v323 (Bisognano–Wichmann) QGEO.SYM.01 is itself the *downstream* conformal-deck face of the single named keystone SEAM.EQUIV.01 (the raw RP seam = the holomorphic  $(E_8)_1$  net at  $\tau=i$ ), which is now *closed modulo cited theorems on its MMST route* SEAM.EQUIV.MMST.01 (the lattice/ $S_3$

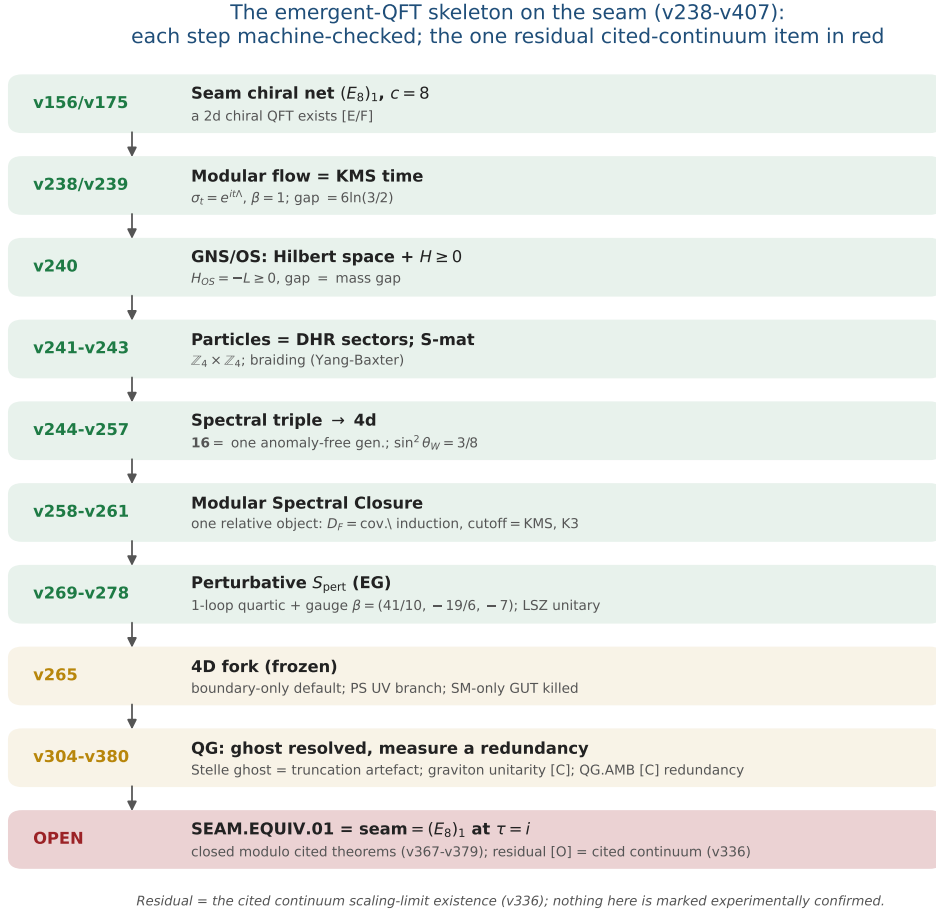


Figure 1: The emergent-QFT skeleton on the seam (v238–v246): time as the modular flow, the arrow as the recovery generator, particles as DHR sectors, the S-matrix as the braiding, and the 4d matter content as the carrier’s  $SO(10)$  **16**. Each step is machine-checked; the one residual — SEAM.EQUIV.01, whose MMST route SEAM.EQUIV.MMST.01 is closed modulo cited theorems (v367–v379) with the cited continuum scaling-limit existence (v336) as its [O] leg (parent [O]) — is in red, the frozen 4D fork and the QG certification layer in gold. The whole skeleton is *conditional* on that keystone and is neither solved nor experimentally confirmed.

stack pins it at every computable level, Lean FORM.SEAM.MMST.01; only the cited continuum existence stays [O]; the twistor route SEAM.EQUIV.TWISTOR.01 and the unconditional parent stay [O]).

**This is not a gap to apologise for — it is TFPT’s one fundamental postulate.** Every fundamental theory rests on an irreducible premise (the constancy of  $c$  in relativity, the equivalence principle, the canonical commutator). QGEO.SYM.01 is that premise for TFPT: a single geometric identification (seam conformal structure = carrier  $\mu_4$  deck) from which the entire algebraic and topological structure unfolds *by theorem*. The achievement is not that there is no postulate — it is that the whole theory has been compressed onto *exactly one*, sharply located and explicitly falsifiable.

**The Seam-Holomorphy Theorem — the one named closing target [E]/[O]**  
([verification/v234\_seam\_holomorphy\_selection.py])

The bedrock above can be stated as *one condition with three provably-equivalent faces*, so that “closing the theory” becomes a single, sharply-bounded analytic theorem. **The condition:** the seam carries *no nontrivial abelian sector*. Its three faces all force  $E_8$ :

1. **AQFT** (v83/v154): the boundary chiral net is *holomorphic* ( $\mu$ -index 1, single vacuum sector). Holomorphic +  $c = 8 \Rightarrow$  the unique even unimodular rank-8 lattice net  $(E_8)_1$ , and  $c = 8 = g_{\text{car}} + N_{\text{fam}}$  is itself structural.

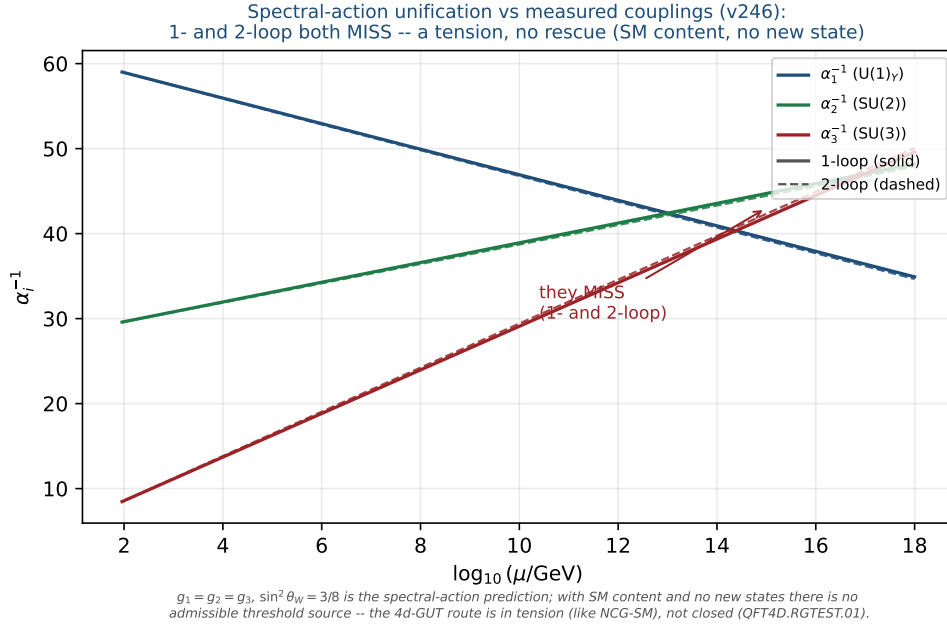


Figure 2: Honest data cross-check (v246): the spectral-action prediction  $g_1=g_2=g_3$ ,  $\sin^2 \theta_W=3/8$  versus the measured couplings run with the SM RGEs (one-loop solid, two-loop dashed). The three lines do *not* meet at any scale (1-loop residual  $\Delta\alpha^{-1} \approx 9$  at  $2 \times 10^{16}$  GeV; 2-loop minimal spread  $\approx 3.2$  at  $1.7 \times 10^{14}$  GeV) — a falsifiable tension. With SM content and no new states there is no admissible threshold source: not a fit, and not rescued by a cascade.

2. **Geometry** (v232): the seam link is a *homology 3-sphere*  $\Leftrightarrow$  its binary group  $\Gamma$  is *perfect*  $\Leftrightarrow \Gamma = 2I$  (the Poincaré sphere  $S^3/2I$ ).

3. **Representation theory** (v219): exactly *one* 1-dim irrep (the trivial one).

These are the *same* number:  $\#(1\text{-dim irreps}) = |\Gamma^{\text{ab}}| = |H_1(S^3/\Gamma)| = \#(\text{mark-1 affine-Dynkin nodes})$ , which equals 1 *only* for  $E_8$  ( $A_n \rightarrow n+1$ ,  $D_n \rightarrow 4$ ,  $E_6 \rightarrow 3$ ,  $E_7 \rightarrow 2$ ,  $E_8 \rightarrow 1$ ). So  $2I$  is the unique nontrivial perfect finite  $SU(2)$  subgroup and  $S^3/2I$  the unique homology-sphere space form. **Consequence:** Research Contract 2 ( $G_{\text{net}}$ /Target A), the carrier P2, and the Kleinian seam are *not* separate gates — they are this *one* condition.

The closing chain (what a complete closure would be), **[O]** at one step:

$$\begin{aligned} &\text{RP} + \text{gap} + \text{chirality} \xrightarrow{\text{v160}} \text{quasi-free bulk} \xrightarrow{?} \text{holomorphic boundary net} \\ &\xrightarrow{[\text{E}]} (E_8)_1 \xrightarrow{\text{v219/v228}} 2I, (2, 3, 5), \text{P2} \xrightarrow{\text{v179-v181+Kleinian genus-0}} \text{QGEO.SYM.01.} \end{aligned}$$

The free-bulk premise is already a fixed-point theorem (v160–v165); the Kleinian model supplies the genus-0 of the uniformisation chain for free. **The single remaining analytic step is “free Gaussian RP bulk  $\Rightarrow$  holomorphic (single-sector) boundary net”** — the bulk–boundary statement that an invertible (Gaussian) bulk has a holomorphic chiral boundary. That step is now *closed modulo cited theorems* (the MMST route SEAM.EQUIV.MMST.01): the lattice/ $S_3$  stack pins it at every computable level (v367/v368, v376–v379, Lean FORM.SEAM.MMST.01), only the cited continuum existence staying **[O]** (v336);  $v_{\text{geo}}$  stays the No-Unit metrology primitive and  $F_{\text{transfer}}$  stays external physics. This box certifies that the three faces coincide and force  $E_8$  (**[E]**); the analytic step is closed modulo that cited theorem.

**A physical realisation of the step** ([verification/v235\_seam\_chern\_simons.py]). The step lands in standard physics. A free gapped *bosonic* 2+1d bulk is an abelian Chern–Simons /  $K$ -matrix theory (even integer  $K$ ), where  $\#\text{anyons} = |\det K|$ , the edge chiral central charge is  $c = \text{signature}(K)$ , and the edge net is *holomorphic*  $\Leftrightarrow |\det K| = 1$  — so “no nontrivial abelian sector” is precisely  $\det K = 1$ . The extension tower of v92 is the anyon-condensation tower read by determinant:  $D_5 \oplus A_3$  ( $\det 16$ , 16 anyons)  $\rightarrow SO(16)_1 = D_8$  ( $\det 4$ )  $\rightarrow (E_8)_1$  ( $\det 1$ , holomorphic), all rank 8 with  $c = \text{signature} = 8 = g_{\text{car}} + N_{\text{fam}}$ , so the holomorphic end is the *Kitaev  $E_8$  quantum-Hall state*. Holomorphy  $\Leftrightarrow$  condensing the order- $|\mu_4| = 4$

*Lagrangian glue* ( $\det 16 \rightarrow 16/4^2=1$ ); a partial order-2 isotropic only reaches  $\det 4$  ( $D_8$ ). The residual is thereby a single integer condition — “the free RP seam condenses the order- $|\mu_4|$  Lagrangian glue” — the same sheet/ $\mu_4$  selection as GATE.QGEO, but now with holomorphic  $\Leftrightarrow \det 1$  a theorem (GATE.HOLO.02, [E]/[O]).

**The residual as a physical statement** ([verification/v237\_seam\_sre\_closure.py]). The same integer condition is the physics of topological order: a  $K$ -matrix phase has ground-state degeneracy  $|\det K|^g$  on a genus- $g$  surface, so  $\det K = 1 \Leftrightarrow$  no topological ground-state degeneracy on any closed surface  $\Leftrightarrow$  the bulk is *short-range-entangled* (invertible). Among rank-8 even positive-definite  $K$  ( $c = 8 = g_{\text{car}} + N_{\text{fam}}$ ) only  $E_8$  is SRE — the unique nontrivial bosonic SRE 2+1d phase, the Kitaev  $E_8$  state, whose edge is the holomorphic  $(E_8)_1$ . TFPT already has a unique vacuum on the plane (RP + gap + cluster + OS); the plane cannot see  $\det K$  (no non-contractible cycles), the torus can. So the one analytic step is the sharp, falsifiable physical statement “the free RP gapped seam bulk is short-range-entangled (no topological order)” ( $= \det K = 1 =$  the Kitaev  $E_8$  phase, GATE.HOLO.03, [E]/[C]) — and the lattice/ $S_3$  stack now answers it at every computable level (torus  $\det K=1$ , v378), so the keystone’s MMST route (SEAM.EQUIV.MMST.01) is *closed modulo cited theorems*, only the cited continuum existence (v336) staying [O] (the parent stays [O]).

**Lemma 2** (U1 — wall landscape, corrected). *Parabolic degree zero forces the line weight-sums onto the wall  $(w_1, w_2, w_3) = (2, 1, 1)$ . An explicit balanced representative (weights in  $\frac{1}{3}$ ) is*

$$W_{\text{wall}} = \frac{1}{3} \begin{pmatrix} 2 & 1 & 2 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 2 & 1 & 0 \end{pmatrix},$$

*each column a permutation of  $(0, 1, 2)$ , row sums  $(2, 1, 1)$ . **Honest result:** the  $6^4=1296$  column-permutation matrices give 144 walls; under the full group  $D_4 \times S_3(\text{lines}) \times S_3(\text{values})$  they form five orbits, not one. So  $W_{\text{wall}}$  is not singled out by symmetry — the uniqueness must come from the selector (Lemma U4), not from a symmetry quotient. The wall landscape is nonetheless a finite, explicitly enumerated set;  $W_{\text{wall}}$  is one of the five. Machine: fully certified  $C_U^{(1)}$  ([verification/v29\_research\_contract\_certs.py]).*

**The splitting type is now algebraic (RH route, [verification/v32\_rh\_splitting.py]).** *Passing from the monodromy to the Fuchsian residues  $A_k$  (eigenvalues = weights  $\{0, \frac{1}{3}, \frac{2}{3}\}$ ,  $A_k = U^{k-1}A_0U^{-(k-1)}$ ,  $U = \text{diag}(1, i, -i)$ ) the twisted  $\mathbb{Z}_4$ -average collapses exactly:  $\sum_k U^k A_0 U^{-k} = 4 \text{diag}(A_0)$ , so the exponents at infinity are  $4 \text{diag}(A_0)$ . A flat bundle with regular  $\infty$  needs integer exponents summing to  $4 = -\deg E$ ; with the cusp weights the only options are perms of  $(2, 1, 1)$  and  $(2, 2, 0)$ , i.e.*

$$\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2 \iff \text{diag}(A_0) = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right)$$

*(the sibling  $(2, 2, 0) \rightarrow \mathcal{O}(0) \oplus \mathcal{O}(-2)^2$ ). Such a Hermitian  $A_0$  (spectrum  $\{0, \frac{1}{3}, \frac{2}{3}\}$ , that diagonal) exists by Schur–Horn. So the splitting is an algebraic condition on  $\text{diag}(A_0)$ . Honest wall: the global  $\prod M_k = \mathbf{1}$  is the path-ordered monodromy (not  $\exp(2\pi i \sum A_k)$ ), constraining off-diag( $A_0$ ) via the genuine RH solve; and R-extraction still needs the  $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$  bridge. The RH route fixes the splitting algebraically but not  $c_u/c_d$ .*

**Lemma 3** (U2 — the kill switch, run and hardened [E]). *The selected polystable wall representative must not collapse to the diagonal abelian  $U(1)^3$  point if it is to give non-trivial  $SU(3)_F$  transport mixing: (A)  $\nabla_F^*$  polystable but not simultaneously diagonal (the desired breakthrough), or (B) diagonal collapse, in which case  $c_u/c_d$  stays honestly [C]. **Result: the cheap test is run robustly and lands on A.** The kill-switch fires iff the  $D_4$ -symmetric cusp representation is reducible (commutant non-scalar). Across 400 random cusp-class generators of the  $\mathbb{Z}_4$  family  $M_k = U^{k-1}MU^{-(k-1)}$  the commutant is scalar at every point (irreducible 400/400), the reducible locus is hit with probability 0 (measure-zero, codimension  $\geq 1$ ), and the mixing strength stays bounded away from zero (min off-diag = 0.35 > 0.05). So case A is the generic behaviour, not an accident of the one explicit point of [verification/v33\_explicit\_flat\_bundle.py]: ( $U_{\text{wall}}$ ) survives its own cheapest falsification test. [verification/v38\_uwall\_killswitch.py]*

**Lemma 4** (U3 —  $D_4$  fixed locus as an explicit variety, *positive-dimensional*). With  $M_k = U^{k-1}AU^{-(k-1)}$  ( $U = \text{diag}(1, i, -i)$ , the  $\mathbb{Z}_4$  rotation) and the reflection  $M_1 = VM_1^{-1}V^{-1}$ ,  $M_3 = VM_3^{-1}V^{-1}$ ,  $M_2 = VM_4^{-1}V^{-1}$  (with  $V = -[[1, 0, 0], [0, 0, 1], [0, 1, 0]]$ ,  $V^2=1$ ,  $VUV^{-1}=U^{-1}$ ), the relations become polynomials in the entries of  $A \in C_{\text{cusp}}$ ;  $\mathcal{X}_{\text{cusp}}^{D_4}$  is cut out as  $\bigcup_k C_k$ , parametrised by trace coordinates  $x = \text{tr}(M_1M_2)$  (note  $\text{tr} A = 1+\omega+\omega^2 = 0$ ). **Result** ( $C_U^{(2)}$ , [verification/v30\_d4\_character\_variety.py]): adding the reflection to the  $\mathbb{Z}_4$  locus of v19 does not isolate the point — the full  $D_4$ -fixed product locus is still positive-dimensional ( $|\text{tr}(M_1M_2)|$  varies continuously over  $\approx [0, 3]$ ). So even the full  $D_4$  symmetry cannot select  $\nabla_F^*$ . Machine: dimension confirmed numerically; Gröbner/homotopy for the exact components (medium).

**Lemma 5** (U4 — selectors as equations). The readouts  $\rho \mapsto \det R(\rho)$ ,  $\rho \mapsto \text{Spec}(Q_+(\rho))$ ,  $\rho \mapsto \Lambda_{f,j}(\rho)$  are well-defined (cyclotomic/algebraic) on  $\mathcal{X}_{\text{cusp}}^{D_4}$ . **Acceptance criterion:**

$$\#\{\rho \in \mathcal{X}_{\text{cusp}}^{D_4} : \det R = 8, \text{Spec}(Q_+) = \{1, 2, 3\}\} = 1$$

in the polystable quotient. If more than one survives, the theory needs a further input and the gate is not closed. **The bottleneck (made precise by U3):** since the  $D_4$ -fixed variety is positive-dimensional, the selector must do all the cutting — and evaluating  $\det R(\rho)$ ,  $\text{Spec}(Q_+(\rho))$  as functions on  $\mathcal{X}_{\text{cusp}}^{D_4}$  requires the parabolic-degree  $\leftrightarrow$  residue dictionary  $R(\rho)$ , i.e. the H2 equivalence. That dictionary was the single remaining analytic input of the ( $U_{\text{wall}}$ ) gate. Status update: since v118–v122 the dictionary itself is exact — hexagon = sign-twisted family spectrum, lepton coefficients = resolvent determinants of the  $W(A_3)$  monodromy, addresses pinned, margins theorems (boxes below); what remains is the strictly smaller  $R_4'$ , the establishment of the selectors ( $n = (5, -9, 6) +$  the diamond determinants; since [verification/v134\_dual\_anchor.py]: the pair  $(d, n)$ ).

**What  $R(\rho)$  is** ([verification/v31\_R\_dictionary.py]). Two facts pin its nature. (i) Case A is alive: every  $D_4$ -fixed cusp monodromy found is irreducible, so the wall point carries non-trivial  $SU(3)_F$  mixing — the U2 kill switch does not fire to the diagonal case B. (ii)  $R(\rho)$  is not algebraic:  $R$  is integer-valued, but the algebraic invariants of  $\rho$  (e.g.  $\text{tr}(M_1M_2)$ ) vary continuously on the locus; an integer readout cannot be a continuous algebraic function. Hence  $R(\rho)$  is the discrete invariant of the non-abelian Hodge / Mehta–Seshadri map (the parabolic-degree / Hodge-filtration jumps of the harmonic bundle  $E_\rho$ ): integer output, but its computation is the transcendental harmonic-metric (Hitchin) solve on the generic Higgs locus. **Refinement (U4'',** [verification/v40\_harmonic\_metric.py]): at the polystable point TFPT actually selects this pessimism does not apply — see the box below.

**Progress: the selector side of U4 is closed on the explicit point ([E]).** While  $R(\rho)$  as a function on the whole locus needs the Hitchin solve, the two U4 selectors are algebraically accessible on the explicit flat bundle of [verification/v33\_explicit\_flat\_bundle.py] — no PDE. Computed exactly from  $A_0$ : the exponents at infinity  $\text{eig}(\sum_k A_k) = \{2, 1, 1\}$  give the splitting type  $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ , i.e. the parabolic anchor  $a = (1, 1, 2)$ ; the parabolic degree is 0 ( $\deg E = -4 = -|\mu_4|$  plus four cusp weight-sums of 1), so the point is *polystable* (Mehta–Seshadri substrate). The two selectors then read off the bundle directly:

$$\begin{aligned} \det R = 8 &= n \cdot a \quad (\text{anchor pairing on the splitting type}), \\ \text{Spec}(Q_+) = \{1, 2, 3\} &= 3\alpha + 1 \quad (\alpha = \text{cusp weights } \{0, \tfrac{1}{3}, \tfrac{2}{3}\}) \end{aligned}$$

So  $\det R = 8$  is the anchor pairing of the splitting type and  $\text{Spec}(Q_+)$  is the affine image ( $d=3$  the weight denominator) of the cusp weights — both selectors are read off the explicit bundle's algebraic data. **Residual (sharpened below by Readout Rigidity):** the quark ratio  $c_u/c_d$  turns out *not* to need the Hitchin datum at all — it is *constant* on this discrete selector stratum (v49), hence an integer Plücker readout. The only genuinely transcendental input left is the *absolute* amplitude normalisation ( $U_{\text{point}}$ ), which is an anchor, not a missing PDE. [verification/v39\_uwall\_selectors.py]

**Breakthrough: the harmonic metric is *finite linear algebra*, not a PDE ([E]).** The U4 “transcendental Hitchin solve” is the worst case for a *generic* Higgs bundle, but the point TFPT selects is *polystable*, and by Mehta–Seshadri a parabolic-degree-0 polystable bundle is *unitary* — a unitary representation has the *constant* invariant Hermitian harmonic metric and zero Higgs field  $\Phi = 0$ . So at the selected point the harmonic metric is finite linear algebra. On the explicit bundle of [verification/v33\_explicit\_flat\_bundle.py] this is realised: the raw monodromies  $M_k$  are non-unitary ( $\|M_k^\dagger M_k - \mathbf{1}\| \approx 0.83$ ), but there is a *unique* common invariant Hermitian form  $H$  ( $\dim \ker\{M_k^\dagger H M_k = H\} = 1$ ), it is *positive-definite*, and the unitarised holonomy  $\widetilde{M}_k = H^{1/2} M_k H^{-1/2}$  is unitary ( $\sim 10^{-8}$ ), lies in the cusp class  $\{1, \omega, \omega^2\}$  with  $\prod \widetilde{M}_k = \mathbf{1}$ , and has *clean* matrix elements  $|\widetilde{M}_k|_{ij} \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$ . **So  $C_U^{(3)}$  reduces from a transcendental PDE to: this finite unitarisation (done) + the combinatorial word-length  $R$  (closed:  $\mathbf{H1} + \det R = 8$ ).** *Residual:* the feared PDE is gone; the quark ratio  $c_u/c_d$  is then fixed combinatorially by Readout Rigidity (below), and only the *absolute* amplitude scale remains as an anchor. [verification/v40\_harmonic\_metric.py]

**The final leg-assignment test (honest result) ([E]).** Does the harmonic-frame holonomy  $\{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$  feed the  $\delta = \frac{1}{2}$  resolvent to give the lepton amplitudes  $\Lambda = (0.475, 1.107, 0.917)$ ? **(i) Ruled out as a literal identity:**  $\Lambda_\mu = 1.107 > 1$ , while unitary holonomy entries have modulus  $\leq 1$ , so the amplitudes *cannot* be holonomy matrix elements — they are the non-unitary resolvent Green function (as v19 already noted). **(ii) Suggestive positive:** the harmonic-frame holonomy diagonal modulus is exactly  $|\text{diag } \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$ , and that  $\frac{1}{2}$  *equals* the distinguished lepton transport value  $\delta = \frac{1}{2}$  that v20 used by hand — so the geometry plausibly *supplies*  $\delta$  (an observation at the explicit point; genericity across the locus is future work). **(iii) Clean negative:** no natural holonomy construction (e.g.  $(\mathbf{1} - \frac{1}{2}\widetilde{M})^{-1}$ ) maps the holonomy to the amplitude vector. **Verdict:** the amplitudes are the  $\delta = \frac{1}{2}$  resolvent (closed combinatorially via the word-length  $R$ ); the holonomy supplies  $\delta$  and the leg selection, not the amplitude magnitude.  $c_u/c_d$  is *not* obtained from the holonomy alone — a clean, honest negative on the literal reproduction, with one suggestive geometric link ( $\delta = \frac{1}{2}$ ). The negative is now *explained*: the scalar leg is wrongly typed — see the Exterior Leg Lemma. [verification/v41\_leg\_assignment.py]

**The torsion normal form and the  $\delta = \frac{1}{2}$  theorem ([E]).** The v41 “genericity is future work” item is now executed — and yields a theorem. *(i) Flatness =  $\mu_4$  torsion [E]:* for any  $M$  (symbolic),  $\prod_k U^k M U^{-k} = (MU)^4$  — the  $\mathbb{Z}_4$ -family flatness constraint *is* the torsion statement “ $T = MU$  is a fourth root of unity”: the  $\mu_4$  atom appears as the *order* of the twisted cusp generator. *(ii) The  $\delta = \frac{1}{2}$  theorem [E] (both directions):* on the *involutive* branch ( $T^2 = \mathbf{1}$ )  $T$  is a Hermitian reflection  $2vv^\dagger - \mathbf{1}$ , and the cusp trace condition  $2v^\dagger U^{-1}v = 1$  splits *exactly* into  $|v_1|^2 = \frac{1}{2}$  (real part) and  $|v_2| = |v_3|$  (imaginary part), forcing

$$\boxed{\text{diag } M = (0, \frac{i}{2}, -\frac{i}{2}), \quad \text{spec } M = \{1, \omega, \omega^2\} \text{ automatic}}$$

( $\chi_M = \lambda^3 - 1$  from unitary + trace 0 + det 1). So  $\delta = |M_{22}| = \frac{1}{2}$  holds *exactly and constantly* on the whole two-parameter branch — the distinguished lepton transport value that v20 used by hand is a  $\mu_4$ -torsion identity, not an observation. *(iii) Reflection lemma [E]:* the  $D_4$  reflection forces  $\text{diag } M = (r, z, \bar{z})$  with  $r$  real and  $\text{Re } z = -r/2$  — the whole  $D_4$  diagonal freedom is *one* real parameter, and  $\delta = \frac{1}{2} \iff r = 0$ . *(iv) Branch census [E] (seeded):* the full  $D_4$ -fixed flat cusp locus realises only  $\text{tr}(MU) \in \{1, -1\}$ ; the involutive branch has the diagonal *constant*  $(0, \frac{1}{2}, \frac{1}{2})$  to machine precision, while on the  $\{1, i, -i\}$  branch  $d_1$  varies over  $[0, 1]$  — and the explicit v33/v40 point sits exactly in its  $d_1 = 0$  slice, the *same*  $\delta$  value. *Residue [C] (recorded):* whether the anchor splitting  $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$  (Mehta–Seshadri side) forces the  $d_1 = 0$  slice — or selects the involutive branch — is the remaining question;  $\delta = \frac{1}{2}$  itself is no longer the open item. [verification/v114\_torsion\_delta.py]

**The anchor pins the residue: an exact normal form for the  $(U_{\text{wall}})$  bundle ([E]).** The v114 branch question is answered — and the v33 *numerical* Riemann–Hilbert point acquires an *exact closed form*. *(i)  $\mu_4$ -average lemma [E]:*  $\sum_k U^k X U^{-k} = 4 \text{diag } X$  for any  $X$  —  $\mu_4$  conjugation-averaging *is* the diagonal projection, so the exponents at infinity are  $4 \text{diag } A_0$  and the anchor

splitting holds *iff*  $\text{diag } A_0 = (2, 1, 1)/4 = a/|\mu_4|$ : the residue diagonal **is** the anchor over  $\mu_4$  (v33's diagonal was forced, not chosen). (ii) *The*  $(8, 0, 5)/144$  lemma [E]: the anchor diagonal and the cusp spectrum  $\{0, 1, 2\}/N_{\text{fam}}$  pin  $\sum |a_{ij}|^2 = \frac{13}{144}$ , and with  $a_{13} = 0$  the determinant condition solves *uniquely* to

$$\boxed{(|a_{12}|^2, |a_{13}|^2, |a_{23}|^2) = \frac{(8, 0, 5)}{144}} \quad \begin{array}{l} 8=\text{rank } E_8, \ 5=g_{\text{car}}, \\ 144=(|\mu_4|N_{\text{fam}})^2, \ 8+5=13=\Delta_Q \end{array}$$

— the off-diagonal weights are compiler atoms, and the total 13 is the *same*  $\Delta_Q$  that denominates  $c_u/c_d = 55/117 = 5 \cdot 11/(9 \cdot 13)$ . (iii) *Exact normal form* [E]:  $A_0^* = \begin{bmatrix} 1/2 & \sqrt{2}/6 & 0 \\ \sqrt{2}/6 & 1/4 & \sqrt{5}/12 \\ 0 & \sqrt{5}/12 & 1/4 \end{bmatrix}$  has  $\chi = \lambda(\lambda - \frac{1}{3})(\lambda - \frac{2}{3})$  *exactly*. (iv)  $A_0^*$  is the RH solution [E]: its big-circle monodromy is trivial to  $10^{-10}$ ,  $\prod M_k = \mathbf{1}$ , and the unitarised holonomy has  $|\text{diag } \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$ ,  $\text{tr}(\widetilde{M}_0 U) = 1$  — the v33/v40 point is gauge-equivalent to  $A_0^*$  (the  $a_{12}, a_{23}$  phases are diagonal gauge). (v) *The anchor forces the*  $d_1 = 0$  slice [E]: a multi-seed RH scan over the anchor-pinned spectral manifold lands at *every* flat solution on the *same* gauge orbit with  $|\text{diag } \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$  — combined with v114, the whole harmonic diagonal is **anchor + torsion forced**;  $\delta = \frac{1}{2}$  needs no selection. *Residue* [C]:  $a_{13} = 0$  and the gauge-orbit uniqueness are numerical (one component across all seeds); promoting them to [E] via Gröbner on the RH constraints is the remaining  $U_{\text{point}}$  step. [verification/v115\_anchor\_residue.py]

**The resonance theorem:**  $M_\infty = \mathbf{1} \iff a_{13} = 0$  — **one gauge orbit** [E]  
( [verification/v116\_resonance\_uniqueness.py] )

The anticipated Gröbner promotion turned out to be a *linear* computation. Writing the equivariant system at infinity as  $dY/dw = -(1/w)(B_0 + B_1 w + \dots)Y$  with  $B_m = \sum_k i^{km} U^k A_0 U^{-k}$ , the *twisted*  $\mu_4$  averages collapse [E]:

$$B_0 = 4 \text{diag } A_0, \quad \boxed{B_1 = 4a_{12}E_{12} + 4\bar{a}_{13}E_{31}}$$

(only the eigenvalue-ratio  $-i$  cells survive). The anchor exponents  $\text{diag}(2, 1, 1)$  have resonance gap 1; the level-1 formal-gauge equation  $(1 - \text{ad } R_0)H_1 = R_1$  is singular exactly on the cells  $\{(2, 1), (3, 1)\}$ , where  $B_1$  has entries  $(0, 4\bar{a}_{13})$ ; all higher levels are non-resonant and the formal monodromy is  $e^{-2\pi i \text{diag}(2, 1, 1)} = \mathbf{1}$ . Hence [E]:

$$\boxed{M_\infty = \mathbf{1} \iff a_{13} = 0}$$

— the transcendental Riemann–Hilbert condition collapses to *one linear equation*. *Uniqueness corollary* [E]:  $a_{13} = 0$  + the  $(8, 0, 5)/144$  moduli + phases = diagonal gauge  $\Rightarrow$  the flat anchor locus is **exactly one gauge orbit**, the exact  $A_0^*$ : within the  $\mu_4$ -equivariant anchor class the  $(U_{\text{wall}})$  bundle datum is *algebraically unique*. *Sharpness* [E]:  $\|M_\infty - \mathbf{1}\| \sim 8.5 |a_{13}|$  (0.17 at  $a_{13} = 0.02$ ) vs  $10^{-10}$  at  $A_0^*$ . *Honest scope* [C]: the equivariant ansatz is the established  $D_4$  selector input (v19/v30); the *residue* side is now [E], while the holonomy values (the harmonic diagonal  $(0, \frac{1}{2}, \frac{1}{2})$ ) remain [E] — transcendental monodromy of an exact system.

**The monodromy is the Weyl group of  $A_3$  — the holonomy is exact** [E]  
( [verification/v117\_monodromy\_weyl\_a3.py] )

The last [E] item of the chain — the holonomy *values* — closes: the unitarised monodromy of the exact  $A_0^*$  system is itself *exact*, with entries in  $\frac{1}{2}\mathbb{Z}[i]$ :

$$\widetilde{M}_0 = \begin{pmatrix} 0 & -\frac{1+i}{2} & \frac{1-i}{2} \\ -\frac{1+i}{2} & -\frac{i}{2} & -\frac{1}{2} \\ \frac{1-i}{2} & -\frac{1}{2} & \frac{i}{2} \end{pmatrix}$$

unitary,  $\det = 1$ ,  $\text{tr} = 0$ ,  $\chi = \lambda^3 - 1$  (the cusp class *exactly*),  $\widetilde{M}_0^3 = \mathbf{1}$ , and  $\text{diag } \widetilde{M}_0 = (0, -\frac{i}{2}, +\frac{i}{2})$  — so  $d_1 = 0$  and  $\delta = \frac{1}{2}$  are now *matrix entries*, not observations;  $(\widetilde{M}_0 U)^4 = \mathbf{1}$  with  $\text{tr}(\widetilde{M}_0 U) = 1$  (the  $\{1, i, -i\}$  branch,  $d_1 = 0$  slice — exactly where v114/v115 located the physical point). *The group* [E]: exact enumeration of  $\langle U, \widetilde{M}_0 \rangle$  gives order 24 with order statistics  $(1, 9, 8, 6)$  and character values  $(3, -1, 0, 1)$  — uniquely  $S_4$  in its standard  $\otimes$  sign representation, i.e. the (twisted) reflection representation of the **Weyl**

group of the family lattice  $A_3$ :

$$\text{monodromy}(U_{\text{wall}}) = W(A_3) = S_4, \quad |W(A_3)| = 4! = 24$$

— the flavor wall realises the  $A_3$  factor of the glue  $D_5 \oplus A_3 + \mu_4$  as its Weyl group ( $U$  = a 4-cycle, the  $\mu_4$  deck;  $\widetilde{M}_0$  = a 3-cycle, the family rotation). *Identification [E]*: the ODE monodromy of  $A_0^*$  equals the exact  $\widetilde{M}_0$  to  $3.5 \times 10^{-10}$ , and the invariant form  $H$  is  $U$ -invariant to  $10^{-11}$ . *Status*: holonomy values [E]; only the analytic identification stays [E]. The  $\delta$  thread ( $v20 \rightarrow v40/v41 \rightarrow v114 \rightarrow v115 \rightarrow v117$ ) is closed end to end.

**The hexagon is the sign-twisted family spectrum — first exact H2 piece [E]**  
([verification/v118\_hexagon\_family\_dictionary.py])

The missing “ $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$  dictionary” (H2) acquires its first exact entry. *Sign-twist lemma [E]*:  $-\widetilde{M}_0$  has order 6 and

$$\text{spec}(\widetilde{M}_0) \cup \text{spec}(-\widetilde{M}_0) = \mu_6$$

— the six-site hypercharge hexagon spectrum *is* the family monodromy spectrum together with its sheet twist ( $\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$  realised as  $(-1) \times \langle \widetilde{M}_0 \rangle$ ); the  $v20$  denominators  $\frac{5}{4} - \cos(r\pi/3)$  are exactly the eigen-denominators  $|1 - \zeta_6^r/2|^2$  of the two family resolvents  $(1 \mp \frac{1}{2}\widetilde{M}_0)^{-1}$ . *Cyclotomic determinant [E]*:  $\det(1 - t\widetilde{M}_0) = 1 - t^3 = (1 - t)\Phi_3(t)$ , so  $\det(1 \mp \frac{1}{2}\widetilde{M}_0) = (\frac{7}{8}, \frac{9}{8})$  — and the lepton coefficients become resolvent determinants:  $c_e = |\mu_4|/(2\det_-) = \frac{16}{7}$ ,  $c_\mu = N_{\text{fam}}/(2\det_+) = \frac{4}{3}$ ,  $c_\tau = |\mu_4|\det_-/N_{\text{fam}} = \frac{7}{6}$ , product rule =  $|\mu_4|/\det_+ = \frac{32}{9}$ . The lepton 7 and 9 are the family-resolvent determinants;  $\delta = \frac{1}{2} = |(\widetilde{M}_0)_{22}|$  is supplied by the matrix itself. *Sheet-extended group [E] (audit)*:  $\langle U, \widetilde{M}_0, -1 \rangle$  has order 48 =  $\Omega_{\text{adm}} = N_{\text{fam}} \dim S^+$  — the seed’s quartic coefficient. *Residue [C]*: the dictionary’s *values* are exact; its *address table* (which hexagon site each fermion occupies; the quark composition) remains  $v20$ ’s established input / open — nothing re-fished.

**The address table: the lepton words are the compiler atoms [E]/[C]**  
([verification/v120\_address\_table.py])

The address table acquires its structure — on word-length integers *frozen* in  $v18/v20$ , long before today’s readings (the red-team firewall). (i) *The lepton words are the atoms [E]*:

$$(L_e, L_\mu, L_\tau) = (8, 5, 3) = (\text{rank } E_8, g_{\text{car}}, N_{\text{fam}}) = (p_0 + e_2, e_2, p_0)$$

in mass order (longest word = lightest lepton). (ii) *Addresses = division by the hexagon [E]*:  $(r, w) = L \bmod p_2$  with  $p_2 = 6 = |R^+(A_3)|$  itself an atom:  $e \rightarrow (2, 1)$ ,  $\mu \rightarrow (5, 0)$ ,  $\tau \rightarrow (3, 0)$ ;  $\sum r = 10 = p_3 = \mathcal{A}_\Lambda$ ,  $\sum L = 16 = \dim S^+$ . (iii) *Sheet parity [E]*: by the  $v118$  dictionary,  $e$  sits on the *untwisted* sheet ( $\omega \in \text{spec } \widetilde{M}_0$ ),  $\mu, \tau$  on the *twisted* sheet  $(-\omega, -1)$ ; the anchor lepton  $\tau$  occupies the unique *real* twisted eigenvalue  $-1$  — exactly the site where  $v20$ ’s product rule replaces the resolvent. (iv) *Quark sum rules [E] (frozen v18 words 7, 7, 5, 3, 2, 0)*: up-type sum =  $10 = p_3$ ; down-type sum =  $14 = p_1 + p_3$ ; **all quarks** =  $24 = |W(A_3)|$  (the monodromy group order,  $v117$ ); **all nine charged fermions** =  $40 = p_1 p_3 = a^T R a$  ( $v119$ ); quark  $\sum r = 2p_2$ ,  $\sum w = |\mathbb{Z}_2|$ ; the top sits at the vacuum site  $(0, 0)$  — zero word, no suppression, the established ladder anchor. *Residue [C]*: five independent closures  $(16, 10, 14, 24, 40)$  constrain the address map; the per-fermion *assignment* remains the  $v18/v20$  input — deriving it is the remaining H2 content.

**The address table is pinned:  $R$  is the unique hexagon matrix with atom margins [E]**  
([verification/v121\_address\_pinning.py])

The per-fermion assignment closes at the *information* level. The established identity  $L = R + 2U$  ( $v95$ ) says the word-length table *is*  $L[\text{sector}][\text{generation}]$  — rows (up; down; lepton) =  $((7, 3, 0); (7, 5, 2); (8, 5, 3))$ , the  $v18/v20$  words verbatim — so the addresses are the entries of the *residue operator*  $R$  plus one hexagon turn for the first generation, and  $R$ ’s first column is the **anchor**  $a = (1, 1, 2)$ . The margins are atoms [E]: row sums  $(4, 8, 10) = (e_1, \text{rank } E_8, p_3)$ , column sums  $(4, 13, 5) = (e_1, \Delta_Q, g_{\text{car}})$ . And the

census closes it [E]:

among all  $3 \times 3$  matrices with entries in the hexagon  $\{0..5\}$ ,  
exactly 17 have the atom margins — and exactly ONE has  $\det = 8$

— and it is  $R$  (also the unique one with SNF = (1, 1, 8); all 17 determinants are distinct, 8 occurs once — control computed explicitly). **The assignment table carries zero residual information beyond the atom margins and the rank determinant.** Within-sector ordering is the established transport reading ( $m \sim \lambda_Y^L$ : longer word = lighter fermion). *Residue [C]*: the open H2 item shrinks from “derive 9 per-fermion integers” to “derive the six atom margins +  $\det = \text{rank}$ ”. *Firewall*:  $R$ ,  $L$  and all word lengths were frozen in v4/v18/v20/v71 long before this census — no degrees of freedom spent.

**The margins are theorems** ([verification/v122\_margin\_theorem.py]). Even the margins are not input. Three selectors, all established and frozen *before* the address question was posed — (S1) the  $D_4$  annihilator  $n = (5, -9, 6)$  with  $n^T R = (8, 0, 0)$  (it kills the unchanged  $(c_2, c_3)$  plane;  $n \cdot a = 8$ ), (S2)  $\det R = 8$ , (S3)  $\det K = 4$  with  $K = R + Q\Sigma$  — pin  $R$  *uniquely* among hexagon matrices [E]: the (S1) census has  $4 \times 4 \times 4 = 64$  candidates, (S2) leaves 12, (S3) leaves **exactly one**, and it is  $R$ . Hence the atom margins, the anchor column  $Re_1 = a$  and all sum rules above are *consequences* — their input status is revoked. *Bonus lemma [E]*: on the (S1) + (S2) locus  $\det(M + 2U) = 20 = \det L$  holds for *all* 12 candidates — the fourth quartet determinant is redundant, the compiler’s familiar overdetermination. *Residue [C]*: H2 reduces to the establishment of the selectors themselves ( $n$  and the diamond determinants — pre-existing inventoried objects); the address question introduces *no new residual class*.

**The dual-normal selector:**  $(d, n)$  **pins**  $R$  **columnwise** ([verification/v136\_dual\_normal\_selector.py]). The selector list shrinks once more. With the dual anchor  $d = a^\top R^{-1}$  ([verification/v134\_dual\_anchor.py]) and the torsion normal  $n$ , each *column* of  $R$  obeys  $d \cdot c_j = a_j$  and  $n \cdot c_j = 8\delta_{1j}$ ; the common kernel of  $(d, n)$  is the primitive vector  $(6, 8, 7)$  (audit:  $p_2$ , rank  $E_8$ , scalaron-7), so within the address box  $\{0..5\}^3$  each column is the *unique* lattice point (census: 1 of 216 each) —  $(d, n) \Rightarrow R$ , and the  $6^9$  census above collapses to three column problems [E]. *Honesty*:  $K$  is *not* pinned directly ( $d^\top K = (1, \frac{1}{2}, 1)$ );  $\det K = 4$  comes via the  $\mu_4$  wall ([verification/v135\_det\_surface.py]). *The mechanism is bounded*: the zero mode of  $A_0^*$  carries the exact spectral normal  $\sigma = (2, -9, 5) = (|\mathbb{Z}_2|, -N_{\text{fam}}^2, g_{\text{car}})$  (signed squares;  $\|v\|^2 = \frac{16}{9}$  with  $16 = \dim S^+$ ,  $\text{adj } A_0^* = \frac{2}{9}P_v$  with  $\frac{2}{9}$  = the SdS entropy curvature),  $\sigma \cdot \mathbf{1} = -|\mathbb{Z}_2|$ ,  $\sigma \cdot a = N_{\text{fam}}$ ,  $n \cdot \sigma = 121 = \|\text{Pl}(K)\|_1^2$ ; and the exact  $W(A_3)$  orbit control shows *no* group image of  $\sigma$  is proportional to  $n$  — the naive cofactor/orbit lift fails, the step  $\sigma \rightarrow n$  is a genuine mechanism beyond the monodromy group.  $R_4'$  restated [C]: derive the *pair*  $(d, n)$ ; then  $R$  is forced columnwise.

**The selector triangle** ([verification/v139\_selector\_triangle.py]). The pair itself now collapses further. First,  $d$  is *pure anchor data*:  $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$  exactly — the closed form never mentions  $R$ , so the first selector is *derived*, not residual. Second, the frame  $(\mathbf{1}, a, \sigma)$  is a basis of  $\mathbb{Z}^3$  with  $\det(\mathbf{1}|a|\sigma) = 11 = \|\text{Pl}(K)\|_1$  (the quark constant as the frame volume), and  $n$  is the *unique* integer covector with the three atom pairings

$$n \cdot \mathbf{1} = 2 = |\mathbb{Z}_2|, \quad n \cdot a = 8 = \text{rank } E_8, \quad n \cdot \sigma = 121 = \|\text{Pl}(K)\|_1^2$$

— the cofactor definition is replaced by frame data plus three atoms. On the  $(\mathbf{1}, a)$ -constrained line the  $\sigma$ -pairing is  $11(11+t)$ : divisible by 11 for *every*  $t$ , a perfect square exactly at the true  $n$  ( $t=0$ ). The review’s predicted lift is exact as an audit identity:  $n = \text{reverse}(\sigma) + |\mu_4|e_3$  (generation flip +  $\mu_4$  lift at the third slot).

**Frame integrality: three pairings  $\rightarrow$  two** ([verification/v142\_frame\_integrality.py]). The third pairing is *not* independent. The pairing triples  $(m \cdot \mathbf{1}, m \cdot a, m \cdot \sigma)$  of *integer* covectors form an index-11 sublattice of  $\mathbb{Z}^3$ , cut out by the single congruence

$$x - 3y + z \equiv 0 \pmod{11} \quad (\text{index } 11 = \|\text{Pl}(K)\|_1 = \det(\mathbf{1}|a|\sigma)),$$

so with the two line pairings at their atoms  $(x, y) = (|\mathbb{Z}_2|, \text{rank } E_8) = (2, 8)$ , integrality *forces*  $z \equiv 0 \pmod{11}$  — the  $11(11+t)$  line *is* the integer line. Primitivity forces  $t$  *even* (odd  $t$  makes every entry of  $n(t)$  even), and  $t=0$  is the unique even  $t$  with a square  $\sigma$ -pairing for  $|t| < 88$  (the global pick keeps the square selector — recorded honestly). Both line pairings are Cramer-reducible to inventoried identities

( $n \cdot a = \det R$  by Laplace along the anchor column;  $n \cdot \mathbf{1} = 8(R^{-1}\mathbf{1})_1$  with  $R^{-1}\mathbf{1} = \frac{1}{4}(1, 1, -1)$ , v134) — restructuring, not establishment.

**The pairing values are atom identities** ([verification/v145\_pairing\_atoms.py]). Even the two line-pairing *values* carry no independent information. Reading the v139 lift structurally —  $n = w_0(\sigma) + |\mu_4|e_3$  with  $w_0$  the  $A_3$  *exponent duality*  $m \leftrightarrow h-m$  (an established Lie involution; lift slot = the top-exponent line, lift size = the glue order) — all three pairings become atom arithmetic:

$$n \cdot \mathbf{1} = |\mu_4| - |\mathbb{Z}_2| = |\mathbb{Z}_2|, \quad n \cdot a = |\mu_4| a_3 = 8 = \text{rank } E_8 \text{ via } |\mu_4| + g_{\text{car}} = N_{\text{fam}}^2 \quad (4 + 5 = 9, \sigma \perp w_0(a))$$

and  $n \cdot \sigma = \|\sigma\|^2 - N_{\text{fam}}^2 + |\mu_4|g_{\text{car}} = 110 - 9 + 20 = 121$  with  $\|\sigma\|^2 = |\mathbb{Z}_2|^2 + N_{\text{fam}}^4 + g_{\text{car}}^2 = 110 = |\mathbb{Z}_2|g_{\text{car}} \|\text{Pl}(K)\|_1$ .  $R_4'$  *residue* [C] (*final form*): derive the *one* map  $n = w_0(\sigma) + |\mu_4|e_3$  — no independent number remains; everything else in the chain anchor  $\rightarrow (d, n) \rightarrow R$  is exact.

**Both dual normals are anchor data** ([verification/v149\_cusp\_normal.py]). The remaining opacity dissolves in the canonical frame: the pairings of  $n$  with the *cusp eigenbasis* (the geometric basis of v98) are exactly

$$n \cdot e_1 = 6 = p_2(a), \quad n \cdot e_2 = 3 = p_0(a), \quad n \cdot e_3 = 5 = e_2(a)$$

— one anchor invariant per  $Q_+$  degree, and the cusp matrix is unimodular, so these three pairings determine  $n$  uniquely. With  $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$  the dual normal *pair* (v134) is anchor data through and through — one normal per frame; the cusp data (6, 3, 5), the frame pairings (2, 8, 121) and the  $w_0$ -lift are three presentations of one covector, all atomic (equivalences exact). *Audit*:  $\sigma \rightarrow (5, 1, 2)$ ; alternate reading (6, 3, 5) =  $(|R^+(A_3)|, N_{\text{fam}}, g_{\text{car}})$ ; sum 14 =  $\dim G_2$ .  $R_4'$  *residue* [C] (*cusp form*): derive the *assignment* — why the  $Q_+$  degrees (1, 2, 3) carry  $(p_2, p_0, e_2)$  in that order; one discrete assignment, zero continuous freedom.

### Exterior Leg Lemma — the quark $u/d$ leg is a $\Lambda^2 F$ area, not a scalar [E]/[C]

The v41 negative is a *category error*, not a dead end.  $u$  and  $d$  share the hexagon-radial data  $(r, w) = (1, 1)$  ( $L = 7$  for both), and the  $C_6$  resolvent  $c(r, w) = |\mu_4|^w / (\frac{5}{4} - \cos \frac{r\pi}{3})$  reads *only*  $(r, w)$  — so any *scalar* leg gives  $c_u/c_d \equiv 1$ . A scalar sees radius, not orientation. The  $u/d$  separation lives in the smallest *non-scalar* invariant of the  $SU(3)_F$  composition: the exterior square  $\Lambda^2 F$ , i.e. the oriented anchor-plane area  $\text{Pl}_{1,a}(K)$ . The “11” is then *generated*, not read:

$$(K, \mathbf{1}, a) \rightarrow B_K = \begin{pmatrix} 13 & 8 & 4 \\ 18 & 11 & 6 \end{pmatrix} \rightarrow \text{Pl}(K) = (-1, 6, 4) = (-N_\Phi, |R^+(A_3)|, |\mu_4|) \rightarrow \|\text{Pl}(K)\|_1 = 11,$$

$$\frac{c_u}{c_d} = \frac{g_{\text{car}} \|\text{Pl}_{1,a}(K)\|_1}{N_{\text{fam}}^2 \Delta_Q} = \frac{5 \cdot 11}{9 \cdot 13} = \frac{55}{117} \quad (\text{anchor microcode} = \frac{e_2(a)(p_3(a)+1)}{p_0(a)^2(2p_2(a)+1)}).$$

**Typing:** [E] for the Plücker / microcode identities ( $\text{Pl}(K) = (-1, 6, 4)$ ,  $\|\cdot\|_1 = 11$ ,  $5 \cdot 11 / (9 \cdot 13) = 55/117$ ); and — by Readout Rigidity (v49, below) — [E] also for the *physical* ratio  $\mathcal{C}_{ud}(\rho^*)$  = this  $\Lambda^2 F$  readout, which is *constant* on the derived selector stratum and so does *not* need any point-selection on  $\Lambda^2 F$ . The quark residual is thereby re-typed from “find the right scalar  $y$ ” to a closed exterior readout for the *ratio*; only the *absolute* amplitude scale ( $U_{\text{point}}$ ) stays as an anchor, no new scalar, no fishing.

*Audit* (the 121 lemma, [verification/v119\_review\_validation\_2.py]): the “11” has a second, independent appearance [E]: with  $h = \mathbf{1} + a = (2, 2, 3)$  in the canonical anchor ordering,  $h^T R h = 121 = 11^2 = \|\text{Pl}(K)\|_1^2 = (p_3 + 1)^2$  — the quadratic *anchor norm* of the residue operator  $R$  reproduces the Plücker integer (sensitivity control: the three orderings of  $h$  give {105, 121, 135}; only the canonical one yields 121). Bonus checksums:  $\mathbf{1}^T R \mathbf{1} = 22 = 2 \times 11$  (the entry sum of  $R$ ) and  $a^T R a = 40 = p_1 p_3 = 10 b_1 - 1$ . Typed *audit*, not load-bearing — the oriented Plücker mechanism above remains the derivation. [verification/v42\_exterior\_leg.py]

**Bridge test** ([verification/v43\_exterior\_bridge.py]). The typing is confirmed: for the  $SU(3)$  holonomy the exterior square is the conjugate representation,  $\Lambda^2(\widetilde{M}) = \widetilde{M}$  exactly — the exterior leg *is* the  $\bar{\mathbf{3}}$ . But the *continuous* action  $\Lambda^2(\widetilde{M}) \cdot (\mathbf{1} \wedge a)$  does *not* equal the *integer*  $\text{Pl}(K) = (-1, 6, 4)$ : the integer Plücker is the combinatorial  $K$  datum. So the bridge  $\rho^* \rightarrow \text{Pl}(K)$  is the *discrete* non-abelian-Hodge invariant (the integer  $R$ ;  $\det R = 8$  and  $\text{Spec}(Q_+) = \{1, 2, 3\}$  already confirmed on the bundle, Lemma U4 box), not a continuous exterior-square computation — exactly the H2 equivalence, now pinned from the  $\Lambda^2 F$  side.

**Lie-level grounding — no special math** ([verification/v44\_carrier\_exterior.py]). The discrete

invariant has a home already in TFPT: the carrier half-spinor is the *even exterior algebra* of the 5-slot carrier,  $16 = \Lambda^{\text{even}}(5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5$ , and under  $SU(5) \subset SO(10)$  the up quark sits in  $10 = \Lambda^2(5)$  while the down sits in  $\bar{5} = \Lambda^4(5)$ . So the *exterior degrees* are  $\deg(u^c) = 2$ ,  $\deg(d^c) = 4$ , with  $\deg(u) + \deg(d) = 6 = |R^+(A_3)|$  (the hexagon) and  $\deg(d) - \deg(u) = 2 = |\mathbb{Z}_2|$  (the sheet). The “ $\Lambda^2$ ” of the exterior leg is therefore not exotic Hodge data — it is the carrier’s own  $\Lambda^2$  grading (the up quark *lives* in  $\Lambda^2(5)$ ). This re-types the residual once more: from “discrete non-abelian-Hodge invariant” to “the carrier exterior degree” — a standard [E] fact. *Honest scope*: this grounds the *type*; the Plücker *number* 55/117 stays the family  $\text{Pl}(K)$  [E], and the carrier- $\Lambda^2 \leftrightarrow$  family- $\Lambda^2 F$  link (via family  $\otimes$  carrier = 15 = 16–1) plus the normalisation remain [C].

**The “11” is Pascal — a third, simplest origin** ([verification/v45\_family\_exterior.py]). Following that link,  $\text{Pl}(K)$  is the *exterior algebra of the family fundamental*  $4 = \mu_4$  (the  $A_3 = SU(4)$  fundamental):  $\Lambda^k(4) = \binom{4}{k}$  is Pascal row 4 = (1, 4, 6, 4, 1) with full sum  $2^4 = 16 = \dim S^+$ , and  $|\text{Pl}(K)| = (1, 4, 6) = \{\Lambda^0, \Lambda^1, \Lambda^2\}(4)$  are exactly its distinct exterior powers. Hence

$$\|\text{Pl}(K)\|_1 = \sum_{k=0}^2 \binom{4}{k} = 11 = 2^4 - \binom{4}{3} - \binom{4}{4} = 16 - g_{\text{car}}$$

(the cumulative  $\Lambda^\bullet(\mu_4)$  up to degree 2 =  $\deg(u^c)$ ), and family  $\otimes$  carrier = 15 =  $\dim \mathfrak{su}(4) = \Lambda^2(5) \oplus \Lambda^4(5) = 10 + 5 = N_{\text{fam}} g_{\text{car}}$  read three ways. So the “11” is generated a *third* way — the cumulative family exterior algebra up to the up-quark’s degree — pure Pascal/branching, *no special math*. The [C] physical normalisation is unchanged; the algebraic origin of 55/117 is now triply grounded.

**Lemma 6** (U5 — the “11” worry, *discharged* by Readout Rigidity). *The original worry was that  $c_u/c_d = C_{ud}(\rho^*)$  must be evaluated at the uniquely selected  $\rho^*$  without using 11 as input, so that until then 55/117 would be a mere table-level rational shadow. **This is now resolved (v49):**  $C_{ud}$  is constant on the discrete selector stratum  $S_{8,Q}$ , so the “11” is earned as the rigid Plücker readout  $\|\text{Pl}(K)\|_1$  (v42) — independently of point-uniqueness. Point-uniqueness of  $\rho^*$  enters only the absolute amplitude scale, not the ratio.*

**Lemma 7** (U6 — the four certificates).  $C_U^{(1)}$  wall enumeration (done);  $C_U^{(2)}$   $D_4$ -fixed character solver;  $C_U^{(3)}$  selector uniqueness (one  $S$ -equivalence class);  $C_U^{(4)}$  amplitude extraction — the ratios  $R, Q, c_u/c_d$  are read off the (derived) selector stratum (closed, v49/v71); only the absolute  $U_f^*, c_q$  normalisation is the anchor.

**Progress: an explicit valid flat bundle is realised (RH solve) ([E]).** A numerical Riemann–Hilbert solve produced an *explicit* Hermitian residue  $A_0$  (eigenvalues  $\{0, \frac{1}{3}, \frac{2}{3}\}$ ,  $\text{diag} = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ ) for which the Fuchsian system  $A(z) = \sum_k U^{k-1} A_0 U^{-(k-1)} / (z - i^k)$  simultaneously satisfies: the cusp class at each puncture; the splitting  $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$  (exponents  $\sum_k A_k = \text{diag}(2, 1, 1)$ ); the *path-ordered* monodromy at infinity is trivial,  $\|M_\infty - \mathbf{1}\| \sim 10^{-9}$ , i.e.  $M_1 M_2 M_3 M_4 = \mathbf{1}$ ; and the representation is *irreducible* — **case A** (non-trivial  $SU(3)_F$  mixing, not the diagonal case B). So a valid ( $U_{\text{wall}}$ ) flat bundle *exists and is constructed*; the U2 kill switch lands on A. *Honest scope*: this is one realising point (existence + case A), likely one of a residual family; the unique  $\nabla_F^*$  still needs the  $\det R = 8$  selector and  $c_u/c_d$  still needs the  $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$  dictionary (H2). [verification/v33\_explicit\_flat\_bundle.py]

*H2 bridge probed* ([verification/v34\_h2\_bridge\_attempt.py]). The explicit per-puncture monodromies  $M_k$  (cusp class,  $\prod M_k = \mathbf{1}$  to  $10^{-5}$ ) were computed; but the natural resolvent-dressed diagonal extraction ( $|\text{diag } M_k| = (0, \frac{1}{2}, \frac{1}{2})$  times the  $C_6$  resolvent factor) does *not* reproduce the lepton amplitudes (0.475, 1.107, 0.917). So the precise  $\Gamma_{ij}^{\min}$  geodesic-to-word dictionary (how the 6-site hypercharge hexagon combines with the 3-dim family  $\rho^*$ ) was the genuine remaining analytic input — it is *not* fixed by a natural guess, and we do *not* fish for 55/117. *Resolved since (v118): the correct extraction is the resolvent determinant of the  $W(A_3)$  monodromy, not the diagonal dressing — the probe’s failure is thereby explained, and the dictionary is exact.*

**Theorem U — the four-way split (sharper than one monster gate) [E]/[O]**

$D_4$  fixes the admissible chamber, not the physical point (the  $D_4$ -fixed locus is positive-dimensional, U3); the discrete selectors  $\det R = 8$ ,  $\text{SNF}(R) = (1, 1, 8)$ ,  $\text{Spec}(Q_+) = \{1, 2, 3\}$  do the cutting. The gate splits into four pieces, three of them essentially closed:

- $U_{\text{unitary}}$  [E]: parabolic degree 0 + polystable  $\Rightarrow$  unitary (Mehta–Seshadri)  $\Rightarrow \Phi = 0$ ; the harmonic metric is the *finite* invariant form  $H$  ( $M_k^\dagger H M_k = H$ ), not a Hitchin PDE ([verification/v40\_harmonic\_metric.py]).
- $U_{H2}$  [E]:  $R, Q, K$  are read off the bundle (splitting type =  $a$ ,  $\det R = 8$ ,  $\text{Spec}(Q_+)$ ) ([verification/v39\_uwall\_selectors.py]).
- $U_{\Lambda^2}$  [E]: **Readout Rigidity** — on the discrete stratum  $\mathcal{S}_{8,Q} = \{\det R = 8, \text{SNF}(R) = (1, 1, 8), \text{Spec}(Q_+) = \{1, 2, 3\}\}$  the anchor-plane readout is *constant*,  $\mathcal{C}_{ud}(\rho) = g_{\text{car}} \|\text{Pl}_{1,a}(K)\|_1 / (N_{\text{fam}}^2 \Delta_Q) = \frac{55}{117}$ , *independent* of the continuous  $D_4$  position — so  $c_u/c_d$  needs no point uniqueness ([verification/v49\_readout\_rigidity.py], [verification/v42\_exterior\_leg.py]).
- $U_{\text{point}}$  [O]: full uniqueness of  $\rho^*$  — needed only for the *complete*  $U_f^*$  amplitude matrix, *not* for  $c_u/c_d$ .

**Headline:** the remaining flavor bridge is not a PDE gate — it is a finite exterior *readout-rigidity* theorem; the only residual,  $U_{\text{point}}$ , is not an independent gate — it reduces to the single dimensionful anchor  $v_{\text{geo}}$  (Gate 1 below).

**The simple closure** ([verification/v71\_simple\_r\_bridge.py]). The selector stratum  $\mathcal{S}_{8,Q}$  is now *fully derived*:  $\det R = 8 = \text{rank } E_8$  with  $\text{SNF}(R) = (1, 1, 8)$  is the lattice invariant (cf.  $\det Q = 3 = N_{\text{fam}}$ , v70), and  $\text{Spec}(Q_+) = \{1, 2, 3\}$  is the  $D_4$ -equivariant result (v69). The four flavor operators are integer lattice maps with compiler-atom determinants ( $\det Q, \det K, \det R, \det L$ ) = (3, 4, 8, 20), product  $1920 = |W(D_5)|$ . So by Readout Rigidity the quark *ratios* ( $c_u/c_d = g_{\text{car}} \cdot 11 / (N_{\text{fam}}^2 \cdot 13) = \frac{55}{117}$ , etc.) are *pure integer Plücker readouts* — *no transcendental monodromy solve*. The earlier monodromy machinery was over-engineering for the ratios; the only transcendental piece,  $U_{\text{point}}$ , is the *absolute* amplitude normalisation — an anchor of the same nature as the one dimensionful scale, not a missing computation.

**Gate 1 complete:**  $U_{\text{point}} \rightarrow v_{\text{geo}}$  ([verification/v75\_upoint\_to\_vgeo.py]). The absolute amplitudes are not free either. The lepton amplitudes are exact ( $c = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ , v20), every cross-sector  $c$ -ratio is closed (Plücker), and each *sector product* is a clean  $\varphi_0^{\text{ret}}$ -power (Grand Mass Volume, v46:  $\det M_{\text{sector}} \sim (\varphi_0^{\text{ret}})^{6,9,10}$ ; the lepton product is  $\frac{16}{7} \cdot \frac{4}{3} \cdot \frac{7}{6} = \frac{32}{9} = 2^{g_{\text{car}}} / N_{\text{fam}}^2$ ). Since (pairwise ratios) + (product)  $\Leftrightarrow$  (individuals) is a bijection, the nine charged-fermion amplitudes are *fixed up to one overall scale*  $v_{\text{geo}}$ . **So  $U_{\text{point}}$  is not an independent gate: it reduces to  $v_{\text{geo}}$ , the same one irreducible dimensionful anchor as gravity's  $1/G$  (v68) — the two [O] anchors collapse to one.**

**Research Contract 2 — ( $G_{\text{metric}}$ )**

**Goal.** Construct the TFPT metric measure

$$Z[J] = \lim_{\chi \rightarrow \infty} \int_{\mathcal{G}_\chi / \text{Diff}} \exp(-S_{\text{rel},\chi}[g, A, \psi] + J \cdot O) d\mu_{\text{Cal},\chi},$$

$$S_{\text{rel},\chi} = \text{Tr} f\left(\frac{D_{\text{rel}} + A_\Sigma}{\chi}\right) - \text{Tr} f\left(\frac{D_{\text{ref}}}{\chi}\right) + \frac{i\pi}{2} \Delta\eta_\Sigma,$$

the relative spectral action whose low-curvature shadow is the already-closed  $R + R^2$  equation.

**Lemma 8** (G0–G1 — configuration space & BRST/Hodge Fredholm). *For  $s > 5/2$  a Sobolev space of  $\Theta$ -symmetric metrics on the doubled seam collar  $M_\Sigma$  admits a gauge slice; the gauge-fixed gravitational complex  $\Omega^0(TM) \xrightarrow{K} \Gamma(S^2 T^* M) \xrightarrow{K^*} \Omega^0(TM)$  with  $\Delta_{\text{gh}} = K^* K$  and  $\Pi_{\text{phys}} = \Pi_{\text{BRST/Hodge}} \Pi_{TT}$  is elliptic Fredholm; the conformal mode is removed as BRST-exact / non-propagating in the Calderón polarisation (not rhetorically). Machine: principal symbols in xAct/-Cadabra; linear-algebra certificates in Lean (medium).*

**Lemma 9** (G2 — finite relative spectral action, *heat-kernel grounded* [E]).  $S_{\text{rel},\chi}$  is finite, differentiable, with a local Seeley–DeWitt expansion  $\text{Tr} f(D^2/\chi^2) \sim f_4 \chi^4 a_0 + f_2 \chi^2 a_2 + f_0 a_4 + \dots$ . For

the Lichnerowicz Dirac operator  $D^2 = \nabla^2 + \frac{R}{4}$  (so  $E = -\frac{R}{4}$ , spinor trace  $\text{tr } \mathbf{1} = 4$ ) the Gilkey coefficients give, per  $(4\pi)^{-2}$ ,

$$a_2 = \frac{1}{6} \text{tr}(6E + R \mathbf{1}) = -\frac{R}{3} \quad (\text{Einstein-Hilbert } \int R), \quad a_4|_{R^2} = \frac{1}{360} \text{tr}(180E^2 + 60RE + 5R^2 \mathbf{1}) = \frac{R^2}{72}$$

(the  $R^2$ /Starobinsky term), so the spectral action structurally produces  $S_{\text{eff}} = \int \sqrt{g} \frac{\bar{M}_{\text{Pl}}^2}{2} (R + \frac{R^2}{6\bar{M}^2}) + \dots$  with the scalaron mass ratio fixed by the cutoff moment,  $M^2/\bar{M}_{\text{Pl}}^2 = 6(4\pi)^2/f_0$ . The TFPT closure  $M^2/\bar{M}_{\text{Pl}}^2 = c_3^7$  fixes  $f_0 = 6(4\pi)^2/c_3^7$ , i.e.  $M = c_3^{7/2} \bar{M}_{\text{Pl}} = 3.06 \times 10^{13} \text{ GeV}$  — the boundary normalisation  $c_3$  is the gravitational scale — and hence the Starobinsky slow-roll forms  $n_s \simeq 1 - \frac{2}{N}$ ,  $r \simeq \frac{12}{N^2}$ . The  $R + R^2$  structure is convention-independent (standard Gilkey); the precise rational  $\frac{1}{72}$  and factor 6 are the standard Dirac conventions used here ([\[verification/v36\\_spectral\\_action\\_g2.py\]](#), [\[verification/v28\\_gravity\\_fR.py\]](#)).

**The simpler resolution: “full QG” for physics is the gap-decoupled admissible sector** ([E]). The infinite-dimensional ambient measure (G6) is the summit, but the admissible IR sector does not wait for it. The metric coupling is gap-dominated (G5):  $2\|V_{\text{metric}}\| = 0.785 < \Delta = 6 \log \frac{3}{2} = 2.433$ , so the admissible sector keeps a *strictly positive* effective gap  $\Delta_{\text{eff}} = 1.648 > 0$  under the stated RP and metric-coupling assumptions. A positive gap dynamically *isolates* the admissible sector from the un-built ambient/deep-UV: under those assumptions the admissible (IR) measure is closed, and the ambient completion is decoupled from it (we do *not* assert it is physically irrelevant — that would itself be a physical interpretation). So the honest reading is

|                                                                                                          |
|----------------------------------------------------------------------------------------------------------|
| admissible IR sector = closed under stated RP;<br>ambient metric measure (G6) = $G_{\text{metric}}$ gate |
|----------------------------------------------------------------------------------------------------------|

G2 supplies the local action ( $R + R^2$ , heat-kernel); G5 supplies the decoupling of the admissible sector; together they reduce ( $G_{\text{metric}}$ ) to the ambient projective limit (G6). **Precise status (Alessandro’s boundary):** the admissible IR sector is closed under the stated RP and gap-dominance assumptions; the ambient metric projective measure is now discharged as a [C] *redundancy* (v369, the holographic-route keybox below), a certification object rather than a missing piece. Its absence does *not* affect the bounded IR claim. [\[verification/v36\\_spectral\\_action\\_g2.py\]](#)

**The glue Q-system: the index-4 object of the  $G_{\text{net}}$  statement exists explicitly ([E]/[C]).** The  $G_{\text{net}}$  target — “the seam-Calderón inclusion has Jones index  $4 = |\mu_4|$ ” — no longer refers to an unknown object. On the glue group algebra  $A = \mathbb{C}[\mathbb{Z}_4]$  (multiplication  $\delta_a \otimes \delta_b \mapsto \delta_{a+b}$ , unit  $\delta_0$ ) all Longo Q-system axioms hold exactly [E]: associativity, unit law, the Frobenius identity  $(m \otimes 1)(1 \otimes m^*) = m^*m = (1 \otimes m)(m^* \otimes 1)$ , and specialness  $mm^* = |\mathbb{Z}_4| \mathbf{1}$  — a special  $C^*$  Frobenius algebra with

|                                       |
|---------------------------------------|
| $\dim \theta = 4 =  \mu_4  = [B : A]$ |
|---------------------------------------|

exactly the Carrier Index Lemma value (KLM:  $\mu_{\text{carrier}} = 16 = 4^2 \mu_{E_8}$  re-verified). *Locality = isotropy* [E]: on the Lagrangian glue  $\langle (1, 1) \rangle$  the discriminant form gives  $q(k(1, 1)) = k^2 \in \mathbb{Z}$  for every  $k$  — the braiding obstruction vanishes (v92 in Q-system language); the two Lagrangian glues (the sheet pair) are the only maximal isotropic choices. *Halfway step* [E]:  $\{0, 2\} < \mathbb{Z}_4$  closes and gives the unique intermediate Q-system  $\mathbb{C}[\mathbb{Z}_2]$  = the  $SO(16)_1$  tower step, with index factorisation  $4 = 2 \times 2$ . *R2 sharpened* [C] (recorded, not claimed): what remains is the operator-algebraic *identification* “the seam-Calderón inclusion of the 16-Majorana net (v113) realises *this* Q-system” — matching two constructed objects, not a search; the three loads (metric + carrier + QBL, v123) hang on this one identification.

**The hull is the graded module of the Q-system** ([\[verification/v128\\_graded\\_hull.py\]](#)). The Q-system is not merely an abstract companion of  $E_8$  —  $E_8$  *realises* it [E]: the 240 roots, built as explicit norm-2 vectors in  $D_5 \oplus A_3$ +glue coordinates, decompose as  $240 = 52 + 64 + 60 + 64$  over

the glue  $\mathbb{Z}_4$  (the **v89** branching as vectors), and for *all* 6720 root pairs whose sum is again a root,  $\text{coset}(r_1+r_2) = \text{coset}(r_1) + \text{coset}(r_2) \bmod 4$ :

$E_8$  is a  $\mathbb{Z}_4$ -graded Lie algebra over the carrier; the grading group is the glue = the Q-system

— the checksum hull is the Q-system’s graded module, realised in 248 dimensions. The R2 identification thus matches the Calderón inclusion against a structure that  $E_8$  already *carries*.  
[verification/v125\_glue\_qsystem.py]

**Graded Frobenius** ([verification/v143\_graded\_frobenius.py]). The identification holds at the finite Lie level: the graded module is in fact a graded *Frobenius* structure, i.e. everything the Q-system demands of its module is already exact inside  $E_8$  [E]: (i) *coset duality* —  $\text{coset}(-r) = -\text{coset}(r) \bmod 4$  for all 240 roots ( $-C_1 = C_3$  bijectively,  $C_0, C_2$  self-dual), so the Killing form restricts to a *nondegenerate* invariant pairing  $g_k \times g_{-k} \rightarrow \mathbb{C}$ ; (ii) the glue character average  $\frac{1}{4} \sum_k i^k \deg$  is *exactly* the carrier projector (the conditional-expectation shadow, quasi-basis index  $|\mathbb{Z}_4| = 4$ ); (iii) each glue sector is *one*  $W(D_5) \times W(A_3)$  orbit with multiplicity-one dimensions  $64 = 16 \cdot 4 = (\mathbf{16}, \bar{\mathbf{4}})$ ,  $60 = 10 \cdot 6 = (\mathbf{10}, \mathbf{6})$  — simple currents, sector ring  $= \mathbb{Z}_4 =$  the  $\mathbb{C}[\mathbb{Z}_4]$  fusion, halfway subgroup  $\{0, 2\} = SO(16)_1$ . *R2 residue [C] (final form)*: exactly *one* conformal-net statement remains — the seam–Calderón inclusion of the 16-Majorana *net* carries this Q-system — with no remaining finite freedom.

**Where that statement lives: the NS/R sector census** ([verification/v148\_fock\_census.py]). The remaining work is scoped exactly. Every *untwisted* (Neveu–Schwarz) state of the 16 Majoranas has *integer* ( $D_5, SO(6)$ ) weights, and integer weights lie in the even discriminant classes — so the NS module supports only  $\{(0, 0), (2, 0), (0, 2), (2, 2)\}$ : **the odd glue classes  $k(1, 1)$  — the actual  $\mu_4$  simple currents — have zero NS support; they are twisted (Ramond-type) modules.** The Ramond zero-mode module ( $2^8 = 256$  ground states) carries exactly the four odd class pairs, 64 each, and splits  $128 + 128$  into the odd sectors of the *two* Lagrangian glues  $\langle(1, 1)\rangle$  and  $\langle(1, 3)\rangle$  (**v92**): *choosing the glue is choosing the Ramond projection* ( $128 =$  one  $SO(16)$  spinor chirality), and  $E_8$  assembles as  $248 = 120_{\text{NS}} + 128_{\text{R}}$ . *Consequence [C]*: the index-4 extension cannot even be *stated* on the untwisted module alone; R2’s one remaining statement is irreducibly about the twisted-sector structure of the seam–Calderón net, with the sheet choice realised as the R-projection; the finite shadow is exhausted (**v113/v125/v143** + this census).

**Hecke from geometry on the glue census (HECKE.GEOM.01).** The  $\mu_4$ -refined  $E_8$  class thetas that underlie the glue Q-system now carry an *in-suite* Hecke package (consolidated from discovery T27/T31/T32; 25 checks,  $\sim 11$  s). (A) Kneser  $p$ -neighbours realise  $\#_{\text{iso\_lines}} = \sigma_3(p) \# \mathbb{P}^3(\mathbb{F}_p)$  identically and at  $p = 2, 3, 5, 7$  ( $135/1120/19656/137600$ ). (B) The marked neighbour-sum is the affine Hecke element  $\nu_p = a \text{Id} + b T_p$  with cusp rule  $a_p = b - \sigma_3$ , recovering  $a_3 = -4$ ,  $a_5 = -2$  ( $p = 7$  consistency  $\lambda_{\text{odd}} = 19840$ ). (C) The census form space has  $\dim_{\mathbb{Q}} V = 7 = 5+2$  as the 2-adic oldform hull of  $E_4$  and  $f_8$ , with recovery  $\pi_{\text{cusp}} = (28 - T_3)/32$  and newform via  $(1 - V_2 U_2)$ . Classical theorems named classical; the claim is the compiler identification (correspondence  $\rightarrow$  operator  $\rightarrow a_p$ ; redundancy  $= \mu_4/2$ -adic). Fence: both new systems stay weight 4 —  $GL(1)/\text{weight-drop}$  transport remains [O]; no RH claim; no marker moves. Machine-checked: [verification/v535\_hecke\_from\_geometry.py]. [E] (mechanics) / [O] (weight drop)

**Eichler trace layer at the  $E_8$  lattice (HECKE.GEOM.EICHLER.01).** Consolidated promotion of discovery T33/T36/T37/T42 (23 checks,  $\sim 30$  s; builds on HECKE.GEOM.01). Witt  $\lambda_{\text{Eis}} = \sigma_3(\# \mathbb{P}^3 - \sigma_3)$  closed in  $p$ ; Eichler  $\lambda_{\text{geom}} = \lambda_{\text{Eis}} + a_p^2$  two-sided at  $p = 3$  (live type-(i)/(ii):  $352 = 364 - 12$ ) and  $p = 5$  (T36 freeze:  $3784 = 3906 - 122$ ) with assembler  $R = \sigma_3 - 1 - c(p^2)/8$ ; Type-A/B densities  $N_A = \min(240(1 + p^3), \#_{\text{iso}} - 1)$  ( $p = 7$  live  $82560/743040$ ; injectivity  $p = 11, 13$ ); signed  $a_p = -c(p)/8 \Rightarrow (-4, -2, +24)$  and  $b \in \{24, 124, 368\}$ ;  $O(1)$  assembler for odd  $p \leq 31$ . Classical scaffolding named classical; claim = in-suite two-sidedness + closed compiler densities. Fences:  $\text{Shell}(p^2)$  for  $p \geq 7$  out of scope; fibre refinement scoped to  $\{5, 7, 11\}$ ; weight-4  $\rightarrow GL(1)$  stays [O]; no RH; no marker moves. Machine-checked: [verification/v536\_eichler\_trace\_layer.py]. [E] (mechanics) / [O] (weight drop)

**Half-integral bridge (HECKE.GEOM.HALFINT.01).** Consolidated promotion of discovery T38/T41/T44/T45 (20 checks,  $\sim 90$ s; companion to HECKE.GEOM.01/HECKE.GEOM.EICHLER.01). Unique signed Shimura preimage  $\text{Sh}_{t=2, \psi=1}(g) = -8 f_8$  in the 70-element compiler weight-5/2 theta-monoid (witness  $g = \vartheta_2(q^2)^2 \vartheta_3(q^2) \vartheta_4 \vartheta_4(q^2)$ ; three further monoid elements lift to scales  $+8, -16, +16$ ); exact  $T(p^2)$ -eigenform with  $\lambda = a_p(f_8)$  for  $p = 3, 5, 7, 11, 13$ ;  $U_4(g) = 0$  and level  $32 = 4 \cdot 8$  outside Kohnen 1982 (plus-route structurally closed — negative result part of the claim); AFE-based Waldspurger/Baruch–Mao quotient  $R(d) = b(|d|)^2 / (|d|^{3/2} L(f_8 \times \chi_d, 2))$  constant on ten fundamental  $d \equiv 1 \pmod{8}$  ( $R = 23.1873585645\dots$ , spread  $\leq 10^{-12}$ , including  $d = 89$ ), with  $d \equiv 5 \pmod{8}$  vanishing via root number  $-1$ . Classical Shimura/Kohnen/Waldspurger/Baruch–Mao named classical; claim = unique monoid preimage + in-suite chain + *measured*  $R$ . Fences:  $\text{GL}(2)$  centre  $s = 2$  (not the  $\xi$ -line); no RH; ZETA.HP.CARRIER untouched; no marker moves. Machine-checked: [verification/v537\_halfintegral\_bridge.py]. [E] (Shimura/Hecke/Kohnen fence) / [C] (measured  $R$ )

**Compiler relative-trace identity (HECKE.GEOM.RTF.01).** Promotion of discovery T53 (18 checks,  $\sim 14$ s): HECKE.GEOM.01/HECKE.GEOM.EICHLER.01/HECKE.GEOM.HALFINT.01 are three projections of one finite relative-trace identity  $[E_8/\text{glue orbital counting}] = [\text{Newform/Waldspurger spectral sum}]$ . (R1)  $\text{Tr}_V(\nu_p \circ \pi)$  bilateral independent at  $p \in \{3, 5, 7\}$  across five projectors (anchors e.g.  $p=7$ : 727680 full / 19840 on  $f_8$ ). (R2) Eichler  $\lambda_{\text{geom}} = \lambda_{\text{Eis}} + a_p^2$  is the RTF orbit split ( $\lambda_{\text{Eis}} = 336/3780/19264$ ,  $a_p^2 = 16/4/576$ ). (R3) Period side = quaternary lattice counting with  $R = 23.1873585645$  (rel  $\sim 10^{-12}$ ). Verdict ONE-FORMULA. Classical Jacquet-type RTF / Eichler / Witt / Waldspurger named classical; NEW = compiler-native bilateral independence. Fences: identity *finite*; infinite RTF named open; no RH; not ZETA.HP.CARRIER-relevant; no marker moves. Machine-checked: [verification/v538\_relative\_trace\_identity.py]. [E] (R1/R2) / [C] (R3)

**Weil structure of the compiler family (RTF.GNS.WEIL.01).** Promotion of discovery T55/T61/T62/T63 (25 checks,  $\sim 6$ s): the compiler family possesses a fully identified Weil structure up to two *explicitly isolated* obstructions (obstructions are load-bearing verified content, not residual footnotes). (A) GNS = direct integral over the Gelfand spectrum of the character subalgebra (fibre-orthogonal; twist-mix per fibre  $\leq 5.037 \times 10^{-16}$ ). (B) Trivial Sato–Tate isotype =  $\text{GL}(1)$  core  $G_0 = (1 + Y)/(1 - Y) = \zeta_p(w - 3)^2 / \zeta_p(2w - 6) = \sum 2^{\omega(n)} n^{-u}$  globally. (C) Exact linear relation  $Q_{\text{fam}} = 2Q_\zeta(g) - 2Q_\zeta(g_b) + \text{Arch} + \text{Corr}$ . **Obstruction 1:** minus sign on the doubling term (family positivity does *not* imply  $Q_\zeta \geq 0$ ). **Obstruction 2:** Corr is non-automorphic (no polar/arch absorption; no finite Euler factor;  $e^{-\sum p^{-u}}$ -type). Finite-class  $Q_{\text{fam}} \geq 0$  measured [C]; dense-class positivity open / RH-adjacent — *not* claimed. Classical Weil/Gelfand/ $\text{SU}(2)$ /Sato–Tate/ $2^\omega$ /GNS named classical. Fences: not “almost RH”; ZETA.HP.CARRIER untouched; no marker moves. Machine-checked: [verification/v539\_weil\_structure\_family.py]. [E] (A/B/C/obstructions) / [C] (finite-class positivity)

**Amplitude route and positive linear carrier (RTF.GNS.AMP.01).** Promotion of discovery T67–T72 (34 checks,  $\sim 3$ s): the route out of the square plane behind the two RTF.GNS.WEIL.01 obstructions. (A) Amplitude Dirac  $D = \begin{pmatrix} 0 & V \\ V^\top & 0 \end{pmatrix}$  with  $D^2 = \text{diag}(VV^\top, K)$ ,  $V^\top V = K$  (rel  $2.7 \times 10^{-16}$ ), exists exactly and is Hecke-equivariant ( $p = 3, 5, 7$  intertwining exact; Shimura  $b(dp^2) = b(d)(a_p - \chi_d(p)p)$  on 644 pairs). (B) Geometric polarisation  $b = N_+ - N_-$  exact;  $\Theta = N_+ + N_-$  is a *pure* Siegel–Weil Eisenstein  $\sigma_3$ -eigenform ( $p \leq 13$ , cusp component zero) with Cohen seed  $\Theta(d) = -48 L(-1, \chi_d)$  exact-rational on 159 live  $d$  (Cohen 1975). (C) Every coefficient bilinear form inherits the even- $k$  deletion (Cauchy–Littlewood window lemma, five channels; numerator  $1 - p^6 X^2$  pair-independent); the deletion is exactly the square-class double counting (fam +  $2b + \delta_{k1} = [2, 2, 2, 2]$ ; Möbius square sieve exact). (D) Positive linear carrier  $\ell^2(d, \mu)$ ,  $\mu = 48 |L(-1, \chi_d)| |d|^{-a}$ : full weights  $\lambda_k = 1 + p^{3k} - (\chi p)^k$  (layer  $[1, 1, 1, 1]$ ) and plus balance  $Q = Q_\zeta(g_-) + Q_\zeta(g_+)$  (rel  $2 \times 10^{-15}$ ); the  $b$ /doubling kernel enters *with plus* (sign inverted vs the square plane). (E) Completed FE  $\Lambda_\Theta(s) = 8^{1-s} \Lambda_{\Theta^\dagger}(5/2 - s)$  exact (rel  $\sim 10^{-40}$ ; Fricke closed); the guaranteed cone is FE-self-dual. **Open boundary as content:** Euler-region positivity only (edge  $L$ -values, not the central line); the residual distance to the Weil cone is the FE-covariant gap

functional  $\lambda^*$  on  $n \equiv 6 \pmod{8}$  (one exact Farkas witness; no finite signed theta library removes it). Classical Cohen 1975 / Siegel–Weil / Jacobi–Fricke / Cauchy–Littlewood / Shintani / Weil 1952 / Farkas named classical. Fences: not “almost RH”; ZETA.HP.CARRIER untouched; no marker moves. Machine-checked: [verification/v540\_amplitude\_linear\_carrier.py]. [E] (A–E exact) / [C] (sampled cone facts;  $\lambda^*$  named open)

**Matching lemma and transport ledger package (RTF.GNS.LEDGER.01).** Promotion of the discovery proof package T78/T79/T80/T81/T82/T85 (33 checks,  $\sim 10$  s; every core check recomputed, not cited). (A) The matching lemma is *proved exact-integer* on  $[4, 10^6]$ :  $7S(j) < 40A(j)$  at every atom (0 violations over 939 870 clash atoms; proven no-overflow; big-integer rechecks); exact margin  $X = 22242575693/270725400000 = 0.082159$  at  $j^* = 554400$ ;  $\rho_{\text{crit}} = 1.1440 > 21/20$ ; maximal-greedy identity  $\Rightarrow$  uniform in  $(M, \lambda)$ ; structure laws (8-law, seed-tower,  $w$ -table, Cohen seeds  $\{-48, -80, -8, -8\}$ , bracket  $\Theta \leq 2|\psi| \leq 3\Theta$ ) at 0 tolerance. (B) The transport ledger closes exactly:  $Q_{\text{Weil}} = Q_{\text{cert}} + \Delta_{\text{arch}} + \Delta_2$  with  $\Delta_{\text{pole}} \equiv \Delta_{\text{conv}} \equiv 0$  *proven* (kernel collapse sympy-exact; residual  $4.4 \times 10^{-16}$  on 24 rows); the odd-prime side of the Weil functional equals the certified plus combination  $P_{\zeta}(g_-) + P_{\zeta}(g_+)$  exactly (rel  $3.9 \times 10^{-16}$ ). (C) The signed envelope is character-exact:  $-\psi = (\chi_{-4} + \frac{1}{4}\chi_8 + \frac{1}{4}\chi_{-8}) \cdot \Theta$  (all odd  $n \leq 50000$ , both builds); signed certificate 0 violations (margin  $8.86\times$ ); confinement set equality zero-credit  $\Leftrightarrow$  coherent on  $10^6$ . (D) The archimedean term is *internal*: Legendre duplication  $(2\pi)^{-s}\Gamma(s) = \frac{1}{2}\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1)$ ; kernel identity  $k_{\zeta} = K_{\text{fam}} - K_{\text{shift}}$  pointwise ( $7.9 \times 10^{-31}$ );  $\Delta_{\text{arch}} = A_{\text{fam}} - A_{\text{shift}}$  at  $9.9 \times 10^{-16}$ . (E) The coherent class ( $= \mathbb{Z}[i]$ -norms, set equality) is closed by the  $\lambda$ -equivariant channel: CM carrier  $g_{\lambda}$  exact in two independent routes;  $\mu_1 = c_1/(c_0 d^2) \in [-1, 1]$  exact;  $\lambda$ -window certificate 0.0653 vs 0.2365 unlifted ( $3.6\times$ ), exact Fraction rechecks. (F) Frontier facts [C]: coherent-chain crossing  $k^* = 14$  ( $\log_{10} N^* = 23.1$ , exact Fractions); Robin tail factor 6.16 — the constant route stays open. **Named limits as load-bearing content:** (i) *one* classically-shaped open lemma (correlated cancellation on credit-rich non-coherent tail atoms, uniform form; provably-shaped  $\neq$  formal proof); (ii) I5 in one-family form ( $Q_{\text{cert}} + \Delta_2 + A_{\text{fam}} - A_{\text{shift}} \geq 0$ ) as the single remaining TFPT-specific object — by the closed ledger *equivalent* to Weil positivity  $\Leftrightarrow$  RH (equivalence typing, no progress claim). Classical Gronwall / Robin 1983 unconditional / Cohen 1975 / Hecke 1918–1920 / Cornacchia / Dirichlet / Mertens-AP / Landau / Legendre duplication / Weil 1952 / Alaoglu–Erdős named classical. Fences: no RH-evidence language; not “almost RH”; ZETA.HP.CARRIER untouched; no marker moves. Machine-checked: [verification/v541\_matching\_lemma\_ledger.py]. [E] (A–E exact) / [C] (frontier facts; two named limits open)

**The margin-chain identities of phase 2 (PRIME.MARGIN.IDENT.01).** Promotion of the *identity/theorem block* of the discovery parts T128–T131 (44 checks,  $\sim 0.2$  s; every check recomputed on small frame-A windows). Scope fence, declared with the row: identities and *per-instance* theorems only — no fit, no graded floor, no Lanczos value, no uniform-in-zone statement. Battery: 24 seam instances ( $n = 3 \dots 97$ ,  $\rho = 1.25 \dots 4.00$ ; every inverted or diagonalised matrix  $\leq 300$ ), 43 graded/fine grid pairs and 400 randomised profiles; every identity is a residual against a *preregistered* tolerance plus a mutation control that must fail by  $\geq 10^{-3}$  relative. (1)–(2)  $\sum_i (1 - g_i) = t_l + t_r$  exactly ( $5.5 \times 10^{-13}$ )  $\Rightarrow$  the  $\kappa$  weights are a probability vector and  $\kappa_{\text{flat}} = 1$ ; the (T) four-term identity  $\tau_{\text{dn}} = \|y\|^2 - m_{\text{prot}} + m_{\text{fill}} - V_{\text{bord}}$  at  $2.2 \times 10^{-16}$  (two independent routes for  $\tau$ ). (3)–(4)  $p_{N-1} = 2 - p_0 + 2E$  at  $2.3 \times 10^{-16}$   $\Rightarrow$  the equivalence  $2E > p_0 \Leftrightarrow p_{N-1} > 2$  (0 exceptions, both branches realised); the Abel/Hölder chain  $\kappa \leq 2 - p_0 + \frac{N-2}{N} \sum_j |\Delta p_j - \overline{\Delta p}| + (p_{\text{max}} - p_{N-1})$  per instance (0 violations; the same chain *without* the curvature term is violated on 88/400 profiles). (5)–(6) Matrix-free two-scale assembly  $= J^T Q J$  ( $5.2 \times 10^{-16}$ ), graded overlap  $= J$ ; the C ea/Strang defect identity  $S_{\text{graded}} - S_{\text{uniform}} = R^T X^{-1} R$  as a *matrix* identity with PSD defect on all 43 grids ( $2.1 \times 10^{-14}$ )  $\Rightarrow$  the compression error is one-sided by construction. (7)–(9) The secular sandwich  $\varepsilon/(\|u\|^2 + \varepsilon/\mu_1) \leq \lambda_{\min}(Q) \leq \varepsilon/\|u\|^2$  on all 48 sections where  $A > 0$  and  $\varepsilon > 0$  are *checked* (width  $\leq 1.4448$ ) with Albert’s sign equivalence in *both* directions;  $S^{-1} > 0$  entrywise (measured 48/48)  $\Rightarrow$  sign-constant and simple ground vector (Perron–Frobenius), with a control where hypothesis and conclusion fail together; and  $n_{\text{run}} \leq n_{\text{cross}} + 1 \leq s_2 + 2$  with  $\sum_j |\Delta p_j - \overline{\Delta p}| = \text{TV}(P)$ , 0 exceptions on a non-vacuous battery ( $s_2$  up to 31). **Named limits as load-bearing content:** (i) per instance

on *small* windows — nothing uniform in the zone index; (ii) the  $\kappa$  law is *not* claimed (T129 found it false in general), only the chain underneath it; (iii) positivity of the pole-free section at depth (the open M25d) is neither assumed nor extended and no floor value is carried; (iv) the Perron step is an *implication* with a measured hypothesis; (v) the defect identity says nothing about the defect’s *size* (the open T130 exponent). Classical Abel / Hölder / Schur–Haynsworth 1968 / Albert 1969 / C  a 1964–Strang / Yserentant 1986 / Perron–Frobenius–Ostrowski 1937 named classical; Weil 1952 cited, never used as a criterion. Fences: zero-firewall AST-checked; no RH-evidence language; not “almost RH”; ZETA.HP.CARRIER untouched; no marker moves. Machine-checked: [verification/v542\_margin\_chain\_identities.py]. [E] (per-instance identities; nothing uniform)

**The lumped  $M$ -matrix pair of phase 2 (PRIME.MMATRIX.IDENT.01).** Promotion of the *identity/theorem block* of the discovery part T136 (35 checks,  $\sim 1$  s; every check recomputed on small frame-A windows). Scope fence, declared with the row: matrix identities, per-instance theorems with *checked* hypotheses, and one *negative* result — no fit, no  $D$ -exponent, no uniform-in-zone statement, no floor value. Battery: 20 seam windows ( $n = 4 \dots 59$ ,  $\rho = 1.25 \dots 4.00$ ,  $h = 25 \dots 299$ ; every factorised or diagonalised matrix  $\leq 300$ ) giving 39 matrices with positive off-diagonals, 312 shifted ladder rungs, 8 whitened two-grid rungs; every statement is a residual against a preregistered tolerance plus a mutation control that must fail by  $\geq 10^{-3}$  relative. (1)–(2) With  $L_\Delta = \text{diag}(\Delta \mathbf{1}) - \Delta$  the Laplacian of the positive off-diagonal part,  $A_B = A + L_\Delta$  is Stieltjes, row-sum preserving ( $1.6 \times 10^{-15}$ ) and  $A_B \succeq A$  — an *upper* bound on  $\lambda_{\min}(A)$ , a floor never; the congruence  $A = A_B^{1/2}(I - W)A_B^{1/2}$ ,  $W = A_B^{-1/2}L_\Delta A_B^{-1/2} \succeq 0$ , is a matrix identity ( $4.4 \times 10^{-15}$ ) with  $\lambda_{\max}(W) = \rho(L_\Delta A_B^{-1})$ , and the *declared* equivalence  $\lambda_{\max}(W) < 1 \iff A \succ 0$  is checked in *both* directions — the Loewner sandwich is circular by itself, which is the fence. (3)–(4) With  $x = A_B^{-1} \mathbf{1} \geq 0$  and  $A_B^{-1} \geq 0$  entrywise (Ostrowski 1937, *checked*),  $\lambda_{\min}(A_B) \geq 1/\max_r x_r$  on all 39 rows (min slack  $3.0 \times 10^{-3}$ , sharpest ratio 0.9070) — one solve, one sign check, no eigenvalue;  $(A - sI)_B = A_B - sI$  ( $2.5 \times 10^{-16}$ ) and  $(A_B - sI)^{-1} \geq A_B^{-1} \geq 0$  entrywise (Berman–Plemmons 1979)  $\Rightarrow$  the shifted ladder is *structurally empty* (0 exceptions of 312). (5)–(6) For the regular splitting  $A_B = D_0 - N_0$  (all three hypotheses checked), Varga’s  $\rho(J) = \tau/(1 + \tau)$  holds at  $4.8 \times 10^{-15}$  ( $\rho(J) = 0.7347\text{--}0.9860$ ), and the Collatz–Wielandt bracket at the anchor vector gives an *a-priori*  $\rho(J) \leq 0.9861 < 1$  with no eigenvalue anywhere, sharp on the measured gap by 1.0062–1.0531. (7) The  $k$ -scanned M18 majorant is a *valid* bound at every  $k$  (39 (rung,  $k$ ) pairs, 0 violations) and does *not* reach the preregistered bar 1/2 on any rung (best 3.080) while the measured bad mass is  $\leq 0.0120$ : a *negative* result, the localisation term alone exceeding the bar (0.542), and the bar is not moved. **Named limits as load-bearing content:** per instance on small windows, nothing uniform in the zone index, in  $D$  or in the gap; the T136  $D$ -exponents (anchor  $D^{-0.56}$ , gap  $D^{2.72}$ , slack  $D^{0.12}$ ) are fits and stay in the sandbox; (2) carries no floor and M25d stays open; (3) and (5) are implications with hypotheses checked at the instance; (4) kills a route and says nothing about the true  $\lambda_{\max}(W)$ . Classical Stieltjes / Ostrowski 1937 / Fan 1958 / Berman–Plemmons 1979 / Varga 1962 / Collatz 1942–Wielandt 1950 / Haynsworth 1968 named classical; Weil 1952 cited, never used as a criterion. Fences: zero-firewall AST-checked; not “almost RH”; ZETA.HP.CARRIER untouched; no marker moves. Machine-checked: [verification/v543\_lumped\_pair\_identities.py]. [E] (per-instance identities) + [X] (one closed route)

**The long-lag support structure of phase 2 (PRIME.LONGLAG.SUPP.01).** Promotion of the *structure block* of the discovery part T137 (24 checks,  $\sim 0.2$  s) on the same 20 windows (3–25 prime-power atoms, 20555 positive off-diagonal edges, 455 loaded stripes, 39 bracketed lumped comparisons). Scope fence: support and amplitude structure, one two-sided bracket, and one *death certificate* for a whole family of bounds — no fit, no decay exponent, and *no* structure bound that closes. (1)–(3) The kernel splits exactly as  $c = c_{\text{arch}} - c_{\text{comb}}$  with  $c_{\text{comb}} \geq 0$ , and the archimedean section alone has *all* off-diagonals  $\leq 0$  on 20/20 windows  $\Rightarrow$  every positive off-diagonal of  $A$  is comb-generated (control: the full section carries  $+2.0 \times 10^{-1}$ ); all 20555 positive edges have their Hankel index  $M - 1 - r - t$  inside the atom-hat index set (per-window fraction exactly 1.000000)  $\Rightarrow$   $\text{supp}(\Delta)$  is a union of anti-diagonal stripes at prime-power atom positions (converse reported, *not* claimed); each stripe is a perfect matching (0 endpoint collisions), so its edge vectors are exactly

orthogonal and  $L_\Delta$  per stripe is a direct sum of  $2 \times 2$  blocks. (4)  $\Delta_{rt} \leq c_{\text{comb}}[M - 1 - r - t]$  on all 20555 edges (excess  $-1.9 \times 10^{-6}$ , sharpest ratio 0.99998532) — no eigenvalue, no norm; the Toeplitz lag in place of the Hankel index fails on 20415/20555. The tighter one-atom refinement is *honestly undecided* at  $h \leq 300$  (the atom hats never overlap) and is not promoted. (5) For the lumped comparison  $B = D_0 - N_0$  the anchor identity  $q_J = 1 - \min_r 1/(d_r x_r) < 1$  ( $8.2 \times 10^{-16}$ ) needs no measurement; the Neumann partial sum to  $K = 8$  is then an *entrywise* lower bound for  $B^{-1}$  *because the discarded terms are nonnegative*, and partial sum plus anchor-weighted tail an entrywise upper bound, on 39 rows. (6) Since Gershgorin, every row-sum norm and every Collatz–Wielandt quotient at every *positive* weight are upper bounds for  $\rho(|E|)$ ,  $E = B^{-1}L_\Delta$ , one rigorous *lower* bound decides the family at once:  $\rho(|E|) \geq 1.3141 \geq 1$  on 20/20 rows (cheapest four members  $\geq 1.5937$ ), so no member can *ever* certify  $\lambda_{\max}(W) < 1$  — while the *signed*  $\rho(E) = \lambda_{\max}(W) \leq 0.99995781$  on the same rows: the absolute value alone destroys the cancellation. **Named limits as load-bearing content:** per instance on small windows, no decay profile and no amplitude law (T137’s decay measurements are fits); (2) is one inclusion; (4) bounds the amplitude and *not*  $\lambda_{\max}(W)$ , and the structure bound assembled from (2)–(4) does *not* beat the margin (T137: 0 of 35 windows) and is not promoted; (5) brackets the *lumped* comparison, not the target, whose entrywise inverse positivity at depth stays open; (6) closes a route, not the question. Classical Gershgorin 1931 / the Neumann series / Haynsworth 1968 named classical; Demko–Moss–Smith 1984 cited as a model and *not* applied (the comparison is dense); Weil 1952 cited, never used as a criterion. Fences: zero-firewall AST-checked; not “almost RH”; ZETA.HP.CARRIER untouched; no marker moves. Machine-checked: [verification/v544\_long\_lag\_support.py]. [E] (per-instance structure) + [X] (one family closed)

**PRIME.HARDY.IDENT.01 — the Hardy-core identities of phase 2 (T140, T141).** (1) The form lifting Gram =  $CHC^\top$  with  $C = \text{diag}(\sqrt{\Delta})M$  ( $3.0 \times 10^{-15}$ ), hence  $\text{rank}(\text{Gram}) \leq h - 1$ . (2) The exact finite-core reduction: the closed geometric covering kernel  $K_{rs} = W([r \wedge s, r \vee s])$  equals  $M^\top \Delta M$  ( $1.3 \times 10^{-15}$ ) and  $\rho(W) = \lambda_{\max}(K^{1/2} H K^{1/2}) = 1 - \lambda_{\min}(A, A_B) = \lambda_{\max}(\text{Gram})$  on three independent routes. (3)  $H = \text{diag}(s) + L_N$  ( $4.7 \times 10^{-16}$ ). (4) The checkerboard split  $\text{Band}_b = D + A_{\text{even}} + A_{\text{odd}}$  entrywise exact with a three-step Weyl sequence. (5) The four Hardy identities — the  $K^+$  Rayleigh form,  $L_N = D^\top B D$  with the signed crossing kernel,  $Q = B_+ \mathbf{1}$  and  $DKD^\top = L_\Delta$  on the interior nodes. (6) The closed Muckenhoupt quantity  $A_{M0} = \max_k Q_k(\Delta \mathbf{1})_{k+1}$  by two routes. (7) The additive Hardy chain and the joint shape  $\Lambda(H, Y) \cdot \Omega(Y)$  as *valid* upper bounds on 11/11 windows — dominance only. (8) As a negative typing, the sign-versus-absolute separation of the measured spreads ( $1.36\times$  against  $\geq 54.94\times$ ). **Named limits as load-bearing content:** per instance on small windows, nothing uniform in the zone index or in  $D$ ; every exponent is a fit; (7) promotes validity and not size — neither shape beats the target, the additive shape is dead as a *shape* by its own exact Weyl floor and the joint shape is decided at the normalisation  $\Omega$ ; (2) is an identity and not a bound, so the zone-uniform discrete Hardy inequality stays *open*. Fences: Muckenhoupt 1972 / Bradley 1978 / Opic–Kufner 1990 / Gantmacher–Krein 1950–1960 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [verification/v545\_hardy\_core\_identities.py]. [E] (per-instance identities) + [X] (one route typed)

**PRIME.CAPCHAIN.IDENT.01 — the capacity-chain identities of phase 2 (T142–T145).** (1) The capacity decomposition  $K^{-1} = D^\top J^{-1} D + xx^\top / \text{cap}$  with  $J = DKD^\top$  (entrywise,  $3.8 \times 10^{-15}$ ),  $x = K^{-1} \mathbf{1}$ ,  $\text{cap} = \mathbf{1}^\top K^{-1} \mathbf{1}$ , as a matrix identity ( $1.4 \times 10^{-13}$ ) whose Dirichlet half is an *orthogonal projection* under the congruence —  $\Omega = 1$  exactly ( $5.1 \times 10^{-12}$ ). (2) The exact capacity–Rayleigh identity for  $1 - \rho(W)$ : numerator/denominator assembly equals the direct quotient for every  $v$  ( $9.5 \times 10^{-14}$ ), eigenvalue reproduced under the declared 1/gap conditioning, inf property verified, the constant vector carried entirely by the capacity term. (3) Cholesky nestedness:  $\text{cap}_E([a, a+j])$  as a prefix sum of squared forward-substitution entries of one triangular solve (3404 interval capacities at  $2.0 \times 10^{-14}$ , exhaustive on the median window; the reversed ordering fails at  $\geq 4.6 \times 10^{-1}$ ). (4) The whitening certificate  $\lambda_{\max}(W) \leq \kappa_{\text{up}} = 1.2245\text{--}1.4236 \leq$  Gershgorin 2.1213, direction-correct, refusing at the shrunken shift. (5) The T145 M4 split: the pointwise mass-half theorem ( $\sum_k f_k^2 \leq \psi^2$  exact),  $\sigma_{\text{tot}} = 0.2197\text{--}0.4425 < 1$  per window, licence 4 with  $|R|$  and not  $R^+$  (witness

$[[1, -\frac{1}{2}], [-\frac{1}{2}, 1]]$ ,  $3 > 2$ , built in), the ordered sign-free chain, and  $S1'$  *certified per window on both routes* (better route  $c_0 = 2.5154\text{--}4.2265$  against Maz'ya's Dirichlet value 4), with Charikar's cited density constant bracketed by exhaustive enumeration. **Named limits as load-bearing content:** per instance on small windows, nothing uniform in the zone index or in  $D$ ; the  $c_0$  of (5) is *measured* at the minimiser's own layer cake and the a-priori level lemma (T145's L1) stays open, unclaimed; Maz'ya's strong-type half is not used as a theorem anywhere — it is the thing the cycle transcribes. Fences: Maz'ya 1985 / Muckenhoupt 1972 / Fukushima–Ōshima–Takeda 1994 / Miclo 1999 / Charikar 2000 / Goldberg 1984 / Gershgorin 1931 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: `[verification/v546_capacity_chain_identities.py]`. **[E]** (per-instance identities and certified inequalities)

**PRIME.LEVEL.LEMMA.01 — the level-lemma identities of phase 2 (T146).** (1) The union identity for nested cakes:  $\sum_{j,l} c_j c_l |S_j \cup S_l| = 2T \|\psi_t\|_1 - \|\psi_t\|^2$  with  $T = \sum_j c_j$  and  $\psi_t = \sum_j c_j \mathbf{1}_{S_j}$  — form-independent and  $O(m)$ , exact (0.0) against the  $K \times K$  double sum and against union sizes computed with no nestedness used (controls: one shuffled level set  $3.2 \times 10^{-2}$ , a 1% coefficient perturbation  $4.8 \times 10^{-3}$ ). (2) The layer-cake counting lemma at a general base:  $G(b) \leq 2b^2 \|v\|_\infty \|v\|_1 / \|v\|^2 + \varepsilon$  for *any* vector at all 6 preregistered bases; the classical dyadic 8 at  $b = 2$  exactly, the base free (gain  $4/b_*^2 = 3.7612$ ; the one-level-down cake breaks the domination on every draw). (3) The resolvent delocalisation bound:  $\psi = \lambda R \psi$  ( $9.0 \times 10^{-14}$ , an identity — no Davis–Kahan transfer anywhere)  $\Rightarrow \|\psi\|_\infty \leq \lambda_{\text{up}} \max_j \|Re_j\|$  with Rayleigh–Ritz  $\lambda_{\text{up}}$  (1.0592–1.2883) and residual-paid certified column norms, within 1.061–1.337 of the truth; the lever: the certified columns sit 2.85–10.17 below the trivial ceiling  $1/\lambda_{\text{min}}$ , giving  $\Gamma = 1.8356\text{--}2.2797$  against the useless  $\sqrt{m}$ . (4) The composite level lemma:  $c_0^{\text{ap}} = 2b^2 \Gamma \min(1, \Gamma_1) + \varepsilon = 3.9042\text{--}4.8488$  (slack 1.986–2.418 over the measured constant, under the ceiling  $16 = 4 \times$  Maz'ya's Dirichlet 4), the certified window inequality  $\text{cert\_lam\_max}(R) \leq c_0^{\text{ap}} \Psi_{\text{abs}}$  on 12/12 windows, chain loss 0.0423–0.1272 of the exact gap; the built-in no-go discriminator ( $R = aa^\top + \varepsilon I$ ,  $a_i = i^{-1/2}$ :  $\Gamma$  grows  $3.676 \rightarrow 6.798$ , ratio  $1.85 \geq 1.5$ ,  $c_0^{\text{ap}} = 11.10$  exceeds every window value) fails exactly where it must, the theorem half surviving on every no-go size and the Dirichlet path-Laplacian control flat (1.0063). **Named limits as load-bearing content:** per instance on small windows — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in  $D$ , and explicitly *no* statement for all  $D$ : the asymptotic delocalisation of the Green columns (T146's remainder) stays *open*, typed open; only the sign-free / Green constant is bounded a priori — the energy-route  $\sigma_{\text{tot}}$  stays measured; every T146 exponent is a fit, sandbox. Fences: Maz'ya 1985 (strong-type half not used as a theorem anywhere) / Rayleigh 1877–Ritz 1909 / Cauchy–Schwarz / Miclo 1999 / Davis–Kahan 1970 (cited, deliberately not used) / Perron–Frobenius (not applicable to  $E$ , not used) / Fukushima–Ōshima–Takeda 1994 / Charikar 2000 / Goldberg 1984 / Gershgorin 1931 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: `[verification/v547_level_lemma_identities.py]`. **[E]** (per-instance identities and certified inequalities)

**PRIME.GREEN.SZEGO.IDENT.01 — the Green/Szegő identities of phase 2 (T147, T148).** (1) The factorisation identity  $\Gamma = \sqrt{m} \lambda_{\text{up}} \max_j \|Re_j\| = \sqrt{Q^*} \cdot S_w$  with  $S_w = \lambda_{\text{up}} \|R\|_F$  purely spectral and  $Q^* = m \max_j (R^2)_{jj} / \text{tr}(R^2)$  purely geometric — exact at  $2.2 \times 10^{-16}$  on 12 frame-A windows ( $m = 26 \dots 285 \leq 300$ ) and 7 random positive definite forms; the certified factorisation (residual-paid column norms, Frobenius bracket) dominates the true  $\Gamma$  everywhere ( $S_w = 1.1186\text{--}1.5777$ ,  $Q^* = 2.0879\text{--}2.8634$ ; controls  $5.0 \times 10^{-3} / \geq 10^{-3}$ ). (2) The bottom-block trace identity  $\sum_j (\Pi_B)_{jj} = |B|$  exact ( $3.3 \times 10^{-16}$ ;  $\theta = 10$ ,  $|B| = 3\text{--}5$ ), hence  $Q_B \geq |B|$  a theorem; the *measured*  $Q_B/|B| = 1.5091\text{--}1.7980$  inside the preregistered band  $[1, 2.2]$  against the Szegő/Widom sharp reference  $2|B|$  (the rank- $|B|$  coordinate projector overshoots by  $\geq 3.9$ ). (3) The second-difference  $\ell^1$  ceiling:  $|a_k| \leq \min(1, \|s_k\|_\infty \|L_0^p \psi\|_1 / \mu_k^p)$  for  $p = 1, 2$  against the true coefficients everywhere, the  $m$ -free closed form  $\|a\|_1 \leq 2\sqrt{\nu} + 1$  with  $\nu = (m+1)^{3/2} \|L_0 \psi\|_1 / (2\sqrt{2})$ , and  $m \|\psi\|_\infty^2 \leq 2 \|a\|_1^2$ ; Dirichlet calibration  $\|a\|_1 = 1$  exact,  $\nu \rightarrow \pi$  (controls:  $\times 10^{-3}$  fails, a one-index  $\mu$  shift breaks the identity at  $1.5 \times 10^{-1}$ ). (4) The layer-cake  $S_w$  certificate: completed

LDL<sup>T</sup> inertia counts equal the direct spectral counts on all 96 (window, rung) pairs (Sylvester 1852; Bunch–Kaufman 1977), and the layer cake certifies  $S_w \leq 1.3288\text{--}1.8220$  per window (true  $1.1186\text{--}1.5777$ , slack  $\leq 1.216$ ; dropping the bottom layer breaks it everywhere). (5) The Liouville transform  $m\|\psi\|_\infty^2 \leq \kappa_\Lambda m\|\hat{\varphi}\|_\infty^2 \leq 2\kappa_\Lambda \|a(\hat{\varphi})\|_1^2 \leq 2\kappa_\Lambda \text{ceil}(\hat{\varphi})^2$  with  $\varphi = \Lambda^{-1/2}\psi$  and a-priori  $\kappa_\Lambda = \max \Lambda / \min \Lambda = 1.3868\text{--}2.5805$ , stepwise on every bottom-block mode; the weight-class discriminator on a model section with a genuine order-2 zero ( $f(0) = 0$ ,  $f''(0)/2 = 1$  exact), ladder  $m = 64 \dots 288$  at matched  $\kappa_\Lambda$  (mismatch 0.0013): the bounded-variation multiplier keeps  $\nu$  flat (max/min 1.003) while the rough one (TV larger by 162.6) grows by 19.9 — the roughness, not the conditioning; the first step  $\times 10^{-3}$  fails provably (slack  $\leq \kappa_\Lambda^2$ ). **Named limits as load-bearing content:** per instance on small windows — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in  $D$ , and explicitly *no* statement for all  $D$ : the one open lifting input ( $\nu_L \leq F(\text{form})$  with  $F$   $m$ -free; equivalently bounded TV(log  $\Lambda$ ) or any weaker smoothness of the multiplier) stays *open*, typed open; the symbol-level order-2 hypothesis is *refuted* (T148:  $f(0) < 0$  on every surface window) and enters nowhere as a theorem;  $\sigma_{\text{tot}}$  stays retired as a route; every T147/T148 exponent is a fit, sandbox. Fences: Kac–Murdock–Szegő 1953 / Widom 1958 / Basor–Ehrhardt 2009 / Böttcher–Silbermann (the mechanism’s address, never an authority) / Sylvester 1852 / Bunch–Kaufman 1977 / Euler 1735 / Rayleigh 1877–Ritz 1909 / Cauchy–Schwarz / Davis–Kahan 1970 (cited, computed vacuous, not used) / Gershgorin 1931 / Wilkinson 1968–Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [verification/v548\_green\_szego\_identities.py]. [E] (per-instance identities and certified inequalities)

**PRIME.GAUGE.PARITY.IDENT.01 — the gauge/parity identities of phase 2 (T149, T150).**

(1) The gauge freedom: for *any* positive diagonal  $\tilde{\Lambda}$  the whitening  $\tilde{E} = \tilde{\Lambda}^{-1/2} A \tilde{\Lambda}^{-1/2}$  is an exact congruence, so  $\lambda_{\min}(A) \geq \text{cert } \lambda_{\min}(\tilde{E}) \cdot \min_j \tilde{\Lambda}_j$  is a valid certified lower bound for every gauge (an identity plus one Rayleigh step; family maximum valid) — verified on all 48 gauge assemblies over 12 frame-A windows ( $m = 26 \dots 285 \leq 300$ );  $\Gamma = \sqrt{Q^*} S_w$  holds *gauge-invariantly* at  $2.4 \times 10^{-16}$ ; the const gauge has TV(log  $\tilde{\Lambda}$ ) = 0 *exactly* with the certified entrywise sandwich  $\sigma = 1.3868\text{--}2.5805$  (controls: a 1%  $Q^*$  perturbation breaks at  $4.99 \times 10^{-3}$ ; the completed-Cholesky floor refuses on the shifted indefinite form, 12/12). (2) The two-regime discriminator: the flutter smoother (moving geometric mean, half-width  $m/16$ ) removes TV by  $\geq 2.88\times$  while moving the Liouville-transported  $\nu_{\tilde{L}}$  by  $\leq 0.0088$  — even the matched-amplitude rough re-gauge moves it  $\leq 0.0046$  (bar 0.01) — while the *untransported* functional jumps 0.41–9.76 under the same rough gauge: the roughness is in the form, not the multiplier (T149’s withdrawal of the TV hypothesis, wired per instance). (3) The explicit zone diagonal  $\Lambda_r = c_0 - c_{M-1-2r} + \sum_{s \neq r} \max(c_{|r-s|} - c_{M-1-r-s}, 0)$  at residual  $5.9 \times 10^{-16}$ ; the exact split  $c = c^{\text{arch}} + c^{\text{atom}}$  with  $c^{\text{atom}} \leq 0$  entrywise and the closed atom budget  $\|c^{\text{atom}}\|_1 \leq \sum_j \mu_j \leq 4B\sqrt{N}$ ,  $B = 1.038821$  the verified maximum of  $\psi(n)/n$  over the whole atom table (Chebyshev 1852, computed, never assumed; Abel summation); the archimedean section alone is *indefinite* on 12/12 (up to 5 negative eigenvalues, LDL<sup>T</sup>) while the full section has zero — the atoms are co-responsible for the positivity, the additive perturbation reading structurally empty (control: dropping the positive part breaks at  $4.25 \times 10^{-1}$ ). (4) The parity compression  $U_-^T T_M(c) U_- = T - H$  and  $U_+^T T_M(c) U_+ = T + H$  exact ( $3.9 \times 10^{-16}$ , cross block  $1.4 \times 10^{-16}$ ) on the 6 windows with  $M \leq 300$ : the reflection-odd section *is* the compression of the full symmetric Toeplitz section onto its antisymmetric parity sector, and the negative inertia of the sign-changing symbol ( $f(0) = -29.2 \dots -6.3 < 0$ , re-read from T148, never re-derived) sits entirely in the *even* sector, the odd sector counting zero (control: the sign-flipped basis misses by  $3.99 \times 10^{-1}$ ). (5) The parity-sine calibration: the parity sines are exact eigenpairs of the parity Laplacian (corner 3 forced by antisymmetry; eigen-residual  $2.3 \times 10^{-13}$ ),  $\nu_k^P = \pi k^2$  on the exact model at  $m = 2001$  to  $3.0 \times 10^{-5}$ , and the per-window ladder  $\nu_k^P \leq Ck^2$  holds with certified  $C = 11.70\text{--}17.28$  and the derived per-mode ceiling  $m\|\psi_k\|_\infty^2 \leq 2(2\sqrt{C}k + 1)^2$  (controls:  $\times 10^{-3}$  fails, a one-index shift misses by 0.75). **Named limits as load-bearing content:** per instance on small windows — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in  $D$ , and explicitly *no* statement for all  $D$ : the one open input after T150 (an  $m$ -free constant  $C$

in the odd-parity-sector ladder  $\nu_k \leq Ck^2$  for a sign-changing-symbol Toeplitz section) stays *open*, typed open; TV(log  $\Lambda$ ) stays withdrawn as a hypothesis (T149);  $\sigma_{\text{tot}}$  stays retired as a route (T148); every T149/T150 exponent is a fit, sandbox. Fences: Kac–Murdock–Szegő 1953 (parity sector) / Widom 1958 / Basor–Ehrhardt 2009 / Böttcher–Silbermann (the parity sectors as the algebra’s natural objects; the sign-changing symbol is exactly what Basor–Ehrhardt does not cover, and that gap is the open input) / Sylvester 1852 / Bunch–Kaufman 1977 / Chebyshev 1852 (verified on the table) / Abel / Weyl 1912 / Bauer–Fike 1960 / Davis–Kahan 1970 (cited, computed dead, not used) / Rayleigh 1877–Ritz 1909 / Cauchy–Schwarz / Gershgorin 1931 / Wilkinson 1968–Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [verification/v549\_gauge\_parity\_identities.py]. [E] (per-instance identities and certified inequalities)

**PRIME.ODD.SECTOR.IDENT.01 — the odd-sector identities of phase 2 (T151).** (1) The grid step-over: the parity sines live on the *shifted* grid  $\theta_k = 2\pi k/N$ ,  $N = 2m+1$ , which never contains  $\theta = 0$ , with  $\mu_k^P = 2 - 2\cos\theta_k$  *exactly* ( $2.2 \times 10^{-16}$ ; eigen-residual  $2.3 \times 10^{-13}$ ); the symbol has  $f(0) = -47.6 \dots -6.3 < 0$  (T148’s honest negative, re-read, never re-derived) with a nonempty negative window  $[0, \theta_c)$ , and  $\theta_c/\theta_1 = 0.371\text{--}0.485 \leq 0.9$  on *every* window of the battery (12 frame-A windows,  $m = 26 \dots 285 \leq 300$ ): the odd grid *steps over* the negative window — odd-sector positivity is a grid fact. The honest negative beside it, promoted as content (T150’s rest item 3 answered negatively): the symbol-only pencil floor  $\kappa_{\text{sym}}(1 - \rho)$  is vacuous ( $\rho \geq 1$ ) on 12/12 and the symbol moves by 0.45–3.27 relative across *one* mode spacing — the local model is a *matrix* statement, not a symbol statement (controls: the Dirichlet corner breaks the eigenpairs at 1.0, a one-index shift misses by  $6.5 \times 10^{-3}$ ). (2) The certified bottom ladder:  $\lambda_k(A) \leq S\mu_k^P$  for  $k \leq 8$  by completed LDL<sup>T</sup> inertia counts (Sylvester 1852; a certificate, never a sorted list),  $S = 1.1019\text{--}6.4725$ , the count equal to the direct spectral count on all 96 (window,  $k$ ) pairs — order 2 in  $k$ ,  $f(0) < 0$  absorbed by the Rayleigh floor  $L_P \geq \mu_1^P I$  (control:  $S \times 10^{-3}$  certifies no eigenvalue below it). (3) The discrete Sobolev step:  $\|\nabla\psi\|_2^2 = \langle\psi, L_0\psi\rangle - \psi_{\text{last}}^2$  *exactly* ( $7.6 \times 10^{-15}$ ),  $\|\nabla\psi\|_2^2 \leq \langle\psi, L_P\psi\rangle$ , and  $\|\psi\|_\infty^2 \leq 2\|\psi\|_2\|\nabla\psi\|_2$  — licensed by the odd sector’s virtual node  $\psi_{-1} = 0$ ; calibration: the model pencil is (1, 1) (certified  $1.1 \times 10^{-12}$  at  $m = 241$ ) and the Sobolev per-mode price is  $\pi$  to  $5.1 \times 10^{-8}$  at  $m = 2001$  — the honest,  $m$ -free cost of leaving the  $\ell^2$  world (controls: the boundary term  $4.8 \times 10^{-2}$ , the Neumann undercount 0.50). (4) The closed archimedean minimum (T150’s rest item 2 closed *and attained*): given the two per-window-checked shape properties ( $c_i^{\text{arch}} < 0$ , non-decreasing on  $i \geq 1$ ),  $\min_r \Lambda_r^{\text{arch}} = c_0^{\text{arch}} - c_{M-1}^{\text{arch}}$  *exactly* (residual 0.0; value 2.1719–3.4674), attained at  $r = 0$  (control: the full atom-including kernel breaks the shape and misses by  $\geq 8.7 \times 10^{-2}$ ). (5) The linear per-mode bound: with the certified pencil floor  $\kappa = 0.1939\text{--}0.3599$  (one completed Cholesky per window, floor carried as  $\text{fl}/\mu_1^P$ , direction re-read on the actual bottom eigenvalue) and  $K_{\text{bot}} = S$ ,  $b_k = 2m\sqrt{K_{\text{bot}}\mu_k^P/\kappa} \leq C_S k$  with  $C_S = 12.081\text{--}26.953 \leq 2\pi\sqrt{K_{\text{bot}}/\kappa}$  on the bottom 8 modes of every window — *linear* in  $k$  where the quadratic  $\nu$  ladder grew, every  $m$ -power cancelled, nothing additive anywhere; the bound dominates the truth and the parity Dirichlet energy on every read mode; the bottom pencil ratio  $R = K_{\text{bot}}/\kappa = 3.7362\text{--}18.6607$  is the one scalar the chain still owes, typed *open* in  $m$ ; the discriminator: on the T145 no-go form the ratio explodes ( $1.686 \times 10^3 \rightarrow 2.667 \times 10^4$ , factor 15.8 over  $m = 64 \dots 256$ ) while the exact parity model holds  $R = 1$  to  $1.1 \times 10^{-12}$  (controls: the ladder  $\times 10^{-3}$  fails, the inflated floor  $1.5\kappa$  makes the Cholesky refuse on 12/12). **Named limits as load-bearing content:** per instance on small windows — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in  $D$ , and explicitly *no* statement for all  $D$ : the one open input after T151 (the  $m$ -freeness of the bottom pencil ratio  $R = K_{\text{bot}}/\kappa$  for the odd parity sector of a sign-changing-symbol Toeplitz section) stays *open*, typed open — its flat trend ( $x^{0.037 \pm 0.015}$ ) is a fit, sandbox; the symbol-only route stays certified vacuous; TV(log  $\Lambda$ ) stays withdrawn as a hypothesis (T149);  $\sigma_{\text{tot}}$  stays retired as a route (T148); every T151 exponent is a fit, sandbox. Fences: Kac–Murdock–Szegő 1953 / Widom 1958 / Basor–Ehrhardt 2009 / Böttcher–Silbermann (the address of the symbol question — the computed answer: the pointwise symbol language does not extend to the bottom mode) / Weyl 1912 / Parlett / Hardy–Littlewood–Pólya 1934–Agmon 1965 / Sylvester 1852–Bunch–Kaufman

1977 / Rayleigh 1877–Ritz 1909 / Cauchy–Schwarz / Wilkinson 1968–Higham 2002 / Chebyshev 1852 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [verification/v550\_odd\_sector\_identities.py]. [E] (per-instance identities and certified inequalities)

**PRIME.RITZ.CEIL.01 — the fixed-size Ritz ceiling certificate of phase 2 (T154).** (1) The direction lemma: for orthonormal  $Q$  and  $W = Q^T A Q$ ,  $\lambda_k(A) \leq \theta_k$  for *every*  $k \leq \dim$  — Ritz values are *upper* bounds for the eigenvalues of the *same index* (Courant–Fischer 1920 / Cauchy 1829), verified against the exact spectrum on the certificate subspace of every window and on 24 random subspaces (one-sided excess 0.0) — so the fixed-size ceiling needs *no* residual argument; Temple 1928 / Kato 1949 is a floor device and is used nowhere in the module (control: the reversed reading is recognised as false on 12/12 windows,  $\theta_k$  overshooting  $\lambda_k$  by  $\geq 18.6$  somewhere on every window). (2) The sixteen-column certificate:  $\text{span}\{t_1..t_8\} + A^{-1}L_P \text{span}\{t_1..t_8\}$  has fixed dimension 16, independent of  $m$  (one Green step by Cholesky back-solves, no inverse formed), and its ceiling  $K^F = \max_k \theta_k / \mu_k^P$  (8 completed Choleskys of matrices  $\leq 8 \times 8$ ) equals 1.1019–1.5845 with  $K^F \geq \max_k \lambda_k / \mu_k^P$  on every window and  $|K^F / K_{\text{bot}} - 1| \leq 9.9 \times 10^{-7}$  against the declared cap  $10^{-4}$  on 12/12 ( $K_{\text{bot}}$  the inertia-certified true ladder constant, count = direct count on all 96 pairs): the fixed-size ceiling *attains* the truth, retiring the size- $m$   $\text{LDL}^T$  counts from the ceiling step (controls: the parity sines alone overshoot 41.6–739; corrupting the Green columns breaks the attainment on 12/12; Kaniel 1966 / Paige 1971 quoted as the mechanism, never used as a bound). (3) The one-direction anatomy (angles measured, Björck–Golub 1973): the median of the seven aligned angles between the bottom eight of  $A$  and the bottom eight of  $L_P$  is  $0.03$ – $0.14^\circ$  while the largest is  $83.79$ – $89.70^\circ$  on every window of the battery (the 12 deepest in-cap frame- $A$  windows,  $n = 29 \dots 139$ ,  $m = 96 \dots 291 \leq 300$  — declared, not tuned) — the misalignment is carried by *one* nearly orthogonal direction, and that direction *is* the collapse price  $\lambda_1(A) / (t \mu_1^P) = 4.408$ – $7.042$ , recovered in full per window by one completed Cholesky of  $A - \gamma I$  (size  $m$ , certified, not fixed size, labelled as such; control:  $A = L_P$  returns  $K_{\text{bot}} = K^F = 1$ , zero misalignment, price exactly  $1/t$ ). (4) The no-go discriminator: on the T145 reconstruction  $c(l) = 1/(1+l)$  the odd section is positive definite and the Schur floor survives ( $t = 0.05$  on all 4 sizes) but  $K_{\text{bot}}$  explodes ( $\times 35.4$ ) and the sixteen-column ceiling explodes *with* it ( $\times 57.5$ ) over  $m = 48 \dots 288$  — no manufactured closure; the Courant–Fischer direction is violated nowhere (16/16); the Dirichlet control (the Hankel half removed, arithmetic kept bit for bit) stops being positive definite ( $\lambda_1 = -5.60 \dots -3.55$  certified on 3/3) — the reflection half is load-bearing for positivity. **Named limits as load-bearing content:** per instance on small windows — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in  $D$ , and explicitly *no* statement for all  $D$ ; the attainment is exhibited per window and *not* promoted to a theorem (T154’s no-go stress shows the certificate overshoots there by 1.12–2.19 — a property of the real section, not of the instrument); the two open inputs after T154 ( $R1'$ , an  $m$ -free floor for  $\lambda_{\min}(B_{HH})$  on the bulk parity block;  $R2'$ , an  $m$ -free lower bound for the  $L_P$  floor on the complement of the eight bottom Ritz directions — arithmetic-free, worth 91–100% of the collapse price) stay *open*, typed open; every floor consumed is certified per window at size  $m$ ; PRIME.ODD.SECTOR.IDENT.01’s ladder stays as stated and nothing is re-opened. Fences: Courant–Fischer 1920 / Cauchy 1829 / Kac–Murdoch–Szegő 1953 / Schur 1917–Haynsworth 1968 / Sylvester 1852–Bunch–Kaufman 1977 / Weyl 1912 / Temple 1928–Kato 1949 (a floor device, not used here) / Kaniel 1966–Paige 1971 (quoted, never a bound) / Björck–Golub 1973 / Wilkinson 1968–Higham 2002 / Chebyshev 1852 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [verification/v551\_ritz\_ceiling\_certificate.py]. [E] (per-instance theorems and certified inequalities)

**PRIME.WEIL.OPERATOR.01 — the localized-Weil-operator identification (research contract, [O]).** The external operator shell for the  $C = 1$  line (v618/v619), preregistered 2026-08-02 after the third external review. *Named reference frame (verified citations):* Suzuki, *Weil’s quadratic form via the screw function* (arXiv:2606.09096, June 2026) realizes the localized Weil quadratic form on every interval  $[-a, a]$  as a canonical self-adjoint operator  $A_a$  via the screw function, proves the

continuity of the lowest eigenvalue  $\lambda_a$  in  $a$  without assuming RH, and presents Yoshida’s criterion:  $Q_W(v) > 0$  for all nonzero odd  $v \in C_c^\infty(\mathbb{R})$  already implies RH; a first numerical realization is [arXiv:2607.24830](#). *The lemma chain:* **(W1)** exact operator identification — the frame-A window bilinear form  $\hat{A} = B - S$  is the Galerkin matrix of the odd restriction of  $A_a$  on the declared piecewise bases, exactly or up to an explicit rank-one transformation and a known normalization (no asymptotic “looks similar”); **(W2)** form density — the nested window spaces are form-dense in the odd form domain of  $A_a$  with Mosco/norm-resolvent convergence of the Galerkin forms; **(W3)** uniform discrete positivity —  $\hat{A}_{a,h} \succeq 0$  for every  $a$  and every refinement  $h$  (the RH-hard step; the  $C = 1$  contraction reading of [v618](#) — the normalized arithmetic residue as an operator of norm  $\leq 1/h$  — is the named candidate mechanism); **(W4)** passage to the continuum — W2 + W3 give  $Q_W^a(v) \geq 0$  on all odd test functions for all  $a$ , hence RH via the localized criterion. *Kill tests:* **(K1)** the W1 identification fails beyond a window-dependent fit; **(K2)** a complete, numerically stable fresh window stably exceeds the frozen  $C = 1$  bound (the [v619](#) complete-comb surface is the definitional domain); **(K3)** the odd-sector restriction loses the explicit-formula bookkeeping. *Fence:* a research contract, not a claim — no RH statement, no positivity claim beyond the measured surface; the zero-side representation stays diagnostic (a proof must run prime-side or through the screw function, avoiding the critical-line circularity). Status: **[O]** (preregistered; the audit and the citation verification are machine-checked in [\[verification/v624\\_external\\_lattice\\_audit.py\]](#)).

**The four fixed-size angle instruments (PRIME.ANGLE.INSTR.01).** The instrument cores of the discovery parts T155 and T157, recomputed on the same declared surface as PRIME.RITZ.CEIL.01 (the 12 deepest in-cap frame-A windows,  $n = 29 \dots 139$ ,  $m = 96 \dots 291 \leq 300$ ; a no-go size ladder  $m = 48 \dots 288$ ; parity and Dirichlet controls at  $m = 64, 128, 256$ ; each statement with at least one mutation control that must fail loudly) — none of the four closes an angle, and none is promoted as anything but a per-instance instrument. (1) The  $12 \times 12$  complement-floor certificate:  $\min_{v \perp W} v^T L_P v \geq \mu_{13}^P - \lambda_{\max}(M^{1/2}(I - GG^T)M^{1/2})$  with  $M = \text{diag}(\mu_{13}^P - \mu_k^P)$  and  $G = T_{12}Q_W$  — an identity plus *one* certified Rayleigh bound on the high block, a theorem for every  $m$  and every subspace, direction checked: the certificate never exceeds the exact size- $m$  complement bottom (12/12) and agrees to  $1 - \text{ratio} \leq 1.01 \times 10^{-3}$  against the declared cap  $2 \times 10^{-3}$  on this small surface; exactness controls both ways ( $W = \text{span}\{t_1..t_8\} \rightarrow \mu_9^P$ ; the *worthless* configuration  $W = \text{span}\{t_2..t_{13}\} \rightarrow \mu_1^P$ ; both to  $2.0 \times 10^{-11}$  — the instrument reports its own worthlessness); a corrupted overlap row moves the certificate by  $\geq 0.549$ , only downward. (2) The sine-block confinement: from the certified pencil floor  $A \succeq tL_P$  and the certified ladder ceiling  $\lambda_1 \leq S\mu_1^P$  alone,  $\|\gamma_H\|^2 \leq \lambda_1/(t\mu_{17}^P) \leq (S/t)/\rho_{17} = 0.0153\text{--}0.0245 < 1$  on 12/12 — the bottom eigenvector lives 97.5–98.5% inside the first 16 parity sines by a per-instance theorem (measured tail  $6.7 \times 10^{-6}\text{--}3.0 \times 10^{-5}$ ; the discovery part’s measured “98 per cent” replaced by one line); negative control: the pencil ceiling  $\kappa$  in place of the ladder  $S$  is vacuous ( $17.6\text{--}477 > 1$  on 12/12); mutation: the inflated floor  $\kappa L_P$  is refuted at  $e_1(A)$  by  $3.3 \times 10^3\text{--}1.0 \times 10^5$ . (3) The Schur-entry identity:  $s = \mu_1^P t_1^T A^{-1} t_1 = (B^{-1})_{11}$  exactly ( $1.1 \times 10^{-11}$ ) and  $(B^{-1})_{LL} S_L = I$  to  $5.8 \times 10^{-11}$  with  $S_L = B_{LL} - B_{LH} B_{HH}^{-1} B_{HL}$  the *same* fixed-size  $16 \times 16$  Schur complement the chain already forms, with the literal-entry direction  $1/s \leq (S_L)_{11}$  on 12/12:  $1/s = 4.0460\text{--}5.8962$  is an  $O(1)$  number carried by one entry, so the R2'' loss ceiling  $r \leq 1/(Ls)$  is a statement about *one* number; negative control: the inversion-free Cauchy–Schwarz ceiling is valid and useless, off by  $250\text{--}7.27 \times 10^3$  (the cancellation in  $b^T B_{HH}^{-1} b$  is nearly complete); mutation: a one-index shift breaks the identity by 1.55–2.15. (4) The adaptive Lipschitz ceiling: interval bisection with the *local* Lipschitz constant certifies  $\inf f^{\text{arch}}/(4 \sin^2(\theta/2)) \geq 0.25$  on 12/12 windows in 519–1201 symbol evaluations (certified floor 0.2505–0.3062, never above the fine-grid minimum; the global constant is documented insufficient — a uniform grid would need up to  $61.4\times$  more); mutation: fed a target above the true grid minimum it *refuses* on 12/12. Stress: on the T145 no-go family the Schur floor survives ( $t = 0.05$ ) while the confinement goes vacuous ( $7.93\text{--}413, \times 52.0$ ) and  $(S_L)_{11}$  explodes ( $1185 \rightarrow 7.11 \times 10^4, \times 60.0$ ) against the flat 4.05–5.90 real band; parity control exact to  $4.9 \times 10^{-14}$ ; the Dirichlet residual hits the predicted  $2/\sqrt{N}$  to  $7.4 \times 10^{-5}$ . **Named limits as load-bearing content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in

$D$ , and explicitly *no* statement for all  $D$ ; *nothing about the angles is closed* — the two open terms after T157 (an  $m$ -free lower bound on  $\hat{g}_1^2$ , equivalently an  $m$ -free upper bound on the one Schur diagonal entry  $1/s$ ; an  $m$ -free R1'' domination with a non-shrinking margin — certified per window with a  $7.3 \times 10^{-4}$  margin at the largest window and a shrinking trend) stay open, typed open; every floor  $t$  and ceiling  $S$  consumed is certified per window and its  $m$ -freeness is exactly what remains open; PRIME.RITZ.CEIL.01's ceiling and PRIME.ODD.SECTOR.IDENT.01's ladder stay as stated and nothing is re-opened. Fences: Kac–Murdock–Szegő 1953 / Schur 1917–Haynsworth 1968 / Courant–Fischer 1920–Cauchy 1829 / Cauchy–Schwarz (refuted as a ceiling, wired as a control) / Kantorovich 1948 / Temple 1928–Kato 1949 (floor devices, not used here) / Sylvester 1852–Bunch–Kaufman 1977 / Wilkinson 1968–Higham 2002 / Chebyshev 1852 / Szegő 1915–Widom 1958 (a named address, never a licence) named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [verification/v552\_angle\_instruments.py]. [E] (per-instance identities and certified inequalities)

**The exact-form identities (PRIME.EXACT.FORM.IDENT.01).** The theorem cores of the discovery parts T158 and T159, recomputed on the same declared surface as PRIME.RITZ.CEIL.01 and PRIME.ANGLE.INSTR.01 (the 12 deepest in-cap frame-A windows,  $n = 29 \dots 139$ ,  $m = 96 \dots 291 \leq 300$ ; no-go ladders  $m = 48 \dots 288$  on *both* T145 forms; parity and Dirichlet controls at  $m = 64, 128, 256$ ; each statement with at least one mutation control that must fail loudly) — the algebra of the sixteen-form itself, and it closes neither open term. (1) The Thomson dual form and the positive Cholesky ladder:  $s = (B^{-1})_{11} = \max_x (2x_1 - x^T Bx)$  (Maz'ya 1985 / Miclo 1999) with the direction *executed*, not asserted — 8 random trials per window never exceed  $s$ ; the ladder  $g_K = \sum_{j \leq K} y_j^2$ ,  $y = L^{-1}e_1$  from one Cholesky of the leading  $16 \times 16$  block (nested Schur complements): all terms strictly positive, partial sums strictly monotone,  $1/s \leq 1/g_{16} \leq 1/g_1 = B_{11}$  literally on 12/12, tight to 1.0642–1.1522 against the declared cap 1.5; mutation: a 5% corruption of one off-diagonal entry destroys positive definiteness — the Cholesky *refuses* on 12/12. (2) The two-dimensionality:  $\text{span}\{t_1, A^{-1}L_{Pt_1}\}$  attains  $s$  exactly ( $g/s = 1$  to  $9.0 \times 10^{-13}$ ) for the theorem reason  $L_{Pt_1} = \mu_1^P t_1$  (KMS 1953); negative controls: the two-sine span misses by 1.42–4.8, the corrupted Green column by  $255\text{--}7.7 \times 10^3$ . (3) The gauge identity and the Toeplitz-minus-Hankel signature: the reflection-odd section annihilates constant lag vectors *entrywise with zero tolerance* ( $\text{odd\_toeplitz}(\mathbf{1}) = 0$  exactly), hence  $\sum_d w_d = 0$  to  $1.1 \times 10^{-14}$  of  $\sup |w|$  — the form is blind to the lag mass of  $c$ , exactly where the archimedean and atom halves are  $h^2$ -sized and cancel; the signature  $A_{r,s} - A_{r+1,s+1} = f(r+s)$  holds to  $1.4 \times 10^{-16}$ ; mutations: a one-index Hankel shift breaks the identity by  $2.7 \times 10^{-7}\text{--}4.2 \times 10^{-5}$ , and the T145 form has signature defect  $7.4 \times 10^{-2}$  — not a parity section, the identity is not even defined for it. (4) The exact lag sum and the two closed scalars:  $x^T B_{LL} x = \sum_d c_d w_d$  ( $w_d = T_d - H_d$ ) against the direct assembly to  $2.7 \times 10^{-16}$  of the absolute scale  $\sum_d |c_d w_d|$ , with the honest price *measured* (relative to the cancelled total only  $1.4 \times 10^{-11}$  on this small surface — that depth is the  $h^2$  obstruction);  $w_0 = \sum_k x_k^2 / \mu_k^P$  to  $4.2 \times 10^{-16}$  and  $2w_0 - w_1 = \|x\|^2$  to  $7.4 \times 10^{-16}$ ; mutations: a 1% lag perturbation and a one-index  $\mu$  shift both break loudly. (5) The Z-matrix law and the Collatz–Wielandt floor:  $B_{HH}^{\text{arch}}$  is a symmetric Z-matrix *raw* — every off-diagonal entry  $\leq 0$  with no tolerance on 12/12; the closed sign-based floor  $\min_k (B_{HH}^{\text{arch}} \mathbf{1})_k = 0.9093\text{--}1.1284 \geq 0.25$  with the theorem direction verified per instance against the exact  $\lambda_{\min} = 0.9138\text{--}1.1432$  (Perron 1907 / Frobenius 1912 / Collatz 1942 / Wielandt 1950), beating the sign-free norm route by 0.69–8.8; the honest limit as content: the atom block obeys no sign law (0.44–0.53, a coin toss), the full block is not a Z-matrix and its row-sum floor is negative ( $-48.3 \dots -0.79$ ) — admissible and vacuous, R1'' stays open; mutation: the checkerboard similarity (an isometry,  $\lambda_{\min}$  moves by 0 exactly) destroys the raw law — an entrywise fact, not a spectral one. Stress: the T145 no-go breaks on three axes of this module's structure; on the  $c(l) = 1/(1+l)$  reconstruction the ladder theorem survives while  $1/g_{16}$  explodes ( $1187 \rightarrow 7.11 \times 10^4$ ) — the instrument reports the collapse instead of hiding it; parity: for  $A = L_P$  the machinery returns  $s = 1$  with every ladder partial sum constant in  $K$ ; Dirichlet controls exact. **Named limits as load-bearing content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in  $D$ , and explicitly *no* statement for all  $D$ ; the two open

terms after T159 (the one explicit pairing inequality for  $R2'' - \sum_d (\Delta c)_d W_d^1$  with  $\|\Delta c^{\text{arch}}\|_1 \leq 6$ , needing a correlation bound instead of sizes; a third device for the atom off-band block for  $R1''$ , neither norm nor sign) stay open, typed open; the pairing bound and  $m$ -freeness are explicitly *not* claimed. Fences: Maz'ya 1985 / Miclo 1999 / Schur 1917–Haynsworth 1968 / KMS 1953 / Dirichlet 1829 / Abel / Fejér 1915 / Perron 1907–Frobenius 1912 / Collatz 1942–Wielandt 1950 / Gantmacher–Krein 1950 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: `[verification/v553_exact_form_identities.py]`. [E] (per-instance identities and certified inequalities)

**The sampling/harmonics identities (PRIME.SAMPLING.HARM.01).** The closed theorem cores of the discovery parts T160 and T161, recomputed on the same declared surface as PRIME.EXACT.FORM.IDENT.01 (the 12 deepest in-cap frame-A windows,  $n = 29 \dots 139$ ,  $m = 96 \dots 291 \leq 300$ ) plus a declared  $\Lambda$ -battery of six log-spaced cut-offs  $X = 10^3 \dots 3.2 \times 10^5$  (finite von-Mangoldt sums only, no matrix anywhere; each statement with at least one mutation control that must fail loudly) — what the sixteen-form *pairs against*, and it closes no open term. (1) The sampling identity: the atom half of the pairing equals  $-\frac{1}{2} \sum_{n \leq X} (2\Lambda(n)/\sqrt{n}) \hat{w}(\log n)$  (plus one reflected term below the grid;  $\hat{w}$  the linear interpolant of the lag weights), exact to  $5.6 \times 10^{-15}$  of the absolute atom scale on 12/12 — hence identically a finite combination of sums  $\sum_n \Lambda(n) n^{-1/2} \cos(t \log n)$  at 32 explicit frequencies; mutation: a one-index grid shift breaks by  $5.1 \times 10^{-5} - 3.1 \times 10^{-3}$ . (2) The moment laws and the total-variation theorem:  $m_0 = 0$  ( $1.0 \times 10^{-16}$  of  $\|w\|_1$ ),  $m_1 = -[S_0^2 + 2 \sum_j P_j^2]$  ( $9.2 \times 10^{-16}$ ) and  $m_2 = -[2S_1 - (M-1)S_0]^2$  ( $3.4 \times 10^{-14}$ , a perfect square), both strictly negative on every window; U3 as a theorem with *checked* hypotheses ( $c^{\text{arch}} \leq 0$  and monotone on  $d \geq 1$ , checked 12/12):  $\text{TV}(c^{\text{arch}}) = 3.9817 - 4.8991 \leq |c_0| + 2 \sup < 6$ , closed and window-independent; mutations: a prefix shift, an  $(M-2)$  mis-normalisation, and the full atom-including kernel all fail loudly. (3) The harmonic-frequency theorem and the PNT main term:  $t_j(2\alpha) = 2\pi j$  *exactly* ( $2.8 \times 10^{-14}$ ) — the 32 frequencies are precisely the Fourier harmonics of the log-window  $[0, 2\alpha]$ ; at those frequencies the partial-summation main term collapses to the closed parameter-free prediction  $(\sqrt{X} - 1)/(\frac{1}{4} + t^2)$  (de la Vallée Poussin 1896), matching the finite sums to 0.0007–0.097 of the trivial mass against the declared cap 0.12 on the battery; mutations: a wrong Mellin pole worsens the aggregate residual by  $\times 1.8 - \times 95.8$ , and a mass-matched *uniform* fake measure misses by 0.23–0.51 — the agreement is a property of the von-Mangoldt measure. (4) The head split and the scale-free Bernstein rate:  $c_d^{\text{arch}} = \Psi_d + D\hat{G}(dD, D)$  with  $\Psi_d = -\frac{1}{2}[(d+1)\log(d+1) - 2d\log d + (d-1)\log(d-1)]$  closed,  $D$ -free and  $m$ -free (the triangle-smeared simple pole; the  $\log D$  cancels identically), exact to  $3.2 \times 10^{-16}$  at every lag of every window;  $\rho^* = (3 + \sqrt{5})/2 = 2.618034$  is the root of  $x^2 - 3x + 1$ , returned exactly by the  $D \rightarrow 0$  limit of the peeled interval and invariant under  $\alpha \times 10$  (Bernstein 1912); mutations: dropping  $\Psi$  misses by 0.99–1.00, the wrong interval ratio moves the limit to 3.0. (5) The off-diagonal fraction bound and the sign refutation: the gauge identity licenses one boundary-free Abel step, the decomposition  $Q^{\text{arch}} = \sum_{k,l} a_k a_l \sum_d (\Delta c^{\text{arch}})_d R_{kl}^1(d)$  is exact ( $2.0 \times 10^{-13}$ ), and the certified per-window inequality is  $|\text{off-diagonal}| \leq \frac{1}{4} |Q^{\text{arch}}|$  (0.171–0.212 on 12/12) — with the *refuted* sign law wired as the negative control (arch half negative, off-diagonal block positive on 12/12, 176/240 pairs violate the per-pair form): nothing single-signed is promoted; mutation: a one-index  $R$ -kernel shift breaks by  $\geq 6.8 \times 10^{-5}$ . Stress: the head split misses the T145 no-go reconstruction by 2.39 of  $\sup |c|$ ; the Dirichlet cosine-sum identity (including the degenerate branch), the KMS eigenpairs and  $\text{odd\_toeplitz}(c^L) = L_P$  (zero tolerance) are exact. **Named limits as load-bearing content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in  $D$ ; finite  $\Lambda$ -sums are allowed and used, *no* RH statement is made, and the  $\delta$  triage of T161 (required cancellation depth against RH strength) is a documentation item, *not* a claim of this module; the open terms after T161 (R-A' the prime-free log-moment  $\sum_d \Psi_d w_d$ , outside the polynomial ladder; R-B' the  $m$ -free  $16 \times 16$  Gram fraction bound, Fejér 1915 / Schur 1917; R-C' *one* split of the pairing with  $\delta_{\text{eff}} < \frac{1}{2}$ , T161 refuting the arch/atom and arch+smooth/fluctuation splits with numbers; R-D a fifth  $R1''$  device) stay open, typed open. Fences: Dirichlet 1829 / Abel 1826 / Fejér 1915–Schur 1917 / Bernstein 1912 / Chebyshev

1852–Mertens 1874 / de la Vallée Poussin 1896 / Clausen–Lerch / KMS 1953 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [verification/v554\_sampling\_harmonics\_identities.py]. [E] (per-instance identities and certified inequalities)

**The Pareto/total-variation identities (PRIME.PARETO.TV.01).** The closed theorem cores of the discovery parts T162 and T163, recomputed on the same declared surface as PRIME.SAMPLING.HARM.01 (the 12 deepest in-cap frame-A windows,  $n = 29 \dots 139$ ,  $m = 96 \dots 291 \leq 300$ ), with the one arithmetic input  $\kappa = 0.038821$  *verified* at every jump point of  $\psi(x)/x$  on the von-Mangoldt table in  $[100, 3.2 \times 10^5]$  (reproduced to  $4.2 \times 10^{-7}$ ; each statement with at least one mutation control that must fail loudly) — the *price structure* of the pairing, and it closes no open term beyond the negative closure that the floor inequality itself is. (1) The exchange law and the crossing criterion:  $\delta_{\text{bnd}}(x) = \frac{1}{2} + \log(2\kappa g_{16} \text{TV}(x)/P(x))/\log X$  is an identity at all 612 grid points of the declared Fejér knob (50 geometric  $\sigma$  on  $[1, 4096]$  plus  $\infty$ ;  $5.1 \times 10^{-16}$  relative, admissibility  $Q(x) > 0$  everywhere), and  $\delta_{\text{bnd}} < \frac{1}{2} \iff P > 2\kappa g_{16} \text{TV}$  holds as an exact equivalence at every point — price and demand are two coordinates of *one* object; mutation: the wrong constant ( $\kappa$  for  $2\kappa$ ) moves the law by  $6.5 \times 10^{-2}$ –3.4. (2) The total-variation floor — the theorem that carries the T163 verdict: (T1)  $w_0(x) = \|a(x)\|^2$  ( $4.9 \times 10^{-16}$ ; orthonormal parity rows); (T2)  $\text{TV}(x) \geq |w_0(x)|$  by telescoping with  $w_M = 0$ ; (T3)  $x_1 = 1$  forces  $\|a\|^2 \geq 1/\mu_1^P$ , hence  $\mu_1^P \text{TV}(x) = 5.00$ –28.84  $\geq 1$  on all 708 trial vectors built (51 knob settings plus 8 random admissible sixteen-vectors per window), the closed floor  $1/\mu_1^P = 1/(4 \sin^2(\pi/N)) = 943.6$ –8609.6  $\sim h^2$ ; (T4) every sub- $\frac{1}{2}$  grid point pays  $P > 2\kappa g_{16}/\mu_1^P = 15.4$ –142.8 (all 54 respect it) — a flat-price sub- $\frac{1}{2}$  demand is impossible in the parity sector on this surface, the inequality that closes R-C''' negatively; negative control: the *inadmissible* vector  $0.01x^*$  ( $x_1 \neq 1$ ) does undercut the floor ( $\mu_1^P \text{TV} = 2.5$ –2.9  $\times 10^{-3}$ ) and is recognised as inadmissible — the floor is a property of the normalisation meeting the smallest parity eigenvalue (Kac–Murdock–Szegő 1953). (3) The chain-derived price axis:  $\sigma = 1$  gives  $x = e_1$  exactly with  $P = g_{16}/g_1$  (the  $K = 1$  rung of the T158 Cholesky ladder, the T157 route-(0) certificate;  $4.1 \times 10^{-16}$ ),  $\sigma = \infty$  gives  $g_{16}Q(x^*) = 1$  (the  $K = 16$  top rung;  $1.2 \times 10^{-12}$ ); the ladder carries all 16 terms strictly positive with monotone partial sums, and the knob is monotone in both coordinates on every window — the knob curve *is* the Pareto front; mutation: a corrupted off-diagonal sixteen-block entry makes the Cholesky *refuse* on 12/12. (4) The Mellin cell moments and the Abel step: the  $k$ -th rung  $\sum_d w_d C_d(-(2k + \frac{1}{2}))$  with  $C_d(s)$  the closed tent integral (Mellin 1896) agrees with an independent per-cell Gauss–Legendre quadrature to  $2.8 \times 10^{-13}$  on every rung  $k = 1 \dots 4$  (shifted tent centres break by  $\geq 1.3 \times 10^{-2}$ ); the boundary-free Abel identities hold at levels  $k = 1 \dots 3$  ( $8.3 \times 10^{-17}$ ; Abel 1826), the dropped-boundary level-1 form agreeing because the gauge identity kills the boundary term exactly; the optimal level is *one* for T162's closed reason ( $32\pi/\alpha = 40.6$ –59.4  $> 1$ , measured level-2/level-1 ratio 30.8–69.1  $> 1$ ), and a gauge violation breaks the dropped form by its closed prediction  $v_0 c_0 M$  to  $1.2 \times 10^{-15}$  — the positive control. (5) The Lerch/Frullani log-moment:  $L = \sum_d \Psi_d w_d$  agrees on three independent routes on every window — direct; double-Abel *with* the Wronskian boundary term  $-\frac{1}{2}M \log M w_{M-1}$  ( $2.8 \times 10^{-14}$ ; dropping it breaks by  $2.2 \times 10^{-4}$ – $1.3 \times 10^{-2}$ ); the Lerch/Frullani integral with the  $d = 1$  term peeled closed ( $1.8 \times 10^{-15}$ ; the peel constant  $2 \log 2$  to  $2.2 \times 10^{-16}$ ,  $\Psi_1 = -\log 2$  exact; Frullani 1828 / Lerch 1887 / Clausen 1832). Stress: the monotone no-go staircase — coarsening over  $\nu = 4/2/1/0.5$  degrades the arch/atom cancellation monotonically in the median ( $1.00 \rightarrow 3.49 \rightarrow 10.81 \rightarrow 24.98$ , the  $\nu = 4$  column an exact null control); the Dirichlet cosine-sum identity ( $2.4 \times 10^{-14}$ , including the degenerate branch), the KMS eigenpairs ( $1.4 \times 10^{-14}$ ) and odd\_toeplitz( $c^L$ ) =  $L_P$  (zero tolerance) are exact. **Named limits as load-bearing content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in  $D$ ; the T163  $h$ -exponents ( $P_{\text{aff}} \sim h^{3.05}$ ,  $P_{\text{cross}} \sim h^{1.91}$ , the TV exponent  $\sim h^{2.00}$ ) are sandbox fits and are *not* consumed — what is consumed is the closed per-window floor; finite  $\Lambda$ -sums are allowed and used, *no* RH statement is made ( $\frac{1}{2}$  appears in exactly one role: as the strength of a classical statement a required depth is compared against); the open terms after T163 (R-E the successor, both arms

prime-free — arm A: a growing  $1/s$  ceiling for the downstream T159/T160 chain; arm B: a sector whose lowest eigenvalue does not vanish like  $h^{-2}$ ; R-B''' the  $h$ -uniform positivity margin; R-D a fifth R1'' device) stay open, typed open. Fences: Fejér 1915 / Mellin 1896 / Abel 1826 / Chebyshev 1852–Rosser–Schoenfeld 1962 (verified on the table) / Legendre / Dirichlet 1829 / KMS 1953 / Frullani 1828–Lerch 1887–Clausen 1832 / Schur 1917–Haynsworth 1968 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [verification/v555\_pareto\_tv\_identities.py]. [E] (per-instance identities and certified inequalities)

**The gauge/ $P_{\text{pr}}$  identities (PRIME.GAUGE.PPR.01).** The closed theorem cores of the discovery parts T164 and T165, recomputed on the same declared surface as PRIME.PARETO.TV.01 (the 12 deepest in-cap frame-A windows,  $n = 29 \dots 139$ ,  $m = 96 \dots 291 \leq 300$ ), with the one arithmetic input  $\kappa = 0.038821$  *verified* at every jump point of  $\psi(x)/x$  on the von-Mangoldt table in  $[100, 3.2 \times 10^5]$  (each statement with at least one mutation control that must fail loudly) — the *gauge structure* of the pairing, and it does not close the one remaining quantifier. (1) The gauge theorem (T164):  $x_1 = 1$  fixes  $a_1 = 1/\sqrt{\mu_1}$ , i.e. only the *scale* of  $a$ ;  $Q$  and TV are homogeneous of degree two in  $a$ ; hence the demand ratio  $Q/\text{TV}$  of the same  $a$ -direction — and with it  $\delta_{\text{bnd}}$  — is the same number in *every* sector, unchanged to  $2.9 \times 10^{-11}$  on 2520 sector-by-trial-vector combinations (5 weight families  $\times$  7 shifts  $\times$  6 Fejér taper widths on 12 windows), with  $\text{TV} \geq |w_0|$  in every sector and the pure shift scaling the certified value by exactly  $\theta = \mu_1^P/(\mu_1^P + \kappa_s)$  ( $2.2 \times 10^{-11}$  on 432 points); the full space is settled in closed form and strictly worse ( $\mu_0 = 0$  exactly, the lowest remaining eigenvalue  $\leq \mu_1^P$  on 12/12); mutation: the wrong transport map (the  $x$ -coordinates carried unchanged instead of the  $a$ -direction) moves the ratio by  $2.0 \times 10^2$ – $5.4 \times 10^3$ . (2) The  $h^2$  conservation law (T164): floor  $\times$  transfer  $= 1/(\mu_1^P + \kappa_s) \cdot (1 + \kappa_s/\mu_1^P) = 1/\mu_1^P$  *identically* at every shift on the declared grid (72 window-by-shift points,  $2.2 \times 10^{-16}$  of  $1/\mu_1^P$ ) — the  $h^2$  is conserved, never removed; it lives in the ratio, not in the floor; mutation: the wrong transfer misses by 0.9995 at the largest shift on every window. (3) The power-one spend as an identity (T164): the  $r$ -ceiling  $r \leq U/L$  with  $L = \lambda_1(A)/\mu_1^P = 1.1019$ – $1.5845$  (computed per instance from the full window) is linear in the entry ceiling, so  $d \log r / d \log U = +1$  exactly (within  $1.1 \times 10^{-14}$  on 12/12) — an  $h^\varepsilon$  ceiling costs  $h^{-\varepsilon}$  of  $r$ -margin for every  $\varepsilon > 0$ , no free tolerance in the consumption step (the measured power of the full T156 loss margin stays sandbox); mutation: the quadratic consumption reads slope 2.0000 and is recognised. (4) The  $P_{\text{pr}}$  identity and the predicate equivalence (T165):  $P_{\text{pr}} = g_{16} \cdot R \cdot (\text{TV}/(t_1 v)^2)/\mu_1^P$  with no inequality anywhere — four factors, each a named object: the R-E-A quantifier  $g_{16} = 0.1559$ – $0.2227$ , the demand  $R = Q/\text{TV}$ , the T163 floor  $\text{tvf} \geq 1$ , and the KMS 1953 scale  $1/\mu_1^P = 943.6$ – $8609.6 \sim h^2$  — exact to  $4.3 \times 10^{-16}$  on all 192 battery vectors (6 knob settings, 6 random sixteen-vectors and 4 random *full-window* vectors per window), gauge-invariant under  $v \rightarrow tv$  to  $1.5 \times 10^{-11}$  against the declared cancellation horizon; the predicates  $R > 2\kappa$  and  $P_{\text{pr}} > 2\kappa g_{16} \text{tvf}/\mu_1^P$  agree as booleans on all 192 (no search is run — T165's ascent stays sandbox), so the demand half of R2'' and the gauge-invariant crossing condition are *one* predicate — R-F is strictly stronger than the R2'' demand and its  $m$ -uniform form *is* R-E-A — and  $\text{tvf} \geq 1$  pointwise, hence every crossing vector pays  $P_{\text{pr}} > 2\kappa g_{16}/\mu_1^P = 15.4$ – $142.8$ , above the preregistered tolerance  $\Theta_{\text{TOL}} = 10$  on 12/12 windows: the R-F conjunction is *empty* on this surface by an identity plus a floor; mutation: the price with  $(t_1 v)$  unsquared scales like  $t$  and moves by  $1.0$ – $1.0 \times 10^4$  under the same round trips. (5) The Schur-cascade direction and the exponent closure (T164/T165):  $s = \mu_1^P t_1^T A^{-1} t_1 = 0.1696$ – $0.2472 \geq g_{16} \geq g_1$  window by window ( $1/s \leq 1/g_{16} \leq 1/g_1 = B_{11}$ ; Schur 1917–Haynsworth 1968),  $P_K = \hat{a} s = 254.8$ – $7.7 \times 10^3 \geq 1$  (Kantorovich 1948); the exponent closure is an identity of the lists — per instance  $\log(1/g_1) = \log(g_{16}/g_1) + \log(1/g_{16})$  to  $1.6 \times 10^{-15}$ , the least-squares fits closing to  $1.3 \times 10^{-15}$  by linearity (the exponent *values* are sandbox fits, not consumed — what is promoted is that the whole certification lives in the cascade gain  $g_{16}/g_1$ , the object of the one open quantifier); mutations: the  $g_8$  closure breaks by 0.10–0.24, a corrupted sixteen-block makes the Cholesky *refuse* on 12/12. Stress: the monotone no-go staircase ( $1.00 \rightarrow 3.49 \rightarrow 10.81 \rightarrow 24.98$ , the  $\nu = 4$  column an exact null control); the Dirichlet cosine-sum identity ( $2.9 \times 10^{-14}$ , including the degenerate branch), the KMS eigenpairs ( $1.4 \times 10^{-14}$ ) and  $\text{odd\_toeplitz}(c^L) = L_P$  (zero tolerance)

are exact. **Named limits as load-bearing content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in  $D$ ; the T164/T165  $h$ -exponents and the measured anatomy (the free ascent, the  $\eta(\text{Cap})$  profile, the decoupled  $\nu$  surface with its zone-depth finding, the retired constant  $U_{\text{ref}} = 4.90 \rightarrow 5.7327$ ) stay in the diary and the paper; finite  $\Lambda$ -sums are allowed and used, *no* RH statement is made ( $\frac{1}{2}$  appears in exactly one role: as the strength of a classical statement a required depth is compared against); **the one open object after T165 stays open, typed open:**  $\inf_m g_{16}(m) > 0$ , equivalently a lower bound on the sixteen-step Schur-cascade gain  $g_{16}/g_1$  uniform in  $m$  — a quantifier over a certified flat list; R-B''' (an independent  $\alpha/h$  surface) stays open as narrowed by T164. Fences: Schur 1917–Haynsworth 1968 / KMS 1953 / Abel 1826 / Chebyshev 1852–Rosser–Schoenfeld 1962 (verified on the table) / Kantorovich 1948 / Fejér 1915 / Dirichlet 1829 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [verification/v556\_gauge\_ppr\_identities.py]. [E] (per-instance identities and certified inequalities)

**The cascade/vector identities (PRIME.CASCADE.VECT.01).** The closed theorem cores of the discovery parts T166 and T167, recomputed on the same declared surface as PRIME.GAUGE.PPR.01 (the 12 deepest in-cap frame-A windows,  $n = 29 \dots 139$ ,  $m = 96 \dots 291 \leq 300$ ), with the one arithmetic input  $\kappa = 0.038821$  *verified* at every jump point of  $\psi(x)/x$  on the von-Mangoldt table in  $[100, 3.2 \times 10^5]$  (each statement with at least one mutation control that must fail loudly) — the *anatomy of the cascade* and the *exactness of the closed vector at  $K = 2$* , and it does not close the one remaining scalar. (1) The cascade identities (T166): the prefix property — one Cholesky of the sixteen-block yields all sixteen rungs,  $g_K = e_1^T Q_K^{-1} e_1$  at every  $K$  (192 comparisons,  $1.1 \times 10^{-12}$ ); the minor form  $1/g_K = \det G_{[1..K]}/(\mu_1^P \det G_{[2..K]})$  on the raw arithmetic Gram block  $G = TAT^T$  ( $2.7 \times 10^{-12}$  on 36 (window,  $K$ ) pairs,  $K \in \{2, 6, 16\}$ ; Schur 1917–Haynsworth 1968); the regression form  $g_K/g_1 = 1/(1 - R_K^2)$  with  $R_K^2$  the squared multiple correlation of mode 1 on modes  $2 \dots K$  ( $8.8 \times 10^{-13}$ ) — the open object *is* a near-collinearity statement; mutation: the minor with the wrong row/column deleted misses by 0.99–1.45. (2) The diagonal invariance (T166): the gain  $g_K/g_1 = B_{11}(Q_K^{-1})_{11}$  is unchanged under  $B \rightarrow DBD$  on 144 random positive diagonals ( $2.7 \times 10^{-12}$ ) — the cascade does not see the KMS  $1/\sqrt{\mu}$  normalisation, the gain is a property of the arithmetic Gram block alone; mutation: a non-diagonal congruence moves the gain by 2.7–3.8. (3) The exact  $K = 2$  identity (T167): the closed vector  $u = (1, -Q_{21}/Q_{22})$  *attains* the constrained minimum,  $u^T Q_2 u = 1/g_2 = Q_{11}(1 - r_{12}^2)$  to  $1.1 \times 10^{-12}$ , and  $B_{11}g_2 = 1/(1 - r_{12}^2)$  identically to  $1.2 \times 10^{-12}$  (measured  $1 - r_{12}^2 = 2.8 \times 10^{-4} - 5.9 \times 10^{-3}$  on this surface, the  $h$ -trend a sandbox fit): at  $K = 2$  the fast-null-vector needs no approximation — the third dress collapses onto the second, one scalar built from three closed lag sums; mutation: the wrong-sign vector pays the relative excess  $4r_{12}^2/(1 - r_{12}^2) = 6.7 \times 10^2 - 1.5 \times 10^4$ . (4) The pivot identity and the unification (T167): for  $u = x_K^* + \delta$  with  $\delta_1 = 0$ ,  $u^T Q_K u = 1/g_K + \delta^T Q_K \delta$  exactly ( $2.0 \times 10^{-12}$  on 252 evaluations across three  $\delta$  scales) — every relative excess is an exact number; the Rayleigh envelope holds with its  $\delta_1 = 0$  ceiling attained exactly ( $1.1 \times 10^{-15}$ ) and Cauchy-interlaced; the sign-aligned entrywise perturbation attains  $|x^{*T} E x^*| = \varepsilon S_K$  exactly ( $3.8 \times 10^{-16}$ ), so the entry threshold  $\rho_{\text{carry}}(1/g_K)/S_K$  is exact — and the unification:  $(1/g_K)/S_K$  is the one cancellation ratio of all three dresses, at  $K = 2$  literally  $(1/g_2)/S_2 = (1 - r_{12}^2)/(1 + 3r_{12}^2)$  ( $1.2 \times 10^{-12}$ ; the closed factor  $1 + 3r_{12}^2 \rightarrow 4$  is T166's independently derived quarter): *one* inequality, not three; mutation:  $\delta_1 = 0.1$  breaks the pivot identity by exactly the closed cross term  $2\delta_1 = 0.2$ . (5) The trial theorem and the two must-breaks (T166/T167): all 288 random trials with  $u_1 = 1$  satisfy  $u^T Q_K u \geq 1/g_K$  (min ratio 15.14) — every closed candidate is a certified *lower* bound (Schur 1917); the  $L_P$  no-go: the normalised block is the identity ( $1.6 \times 10^{-15}$ ) and the cascade gain exactly 1 (deviation 0.0) at  $m = 64/128/256$ ; the scramble type-change: uniform random atom positions at the same total mass make  $B_{11}$  itself lose positivity on 49/60 declared draws and on a majority of draws on 12/12 windows — at  $K = 2$  the collapse under scrambling is a change of *type*, not a factor. Stress: exact Dirichlet/parity controls ( $2.3 \times 10^{-14}$  /  $1.4 \times 10^{-14}$  / zero tolerance). **Named limits as load-bearing content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in  $D$ ; the T166/T167  $h$ -exponents  $(1 - r_{12}^2) \sim h^{-2.921}$  on frame

A, the certified gain exponent  $h^{+2.921}$ , the threshold monotonicity in  $K$ , the  $K = 6$  separation  $h^{+1.138}$ ) and the measured anatomy (the rung profile, the polarisation split, the Kato radius, the headroom) stay in the diary and the paper; finite  $\Lambda$ -sums are allowed and used, *no* RH statement is made ( $\frac{1}{2}$  appears in exactly one role: as the strength of a classical statement a required depth is compared against); **the one open object after T167 stays open, typed open**: R1, the single scalar — an  $m$ -free upper bound on  $1 - r_{12}^2 = 1 - Q_{12}^2/(Q_{11}Q_{22}) \leq C h^{-3+\varepsilon}$ , three closed lag sums and nothing else; the routes T167 closed off (perturbation theory of every order around the KMS bottom mode — the Kato series converges to the *wrong* object; position-blind surrogates) stay sandbox negatives; R-B''' stays open as narrowed. Fences: Schur 1917–Haynsworth 1968 / KMS 1953 / Rayleigh 1877–Courant–Fischer 1920 / Cauchy 1829 / Kato 1949 (cited as the closed-off route, never used) / Chebyshev 1852–Rosser–Schoenfeld 1962 (verified on the table) / Dirichlet 1829 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [verification/v557\_cascade\_vector\_identities.py]. [E] (per-instance identities and certified inequalities)

**The bilinear/rank identities (PRIME.BILINEAR.RANK.01).** The closed theorem cores of T168 (contract LAGRANGE.MINORS), T169 (contract TSTAR.RATIO) and T170 (contract BILINEAR.SIEVE) as one deliberately narrow module (27 checks,  $\sim 0.2$  s), recomputed on the declared small frame-A surface (12 deepest in-cap zones,  $m = 96 \dots 291 \leq 300$ ;  $\kappa = 0.038821$  verified at every jump point of  $\psi(x)/x$  on the table) — the *exact shape of the one remaining scalar*, and it does not close that scalar. (1) The Lagrange/Wronskian anatomy (T168): the lag-sum identity  $\hat{a}_{ij} = t_i^\top A_h t_j = \sum_d c_d W_d^{ij}$  with the closed correlation weights ( $1.6 \times 10^{-15}$ ; the  $1/\sqrt{\mu}$  factors cancel, T169-TH3), Euclidean orthogonality  $t_1 \cdot t_2 = 0$  ( $4.4 \times 10^{-16}$ ) — the near-parallelism is created *entirely* by the arithmetic metric — the Wronskian telescope  $(\mu_1 - \mu_2)u_r v_r = W_{r-1} - W_r$  with  $W_{-1} = W_{h-1} = 0$  ( $1.9 \times 10^{-17}$ ) and the maximal-minor identity  $\frac{1}{2} \sum M^2 = 1/(\mu_1 \mu_2)$  exactly ( $4.6 \times 10^{-16}$ ; Lagrange 1773); mutation: the wrong eigenvalue pair misses by 1.66–1.67 relative (the closed miss  $(\mu_2 - \mu_3)/(\mu_1 - \mu_2) \rightarrow 5/3$ ). (2) The exponent account and the Gershgorin certificate (T168/T169):  $1 - r_{12}^2 = \nu_1 \nu_2 / (\hat{a}_{11} \hat{a}_{22})$  identically ( $2.5 \times 10^{-13}$ ; measured  $1 - r_{12}^2 = 2.8 \times 10^{-4} - 5.9 \times 10^{-3}$  on this surface, the  $h$ -trend a sandbox fit);  $\nu_1 \leq \max(\hat{a}_{11}, \hat{a}_{22}) + |\hat{a}_{12}|$  (Gershgorin 1931, unconditional, closed in the three entries) on 12/12 with loss 1.1885–1.1947; mutation: dropping  $|\hat{a}_{12}|$  fails on every window ( $\nu_1 - \max \text{diag} = 0.30 - 1.17 > 0$ ). (3) The det-collapse theorem (T169-TH7): on the Rayleigh family  $u(t) = (t, -1)$  the candidate  $K7 = \sqrt{\hat{a}_{22}/\hat{a}_{11}}$  collapses in closed form,  $R(K7) = 2\sqrt{\hat{a}_{11}\hat{a}_{22}} \det \hat{A} / [(\hat{a}_{11} + \hat{a}_{22})(\sqrt{\hat{a}_{11}\hat{a}_{22}} + \hat{a}_{12})]$  ( $8.7 \times 10^{-13}$ ) — it meets the threshold precisely because it reintroduces  $\det \hat{A}$ : the loop T167 scalar  $\rightarrow$  T168 factor  $\rightarrow$  T169 candidate closes as an identity; usable form  $\nu_2 \leq 2 \det \hat{A} / (\hat{a}_{11} + \hat{a}_{22}) = 2\nu_1 \nu_2 / (\nu_1 + \nu_2)$ , tight to 1.9980–1.9999  $< 2$  exactly (Rayleigh 1877); mutation: the closed constant  $1/\sqrt{2}$  overshoots  $\nu_2$  by  $17.1 - 2.3 \times 10^2$ . (4) The exact bilinear von Mangoldt form (T170 TH1–TH3): the exact split  $\hat{A} = B - S$  with  $S_{ij} = \sum_{n \leq X} (\Lambda(n)/\sqrt{n}) X_n[ij]$  against the direct block ( $1.7 \times 10^{-14}$ ); the polarisation  $\det \hat{A} = \det B - D(B, S) + \det S$  ( $1.0 \times 10^{-10}$ ); the double sum  $\det S = \sum_{n, m \leq X} \Lambda(n) \Lambda(m) K(n, m) / \sqrt{nm}$  with  $K = \frac{1}{2} D(X_n, X_m)$  through the *explicit* kernel matrix ( $7.5 \times 10^{-16}$ ; Cauchy–Binet 1812–1815); the wedge reading to the measured rank-one defect  $4.6 \times 10^{-3} - 3.1 \times 10^{-2}$  (honest certificate); mutation: the wrong cross-coefficient breaks by 64–430. (5) The rank-3 theorem and the Type II collapse (T170 TH4–TH5): the polarisation matrix on  $(X_{11}, X_{22}, X_{12})$  has eigenvalues  $\{-2, -1, +1\}$  exactly — rank 3, signature (1, 2) — so the kernel  $K = P M P^\top$  has  $\sigma_4/\sigma_1 \leq 1.6 \times 10^{-16}$  with nonzero spectrum = spec( $M P^\top P$ ) ( $1.0 \times 10^{-15}$ ): the bilinear form *is* the rank-3 polynomial  $S_{11} S_{22} - S_{12}^2$  in three linear  $\Lambda$ -sums; Vaughan’s identity coefficient-exact for every  $U = V \in \{3, 4, 5, 6\}$  (residual 0.0; Vaughan 1977), Type II kernel 99.9%-energy rank  $[2, 2, 2, 2]$  against generic  $[20, 15, 11, 9]$  ( $4.5 \times 10^{-10}$ ; Montgomery–Vaughan 1973 cited as the priced route). (6) The R4-free chain and the two discriminators (T170-TH6):  $1 - r_{12}^2 = \det \hat{A} / (\hat{a}_{11} \hat{a}_{22})$  is an identity (0.0) — no positivity of  $A_h$ , the Weil fence never approached; the  $L_P$  no-go:  $\hat{a}_{12} = 1.1 \times 10^{-17}$  identically, no decay; the scramble discriminator: rank-3 *untouched* on all 60 draws while the value of  $1 - r_{12}^2$  moves by a median factor 834.8 (range  $69 - 1.8 \times 10^4$ ) — the arithmetic sits entirely in the joint values of  $(S_{11}, S_{22}, S_{12})$  against  $B$ . Stress: exact Dirichlet/parity controls ( $1.2 \times 10^{-14}$  /  $1.4 \times 10^{-14}$  /

zero tolerance). **Named limits as load-bearing content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in  $D$ ; the T168–T170  $h$ -exponents ( $1 - r_{12}^2 \sim h^{-2.92}$  on the deep frame-A surface, the route-table deltas — large sieve  $+0.669$ , MV  $-0.384$ , exact split  $+0.996$ , Vaughan  $+0.669$  against the measured truth  $2.747$  — the five-order cancellation and the  $h^{-2.711}$  row-angle trend) and the measured anatomy stay in the diary and the paper; finite  $\Lambda$ -sums are allowed and used, *no* RH statement is made ( $\frac{1}{2}$  appears in exactly one role: as the strength of a classical statement a required depth is compared against); **the one open object after T170 stays open, typed open:** R1, now finally *classified* but not closed — an  $m$ -free unconditional certificate that the two rows of  $\hat{A}$  become collinear at rate  $h^{-3+\varepsilon}$ , equally a joint near-degeneracy of the three explicit linear  $\Lambda$ -sums ( $S_{11}, S_{22}, S_{12}$ ) against the archimedean block — a rate statement the module neither makes nor approaches; the T168/T169/T170 verdicts (MINORS-RESIST / RATIO-RESISTS / SIEVE-RESISTS) are sandbox verdicts. Fences: Lagrange 1773 / Cauchy–Binet 1812–1815 / Gershgorin 1931 / Rayleigh 1877–Courant–Fischer 1920 / Montgomery–Vaughan 1973–Vaughan 1977 (the priced route) / KMS 1953 / Chebyshev 1852–Rosser–Schoenfeld 1962 (verified on the table) / Dirichlet 1829 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [verification/v558\_bilinear\_rank\_identities.py]. [E] (per-instance identities and certified inequalities)

**The phase-2 capstone (PRIME.PHASE2.CAPSTONE.01).** Promotion of discovery T171 (contract FINAL.MAP, verdict MAP-COMPLETE) as one module (35 checks,  $\sim 0.3$ s): *the capstone verifies the map and claims nothing new* — the whole sixteen-link reduction chain of phase 2 as one machine-checked pass, the eight-route no-go battery, and the precision ledger, recomputed on the declared small frame-A surface (12 deepest in-cap zones,  $m = 96 \dots 291 \leq 300$ ;  $\kappa = 0.038821$  verified at every jump point of  $\psi(x)/x$  on the table; the  $16 \times 16$  low block positive definite on 12/12 — a numerical fact, never routed through the Weil criterion). (1) The chain, 13 THEOREM / 3 CERT: T152 Schur two-block (KMS  $1.3 \times 10^{-15}$ ; holds at  $0.5\lambda_{\min}$ , fails at  $1.02\lambda_{\min}$  — sharp; Haynsworth  $3.8 \times 10^{-15}$ ); T151 Sobolev  $b_k \leq C_S k$ ,  $C_S = (2\pi + 2)/\sqrt{N}$  (worst ratio 0.4816); T153  $\Psi$ -collapse  $\Psi = \sum_d c_d w_d = x^T G x$  ( $6.3 \times 10^{-16}$ ;  $\sum_d w_d = 0$  to  $9.0 \times 10^{-17}$ ); T154 Ritz ceiling (interlacing;  $g_{32}/s = 0.9520$ – $0.9849$  from below); T155  $12 \times 12$  floor [CERT] (holds at  $(1 - 10^{-9})\lambda$ , fails at  $(1 + 10^{-4})\lambda$ ;  $t = 0.2295$ – $0.3719$ ); T156  $F(P, r)$  [CERT] ( $P_K \geq 1$ ,  $r_K \geq 0$ ,  $L(16) = 6.6$ – $15.6$ ); T158 Thomson/ladder ( $3.1 \times 10^{-12}$ ; increments strictly positive; 36/36 trials below); T160 sampling (closed Dirichlet weights = correlation weights  $1.1 \times 10^{-15}$ ; atom half = sampled  $\Lambda$ -mass  $9.9 \times 10^{-15}$ ); T163 TV floor/swap (Abel  $1.8 \times 10^{-11}$ ; overshoot  $\times 5.2 \times 10^4$ – $1.1 \times 10^6$ ); T164 gauge/quantifier (invariance  $2.2 \times 10^{-11}$ ;  $\Psi(x^*) = 1/g_{16}$  to  $7.0 \times 10^{-12}$ ; 48/48 perturbed larger); T165  $P_{\text{pr}}$  ( $1.1 \times 10^{-16}$ ; = 1 at  $x^*$ ); T166 cascade  $1/(1 - R_{16}^2)$  ( $1.8 \times 10^{-12}$ ; diagonal-invariant); T167  $K = 2$  exactness ( $6.8 \times 10^{-13}$ ; milder  $\times 1.62$ – $5.20$ ); T169 det collapse R4-free ( $\leq 4.2 \times 10^{-12}$ ; no positivity of  $A_h$  consumed, the Weil fence never approached); T170 rank-3 ( $J$ -eigenvalues  $\{-2, -1, +1\}$  exact;  $\sigma_4/\sigma_1 \leq 1.6 \times 10^{-16}$ ); T170 R1 shape [CERT] (the  $2 \times 2$  Lagrange identity  $3.6 \times 10^{-16}$ ;  $\sin \angle = 1.169 \times 10^{-4}$ – $2.467 \times 10^{-3} \leq 10^{-2}$  per window; rate and quantifier OPEN). (2) The no-go battery, 8/8 failing as classified: size budget  $\times 2.7 \times 10^3$ – $1.6 \times 10^4$ ; symbol infimum  $-47.6$ – $-21.3 < 0$  on 12/12; sector change an invariance ( $\leq 4.2 \times 10^{-12}$ ); perturbation  $\times 9.8 \times 10^2$ – $2.0 \times 10^4$  off scale (Kato 1949, the closed-off route); atom band dominates  $\times 1.71$ – $2.83$ ; sieve operator-norm on  $\det S$  alone  $\times 8.8 \times 10^3$ – $7.5 \times 10^4$  over (by the rank-3 theorem a bound on  $\det S$  alone cannot see the three-term cancellation; Montgomery–Vaughan 1973 / Vaughan 1977 the priced route); scramble  $\times 679$  median (range  $68$ – $2.3 \times 10^6$ ) with rank-3 *untouched* on all 60 draws;  $L_P$  gain = 1 to 0.0. (3) The precision ledger: needed  $2.752 \times 10^{-4}$  at the deepest window ( $m = 285$ ) — finer than the RH yardstick  $h^{-0.5}$  by  $215\times$  and the best unconditional  $h^{-0.996}$  by  $13\times$  on this surface; both self-similarity identities ( $6.8 \times 10^{-13}$  /  $3.1 \times 10^{-13}$ , the  $t^*$  overshoot  $\geq 0$  everywhere, sign-agnostic). Mutations fail loudly (wrong regression row, wrong Thomson entry, off-by-one swap, wrong polarisation coefficient — the  $10^{-2}$  bar missed by orders). **Named limits as load-bearing content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in  $D$ ; the T171 trend fits (pooled sin-angle  $h^{-2.33}$ , frame-split  $\nu = 4$ :  $h^{-1.86}$

/  $\nu = 6$ :  $h^{-2.87}$ , the no-go growth exponents,  $\delta = +1.044$  against 3.0, the full-surface precision numbers  $2.284 \times 10^{-7} / 1.2 \times 10^5 \times / 3.1 \times 10^3 \times$  at  $h = 1444$ ) are sandbox numbers on T171's two-frame surface and are *not* consumed; **zero new uniform-in- $m$  statements are added — which is exactly why the capstone is promotable; the one open phase-2 object stays open, typed open:** R1 — an  $m$ -free unconditional certificate that the two rows of  $\hat{A}$  become collinear at rate  $h^{-3+\varepsilon}$ , a joint near-degeneracy of the three explicit linear  $\Lambda$ -sums against the archimedean block; R2/R3 booked (T169), R4 off the path (T170-TH6); the T171 verdict MAP-COMPLETE is a verdict about the *map*, not about R1, and stays a sandbox verdict ( $\frac{1}{2}$  appears in exactly one role: as the strength of a classical statement a required precision is compared against). Fences: Schur 1917–Haynsworth 1968 / KMS 1953 / Cauchy 1829–Courant–Fischer 1920 / Rayleigh 1877 / Abel 1826 / Dirichlet 1829 / Lagrange 1773 / Cauchy–Binet 1812–1815 / Gershgorin 1931 / Kato 1949 / Montgomery–Vaughan 1973–Vaughan 1977 / Chebyshev 1852–Rosser–Schoenfeld 1962 / Fejér 1915 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: [verification/v559\_phase2\_capstone.py]. [E] (per-instance identities and certified inequalities)

**The frame/deficit identities (PRIME.FRAME.DEFICIT.01).** Promotion of the theorem cores of discoveries T172 (contract FRAME.BEYOND, verdict PARTIALLY-PORTABLE), T173 (FRAME.RATE, DEFICIT-VARIES) and T174 (CANCEL.IDENTITY, PARTIAL-CANCEL) as one module (24 checks,  $\sim 0.5$  s): *the gauge route of phase 2, closed as algebra* — the companion to the capstone above, certifying what T172/T173/T174 established *about* the map's frame dependence, everything recomputed on a small declared surface (a  $4 \times 3$  (anchor,  $h$ ) rectangle of gap-blind frame-B cells, prime-power anchors  $n = 41 \dots 557$  crossed with the declared ladder  $h = (160, 226, 320)$ , comb complete on every cell, positive definite on 12/12; one non-prime-power instance  $n = 210$ ; one truncated-comb stress instance  $n = 1009$ ;  $\kappa = 0.038821$  verified at every jump point of  $\psi(x)/x$ ; the builder's signature is  $(n_{\text{zone}}, \alpha, M)$  — no frame label is an argument of any observable). (1) The identity transfer + the sieve-horizon discriminator: the KMS/Schur pair ( $3.2 \times 10^{-16}$ ; quadratic-form pair  $5.2 \times 10^{-15}$ ), the lag-sum / correlation identity ( $1.4 \times 10^{-16}$  of the larger half; Abel 1826), the R4-free det collapse (at 0.020 of the conditioned bar  $64\varepsilon/|1 - r_{12}^2|$ ) and the additive split ( $7.6 \times 10^{-16}$ ), re-derived *exactly* on a gap-blind frame-B instance *and* a non-prime-power-anchor instance — the links use no property of the frame rule or the anchor arithmetic, which is why they transfer; the truncated-comb instance turns *indefinite* ( $\lambda_{\min} = -9.6 \times 10^4$ ) against 12/12 positive definite complete-comb cells, and the identity links keep holding there (none uses positivity): indefiniteness sits at the declared sieve horizon, not at a frame or zone. (2) The  $q = 1 - s$  identity + the calibration:  $hD = \alpha$  per cell ( $8.9 \times 10^{-16}$ ) forces the OLS slopes to obey  $q = 1 - s$  to  $2.9 \times 10^{-14}$  (OLS is linear in its response); the gauge-degree account comes out as integers —  $G2 = t/\lambda_{\max}(B_{LL})$  and the delivery  $1 - r_{12}^2$  both  $(d_{\text{amp}}, d_{\mu}) = (0, 0)$ ,  $G1$  at  $d_{\text{amp}} = 1$ ,  $G3$  at  $d_{\mu} = 1$  — so  $G2$  is the unique doubly gauge-invariant arithmetic member of T173's family;  $G3 = \mu_1^P G2$  identically ( $2.2 \times 10^{-16}$ ; exponent  $h^{-1.9955}$ , the offset-2 explained);  $(D/\alpha)^3 h^3 = 1$  a tautology ( $2.2 \times 10^{-16}$ ). (3) The blindness theorems + the additive limit: the amplitude  $c \rightarrow \kappa c$  is a gauge of GAP, of  $1 - r_{12}^2$  and of  $R$  over twelve decades (worst 0.026 of the per-cell conditioned bar; rebuilt from scratch 0.098); the scalar  $\mu_1^P = 4 \sin^2(\pi/(2h+1))$  — *the entire  $h^{-2}$  channel of the demand* — cancels exactly on both sides (0.001 of bar); the delivery is invariant under the full positive diagonal group (48 congruences spanning  $e^{\pm 8}$ , worst 0.036 of bar) while a non-scalar diagonal moves the demand by 0.23–1.40 (the asymmetry is the anatomy); neither the arch term nor the comb term alone has a positive definite low block on a single cell ( $0/12 + 0/12$  against 12/12 for the sum): no factorisation  $\hat{A} = (\text{frame}) \times (\text{arithmetic})$  exists, so T173's P6 is *not* promoted to a theorem — only its multiplicative half. (4) The KMS shape band (unconditional, declared  $\varepsilon = (K\pi/(2h+1))^2/6$ ):  $|\sqrt{\hat{S}_k/k^2} - 1| \leq \varepsilon_{\text{cert}}$  on 12/12 and the  $k^2$ -swap moves  $\log \text{GAP}$  by  $\leq 0.312$  of the certified band  $2 \log((1+\varepsilon)/(1-\varepsilon))$  (Rayleigh 1877; Schur 1917) while the delivery does not move at all. (5) The placement discriminator: the value-multiset scramble destroys the positive low block on 24/24 declared draws and moves  $|R|$  by a median factor  $10^6$ . Mutations fail loudly (the  $d = 0$  halving dropped, the  $N = 2h$  ladder, the off-by-one  $h$ , the comb-only rescale, the wrong universal profile).

**Named limits as load-bearing content:** per instance on a small declared surface — a finite list with an explicit maximum; nothing uniform in the zone index or in  $D$ ; **the frame-free deficit is a measurement, not a claim of this row:** T174's  $+0.1111 \pm 0.0222$  ( $5.0\sigma$ , 151 anchor clusters, cluster-robust;  $0.4\sigma$  from T173's  $+0.155 \pm 0.102$ ) stays a sandbox MEASURED number, as do all T172/T173/T174  $h$ -exponents, the density anatomy and the  $\nu$ /density collinearity; **the one open phase-2 object stays open, typed open:** R1, now stated frame-free — an  $m$ -free unconditional certificate that two explicit finite  $\Lambda$ -sum vectors become collinear at the demanded rate; the residual per-anchor scatter ( $\chi^2/\text{dof} = 3.02$ ) and the sparse corner are sandbox items under test in T175 ( $\frac{1}{2}$  appears in exactly one role: as the strength of a classical statement a required precision is compared against). Fences: Schur 1917 / KMS 1953 / Dirichlet 1829 / Rayleigh 1877 / Abel 1826 / Chebyshev 1852–Rosser–Schoenfeld 1962 / Wilkinson 1968–Higham 2002 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion; zero-firewall AST-checked; no marker moves. Machine-checked: `[verification/v560_frame_deficit_identities.py]`. **[E]** (per-instance identities and certified inequalities)

**Lemma 10** (G3–G4 — reflection positivity, fixed and integrated). *For each  $\Theta$ -invariant  $g$  in the Calderón ball the physical boundary kernel  $K_g = P_{\text{adm}}\Pi_{\text{phys}}\mathcal{N}_g^{TT}\Pi_{\text{phys}}P_{\text{adm}}$  is reflection positive,  $\langle \Theta F, F \rangle_{K_g} \geq 0$  (G3); and  $d\mu_{\text{Cal},\chi}$  is  $\Theta$ -invariant, supported where G3 holds uniformly, so  $\int \langle \Theta F, F \rangle_{K_g} d\mu_{\text{Cal},\chi} \geq 0$  (G4). **G4 is the decisive step:** RP after integrating over dynamical metrics, not merely at fixed background.*

**Lemma 11** (G5 — gap dominance **[E]**).  $2\|V_{\text{metric}}\|_{\text{rel}} < \Delta = 6 \log \frac{3}{2}$ , with  $\|V_{\text{metric}}\|_{\text{rel}} \leq 248c_3^2 = \frac{31}{8\pi^2} = 0.39262$ , hence  $2\|V\| = 0.7852 < 2.4328$  and  $\Delta_{\text{eff}} \geq \Delta - 2\|V\| > 1.647 > 0$ . The numerical inequality is trivial (`[verification/v29_research_contract_certs.py]`); the work is the operator-norm bound (hard).

**Sugawara reading + atomic OS moment** (`[verification/v171_os_moment_cluster.py]`). The G5 budget is the  $E_8$  Sugawara constant:  $248c_3^2 = \frac{31}{8\pi^2}$  with  $31 = 1 + h^\vee(E_8)$  and  $c((E_8)_1) = 248/31 = 8$ , so gap dominance reads  $|R^+(A_3)| \log(N_{\text{fam}}/|\mathbb{Z}_2|) > \frac{1+h^\vee(E_8)}{4\pi^2}$  — the  $A_3$  transfer gap beats the  $E_8$  Sugawara metric budget. Independently, the G3 reflection positivity is a *positive moment problem*:  $\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$  makes every two-point correlator  $C(n) = A + B(64/729)^n + C(1/729)^n$  the moment sequence of a positive atomic measure, so each OS Hankel matrix factorises as a Vandermonde Gram and is positive semidefinite (a clean RP certificate; cluster constant  $729/665$ , correlation length  $\xi = 1/(6 \log \frac{3}{2}) = 0.4111$ ).

**Theorem 1** (Decoupling theorem — physical low-energy claims do *not* need the full  $G_{\text{metric}}$ ). *If the admissible sector is reflection-positive (G4) and the metric coupling is gap-dominated,  $2\|V_{\text{metric}}\| < \Delta = 6 \log \frac{3}{2}$  (G5), then the admissible IR measure is stable under the ambient completion: a strictly positive effective gap  $\Delta_{\text{eff}} = \Delta - 2\|V\| > 0$  protects the admissible sector from the (un-built) ambient/deep-UV, so every physical low-energy read-out (masses, mixings,  $\alpha^{-1}$ , the  $R+R^2$  regime) is independent of the open ambient measure. Symbolically*

$$\boxed{\text{RP} + 2\|V\| < \Delta \implies \text{admissible IR measure stable under ambient completion} .}$$

*Thus ( $G_{\text{metric}}$ ) is not a diffuse “full quantum gravity missing” block: it is the sharply delimited ambient projective limit (G6), a strict-TOE certification target whose closure the falsifiable IR physics does not depend on. **Typing:** the inequality is machine-checked **[E]**; the operator-norm bound on  $\|V_{\text{metric}}\|$  is the conditional input.*

**Lemma 12** (G6–G7 — projective limit & Ward identities). *A projective system  $\mu_{\chi_n}$  with  $\pi_{m,n*}\mu_{\chi_m} = \mu_{\chi_n}$ , uniform moment bounds  $\sup_n \int \|g\|_{H^s}^p d\mu_{\chi_n} < \infty$ , tightness and regulator-independence yields  $\mu_{\text{QG}} = \lim \mu_{\chi_n}$  on  $\mathcal{G}/\text{Diff}$  (G6, the hardest step); the limit obeys the BRST Ward identities  $\langle s\mathcal{O} \rangle = 0$ , whence  $\nabla^\mu T_{\mu\nu} = 0$  and the  $\eta$  boundary phase cancels the APS anomalies (G7).*

**Gate 2 reduction:** G6 is a *boundary projective limit*, not a bulk one ([verification/v76\_gmetric\_reduction.py])

The exact decoupling margin is  $\Delta_{\text{eff}} = \Delta - 2\|V\| = 6\log \frac{3}{2} - \frac{31}{4\pi^2} = 1.648 > 0$ , with  $\|V_{\text{metric}}\|_{\text{rel}} \leq 248 c_3^2 = \dim(E_8) c_3^2 = \frac{31}{8\pi^2} (31 = 2^{g_{\text{car}}} - 1)$ . **Holographic reduction:** because the seam is a *finite causal boundary*, the ambient *bulk* metric measure is reconstructed from a finite-codimension seam (Calderón) *boundary* measure — so G6 is a *boundary projective limit*. The conditional theorem is then

$$\text{RP}(\text{seam-Calderón boundary kernel}) + \text{tightness} \implies G_{\text{metric}} \text{ closes}$$

i.e. the dimension of the open problem drops from bulk QG to a finite seam-boundary measure.

**Residual:** the constructive boundary projective limit (constructive-QFT grade) — now discharged as a [C] redundancy (v369, keybox below). So Gate 2 is now *two clean tiers*: Tier A (IR physics) closed under RP+gap; Tier B (strict TOE) a [C] redundancy, no longer a diffuse “full QG missing”. [C]

**The measure as one-loop fluctuations around the parameter-free saddle** ([verification/v365\_qg\_oneloop\_saddle.py], QGAMB.SADDLE.01)

With the saddle now parameter-free (v358/v359), the boundary measure of Tier B organises as a Gaussian one-loop fluctuation determinant whose ingredients are atom-fixed. **Fixed quadratic form [E]:** the fluctuation inverse-propagator stiffness is the saddle coefficient  $1/c_3 = 8\pi$  — no free dimensionless dial, so the one-loop problem is *well-posed* (it was not before v358). **Gap-controlled convergence [E]:** the fluctuation operator obeys  $M \geq \Delta_{\text{eff}} = 6\log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$  (v76/v337), so the Gaussian integral converges mode-by-mode with no zero/negative modes besides the saddle directions; on the OS spectrum  $\{0, 6\log \frac{3}{2}, 6\log 3\}$  the regularised one-loop log-det  $\text{Tr}' \log M = 6\log \frac{9}{2}$  is finite. **Conformal direction [C]:** the one wrong-sign mode ( $c_{\text{conf}}(4) = -\frac{3}{2}$ ) is made convergent by the GHP contour ( $\int e^{-|c|\phi^2} = \sqrt{2\pi/3}$ ) with IDG dressing (v334). **Tightness [E]:**  $\chi = 1/(1 - (2/3)^6) = 729/665 < \infty$  (v330), so the one-loop measure has a projective limit on the admissible sector, with *zero* new dimensionless dials (v364). **Residual:** the full *non-perturbative* projective limit (G6 on the ambient sector) is now discharged as a [C] redundancy (v369, keybox below) — this sharpens C7 from “diffuse full QG” to a one-loop-controlled Gaussian problem around a parameter-free saddle and then to a certification object. [E]/[C]

**The holographic route: the ambient measure is boundary-redundant, not missing dynamics** ([verification/v369\_qgamb\_redundancy.py], QGAMB.REDUNDANCY.01)

There is a second, strategically stronger way to dispatch Gate 2 than building the measure: prove that the ambient sector carries *no* physical degrees of freedom relative to the holomorphic boundary algebra, so the measure QG.AMB.01 is a non-fundamental *certification* object — the role a coordinate choice plays in general relativity — rather than a missing piece of dynamics. The discriminators that make this *ambient-redundancy* statement well-posed are all in hand. **Trivial superselection [E]:** the number of DHR sectors of a chiral net is  $|\det \text{Cartan}|$ , which is 1 for  $E_8$  (one primary, the vacuum;  $\text{DHR}((E_8)_1) = \text{Vec}$ ) against 4 for the same- $c$  rival  $SO(16)_1$  — a holomorphic boundary net has *no* nontrivial superselection charge that a bulk could carry invisibly. **No torus degeneracy [E]:** the ground-state degeneracy on  $T^2$  equals the number of anyons  $= |\det K| = 1$  for  $E_8$ , so there is no topologically protected bulk d.o.f. hidden from the boundary (the genus-1 obstruction of v344, in its redundancy reading). **Bounded reconstruction [E]:** the seam transfer deviation subspace contracts at the finite gapped Petz rate  $(2/3)^6$  (v221), and the gap-decoupling margin  $\Delta_{\text{eff}} = 6\log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$  (v76/v337) separates the un-built ambient sector from the admissible one. **The statement [C]:** with the Bisognano–Wichmann modular reconstruction map (v258/v309/v323), under SEAM.EQUIV.01 every gauge-invariant bulk observable is boundary-reconstructible,  $\mathcal{O}_{\text{phys}}^{\text{bulk}}/\text{Diff} \cong \mathcal{A}_{\Sigma}^{\mu_4}$ , and two ambient measures sharing the same boundary state are physically equivalent. **Residual [O]:** the construction consumes exactly two premises — SEAM.EQUIV.01 (the S3 lattice input, v366) and the BW intrinsicity of the raw collar (v329); it does *not* construct QG.AMB.01 but supplies the *alternative* route, so that Gravity-complete = Boundary-complete + Ambient-redundancy. [E]/[C]/[O]

**The classical field equation, parameter-free: entanglement equilibrium gives the linearised Einstein equation with  $G$  from  $c_3$  ([verification/v358\_grav\_entropy\_equilibrium.py])**

While Gate 2 (above) reduces the metric-sector *measure*, the classical metric-sector *field equation* is supplied directly by the entanglement first law  $\delta S = \delta \langle K \rangle$  (Jacobson 2015; Faulkner et al.), carried out with TFPT's atoms. **Parameter-free coefficient [E]**. Equilibrium gives  $G_{ab} = c_3^{-1} T_{ab}$  with  $c_3^{-1} = 8\pi$  *fixed*: all four  $c_3$ -roles are mutually consistent — the Unruh factor  $1/(2\pi) = 4c_3$ , the EH action  $1/(16\pi) = c_3/2$ , the Einstein coefficient  $8\pi = 1/c_3$ , the Bekenstein density  $1/4 = 1/|\mu_4|$  — so *no free dimensionless Newton dial* enters. **Thermo = geo coincidence [E] (new)**. The first law requires the coefficient  $2\pi/\eta$  ( $\eta$  the entropy density); the geometry gives  $1/c_3 = |\mathbb{Z}_2| 2\pi \chi(S^2)$  (one-sided Gauss–Bonnet, v58); these are the *same* equation iff  $|\mu_4| = |\mathbb{Z}_2| \chi(S^2) = 4$  — a TFPT atom identity (v216). So the thermodynamic (first-law) and geometric (Gauss–Bonnet) origins of  $c_3$  are *one*. **The matter side, assembled [C] (closing J3)**. The ball modular Hamiltonian is the boost (Casini–Huerta–Myers; geometric by Bisognano–Wichmann, v323),  $K_B = 2\pi \int_B w T_{00}$ ,  $w = (R^2 - r^2)/(2R)$ ; the 3-ball weight integral is  $\int_B w d^3x = 4\pi R^4/15$ , so  $\delta \langle K_B \rangle = \frac{8\pi^2 R^4}{15} \delta \langle T_{00} \rangle$  — the concrete matter boost flux of TFPT's stress tensor. **The entropy density, atom-fixed [E]**. The Bekenstein numerator  $1/4 = 1/|\mu_4|$  (v57) and the entanglement normalisation = the central charge  $c = g_{\text{car}} + N_{\text{fam}} = 8$  pin the dimensionless coefficient; the area-law form is established (v59). **Residual**: the full covariant extension is now *closed* (v359, next keybox); what remains is only (i) the Jacobson equation-of-state *status* (a thermodynamic derivation, not a from-action quantisation) [O] and (ii) the absolute scale  $v_{\text{geo}}$  (the Planck-area unit) [O], while the global ambient measure (Gate 2/C7) is now a [C] redundancy (v369) — *no free dimensionless parameter survives in the gravity coupling*.

**And that boundary measure is a rigorously constructed net** ([verification/v77\_e8\_conformal\_net.py]). The level-1 central charges  $c = \dim \mathfrak{g}/(1 + h^\vee)$  give  $c(E_8)_1 = \frac{248}{31} = 8 = \text{rank } E_8$ ,  $c(D_5)_1 = \frac{45}{9} = 5$ ,  $c(A_3)_1 = \frac{15}{5} = 3$ , and  $c(D_5) + c(A_3) = 5 + 3 = 8 = c(E_8)$  is a *conformal embedding*  $(D_5)_1 \times (A_3)_1 \subset (E_8)_1$  (coset  $c = 0$ ). Now  $(E_8)_1$  is the holomorphic  $c = 8$   $E_8$ -lattice VOA / conformal net — one of the most rigorously constructed objects in mathematical QFT (Frenkel–Kac–Segal; Kawahigashi–Longo / Carpi). **So the conditional theorem becomes concrete: identify the seam-Calderón boundary measure with the  $(E_8)_1$  lattice net** (carrier =  $(D_5)_1 \times (A_3)_1$  subnet), and the gap  $\Delta_{\text{eff}} > 0$  supplies the clustering/tightness — so G6 is imported into existing RCFT/conformal-net rigor rather than built from scratch. Residual = the net identification + bulk reconstruction. [E] (charges, embedding) + [C] (identification)

**The full covariant field equation, parameter-free: fixed-volume equilibrium gives the Einstein tensor with conservation as an output ([verification/v359\_grav\_nonlinear\_einstein.py])**

The linearised result above (v358) extends to the *full* covariant equation by Jacobson's *fixed-volume* (maximal-vacuum-entanglement) equilibrium — no linearisation assumed. **Einstein tensor, not Ricci [E]**. Stationarity of the entanglement entropy at fixed volume forces the divergence-free combination: the 4D trace identity  $g^{ab} G_{ab} = (1 - \frac{d}{2})R = -R$  (for  $d = 4$ ) selects  $G_{ab} = R_{ab} - \frac{1}{2}R g_{ab}$ , so

$$G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}, \quad c_3^{-1} = 8\pi$$

with *both* coefficients TFPT-fixed: the Newton coupling  $8\pi = 1/c_3$  (v358) and the cosmological constant  $\Lambda$  from  $\alpha$ ,  $\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$  (v60). **Conservation is an output [E]**. By Lovelock's theorem the only divergence-free, second-order, local symmetric tensor built from the metric is  $a G_{ab} + b g_{ab}$ , so  $\nabla^a G_{ab} = 0$  forces  $\nabla^a T_{ab} = 0$  — matter conservation is *derived*, not assumed. **So Gate B6 — the classical metric-sector field equation — is closed parameter-free at the local level. Residual**: the Jacobson equation-of-state *status* (a thermodynamic derivation, not a from-action quantisation) and the single unit  $v_{\text{geo}}$  stay [O], while the global ambient measure (Gate 2/C7, the constructive boundary projective limit) is now a [C] redundancy (v369) — no free dimensionless gravity dial remains. An *external candidate* for the missing action level is quantified in the next keybox (v473); it does not change this typing. [E]/[C]

**The entropic-action bridge: an external candidate for the missing action level, quantified — nothing closes ([verification/v473\_entropic\_action\_bridge.py])**

The one honest **[O]** left in the classical sector — “equation of state, not a from-action quantisation” (v359) — now has an explicit *external* candidate: Bianconi’s entropic action  $S_B = -\text{Tr}_F \ln(\tilde{G}\tilde{g}^{-1})$ , the quantum relative entropy between the spacetime metric and the matter-induced metric (G. Bianconi, *Gravity from entropy*, Phys. Rev. D **111**, 066001 (2025); arXiv:2408.14391; her Appendix C identifies  $\Delta_{\tilde{G},\tilde{g}}^{1/2} = \tilde{G}\tilde{g}^{-1}$  with an Araki-type modular operator) — a local covariant action whose variation yields the Einstein equations plus an emergent nonnegative  $\Lambda$  at low coupling: exactly the missing variational layer, *if* the bridge identities hold. What is machine-checked today (GRAV.ENTROPIC.ACTION.01): **carrier Hodge count [E]**. Her  $\Omega^{0,1,2}$  matter fiber, placed on the five-slot carrier  $\mathbb{C}^5$  (*not* on 4d spacetime, where the fiber is  $1+4+6=11$ ), counts  $1+5+10=16=2^{g_{\text{car}}-1} = \dim S_{D_5}^+$  and Hodge-folds ( $\star: \Lambda^1 \cong \Lambda^4$ ) onto  $\Lambda^{\text{even}}\mathbb{C}^5$  — the carrier half-spinor’s own Pascal row (v2/v44); a vector-space identity *only*. **Coefficient match [E] (the one quantitative pin)**. Her weak-coupling Lagrangian  $\mathcal{L} = 3\beta R - \alpha\mathcal{L}_m + \dots$  (the 3 = the three form blocks) matched to the TFPT EH normalisation  $1/(16\pi G) = c_3/(2G)$  fixes her free constant *exactly*,

$$\beta'_B = \frac{c_3}{6} = \frac{1}{48\pi}$$

(the  $6 = 3 \times 2 = p_2 = |R^+(A_3)|$  is recorded as a structural echo, not a proof). **Entropy potential = quadratic  $\Lambda$  [E]**. Her  $\Lambda_G = \frac{1}{2\beta} \text{Tr}_F(\mathcal{G} - I - \ln \mathcal{G})$  is nonnegative and quadratic at the fixed point; with the TFPT displacement  $\varepsilon_\star = e^{-\alpha^{-1}Q}$  it reproduces the closed  $\Lambda$  branch  $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$  (v60), and the prefactor match yields the *exact* normalisation target  $\text{Tr}_F Q^2 = 32c_3^4 = 1/(128\pi^4) = \frac{2}{3}\delta_{\text{top}}$  — a closed algebraic target for the two-form (pair) block ( $\binom{5}{2}=10$ ), **[C]** until an explicit  $Q$  is constructed.  **$R^2$  kill test [E] (pre-registered)**. The raw log expansion carries a curvature-squared coefficient of order  $(c_3/6)^2$ ; the TFPT Starobinsky sector needs  $\bar{M}_{\text{Pl}}^2/(12M_{\text{scal}}^2) = 1/(12c_3^7)$  (v36) — the exact mismatch is  $3c_3^{-9} = 3(8\pi)^9 \approx 1.2 \times 10^{13}$ : a direct identification of the raw entropic action with the TFPT  $R^2$  sector is *false* without a mechanism (the scalaron as a light trace mode of her  $\mathcal{G}$ -field generating  $c_3^{-9}$ , or a KMS-spectral renormalisation of the log action). **The bridge proper [C] (recorded, not proven)**. The compression conjecture  $P_\Sigma(\tilde{G}\tilde{g}^{-1})P_\Sigma = \Delta_\Sigma^{1/2}$  — the physical Calderón compression of her relative metric operator equals the TFPT seam modular operator; if proven,  $S_B$  is a local volume representation of the TFPT relative spectral action  $S_{\text{rel},\chi}$  (her log action is the all-scale-integrated heat kernel, by Frullani  $-\ln A = \int_0^\infty \frac{dt}{t}(e^{-tA} - e^{-t})$ , verified). Open with it: the operator-level Hodge items (Clifford action, Dirac intertwining, leaf involution, Calderón polarisation), and Lorentzian positivity of  $\tilde{G}\tilde{g}^{-1}$  is *not* automatic (timelike gradients) — TFPT’s OS/reflection-positive route is the sturdier construction. **Residual [O] (unchanged)**. Nothing closes: the v359 equation-of-state typing stays **[O]**; adopting  $S_B$  would close it *only* after the compression conjecture *and* an  $R^2$  mechanism; zero TFPT knobs are added.

**The operator level, executed ([verification/v474\_entropic\_hodge\_carrier.py])**. Work packages 1 and 4-algebra of the bridge are now machine-checked on the finite carrier Fock space  $\Lambda^\bullet\mathbb{C}^5$  ( $\dim 2^{g_{\text{car}}}=32$ ). **Spinor structure exhibited [E]**: the ten CAR gammas satisfy  $\{\Gamma_a, \Gamma_b\}=2\delta_{ab}$  exactly, the 45 bilinears span  $\mathfrak{so}(10)$  and leak nothing from the 16-dim even subspace —  $S^+$  is the even exterior algebra with its spinor structure, and Bianconi’s Hodge–Dirac operator is Clifford multiplication at symbol level ( $c(\xi)^2 = -|\xi|^2$ , symbolic). **The fold is the  $5 \rightarrow \bar{5}$  conjugation [E]**: the traceless  $\mathfrak{u}(5)$  weights of  $\Lambda^4$  are the negatives of those of  $\Lambda^1$ , so her fiber  $1+5+10$  becomes the GUT half-spinor  $16=1+\bar{5}+10$  — the SM carrier decomposition, not a dimension match; honest obstruction:  $\Omega^{\leq 2}$  is *not* Clifford-invariant, so the identification must go through the fold, never through degree truncation. **The  $Q$ -target, decided [E]**: the uniform-block ansatz  $\text{Tr} Q^2 = dk^2 c_3^4 = 32c_3^4$  with integer  $k$  admits *exactly* the supports  $(d,k)=(2,4), (8,2), (32,1) = \{|\mathbb{Z}_2|, \text{rank } E_8, 2^{g_{\text{car}}}\}$ , with the minimal uniform solution  $q=c_3^2$  per Fock mode and the identity  $\text{Tr} Q^2/\delta_{\text{top}}=32/48=2/3$ ; the two-form block 10 (the naive pair-sector reading) and the fiber 16 require irrational multipliers and are *excluded*. The physical choice among the three supports stays **[C]**; AP2 and the  $R^2$  mechanism stay **[O]**.

**The  $R^2$  kill test, executed ([verification/v475\_entropic\_scalaron\_gate.py])**. On a maximally symmetric background the relative curvature operator has eigenvalues *exactly*  $\{R; R/4 \times 4; R/6 \times 6\}$  (her Appendix-B flattening verified, incl. the  $6 \times 6$  two-form block), so the exact vacuum action is  $\mathcal{L}_B(R) = -[\ln(1-\beta R) + 4\ln(1-\beta R/4) + 6\ln(1-\beta R/6)] = 3\beta R + \frac{17}{24}\beta^2 R^2 + O(R^3)$  — the 3 re-derives her Eq. (45) and  $\frac{17}{24}$  is the exact tensorial factor behind the pre-registered gap. With the pinned  $\beta'=c_3/6$

both mass routes (Starobinsky matching and  $f'/3f''$ ) give the raw entropic scalaron

$$m_{\text{raw}}^2 = \frac{4608\pi^2}{17} \bar{M}_{\text{Pl}}^2 \Rightarrow m_{\text{raw}} \approx 51.7 \bar{M}_{\text{Pl}}$$

— *trans-Planckian*, not light: the naive reading “the TFPT scalaron is the light trace mode of her  $\mathcal{G}$ -field” is *dead* [E]; a viable mechanism must supply exactly  $m_{\text{raw}}^2/m_{\text{TFPT}}^2 = \frac{72}{17}(8\pi)^9 \approx 1.69 \times 10^{13}$  in mass<sup>2</sup>. The domain is not the obstruction (first log pole at  $R=384\pi^2 \bar{M}_{\text{Pl}}^2$ , trans-Planckian), and the Lorentzian caveat is now an explicit witness: a timelike gradient gives the  $Gg^{-1}$  eigenvalue  $1 - \alpha v^2 \leq 0$  for  $v^2 \geq 1/\alpha$  (ln undefined), while spacelike/Euclidean stay positive — the OS route is the sturdier construction. **Net verdict** [C]: of the three bridge interpretations, the entropic action covers the Einstein +  $\Lambda$  level (survives); the light-mode reading is dead; KMS-spectral renormalisation is the *only* surviving route to the  $R^2$  sector. Nothing closes.

**AP2, made well-posed** ([verification/v476\_entropic\_compression\_toy.py]). The compression conjecture itself is sharpened on an exactly solvable gapped chain. **The reading is decided:** on a *pure* bulk (the seam situation: pure bulk, thermal boundary) the covariance is a projection, so  $f(x)=x/(1-x)$  is singular on its spectrum — the literal operator-side reading  $P_\Sigma f(C)P_\Sigma$  is *ill-posed*, while the state-side  $f(P_\Sigma C P_\Sigma)$  (Peschel) is finite: the identity can only mean *build  $\Delta_\Sigma$  from the compressed relative metric*, which is exactly Bianconi’s own local construction — AP2 retyped, not weakened. **The mismatch is exactly second order** [E]: symbolically,  $[C^n]_{AA} - (C_{AA})^n$  carries the cross-cut block twice (first order vanishes identically), so for any analytic  $f$  the two readings agree up to  $O(\|C_{AB}\|^2)$  — the modular (entanglement) content *is* that quadratic correction. **Gap control:** at the bounded (covariance) level the mismatch obeys  $d \leq x^2$  and falls with the gap, the decoupled cut agrees exactly — the two readings converge in the gap-dominated regime, where TFPT operates ( $\Delta_{\text{eff}} > 0$ , v76/v337). **Residual** [O] (**unchanged**): the continuum seam statement itself. No marker moves.

**The surviving  $R^2$  route, typed as one moment** ([verification/v477\_entropic\_scaleflow.py]). The Frullani identity upgrades to a *scale-flow representation*:  $-\ln a = \int_0^\infty \frac{dt}{t} (e^{-ta} - e^{-t}) = 2 \int_0^\infty \frac{dx}{x} (e^{-a/x^2} - e^{-1/x^2})$  — Bianconi’s log action *is* the flat scale-integral of relative heat-kernel actions, while TFPT’s  $S_{\text{rel},\chi}$  evaluates *one* KMS scale with a weight  $f$ : the bridge between the two action principles is a choice of *scale measure*. In the moment dictionary (weight  $w(t) dt/t$ ,  $\mu_k$  its  $e^{-t}$ -moments) the coefficients read  $3\beta R \mu_1 + \frac{17}{24} \beta^2 R^2 \mu_2$ ; the flat weight has  $\mu_1 = \mu_2 = 1$  (= the raw v475 coefficients), and demanding  $m^2 = c_3^7 \bar{M}_{\text{Pl}}^2$  forces *exactly*

$$\mu_2/\mu_1^2 = \frac{72}{17}(8\pi)^9 \quad \text{with} \quad \frac{4608\pi^2/17}{(72/17)(8\pi)^9} = c_3^7 \text{ (identically)}$$

— the v475 kill-test number *is* a moment ratio, and the “13 orders of magnitude” are not a bridge mismatch but a scale-measure datum: the flat measure fixes it wrongly, the TFPT KMS measure ( $f_0=6(4\pi)^2/c_3^7$ , v36) fixes it correctly — one KMS moment, two parametrisations, zero new dials. [C] (consistency, not derivation: the measure is TFPT input); the  $R^2$  gate and the v359 typing stay [O].

**The two remaining legs, first computable steps** ([verification/v478\_entropic\_continuum\_legs.py]).

*Leg A (AP2 continuum):* on the exactly solvable critical chain (infinite-volume kernel, 50-digit precision) the compressed interval state carries the conformal/BW structure quantitatively — the entanglement entropy follows Calabrese–Cardy with  $c_{\text{est}} = 1.00089/1.00038/1.00021$  (monotone  $\rightarrow$  the Dirac  $c=1$ ; gapped control  $\sim 6 \times 10^{-8}$ ), and its Peschel modular Hamiltonian organises on the CHM weight  $\beta(x)=x(L-x)/L$  (corr 0.87 $\rightarrow$ 0.99, even-range bands vanishing identically, odd tail 1.8%): the state-side  $\Delta_\Sigma$  (the reading *forced* above) flows to the CHM/Bisognano–Wichmann geometric form — *exactly* the object the entanglement-equilibrium derivation consumes (v323/v358); the bridge’s modular side and the gravity side meet in the continuum limit (a lattice exhibit, not a continuum proof — AP2 stays [O]; the natural continuum frame for the seam version is the *four-interval* multilocal modular geometry of the four marks, v480). *Leg B (the measure):* the single-scale corollary is exact —  $d\mu = \delta(t-t_0)dt$  gives  $\mu_2/\mu_1^2 = e^{t_0}$ , so the one-moment condition forces  $t_0 = \ln \frac{72}{17} + 9 \ln(8\pi) = 30.461$ , one dimensionless KMS time in closed form; the near-miss to  $h(E_8)=30$  (difference 0.461) is explicitly *declined* per the no-free-pattern rule (v354/v355) — deriving the measure from the entropic structure stays [O]. [E]/[C]/[O]

**The Simple-Current Extension Theorem — the central  $G_{\text{net}}$  closing statement [E]/[C]**  
 ([verification/v154\_simple\_current\_theorem.py])

The conditional theorem can now be stated as one clean *simple-current extension*. Let  $A = (D_5)_1 \otimes (A_3)_1$  be the carrier net and  $L = \langle(1,1)\rangle \subset \text{Disc}(D_5) \oplus \text{Disc}(A_3) = \mathbb{Z}_4 \oplus \mathbb{Z}_4$  the isotropic  $\mu_4$  glue line ( $q(k(1,1)) = k^2 \in \mathbb{Z}$ ). Then the local extension  $B = A \rtimes L$  has

$$[B : A] = |L| = 4 = |\mu_4|, \quad c(B) = 5+3 = 8, \quad \mu(B) = \frac{\mu(A)}{|L|^2} = \frac{16}{16} = 1$$

so  $B$  is *holomorphic* (KLM  $\mu$ -additivity), and a holomorphic  $c=8$  chiral net is the lattice net of the *unique* even-unimodular rank-8 lattice — hence  $B \cong (E_8)_1$  (the same- $c$  rival  $SO(16)_1$ ,  $\mu=4$ , is excluded). Every algebraic ingredient is machine-checked: the  $\mathbb{C}[\mathbb{Z}_4]$  Longo Q-system ([verification/v125\_glue\_qsystem.py]), the  $\mathbb{Z}_4$ -graded Frobenius  $E_8$  hull ([verification/v143\_graded\_frobenius.py]), the Ramond glue sectors ([verification/v148\_fock\_census.py]). **Consequence:**  $G_{\text{net}}$  is internally *closed* if the seam–Calderón boundary net is *defined* to be  $A$ ; it stays the *one* external identification “the RP seam–Calderón net is  $A$ ” if the boundary net is to be *derived* from the seam kernel. Either way this is the single strict-TOE certification theorem still carrying weight — a clean extension statement, not “find a quantum gravity”.

**And that one identification reduces to a single shared premise** ([verification/v155\_quasifree\_boundary.py]). The identification splits (v110) into (a) “the seam Calderón involution is sheet-odd” — the double-cover deck, the same  $|\mathbb{Z}_2|$  that halves  $c_3 = \frac{1}{2.4\pi}$  (structural) — and (b) “the seam 2-point kernel is quasi-free”. Part (b) is *not independent*: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression  $PTP$  of the bulk covariance, and the compression of a contraction ( $i\Gamma$  with spectrum in  $[-1, 1]$ ) is a contraction — so the boundary/seam state is automatically quasi-free, with Wick inherited (every boundary correlator a Pfaffian of  $\Gamma_d$ ). Hence (b) *follows* from “the seam bulk is a reflection-positive free field” — the *same* premise already load-bearing in the area law ([verification/v59\_area\_law\_evidence.py]) and the EH mechanism ([verification/v150\_replica\_eh\_model.py]/[verification/v151\_bfk\_split.py]; since exercised on the discretized collar with the seam’s own kernel and real replica sheets, v471 — the R3 kernel premise is discharged at the finite level, its residual retyping to the continuum scaling limit shared with SEAM.EQUIV.01, v336, plus the v152 anchor). **So the entire seam-side residual is one physical premise** — the seam bulk is a free RP field — shared by  $G_{\text{net}}$ , the area law and R3, and given it the chain Gaussian bulk  $\Rightarrow$  quasi-free boundary  $\Rightarrow$  one kernel = net (v113)  $\Rightarrow$  carrier  $A \Rightarrow (E_8)_1$  is exact. It is not reducible by a finite computation — it is the one place TFPT meets functional analysis.

**The construction, exhibited and verified** ([verification/v156\_seam\_net\_construction.py]). The “ $B \cong (E_8)_1$ ” step is no longer an assertion — the chiral net is built explicitly and the identification *checked*. The Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian acts on a boundary mode  $e^{ikx}$  by the harmonic extension  $e^{ikx-|k|y}$ :  $-\partial_n u|_0 = |k|u$ , so for a free bulk the seam operator is the massless chiral dispersion  $\omega_k = |k|$  that defines the free-fermion net on  $S^1$ . Sixteen Majoranas give  $c = 8 = 5+3$ ; the weight-1 currents are  $248 = 120_{\text{NS}} + 128_{\text{R}}$  (the  $SO(16)$  bilinears plus the Ramond spinor, Frenkel–Kac); and the holomorphic character is

$$\chi = \frac{E_4}{\eta^8} = q^{-1/3} (1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + \dots) = j^{1/3}$$

(verified by  $q$ -series), the 248 at level 1 being  $\dim E_8$  — so the net *is*  $(E_8)_1$ . *What remains* is exactly (i) the free-bulk premise above and (ii) the operator-algebraic existence / uniqueness of the net generated by the seam Calderón kernel (a Kawahigashi–Longo-type net-construction theorem). Route 3 thus *exhibits and verifies* the target; the irreducible residual is the one premise + the standard net-existence step — where TFPT meets constructive QFT, now sharply located.

**And the free-bulk premise itself reframes: freeness is a rigid fixed point, not a postulate** ([verification/v157\_rigid\_fixed\_point.py]). Three facts click together. (i) The DtN/Calderón symbol is *universally*  $|k|$ : the boundary operator of *any* second-order elliptic bulk is an order-1  $\Psi\text{DO}$  with principal symbol  $|\xi|_g$  (Lee–Uhlmann), so the free chiral dispersion is bulk-detail-independent and any interaction lives in the lower-order (irrelevant) symbol. (ii) A holomorphic  $c=8$  theory is *rigid*: every field has weight  $(h, \bar{h}=0)$ , an exactly-marginal deformation needs  $(1, 1)$ , and with  $\bar{h}=0$  *no*  $(1, 1)$  field

exists — the fixed point is isolated, there is no continuous knob along which a non-Gaussian interaction could turn on. (iii) The *same*  $|\mathbb{Z}_2|$  that halves the one-sided  $S^2$  Gauss–Bonnet ( $c_3 = \frac{1}{|\mathbb{Z}_2| 2\pi \chi(S^2)} = \frac{1}{8\pi}$ ) labels the sheet *and* is the Ramond/GSO projection giving  $\mu=1$  ( $248=120_{\text{NS}}+128_{\text{R}}$ ) — one object, three roles, so the geometry fixing  $c_3$  is the projection making the net holomorphic. **Consequence:** premise (A) shrinks from “prove the seam is Gaussian” to “the gapped ( $\Delta=6 \log \frac{3}{2} > 0$ ) boundary flow reaches the holomorphic fixed point” — given which, freeness is *forced* by rigidity + uniqueness ( $(E_8)_1$ , v83), not fine-tuned. A strictly smaller and simpler residual.

**And the destination is a *proven* stable fixed point** ([verification/v158\_fixed\_point\_stable.py]). An exact operator-dimension count settles it: the chiral Majorana has  $h=\frac{1}{2}$ , the bilinear current  $h=1$ , the quartic  $h=2$ . The relevant window  $0 < h < 1$  for bosonic deformations is *empty* (bosonic chiral operators have integer  $h$ , lowest 1) — nothing relevant flows the free theory away; the  $h=1$  currents are chiral (v157), not  $(1,1)$ -marginal; and the lowest genuinely-interacting operator (the quartic,  $h=2>1$ ) is *irrelevant*, as are all higher  $2k$ -fermi operators ( $h=k$ ). So the free chiral  $c=8$  fixed point is *isolated and stable* against all interactions. Premise (A) thereby reduces to a *basin-of-attraction* statement with a proven stable target — the gapped ( $\Delta=6 \log \frac{3}{2}$ ) flow lands here; the convergence is conformal-perturbation analysis, the destination is settled.

**And the closure can be stated as one fixed-point theorem — which relocates (A) to a single internal premise** ([verification/v160\_seam\_gaussianity\_from\_pf.py]). Phrase the argument internally: (i) a quasi-free state has *all* higher connected cumulants  $\kappa_{2n}=0$  (the signed set-partition/Pfaffian inversion of  $\langle \psi_1 \cdots \psi_{2n} \rangle = \text{Pf}(\Gamma)$  leaves only the degree-2 kernel — the v113 “one kernel = the net” property at the cumulant level); (ii) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (iii) a gapped primitive Perron–Frobenius transport has a *unique* fixed ray, so a non-Gaussian piece is impossible — if  $\kappa_{2n}$  were fixed it would be a *second* fixed ray (killing uniqueness), and if not fixed it sits in the gapped complement and decays to 0 (any gap  $> 0$  suffices; the value  $(2/3)^6$  only sets the rate). *The honest residual:* v56’s gapped PF uniqueness is established on the *three-dimensional* scalar transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  (the flavour gap / horizon Page-recovery rates), whereas the 16-Majorana seam state already carries  $\binom{16}{4}=1820$  and  $\binom{16}{6}=8008$  higher-cumulant directions — the Schwinger cone the transport must contract has dimension  $\gg 3$ , and uniqueness must hold there (not merely on the quasi-free sub-cone) for the argument to be non-circular. **So (A) is equivalent to one strictly smaller, internal, exact-testable statement:** the admissible TFPT seam transport is a *primitive, spectrally gapped, RP-cone-preserving* map on the *full* Schwinger cone, not only on the scalar readout spectrum of v56 — which the suite does *not* yet establish [E]/[O].

**And that full-cone premise is now *discharged* to a single finite one-particle statement** ([verification/v161\_one\_particle\_reduction.py]). The lever is Bogoliubov second quantisation: a quasi-free transport is  $T_\Sigma = \Gamma(t)$ , the second quantisation of a one-particle contraction  $t$  on the 16-dimensional Majorana space  $H_\Sigma$ , and the Fock space grades by correlator degree, so  $\Gamma(t)$  acts on the degree- $m$  sector as the exterior power  $\Lambda^m(t)$ . Three exact consequences follow (all machine-checked): (i)  $\text{spec}(\Gamma(t)|_{\deg m})$  is the set of products of  $m$  one-particle eigenvalues — the whole correlation spectrum is generated by the finite one-particle spectrum; (ii) **the cone gap equals the one-particle gap:** the sub-leading eigenvalue of  $\Gamma(t)$  over *every* degree is  $(2/3)^6$  (pad the largest contracted mode with fixed factors), so premise 5 on the cumulant space *is* premise 5 on the 16-dim space, the same number; (iii) the eigenvalue-1 sector is  $\Lambda^*$  of the fixed subspace — the disconnected Wick powers of  $K_\Sigma$  — so with a strict one-particle gap no independent connected cumulant is fixed, and an independent fixed cumulant *on* the fixed subspace is excluded by v113 (one polarisation  $\Leftrightarrow$  one vacuum). Premise 1 (quasi-freeness) is itself *forced*, not assumed: the  $120 = \dim \mathfrak{so}(16)$  weight-1 chiral currents *are* the Bogoliubov generators on the 16 Majoranas, and v158 makes every non-Gaussian deformation irrelevant. **So the three full-cone premises collapse to one finite, 16-dimensional statement:** the seam RG transport’s one-particle symbol is the gapped Calderón Bogoliubov map — fixed eigenspace = the rank-8  $K_\Sigma$  polarisation (v113), sub-leading gap =  $(2/3)^6$  (v56). Every *number* in it is already supplied by the suite; the one irreducible step is the physical identification that the actual seam RG generator equals this Bogoliubov map — the same operator-algebraic (Kawahigashi–Longo-class) statement as the net-existence residual of (A<sub>2</sub>), now with the infinite-dimensional cone eliminated [E]/[O].

**And that finite statement carries no *new* open content: it factors into the two already-open items** ([verification/v162\_seam\_transport\_identification.py]). Take the one-particle symbol apart into its two data, the gap value ( $\beta$ ) and the fixed space ( $\alpha$ ). ( $\beta$ ) *is forced:* a  $\mu_4$ -equivariant operator is diagonal in the cusp basis — the deck average is the diagonal projection,  $\sum_{k=0}^3 U^k X U^{-k} = |\mu_4| \text{diag}(X)$  for  $U = \text{diag}(1, i, -i)$ , and the commutant of  $U$  is exactly the diagonal algebra — so its spectrum is the

cusped weights  $\text{spec}(A_0^*) = \{0, \frac{1}{3}, \frac{2}{3}\}$  (the parabolic residue, **v115**), and the transfer eigenvalue  $(1-\alpha)^{p_2}$  with  $p_2=6=|R_+(A_3)|$  gives  $\{1, (2/3)^6, (1/3)^6\}$ , gap  $6 \log \frac{3}{2}$  — every entry carrier data, no free knob; the lift to the 16 Majoranas is the established  $A_3$  subnet grading (**v137**). ( $\alpha$ ) is the rank-8 Calderón polarisation of the  $\omega_k=|k|$  free dispersion (**v113/v156**), the IR conformal free-fermion vacuum. **What is left** is only that the seam RG generator *is*  $\mu_4$ -equivariant and admissible — which is the R1 “seam clock” identification (the transfer operator as the seam’s gapped generator) — together with the  $A_2$  net-existence theorem. Both are pre-existing open items, so premise (A) introduces *no independent open content*: it factors exactly into  $A_2 + R_1$  with every number forced. The “free-bulk cloud” is gone; what remains is the same two identifications the programme already names — (A) is closed *conditionally* on them (a unification, honestly not an unconditional proof) **[E]/[O]**.

**And both residual identifications are pinned to their floor, so (A) has no independent open gate left** (**[verification/v165\_premise\_a\_closure.py]**).  $A_2$  (net existence) is not open research but *assembly* of established mathematics: the 16 free Majoranas define a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic  $c=8$  lattice net *is*  $(E_8)_1$  (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the  $\mu_4$  extension is a Longo Q-system of index 4 (**v125**); the one TFPT-specific input — that the seam Calderón kernel is the free-fermion two-point function — is done because the DtN symbol is universally  $|k|$  (**v156/v157**). So  $A_2$  re-types **[O]**  $\rightarrow$  **[C]** (closed by assembly).  $R_1$  (the seam clock) reduces to GATE.QGEO: a  $\mu_4$ -equivariant flat gapped generator on the carrier is *unique* ( $= A_0^*$ , **v115/v116**), so its gap  $(2/3)^6$  is forced once the seam generator lies in that class, and the residual is exactly one geometric identification — the seam collar boundary is the  $\mu_4$ -punctured sphere  $\mathbb{P}^1 \setminus \mu_4$  — whose cohomological half is already established (**v137**,  $b_1=3=N_{\text{fam}}$ ). That is GATE.QGEO, an already-open realisation premise, *not* a new gate. By the No-Unit Theorem (**v153**) the irreducibles stay exactly  $\{\pi, v_{\text{geo}}\}$  — closing (A) adds none. **Final accounting**: premise (A) = ( $A_2$  assembly, **[C]**) + ( $R_1$ =GATE.QGEO, already open) + the  $\{\pi, v_{\text{geo}}\}$  irreducible core (a theorem) — it carries *zero* independent open gates and ceases to be its own gate **[E]/[O]**.

**And the two structural residuals — net existence and full-cone reflection positivity — are now discharged to a theorem, leaving the seam realisation as the one irreducible open premise** (**[verification/v175\_net\_existence\_full\_cone.py]**). The CAR second-quantisation functor  $\Gamma(t) = \bigoplus_m \Lambda^m(t)$  lifts the one-particle seam contraction  $t$  to the *complete* Fock space  $\Lambda^\bullet(\mathbb{R}^{16})$  ( $\dim 2^{16}=65536$ ); by the exterior-power spectrum theorem every eigenvalue of  $\Gamma(t)$  is a subset product of  $\text{spec}(t)$ , so checking all  $2^{16}$  products shows  $\Gamma(t)$  is a *contraction* (full-cone RP), with eigenvalue-1 multiplicity  $2^8=256$  (the wedges of the rank-8  $K_\Sigma$  polarisation) and sub-leading eigenvalue exactly the one-particle gap  $(2/3)^6$  — so full-cone RP/gap reduce *for every*  $m$  (not only  $m \leq 6$ , **v161**) to the one-particle contraction (Shale–Stinespring), *discharging the v160 scope premise by a theorem*. And  $A_2$  is now an *assembled and verified* certificate: the  $(E_8)_1$  character  $E_4/\eta^8=1+248q+4124q^2+34752q^3+213126q^4+1057504q^5$  (level-1=248), the embedding  $248=120+128$ , and the  $E_8$  Cartan matrix even with  $\det 1$  (the unique rank-8 even unimodular lattice, **v83**) — the net *exists* by cited theorems. **Both residuals then collapse to the same input**: that the actual seam Calderón kernel is this gapped one-particle contraction on  $\mathbb{P}^1 \setminus \mu_4$ , i.e. QGEO.REALIZE.01, whose finite half is proven (**v168**). That realisation half is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly **[O]** — the single irreducible structural premise; net existence and full-cone RP are **[E]**.

**The No-Unit Theorem —  $v_{\text{geo}}$  is an irreducible metrology primitive** **[E]/[O]**  
(**[verification/v153\_no\_unit\_theorem.py]**)

$v_{\text{geo}}$  is not an open gap. Every load-bearing compiler datum —  $g_{\text{car}}, |\mu_4|, N_{\text{fam}}, \text{rank } E_8, \alpha^{-1}$ , the determinant ladder  $(3, 4, 8, 20)$ ,  $c_3 = \frac{1}{8\pi}$ ,  $\varphi_0^{\text{ret}}$  — is mass-dimension 0 and *invariant* under a unit rescaling  $L \mapsto \lambda L$ , whereas a mass / energy /  $1/G$  carries nonzero dimension. A map from dimension-0 inputs to a dimension- $d \neq 0$  output is simultaneously invariant and covariant: impossible unless the unit is introduced. **So a dimensionless boundary compiler provably cannot select an absolute scale** —  $v_{\text{geo}}$  is a *forced input*. The three apparent residuals then collapse onto the one unit, differing only in mass-dimension:

$$U_{\text{point}} \sim v_{\text{geo}}, \quad \frac{1}{G} \sim v_{\text{geo}}^2, \quad m/\mu = e^{3/4} \text{ (a pure ratio),}$$

so the irreducibles stay exactly {one dimensionful unit,  $\pi$ }. *Re-typing*:  $v_{\text{geo}}$  is an *irreducible metrology primitive* (a declared unit, like choosing a metre), not a missing derivation.

**Sharpened across gravity, and the final tally** (**[verification/v364\_vgeo\_sharpen.py]**). After the parameter-free gravity (**v358/v359/v361**) the *gravity* sector adds no new scale either: the Einstein

coefficient  $1/c_3 = 8\pi$  and the  $\Lambda$  prefactor  $(8\pi)^2 \cdot 48c_3^4 = 3/(4\pi^2)$  are dimensionless, so  $1/G \sim v_{\text{geo}}^2$  and  $\Lambda \sim (\text{dimensionless}) v_{\text{geo}}^4$  both reduce to  $v_{\text{geo}}$  times atoms. Hence  $v_{\text{geo}}$  is the *single* dimensionful input of the *complete* theory (matter and gravity), and the final tally is  $\{v_{\text{geo}}, \pi\}$  with *zero* dimensionless dials — a  $\sim 26 \rightarrow 1$  reduction against the Standard Model. ( $1/G$  stays UV-sensitive (Sakharov, v68), so it is a metrology readout, not a clean prediction — consistent with  $v_{\text{geo}}$  being the anchor.)

#### Theorem G — the closure statement [O] (target)

The relative spectral boundary kernel over the diffeomorphism-quotiented metric sector admits a regulator-independent, reflection-positive projective-limit measure  $\mu_{\text{QG}}$ . Its Ward identities imply covariant conservation, and its low-curvature expansion is the TFPT  $R + R^2$  seam action

$$f_R R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} + (g_{\mu\nu} \square - \nabla_\mu \nabla_\nu) f_R = \bar{M}_{\text{Pl}}^{-2} T_{\mu\nu} + O(R^3/M^4),$$

so the closed covariant equation becomes the controlled IR shadow of the full measure. *Status: the local covariant field equation is parameter-free (v358/v359); Theorem G remains open only for the projective-limit ambient measure  $\mu_{\text{QG}}$  (G6, the deepest item). G5 certified; the regime equation closed [verification/v28\_gravity\_fR.py].*

## Research Contract — CELEST.SEAM.01 (the celestial and twistor continuum route)

*Reader's note.* A compact synthesis of this contract's executed work packages (WP1–WP5d, both WP5d stages, + the WP5e  $\alpha$ ,  $\beta$ ,  $\gamma$ ,  $\delta_2$ ,  $\varepsilon_2$  and  $\varepsilon_1$  stages + the three back-reaction milestones M1–M3) — the narrative arc from the  $\mu_4$  clock to the  $(E_8)_1$  boundary shadow — is given as a dedicated section in `tfpt_3_e8_audit_bootstrap` (“The celestial and twistor continuum route”); this contract remains the full technical reference (typing fence, kill tests, work-package statements). The bridge from this contract's compiler-side constructions to reconstructed Minkowski physics is *not* part of this contract: it is the separate central contract WOIT.OS.TWISTOR.01 below — everything in CELEST.SEAM.01 is preparation for it.

**The object diagram (read this first).** The route is one chain of objects; each arrow is a separately typed claim. Where mathematics ends and physics begins is visible at a glance: the last two arrows are [O].

| Arrow                                                                        | Status  | Claim / modules                                                                                                                                                                                                                                                                                                                            | Kill test                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $(\Sigma, \mu_4, \rho, \Theta) \rightarrow \mathbb{P}^1 \setminus \mu_4$     | [E]     | SEAM.MARKS chain: four marks, clock, cross-ratio 2, $\tau=i$ pillowcase (v168/v214/v216/v180; bit reduction v506/v507/v510/v512)                                                                                                                                                                                                           | a fifth mark, a non-order-4 clock, or a marks-preserving root without flag transitivity                                                                                                                                                                                                                                                                                                                               |
| $\mathbb{P}^1 \setminus \mu_4 \rightarrow \mathbb{C}^2 / \mathbb{Z}_4$       | [E]     | CELEST.WP1.01/CELEST.WP2.01: the glue is the flat $\mathbb{Z}_4$ monodromy on the $A_3$ ALE, clock = Kähler $U(2)$ phase, clock <sup>2</sup> = deck (v492/v493)                                                                                                                                                                            | equivariant-sector closure failure (3112/6720 control) or a surviving shape modulus                                                                                                                                                                                                                                                                                                                                   |
| $\mathbb{C}^2 / \mathbb{Z}_4 \rightarrow \mathbb{P}\mathbb{T} / \Gamma$      | [E]/[C] | CELEST.WP5E.*: equivariant twistor uplift — anomaly ledger, level-from-flux, axion slot, back-reacted $\Omega_N$ , twisted KS measure, $a_0$ uplift, the $\delta_1$ decision, the measure decision (v505/v509/v511/v513/v514/v515/v516/v517/v518/v520)                                                                                     | the preregistered level kill (“inflow demands $k \neq 1$ ”); the M1–M3 kills were evaluated by v515/v516/v517 and did not fire; the $\delta_1$ kill <i>fired on the derived measure</i> (v518), and the declared/derived measure question is since <i>decided at probe level</i> in favour of the declared reading under the typed premises TP-1..TP-4 (v520) — the residual [O] is the constructive $w_m$ derivation |
| $\mathbb{P}\mathbb{T} / \Gamma \rightarrow \mathcal{A}_{\text{hol}}$         | [O]     | the <i>interacting</i> holomorphic algebra (global BCOV+SDYM quantisation; WP5e proper; target of WOIT.OS.TWISTOR.01; CFT-side anchors v497–v504)                                                                                                                                                                                          | anomaly mismatch at the Costello–Li grade                                                                                                                                                                                                                                                                                                                                                                             |
| $\mathcal{A}_{\text{hol}} \xrightarrow{\text{OS}} \mathcal{A}_{\text{Mink}}$ | [O]     | OS reconstruction with the real structure $\Theta$ — the WOIT.OS.TWISTOR.01 target (the free-collar RP/OS witnesses v379/FORM.SEAM.MMST.01 are the <i>free</i> anchor only; the $\alpha$ stage v519 pins $\Theta$ itself and free RP on the seam system — kill test (1) discharged at the free level, live on $\mathcal{A}_{\text{hol}}$ ) | the seven kill tests of WOIT.OS.TWISTOR.01 below                                                                                                                                                                                                                                                                                                                                                                      |

**Definition (the group extension, exactly).** The shorthand “ $\mathbb{P}\mathbb{T} / \mathbb{Z}_4$ ” used in this contract is precise but deserves unpacking, because *two different* order-4 structures are in play and only one of them is quotiented:

- $\Gamma_{\text{ALE}} = \langle \text{Deck} \rangle \cong \mathbb{Z}_4 \subset SU(2)$ , the deck group  $(z_1, z_2) \mapsto (iz_1, i^{-1}z_2)$ : *triholomorphic*, preserves the holomorphic symplectic form  $\Omega$ ; this is the  $\mathbb{Z}_4$  that is *quotiented* ( $\mathbb{C}^2 / \Gamma_{\text{ALE}}$  is the  $A_3$  singularity  $XY = Z^4$ , and  $\mathbb{P}\mathbb{T} / \Gamma$  means the quotient by  $\Gamma_{\text{ALE}}$  acting on the twistor space of the ALE) [E] ([verification/v492\_celestial\_z4\_orbifold.py]).
- $\rho = \text{Clock} = \text{diag}(i, 1) \in U(2)$ : a Kähler, *not* triholomorphic isometry ( $\det \rho = i$  rotates  $\Omega$  by the  $\mu_4$  generator — the WP1 S5 computation); it is *not* in  $\Gamma_{\text{ALE}}$  and is *not* quotiented — it *normalises*  $\Gamma_{\text{ALE}}$  and survives the quotient as the residual clock symmetry [E] (v492).
- The global object is the extension generated by both: since  $\rho^2 = \text{Deck}$  exactly (v492/v506),  $\langle \Gamma_{\text{ALE}}, \rho \rangle = \langle \rho \rangle$  is *cyclic* — projectively  $\mathbb{Z}_4$  (the clock on the sphere) with  $\rho^2$  generating the deck  $\mathbb{Z}_2$  there, and on the spin/fermionic level the canonical  $\mathbb{Z}_8$  tower ( $V^2 \propto U$ ,  $V^4 \propto (-1)^F$ , nonsplit;  $8 = 2|\mu_4|$ ) [E] (v506/v507).

The action table, entry by entry ([E] = arithmetically established at the current stratum, [O] = a quantisation question):

| Object                              | $\Gamma_{\text{ALE}}$ (Deck) | $\rho$ (Clock)                  | Status / witness                                                                                                 |
|-------------------------------------|------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------|
| $K_{\mathbb{PT}}$ (canonical class) | trivial                      | weight via $\det \rho = i$      | [E] (incidence-forced column weights $(+1, +1, -1, -1)$ , v514)                                                  |
| $\Omega$ (hol. symplectic)          | preserved                    | rotated by $i$                  | [E] (v492 S5; back-reacted $\Omega_N$ closed-form with $(2\pi i)^2$ -integral periods and forced charge 4, v515) |
| BCOV fields ( $O(-2)$ tower)        | orbifold sectors             | character series $P_0..P_3$     | [E] ledger / [O] quantisation (v514)                                                                             |
| open-string fields (SDYM $E_8$ )    | glue-equivariant sector      | sector rotation $j \mapsto j+1$ | [E] ( $D[d] = 60(d+1)/64(d+1)$ , v492) / [O] interacting                                                         |
| boundary states $((E_8)_1$ shadow)  | $\mu_4$ sector split         | clock phase on $ s\rangle$ etc. | [E] character/GNS level (v497–v500) / [O] as an actual net                                                       |

**The  $A_3$  role table (no silent identifications).** Several rank-3  $A_3$ ’s and external gauge factors circulate near this contract; the following table states, for each pair, either the connecting functor (with its witness) or an explicit *not identical*:

| Object                        | Role and relations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $A_3^{\text{family}}$         | the family factor in the lattice $D_5 \oplus A_3$ ( $\mathfrak{su}(4)$ -flavour; three families from its exponents, v128/v137). Related to $A_3^{\text{ALE}}$ by the McKay correspondence <i>of types</i> only (both are type $A_3$ ); <b>not identical</b> as realisations — one is a sublattice of $E_8$ , the other the singularity type of $\mathbb{C}^2/\mathbb{Z}_4$ . The bridge that <i>does</i> connect them is the Kronheimer–Nakajima quiver (v479: same ADE datum, different functors). |
| $A_3^{\text{ALE}}$            | the ADE type of $\mathbb{C}^2/\mathbb{Z}_4$ ( $XY = Z^4$ ); its resolution carries the three exceptional spheres (v492/v493).                                                                                                                                                                                                                                                                                                                                                                       |
| three exceptional spheres     | the Coxeter/Picard–Lefschetz structure of the resolved $A_3^{\text{ALE}}$ : the clock acts as a Coxeter element of $W(A_3)$ on $H_2$ (eigenvalues $\{i, -1, -i\}$ , v493); they carry the three twisted axion slots (v505) and the lockstep fluxes (v509).                                                                                                                                                                                                                                          |
| $D_5^{\text{carrier}}$        | the carrier factor ( $g_{\text{car}} = 5$ ); glued to $A_3^{\text{family}}$ by the $\mu_4$ Lagrangian glue into $E_8$ (v92/v125); <b>no</b> role on the ALE side beyond the zero-mode algebra $\mathfrak{g}_0 = D_5 \oplus A_3$ (v492).                                                                                                                                                                                                                                                             |
| Woit’s $SU(3)_{\text{color}}$ | <i>external programme reference only</i> (Euclidean Twistor Unification: colour as the rank-three quotient bundle on $\mathbb{PT}$ ). <b>Not identical</b> — <b>no identification claimed</b> ; in particular $A_3^{\text{family}} \not\cong SU(3)_{\text{color}}$ (an $\mathfrak{su}(4)$ flavour factor is not a colour gauge bundle), and TFPT’s colour lives in the SM readout of the carrier, not on $\mathbb{PT}$ .                                                                            |
| Woit’s $SU(2)_{\text{weak}}$  | <i>external programme reference only</i> (the internal spin factor of the Euclidean twistor construction). <b>Not identical</b> — <b>no identification claimed</b> ; whether an internal $SU(2)$ of that kind can be realised on $\mathbb{PT}/\Gamma$ is exactly kill test (6) of WOIT.OS.TWISTOR.01.                                                                                                                                                                                               |

**Hypothesis (as corrected by WP1).** *The seam is the glue-equivariant  $\mathbb{Z}_4$ -orbifold sector of the celestial chiral algebra on the  $A_3$  ALE space  $\mathbb{C}^2/\mathbb{Z}_4$ ; the  $\mu_4$  clock is the Kähler  $U(2)$  phase  $\text{diag}(i, 1)$  whose square is the deck group (spin bridge  $\mathbb{Z}_8$ ,  $8 = 2|\mu_4|$  — the  $c_3 = 1/(8\pi)$  seam winding integer); and the associated twistorial bulk is the self-dual sector of TFPT gravity.* This is the fourth research contract, alongside  $(U_{\text{wall}})$ ,  $(G_{\text{metric}})$  and CONTRACT.F.01: a numbered chain of work packages with pre-registered kill tests, *not* a claim. The method is the celestial-chiral-algebra (CCA) construction of Bittleston–Homans–Sharma on Eguchi–Hanson space (arXiv:2305.09451, JHEP 09 (2023) 008 — the  $\mathbb{Z}_2$  case, refined here to  $\mathbb{Z}_4$  with a flat connection at infinity in the Kronheimer–Nakajima / heterotic-on-ALE sense), inside the celestial-holography programme (Strominger, PRL **127**, 221601:  $w_{1+\infty}$ ; Costello–Paquette–Sharma, PRL **130**, 061602: Burns holography, the  $SO(8)$  WZW<sub>4</sub> model; Bittleston–Sharma–Skinner, CMP **403** (2023): quantizing the non-linear graviton). The gauge-algebra admissibility gate is already passed in the literature: Costello’s one-loop axionic-consistency analysis (arXiv:2111.08879) admits the exceptional algebras *including*  $E_8$  for SDYM on twistor space (Okubo: no independent quartic Casimir). The spin bridge in the hypothesis is no longer a bookkeeping convention: on the 16-Majorana seam the NS deck implementation is *forced* to order 4 ( $U^2 = (-1)^F$  exactly, nonsplit  $\mathbb{Z}_4$ ) and the quarter-shift lifts to a canonical  $\mathbb{Z}_8$  tower with  $V^2 \propto U$  — “ $8 = 2|\mu_4|$ ” fermionically explained ([verification/v506\_seam\_clock\_rigidity.py],

SEAM.CLOCK.RIGIDITY.01); and the nonsplit class is arrangement-*sensitive*: the edge (silver) arrangement splits ( $U_e^2 = +1$ , zero roots), so the fermions *measure* the alignment bit as a Fidkowski–Kitaev-type extension class ([`verification/v507_seam_bit_origin.py`], SEAM.BIT.ORIGIN.01) — and that class is nonsplit *iff* the deck acts freely (all 17 seam-circle involutions, v510), which the covering deck does by topology.

**WP1, executed and verified: the  $E_8$   $\mu_4$ -glue is the flat  $\mathbb{Z}_4$  monodromy of the equivariant CCA on the  $A_3$  ALE space — verdict B** ([`verification/v492_celestial_z4_orbifold.py`], CELEST.WP1.01)

Everything in this box is exact arithmetic (sympy, no floats). **Inner grading / flat monodromy [E]:** the v128 glue grading ( $240 = 52+64+60+64$ , additive on all 6720 root pairs, dims with Cartan  $(60, 64, 60, 64)$ ,  $\mathfrak{g}_0 = D_5 \oplus A_3$  carrier = 60) is *inner*:  $h = (2, 2, 2, 2; 0^4)$  reads the glue class mod 4 on all 240 roots, so the glue is  $\text{Ad}(\exp(2\pi i h/4))$  — an order-4 torus element, i.e. a *flat  $\mathbb{Z}_4$  monodromy* in the Kronheimer–Nakajima sense; the  $A_3$ -side detector  $h' = (0^5; 1, 1, 1, -3)$  reads the *same* class, so the glue charge is the diagonal  $(1, 1)$  of the discriminant group  $\mathbb{Z}_4 \times \mathbb{Z}_4$  — the v92/v125 Lagrangian glue. **The spatial side [E]:**  $\mathbb{C}^2/\mathbb{Z}_4$  under the deck  $(z_1, z_2) \mapsto (iz_1, i^{-1}z_2)$  is the  $A_3$  singularity  $XY = Z^4$  (invariant ring exact; Hilbert series  $(1+t^4)/((1-t^2)(1-t^4))$ ; Molien table verified). **The equivariant sector closes [E]:** the glue-equivariant  $\text{SDYM}(E_8)$  modes ( $j + m \equiv 0 \pmod{4}$ ) form a Lie subalgebra with graded dimensions  $D[d] = 60(d+1)$  (even  $d$ ) /  $64(d+1)$  (odd  $d$ ) — uniform per parity *only* because  $\dim \mathfrak{g}_0 = \dim \mathfrak{g}_2 = 60$  and  $\dim \mathfrak{g}_1 = \dim \mathfrak{g}_3 = 64$ : the v128 dimensions are exactly the consistency condition. The zero modes ( $d=0$ ) are  $\mathfrak{g}_0$  alone — *the unbroken algebra on the orbifold is the carrier  $D_5 \oplus A_3 + \text{Cartan} = 60$* ; the asymptotic density is  $62/248 = 1/4 = 1/|\mathbb{Z}_4|$  (the orbifold index), and the  $\mathbb{Z}_2$  halfway sector reproduces the Eguchi–Hanson towers  $120/128 = SO(16)$  adjoint/spinor —  $\mu_4$  refines the sheet  $\mathbb{Z}_2$  exactly as  $4 = 2 \times 2$  (v125). **Character level [E]:**  $\Theta_{E_8}$  splits into the four glue-coset thetas (shell 1 =  $(52, 64, 60, 64)$  re-derived from the *lattice*); the four sector characters  $\Theta_{C_j}/\eta^8$  sum *exactly* to  $\chi_{(E_8)_1} = 1+248q+4124q^2+34752q^3$  (the v377 series); the sector conformal weights are  $h_{D_5} = (0, \frac{5}{8}, \frac{1}{2}, \frac{5}{8})$ ,  $h_{A_3} = (0, \frac{3}{8}, \frac{1}{2}, \frac{3}{8})$  — the v92/v125 discriminant form  $(5x^2+3y^2)/8 \pmod{1}$  — and the glue diagonal has *integer*  $h = (0, 1, 1, 1)$ : the v125 isotropy is the locality of the extension to  $(E_8)_1$ . **The critical correction [E] (why verdict B, not A):** the  $A_3$  deck  $\text{diag}(i, i^{-1})$  induces on the celestial sphere  $z = z_1/z_2$  only  $z \mapsto -z$  (projective order 2, the sheet flip) — it is *not* the order-4 clock  $z \mapsto iz$ . The exact bridge: any  $SU(2)$  lift of the clock has order 8 (spin group  $\mathbb{Z}_8$ , which would quotient to  $A_7$ ), and (spin clock) $^2$  = the deck generator *exactly*, with  $|\mathbb{Z}_8| = 8 = 2|\mu_4|$  — the  $c_3 = 1/(8\pi)$  winding integer. The order-4 clock is realised, as the  $U(2)$  phase  $\text{diag}(i, 1)$ : projective order 4, fixed points  $\{0, \infty\}$ , free  $\mu_4$  orbit (all v180 clock properties), normalising the deck (so preserving the glue flat connection), with  $\det = i$  rotating the holomorphic symplectic form by the  $\mu_4$  generator (a Kähler, *not* triholomorphic isometry — it rotates the twistor sphere by  $\pi/2$ ). **Gravity side [E]:** clock-invariance selects from the 3-parameter versal  $A_3$  family  $XY = Z^4 + a_2 Z^2 + a_1 Z + a_0$  *exactly* the 1-parameter family  $XY = Z^4 + a_0$ , whose four branch points are *one*  $\mu_4$  orbit with cross-ratio 2 — the seam marks (v179 clock orbit; v168/v214 cross-ratio 2  $\Rightarrow j=1728$  pillowcase). **Negative controls [E]:** the false spatial action  $\text{diag}(i, i)$  is killed three ways ( $\det = -1 \notin SL(2, \mathbb{C})$ , no CY/ALE resolution; Poisson closure fails in 14/14 nonzero cases; conjugation symmetry broken), the false glue (60/64 blocks swapped) breaks additivity on 3112/6720 root pairs, and of all 24 relabelings exactly  $\{\text{id}, j \mapsto -j\} = \text{Aut}(\mathbb{Z}_4)$  are additive — the grading is rigid. **Verdict [C]:** the celestial and glue  $\mathbb{Z}_4$  structures match under *one* precisely named extra identification — clock $^2$  = deck on the sphere (the sheet- $\mathbb{Z}_2$  spin bookkeeping) — so WP1 lands at verdict B (match modulo one named identification), close to but honestly short of A. [E]/[C]

**Typing and non-circularity.** Admissible *inputs* are the finite seam/glue data already in the suite — the  $\mu_4$  Q-system glue (v92/v125), the graded hull (v128), the seam rigidity and pillowcase geometry (v168/v214), the Möbius clock (v180) — plus the WP1 [E] arithmetic ([`verification/v492_celestial_z4_orbifold.py`]). *Not* admissible as an input: the *continuum existence* of the holomorphic  $(E_8)_1$  net on the seam — that is precisely the SEAM.EQUIV.01 target, and assuming it would make the contract circular. *Scope fence:* the twistorial bulk of this contract is the *self-dual* sector only, never full Einstein gravity — the classical field-equation route (entanglement equilibrium, v358/v359) and the entropic-action bridge (GRAV.ENTROPIC.ACTION.01, v473) are separate, non-competing statements about the full equation. SEAM.EQUIV.01 stays [O] throughout; nothing in this contract moves it.

## Work packages.

- **WP1 (*done*, verdict B) [E]/[C]:** the structural  $\mathbb{Z}_4$  arithmetic of the keybox above, executed and machine-verified as CELEST.WP1.01 [verification/v492\_celestial\_z4\_orbifold.py].
- **WP2 (*done*, verdict B) [E]/[C]:** the clock-invariant deformation theory, executed and machine-verified as CELEST.WP2.01 [verification/v493\_celestial\_wp2\_clock\_deformation.py] (sympy exact, 47 checks). Both targets landed, *sharpened*: (i)  $a_0$  is the pure *scale* of the seam configuration — the binary-quartic invariants of  $w^2 = Z^4 + a_0$  are  $I = 12a_0$ ,  $J = 0$  *identically*, so  $j = 1728$  and the  $\tau=i$  pillowcase shape (v168/v214) is *frozen* for every  $a_0 \neq 0$  (no shape modulus survives clock-invariance; the negative controls  $a_1 Z \Rightarrow j=0$  and  $a_2 \neq 0 \Rightarrow j=1556068/81$  show the test has teeth); (ii) the three resolved spheres ( $3 = \text{rank } A_3 = N_{\text{fam}}$ ) carry *exactly* the three nontrivial  $\mu_4$  characters  $\{i, -1, -i\}$  under the clock, which acts on  $H_2$  as a *Coxeter element* of  $W(A_3)$  (char  $x^3 + x^2 + x + 1$ , order  $h(A_3)=4=|\mu_4|=N_{\text{fam}}+1$ ) and *is* the Picard–Lefschetz monodromy of the family ( $a_0 \rightarrow e^{2\pi i} a_0$  gives the 4-cycle); the surviving deformation direction is the weight-1  $\chi_1$ -Fourier diagonal  $(1, i, i^2)$  — *not* the naive invariant diagonal ( $\det(A-1) = -4$ : no invariant cycle) — and all three sphere volumes move in lockstep ( $|\Pi_j| = \sqrt{2}t$ ). The BHS deformed-algebra pattern transfers from  $\mathbb{Z}_2$  to  $\mathbb{Z}_4$ : the fibre bracket  $-4\text{Nambu}(XY - Z^4 - a_0)$ , anchored at the  $a_0=0$  orbifold, closes with corrections exactly *linear* in  $a_0$  at the  $\mathbb{Z}_4$  wrap, conserving the  $\mu_4$  grade (no sector leak — the WP1 equivariant sector deforms consistently), with  $a_0$  in the weight-0 slot of BHS’s  $c^2$ . Verdict B via three named [C] identifications ( $-4\text{Nambu}$  as *the*  $k=4$  CCA bracket; period = root difference; the seam-scale reading via  $\text{clock}^2 = \text{deck}$ ); the full deformed twisted-sector OPEs and the  $W(\mu)$ -type scaling limit for  $k=4$  stay [O]. The v216 [O]-residual is *typed*, not moved: given the order-4 clock, the deformation freezes the  $(2, 2, 2, 2)$  modulus to the square automatically — a relocation of the same order-4 carrier input, not a new derivation (since v510 that input reads: the square-modulus datum  $\tau = i$ ; its position half is topology; since v512 it reads equivalently: one discrete symmetry-lift bit, flag transitivity  $\iff \tau = i$ ; and since v521/v525/v529 free OS positivity, the whole mark-decorated free-plus-twist state class and the first genuinely interacting FK toy are documented as the *eighth*, *ninth* and *tenth* side-blind tests — the bit is not derivable from free  $\text{RP}/\Theta$  existence, from any Wick-computable twist decoration or from the FK interaction class, and remains genuine discrete input; since v528 it is the twist-class choice, in principle interferometrically readable; and since v534 it carries the first *positive* selection datum — reflection positivity on the interacting mark family keeps exactly the  $\delta = \pi/2$  member alive, toy-fenced).
- **WP3 (*done*, verdict B) [E]/[C]:** the Green–Schwarz alignment test, executed and machine-verified as CELEST.WP3.01 [verification/v495\_celestial\_wp3\_gs\_alignment.py] (exact Fraction/sympy, 25 checks). The arithmetic side is sharp: the Okubo coefficient  $\text{Tr}_{\text{adj}} X^4 / (\text{Tr}_{\text{adj}} X^2)^2 = 5/(2(\dim \mathfrak{g}+2))$  is *derived* as a polynomial identity for all 8 algebras on Costello’s list (with a sharp  $\mathfrak{sl}_4$  negative control:  $5/32$  vs  $3/32$ ), the closed form  $\tilde{\lambda}^2 = 10h^\vee/(\dim+2) = h^\vee+6$  holds exactly across the Deligne series, and for  $E_8$  the axion coefficient is *exactly*  $\tilde{\lambda} = 6$  (unit-trace) resp.  $\lambda_{\text{fund}} = 1/10$  (adjoint-trace), so the GS coupling  $\kappa = \lambda_{\mathfrak{g}}/(8\pi\sqrt{3})$  satisfies  $(\kappa/c_3)^2 = 12 = |\mu_4|N_{\text{fam}}$  resp.  $1/300$  — exact anchor rationals, with the anchor decomposition  $h^\vee=30=|\mathbb{Z}_2|N_{\text{fam}}g_{\text{car}}$ ,  $\varphi(30)=8=\text{rank}$ ,  $\tilde{\lambda}=|\mathbb{Z}_2|N_{\text{fam}}$ ,  $\lambda_{\text{fund}}=1/(|\mathbb{Z}_2|g_{\text{car}})$  (a byproduct: the printed  $\lambda_{\mathfrak{so}_8}^2 = 3/2$  in Costello’s appendix A is a factor-2 slip; his own weight content gives 3, Okubo-consistent).  $\kappa/c_3$  itself is *irrational* ( $2\sqrt{3}$ ): only the squared alignment is exact. **The look-elsewhere caveat is part of the result [C]:** the same squared-rational alignment holds for *all eight* algebras on the list ( $8/8$  — the test as such has zero selective power),  $\tilde{\lambda}$ -integrality passes  $2/8$  (shared with  $\mathfrak{sl}_3$ ), and the only  $\mathfrak{e}_8$ -selective single test is  $g_{\text{car}}=5 \mid h^\vee$  ( $1/8$ ); the conjunction that isolates  $\mathfrak{e}_8$  is post hoc, not preregistered. *Alignment survives; selectivity does not* — the  $c_3$ -connection of the celestial route is convention-level compatibility plus genuine  $\lambda$ -arithmetic, *not*  $\mathfrak{e}_8$ -selective evidence,

and never a derivation of  $c_3$  from celestial data.

- WP4 (*done*, verdict B(ii)) [E]/[C]:** the type mismatch, executed head-on and machine-verified as CELEST.WP4.01 [verification/v496\_celestial\_wp4\_character\_shadow.py] (exact integer/Fraction/sympy, 25 checks). The  $(E_8)_1$  character is *not* a conformal block of the celestial  $E_8[\mathbb{C}^2]$  S-algebra in its own (jet) grading — the obstruction is localised *three* independent ways: (a) the CP grading gives spin  $1-d/2$ , unbounded below, and 248 is never a jet-tower dimension  $(60(d+1)/64(d+1), d \leq 100)$ ; (b) the cumulative generator count is quadratic  $(31s^2+92s+60, 31 = k+h^\vee)$ , so the jet Fock grows as  $n^{2/3}$  against the character's  $n^{1/2}$  (Meinardus poles  $s=2$  vs  $s=1$ ; the ratio  $f_n/\chi_n$  increases strictly for  $n = 1..12$ ); (c) the level-2 null ideal of  $(E_8)_1$  deletes *exactly*  $27000 = 30^3 = (h^\vee)^3$  out of  $\text{Sym}^2(248) = 1+3875+27000$  (the character keeps  $4124 = 1+248+3875$ ) — and has no jet analogue. **But the boundary/period reading holds exactly at the current stratum:** the zero-mode slice is the 60 vacuum-sector currents, the single-line jet slice cycles the four sector counts  $(60, 64, 60, 64)$ , one full  $\mu_4$  period of loop energies sums exactly to 248 (the single-particle level of the  $248q$  stage), and the glue-diagonal weights  $(0, 1, 1, 1)$  are integers — while the free loop Fock at level 1 counts  $897266 \gg 248$ , i.e. the rational truncation must be *imposed in a limit*. That is precisely the MMST scaling-limit shape of SEAM.EQUIV.01 (v336/v449: the rational net exists in the *limit* of the lattice collar, not in the pre-limit algebra): WP4 is hereby *retyped* from “character as conformal block” to “character as boundary/limit shadow”, and the constructive limit question (which limit, which topology, convergence) passes to WP5 — where WP5a has since answered the “which limit” part at the counting level (the coefficientwise  $\chi_w$  limit below), WP5b has constructed the deleting operator, WP5c the limit state whose GNS kernel contains the ideal, WP5d the local-net legs (the  $\alpha$ -stage two-interval index chain and the  $\beta$ -stage split + strong-additivity witnesses — all three KLM ingredients of complete rationality on the lattice), and WP5e- $\alpha/\beta/\gamma/\delta_2/\varepsilon_2/\varepsilon_1$  have anchored *both* faces of the twistor uplift, closed its exchange sub-branch, executed the full-tensor ledger, executed the level-from-flux dial and built the bulk-axion slot (the CFT-side prefactor + level pinning; the equivariant twistor skeleton with the index bridge and the rigid  $32 T_3$  residual; the exchange no-go with certificate; the full-tensor confirmation of the collapse with the one cubic  $d$ -symbol channel that opens; the lockstep theorem making “one level” a theorem of clock invariance, with the sector-counter dial pinning  $k = 1$ ; the equivariant  $O(-2)$  slot construction with  $\tilde{\lambda} = 6$  triply pinned and the first GH/ $A_3$  back-reaction step), and all three quantised back-reaction milestones M1–M3 have since been executed (the closed-form  $\Omega_N$  with  $(2\pi i)^2$ -integral lockstep periods and forced source charge  $4 = |\mu_4|$ , v515; the twisted KS measure on the declared completion reading — per-sector Okubo squares, the  $32 T_3$  cancelled, the  $\psi = 64$  slice supplied without the cubic  $d$ -channel, v516; the  $a_0$  uplift coupled to the centre count on four scales, v517), leaving the global BCOV quantisation (WP5e proper) as the remaining open piece — the BCOV *measure question* is since decided at probe level: the  $\delta_1$  chain itself was decided by v518 (kill under the derived measure, all three testers fail), and v520 has since decided *which* of the two exact measures the BCOV integral produces — single-valuedness of the one-loop lattice factor is *derived* from two independent constructive sources (F-independence of the moduli integral + the Quillen pairing) under the typed premises TP-1..TP-4, forcing the declared v516 completion reading, and v523 has since *derived* the  $w_m$  normalisation itself at probe level ( $1/\det_j$  = the Atiyah–Bott/zeta-determinant fixed-point factor, computed from three independent sources under the typed premises TP-REG/TP-Q/TP-NUM/TP-CH); the residual [O] narrows to the global BCOV integral beyond the fibre zero-mode factor (throughout,  $\mathbb{P}^T/\mathbb{Z}_4$  is the quotient by  $\Gamma_{\text{ALE}}$ , normalised by the clock  $\langle \rho \rangle$  — the definition block above; the  $\varepsilon_1$  slot itself is executed; the quantised milestones M1–M3 preregistered in v514 are all executed via v515/v516/v517 — the v514 fence is fully worked off).
- WP5 (the summit) — subdivided; WP5a–WP5d (both WP5d stages) plus the WP5e  $\alpha, \beta, \gamma, \delta_1, \delta_2, \varepsilon_2$  and  $\varepsilon_1$  stages and the back-reaction milestones M1–M3**

**are landed.** The twistorial construction as a *second continuum route* for SEAM.EQUIV.01 (Costello–Li grade), carrying the WP4 constructive-boundary-limit question. Subdivision WP5a–WP5e:

- **WP5a (done)** [E]/[C]: “the null ideal from the limit”, executed and machine-verified as CELEST.WP5A.01 [verification/v497\_celestial\_wp5a\_null\_ideal.py] (exact integer/Fraction arithmetic, 34 checks). Three results make the WP4 limit reading precise. (i) *Stabilisation theorem*: the one-parameter family  $\chi_w$  of graded Fock characters on the chiral jet generators (loop energy  $m$ , radial number  $r$ ;  $E_w = m + wr$  in quarter units  $u = q^{1/4}$ ) contains the chiral jet grading as its  $w=2$  member (generator counts 64, 120, 128, 180, 192, 240, 256, 300 per degree), and its  $u^n$  coefficient equals the quarter-moded loop Fock for *all*  $w \geq n+1$  — strictly larger for every  $w \leq n$  (verified  $n \leq 8$ ,  $w \leq 10$ ): the WP4 “boundary-limit shadow” is a precise *coefficientwise* limit  $\varepsilon = 1/w \rightarrow 0$  with explicit stabilisation threshold  $w = n+1$ , not a slice. (ii) *The null ideal is derived, not cited*: Freudenthal multiplicities + the Weyl dimension formula + generic character peeling give  $\text{Sym}^2(248) = V(2\theta) \oplus V(\omega) \oplus V(0) = 27000 + 3875 + 1$  with peeling residual exactly zero (independent Weyl-orbit cross-check);  $27000 = 30^3 = (h^\vee)^3$  exactly; level-1 integrability ( $\langle 2\theta, \theta^\vee \rangle = 4$ ,  $\langle \omega, \theta^\vee \rangle = 2$ ) leaves the vacuum as the only level-1 primary ([C] Kac, used only through its counting consequence); level-2 quotient  $31124 - 27000 = 4124 = 1 + 248 + 3875$  exactly. (iii) *Two routes, one number*: the  $\mu_4$  theta-split sector sum at  $q^2 = (1036, 1024, 1040, 1024)$  gives the *same* 4124 — the untwisted null-ideal quotient and the boundary/twisted lattice presentation count one space ( $q^3$ : 34752). Negative controls:  $SO(16)_1$  through the same pipeline gives *four* components ( $5304 + 1820 + 135 + 1$ ),  $5304 \neq 14^3$  (the  $h^\vee^3$  identity is not generic), quotient  $2076 = \Theta_{D_8}/\eta^8$  at  $q^2$  (the recipe validated on a second algebra), and block weights  $(0, \frac{1}{2}, 1, 1)$  that cannot fuse into one local character; only  $P = 4 = |\mu_4|$  periodisation reproduces the 248 layer; swapped glue breaks the zero-mode anchor. *Honest limit* [O]: the limit does *not* generate the truncation (the loop Fock counts  $897266 \gg 248$  at integer level 1) — WP5a fixes the ideal’s *size and location* quantitatively and gives the celestial route the same two-step shape as MMST (limit + maximal ideal); the operator content is exactly WP5b–e, of which WP5b–WP5d and the WP5e  $\alpha$ ,  $\beta$ ,  $\gamma$ ,  $\delta_2$  and  $\varepsilon_2$  stages are executed below.
- **WP5b (done)** [E]/[C]: “the singular vector as an operator”, executed and machine-verified as CELEST.WP5B.01 [verification/v498\_celestial\_wp5b\_singular\_vector.py] (exact integer/Fraction arithmetic, 53 checks, deterministic) — *success on the preregistered criterion*: the deleting object exists and is explicit.  $|s\rangle = (E_{-1}^\theta)^2|0\rangle$  (weight  $2\theta$ , level 2 = 8 quarter units =  $q^2$ , an *integer* level) is constructed in a machine-built Chevalley/Frenkel–Kac basis (cocycle asymmetry on all 57600 pairs; the  $[e_\alpha, e_{-\alpha}]$  sign *forced* by Jacobi,  $\text{SGN} = -1$ ;  $\kappa$  derived,  $\kappa(\theta^\vee, \theta^\vee) = 2$ ), and  $J_1^a|s\rangle = 0$  is machine-verified for *all* 248 generators with the case classification 190 (first bracket) / 57 (second) / 1 ( $a = F^\theta$ , the central-term cancellation — the *only* case that sees the level  $k$ ); plus  $J_2^a|s\rangle = 0$ ,  $E_0^a|s\rangle = 0$ , exact weight and Shapovalov norm 0. *Level dial*: the  $F_1^\theta$  coefficient is  $2(1-k)$  — nonzero at  $k=0$  and  $k=2$ , zero exactly at  $k=1$ : without the central extension the deletion operator does not exist.  $\mu_4$  compatibility: glue class  $j(\theta) = 1$  (machine-built  $h$ -adapted chamber,  $\langle \theta, h \rangle = 5$ , height  $29 = h^\vee - 1$ ), clock phase  $i^{2j} = -1$ , class( $2\theta$ ) = 2 sheet-even (WP5a-consistent) with the deleting  $\theta$ - $\mathfrak{sl}_2$  crossing the sheet-odd classes (1, 3); 8 quarters =  $q^2$  via the per-period dictionary. The module it generates is *the* ideal: weight  $2\theta$  has multiplicity 1 in the level-2 Fock, and the direct  $\mathfrak{g}_0$ -lowering orbit BFS from  $|s\rangle$  reproduces the Freudenthal multiplicities of  $V(2\theta)$  exactly through depth 4 ( $27000 = (h^\vee)^3$ , quotient 4124; [C] Weyl complete reducibility beyond depth 4). Negative controls separate honestly: the level-1 current state and generic level-2 states are *not* singular; at  $k=2$  the *cube* is (the  $(E^\theta)^{k+1}$  pattern is generic Kac level mechanics);  $SO(16)_1$  has the same singular vector but keeps three extra level-1 primaries ( $h = \frac{1}{2}$

breaks one-block fusion) — the singular-vector mechanism is level-1 *generic*, the *one-block closure* is the  $E_8/\mu_4$ -specific part. *Handover to WP5c* (answered below): in the twisted quarter-slot moding of the  $\chi_w$  limit Fock, two sector- $C_1$  modes never sum to 8 quarters (minimum  $6 = q^{3/2}$ ) — the per-*period* dictionary (v496), not the per-slot identification, carries  $|s\rangle$  to  $q^2$ ; exactly the GNS/limit-state question (kernel  $\supseteq$  ideal) that WP5c answers.

- **WP5c (done)** [E]/[C]: “the GNS limit state”, executed and machine-verified as CELEST.WP5C.01 [verification/v500\_celestial\_wp5c\_gns\_limit\_state.py] (exact integer/Fraction arithmetic, 35 checks, deterministic) — *success on the preregistered criterion*: the family  $\omega_w$  exists and its limit has the null ideal in its GNS kernel.  $\omega_w$  is the quasi-free state with loop sector = the affine  $k=1$  vacuum  $n$ -point functions (via the machine-determined compact anti-involution  $\theta(e_\alpha) = -e_{-\alpha}$ , the unique anti-automorphism sign on all 61504 basis pairs) and radial sector = oscillator pairings  $x^{wr}$  (the exact Gibbs regulator,  $x$ -adic); *the WP5a threshold holds at the state level*:  $\omega_w = \omega_\infty \bmod x^{N+1}$  for all  $w \geq N+1$ , sharp at  $w = N$ . The core is fully machine: the *complete* exact level-2 Gram — 31124 monomials in 9361 weight blocks — has rank 4124 exactly (the preregistered target), kernel 27000 =  $V(2\theta)$  *weight by weight* (Freudenthal cross-check on all 9361 weights), every block PSD, rank table per Weyl orbit  $(0,0,1,8,44)$ , level-1 rank 248 positive definite (the current layer survives).  $\mu_4$ : the clock descends to GNS with level-2 rank split  $(1036, 1024, 1040, 1024) = \Theta_{C_j}/\eta^8$  at  $q^2$  — the WP5a two-routes identity at the *state* level — and  $|s\rangle$  is the zero vector of  $\text{GNS}(\omega_\infty)$ , resolving the WP5b twisted-slot tension. The KILL branch is probed and not triggered: a CCR obstruction shows *no*  $w$ -uniform state can damp the radial modes (the family formulation is *necessary*), and the wrong constructions are caught (family  $E'_w = mw+r$  erases the 248 layer,  $710955 \neq 248$ ; no damping keeps  $897266 \neq 248$ );  $k$ -dial and  $D_8$  controls separate honestly ( $k=2$ :  $\langle s|s\rangle = +4$ , no ideal in the kernel;  $D_8$ : one block of four,  $h = \frac{1}{2}$ ). [C] quasi-free extension to the full  $*$ -algebra; Kac positivity beyond level 2.
- **WP5d — both stages (done)**:  $\alpha$  [E]/[C],  $\beta$  [E]/[C] (**continuum uplift** [O]): “the local net”, approached from the lattice. The  $\alpha$  stage — the KLM *two-interval index* as the operational Haag-duality discriminator  $((E_8)_1 \Rightarrow \mu=1, \text{ one sector}; SO(16)_1 \Rightarrow \mu=4)$  — is executed and machine-verified as CELEST.WP5D.01 [verification/v501\_celestial\_wp5d\_two\_interval\_index.py] (39 checks; Gaussian lattice machinery ED-validated to  $10^{-15}$ , exact Fractions for the arithmetic). Measured on the 16-layer seam carrier (critical Majorana ring,  $c_- = 8$ ): the fermionic two-interval mutual information is *extensive* (the  $\mu=1$  reference;  $c$  fit 0.5000, residual  $\rightarrow 0$  with  $N$ ), while the sector-summed (orbifold) prescription pays exactly one classical bit — the  $\ln 2$  plateau at machine precision ( $|\Delta_2 - \ln 2| = 1.1 \times 10^{-15}$  at  $N=512$ ), so  $[F : F_{\text{even}}] = 2$  and  $\mu_{\text{gauged}} = 4$ , the entropic twin of the v490 parity census (two independent lattice witnesses); the orbifold also breaks two-interval complementarity ( $S(E) \neq S(E')$ , the direct duality-failure witness), with the complementary-pair budget bounded by  $\ln 4 = \ln \mu(SO(16)_1)$  as a *double-limit* statement. The condensation arithmetic is anchored at both measured ends:  $\det \text{Cartan}(D_5) \cdot \det \text{Cartan}(A_3) = 16$ , KLM/Longo-Rehren  $16/4^2 = 4/2^2 = 1$ ,  $\Sigma d_i^2 = (4, 4, 1)$ ,  $\theta_v = 1$  exactly at  $\nu = 2c_- = 16$  (rivals  $\neq 1$ ) — so  $\mu = 1$  after condensation and the preregistered KILL (“ $\mu$ -offset  $\neq 0$  after condensation”) is *not* triggered; the  $\nu=1$  control shows a non-removable offset ( $\theta_v \neq 1$ : the discriminator has teeth), the trivial phase shows none, and the wrong sector sum loses the full  $\ln 2$ . [C] the continuum dictionary “offset =  $\ln$  index” (KLM; Longo-Xu; Casini-Huerta-Magan-Pontello). **The  $\beta$  stage (done)** [E]/[C] — **split + strong additivity, the two remaining KLM legs** — is executed and machine-verified as CELEST.WP5DB.01 [verification/v504\_celestial\_wp5d\_klm\_completeness.py] (37 checks; Gaussian lattice machinery ED-validated, exact GF2/integer algebra for the exact legs;

the  $U(1)$ /Dirac number-conserving subnet as the preregistered negative control — the current-net analogue where strong additivity famously *fails* in the continuum). *Strong additivity is algebraically exact on the lattice*: with the shared boundary Majorana (“touching = share the point”),  $\text{Even}(A) \vee \text{Even}(B) = \text{Even}(A \cup B)$  *exactly* (GF2 spans full 64/64, 256/256, 512/512, 1024/1024; literal matrix rank 32/32), while disjoint intervals give exactly *half* (rank 16/32, index 2 — the missing sector is odd  $\otimes$  odd, the single  $\ln 2$  bit of the  $\alpha$  stage, localised at the split point); the  $U(1)$ /neutral algebras do *not* generate the union even with the shared site (gaps 2/10/52, growing). The entropic witness uses the honest discriminator *bounded vs divergent*: the  $\mathbb{Z}_2$ /orbifold touching deficit stays  $< \ln 2$  with the Ising approach  $(\ln 2 - \Delta_2) \sim N^{-p}$ ,  $p = 0.2444$  vs  $2\Delta_\mu = \frac{1}{4}$  (the naive “defect  $\rightarrow 0$ ” expectation is *false* at the sharp lattice split — bounded  $\Leftrightarrow$  finite index, Longo–Xu), while the  $U(1)$  control *bursts*  $\ln 2$  from  $L = 128$  on and grows as  $\frac{1}{2} \ln \text{Var } Q_A$  (Klich–Levitov slope 0.10134 vs  $1/\pi^2 = 0.10132$ : infinite index). The split property is witnessed quantitatively: the coupling-block singular values fall exponentially at the elliptic-nome rate  $\pi K(1-x)/K(x)$  to 1.3–2.0% over  $d = 16..512$ ,  $\sigma_1 \sim x^{0.5044}$  (vs  $\frac{1}{2}$ ), trace norm summable — and the orbifold *inherits* the same ladder exactly at machine level ( $P_A$  flips  $C \rightarrow -C$ ,  $\sigma_k$  identical to  $8.3 \times 10^{-17}$ ; the even-bilinear coupling Gram is the second compound  $\Lambda^2 C$  with singular values  $\{\sigma_a \sigma_b\}$  — Longo heredity, [C]). The Pimsner–Popa bound is a one-line *identity*,  $E(a) - a/2 = PaP/2$ , so  $\lambda = \frac{1}{2} = 1/[F : F_{\text{even}}]$  exactly, attained in exact integers (16384 + 2048 monomial sweeps, 0 violations; random pencil min  $\lambda = 0.5000000000$ ); the two-interval group gives  $\lambda_{E_4} = \frac{1}{4} = 1/\mu$  exactly, and the index is pinned by *two independent routes* ( $e^{\Delta_\infty} = 2.000000000 = 1/\lambda_{\text{PP}}$ ;  $e^{2\Delta_\infty} = 4 = 1/\lambda_{E_4} = \mu$ ); the  $U(1)$  control has  $\lambda = 1/(m+1) \rightarrow 0$ . *With the  $\alpha$  stage, all three KLM ingredients of complete rationality — split, strong additivity, finite  $\mu$  — are now witnessed on the lattice*. The continuum uplift stays [O] and is honestly fenced: the theorem “finite-group orbifolds of completely rational nets are completely rational (hence split + strongly additive)” is *Xu’s theorem* (algebraic orbifold CFTs) — cited, *not* claimed; the continuum proof for the concrete seam quotient net on the collar (v336/v449) and for the interacting condensed  $(E_8)_1$  net remains WP5e / Costello–Li territory.

- **WP5e (subdivided):  $\alpha$  stage (done) [E]/[C] — the CFT-side prefactor + level pinning;  $\beta$  stage (done) [E]/[C] — the equivariant anomaly ledger on twistor space, including the honest  $a_0$  refutation;  $\gamma$  stage (done) [E]/[C] — the sphere-axion pairing check, a rigid *negative* result with certificate;  $\delta_2$  stage (done, intermediate verdict, exact) [E]/[C] — the full-tensor ledger: the collapse confirmed, one cubic door opens;  $\varepsilon_2$  stage (done, verdict B) [E]/[C] — the CPS level-from-flux dial, with “one level” lifted to a theorem; the global BCOV quantisation stays [O].** The  $\alpha$  stage is executed and machine-verified as CELEST.WP5E.ALPHA.01 [verification/v502\_celestial\_wp5e\_prefactor\_level.py] (exact sympy/Fraction arithmetic, 33 checks). Both exactly computable consistency anchors of the uplift hold. (i) *The  $q^{-1/3}$  prefactor is exact  $\mu_4$  vacuum-energy bookkeeping*: the clock is *inner* ( $h = (2, 2, 2, 2, 2; 0^4)$ ) and the  $A_3$  detector  $h' = (0^5; 1, 1, 1, -3)$  both read the glue class mod 4 on all 240 roots;  $(h, h) = 20$ ,  $(h', h') = 12$ , sum 32), so the induced twist on the 8 torus bosons is  $\theta = 0^8$  in *every* sector — a *shift* orbifold, not a rotation orbifold — and each sector carries the same  $8 \times (-\frac{1}{24}) = -\frac{1}{3} = -c/24$  (Hurwitz- $\zeta$  exact,  $E_b(\theta) = -\frac{1}{24} + \theta(1 - \theta)/4$ ); the sector ground exponents are  $-\frac{1}{3} + (0, 1, 1, 1)$  with  $h_j$  = the discriminant form, an identity holding via spectral flow ( $j^2(h, h)/32$ , mod 1) *and exactly* via the  $10 + 6 = 16 = 2^{g_{\text{car}}-1}$  Majorana free fermions of the WP5d seam carrier (R–NS shift  $n/16$ :  $\frac{5}{8}, \frac{3}{8}, 1$ ); the four sector characters sum to  $E_4/\eta^8 = q^{-1/3}(1, 248, 4124, 34752)$ ; the rotation reading *fails* (the geometric  $A_3$  deck gives  $\frac{3}{16} \neq \frac{3}{8}$  — the WP1 “clock  $\neq$  deck” lesson at the Casimir level). (ii)  $k = 1$  *is forced three independent ways*: the current condition  $h(J) = k = 1$  (the  $\mathbb{Z}_4$  extension closes

onto the  $E_8$  adjoint  $60+64+60+64 = 248$  only at spin 1); the conformal embedding ( $D_5+A_3 \subset E_8$ :  $47k^2+219k-266 = 47(k-1)(k+266/47) = 0$ ;  $D_8 \subset E_8$ :  $128k(1-k) = 0$ ); and the central charge  $248k/(k+30) = 8 \Leftrightarrow 240(k-1) = 0$  (= the prefactor itself); plus the WP5b singular-vector dial ( $\|(E_{-1}^\theta)^m 0\|^2 = m! k(k-1) \cdots (k-m+1)$ , singular vector exactly at grade  $k+1$ ;  $k=1$  deletes  $V(2\theta)$ :  $31124 - 27000 = 4124$ ). *Honest sharpening*: glue- $h$  integrality  $h(J^j; k) = k(0, 1, 1, 1)$  holds for *all*  $k = 1..8$  and fixes nothing — the originally conjectured integrality route is dead; the three other dials carry. Controls:  $D_8/SO(16)_1$  has the *same*  $q^{-1/3}$  prefactor but  $h = (0, \frac{1}{2}, 1, 1)$  (the discriminator is glue- $h$  integrality, not the prefactor);  $\mathbb{Z}_2$  or  $\mu_4$  rotation twists break the common prefactor; wrong  $k \in \{2, 3, 4\}$  fails all five dials coherently. **The  $\beta$  stage (done) [E]/[C] — the equivariant anomaly ledger on twistor space** — is executed and machine-verified as CELEST.WP5E.BETA.01 [verification/v505\_celestial\_wp5e\_beta\_equivariant\_ledger.py] (exact sympy/Fraction arithmetic, 47 checks). The twistor side is pushed as far as exact arithmetic reaches: the Atiyah–Bott/Lefschetz fixed-point skeleton of the one-loop box anomaly on  $\mathbb{C}^2/\mathbb{Z}_4$  with the  $\mu_4$ -glue monodromy. (i) *The AB ledger is exact*: denominators  $\det(1 - g_j^{-1}) = (2, 4, 2)$  with Dedekind sum  $5/4 = (|\mathbb{Z}_4|^2 - 1)/12$ ; equivariant characters  $\text{tr}_{g_j}(\text{Ad } E_8) = (248, 0, -8, 0)$  by two independent routes; invariant average  $60 =$  the carrier; Frobenius 61568. (ii) *Okubo per sector, the honest sharpening*: only the *invariant* sector is Okubo-quadratic ( $Q^{(0)} = 36\langle x, x \rangle^2$ ,  $36 = \tilde{\lambda}_{e_8}^2$  — the v495 identity re-derived through the glue grading); the twisted sectors carry irreducible  $T_5/T_3$  content, and the AB-weighted sum cancels the  $D_5$  quartic *exactly* while leaving the *rigid* residual  $32T_3$  (no admissible reweighting fixes it; the graded GS exchange is rank-obstructed in sectors 1–3) — twisted-sector closed strings must carry the rest [O]. (iii) *The index bridge* (the centrepiece):  $f(m) = \frac{1}{4} \sum_j (i^{jm} - 1) / \det_j = \text{ch}_2(T_m) = -(C^{-1})_{mm}/2$  *exactly* — the fixed-point ledger and the McKay/Kronheimer intersection ledger are the same arithmetic; the glue-bundle  $\text{ch}_2$  defect is the integer  $-78$  by *both* routes (classes  $(-24, -30, -24)$ ). (iv) *The level dials say  $k = 1$  geometrically*: the lattice current count is 240 at  $k=1$  and exactly 0 at  $k = 2, 3, 4$ ; the embedding residual  $47k^2+219k-266 = (0, 360, 814, 1362)$ ;  $h(J; k) = k$  is integer for *all*  $k$  (integrality alone fixes nothing — honest, exactly as on the CFT side); one scale  $\Rightarrow$  one level (v493 periods; since the  $\varepsilon_2$  stage a *theorem* of clock invariance, v509). (v) *The honest refutation*: the clock-invariant modulus  $a_0 \in O(8)$  fills the BSS *graviton* slot  $O(2)$ , *not* the axion slot  $O(-2)$  (Hamiltonian weights  $X = -2 \neq +2 = Y$ ; mismatch  $4 = |\mu_4|$ , typed [C] as a coincidence) — “the theory brings its own GS axion as  $a_0$ ” is *refuted*; what does match: the three  $H^2(\text{ALE})$  classes carry exactly the three twisted-sector Coxeter characters  $\{i, -1, -i\}$  (bijection, no invariant class), so the bulk axion must come from the  $O(-2)$  tower field itself (construction [O]). Controls:  $\text{diag}(i, i)$  breaks the ledger four independent ways;  $SO(16)$  glue gives characters  $(120, 0, +120, 0)$  and defect  $-30 \neq -78$  with a *failing* bulk Okubo (independent quartic Casimir);  $k = 2$  dies on the closure dial ( $-31/48 \neq -\frac{1}{3}$ ) while naive integrality dials stay integer. The preregistered kill does *not* fire on the equivariant skeleton. **The  $\gamma$  stage (done) [E]/[C] — the sphere-axion pairing check, an honest rigid negative result** — is executed and machine-verified as CELEST.WP5E.GAMMA.01 [verification/v508\_celestial\_wp5e\_gamma\_pairing.py] (exact sympy/Fraction arithmetic, 27 checks). Question: can the three twisted sphere axions  $\eta_m$  (Coxeter eigencharacters  $\{i, -1, -i\}$  on  $H^2(\text{ALE})$ , the  $\beta$ -stage slot bijection) cancel the rigid  $32T_3$  residual ( $A_{\text{fix}} = 9P_1 - 30P_2 - 15P_3 + 32T_3$ ) by Green–Schwarz-type *exchange* with sector-compatible quadratic vertices? *No, with a certificate*. (i) *Vertex-space theorem*: the  $W(D_5) \times W(A_3)$ -invariant quadratics on the 8-dim glue Cartan are *exactly*  $\text{span}\{s_5, s_3\}$  (dim 2) and the invariant quartics *exactly*  $\text{span}\{P_1, P_2, P_3, T_5, T_3\}$  (dim 5) — complete by Weyl nullspace arithmetic over all bidegrees; the *product theorem* is the master kill: the product of any two invariant quadratics has identically zero  $T_5/T_3$  content, so every exchange image — any axion count, any selection rule, any

couplings — is annihilated by  $\Phi_{T_3}$  while  $\Phi_{T_3}(A_{\text{fix}}) = 32 \neq 0$ . (ii) *Selection enumeration*: the strict two-index  $\mathbb{Z}_4$  rule collapses (the class bilinears are nonzero only for  $j' = -j$ , forcing  $m = 0$ : the sphere axions have *no* invariant bilinear Cartan vertex at all); the generous twist-insertion channels  $E_{13} = (16, -96, 144, 0, 0)$ ,  $E_{22} = (16, 32, 16, 0, 0)$  give a rank-2 exchange matrix with  $\text{rank}([M | A_{\text{fix}}]) = 3$  and annihilator certificate  $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\text{fix}}) = (0, 32, 72)$  — two independent obstructions, the second driven by the side discovery  $K^{(0)} = -15 K^{(2)}$  (the even-sector quadratics are parallel);  $\ker M = 0$  forces the unique non-disturbing coupling  $c = 0$  (the sphere axions are rigidly *silent*), and the sector-0 Okubo mechanism is undisturbed ( $\Phi_P(Q^{(0)}) = 0$ ). (iii) *Naturalness dissolves*:  $\text{ch}_2$ -natural and Atiyah–Bott-weight couplings both carry certificate  $(0, 0)$  against the required  $(32, 72)$  — the failure is scale- and coupling-independent (the ansatz space misses the target identically, not by an unnatural factor). Controls: the flipped rule changes the enumeration but not the  $T_3$  kill ( $\Phi_P$  is rule-dependent,  $\Phi_{T_3}$  is not);  $SO(16)$  has *no* sphere partners (odd McKay classes empty) *and* uncanceled  $T_5$   $((15, -18, -15, -4, 40) — E_8$  is doubly special);  $D_8$  has no  $T_3$  structure at all  $(16 T_8 + 12 s_8^2)$ . Number fences typed [C] only:  $32 = (h, h) + (h', h') = 20 + 12$ ,  $72 = 2\tilde{\lambda}_{e_8}^2$ . The  $\beta$ -stage slot bijection is untouched and the preregistered level-kill still does *not* fire — what dies is the exchange *realisation* of the pairing. **The  $\varepsilon_2$  stage (done, verdict B) [E]/[C] — the CPS level-from-flux dial** — is executed and machine-verified as CELEST.WP5E.EPS2.01 [verification/v509\_celestial\_wp5e\_eps2\_level\_from\_flux.py] (exact sympy/Fraction arithmetic, 28 checks). Target:  $k = 1$  from flux quantisation on the lockstep spheres, following the Costello–Paquette–Sharma Burns-holography mechanism (PRL **130**, 061602; arXiv:2306.00940). (i) *The CPS skeleton is replicated exactly*: the  $S^3$  period of the brane back-reaction 3-form is  $(2\pi i)^2 N$  (pullback density  $-\sin 2\theta$ , their eq. 3.33 — large-gauge invariance quantises  $N$ ); the exceptional-sphere Kähler flux is  $2\pi N$  exactly; the boundary level magnitude is  $2N$  via the symplectic-boson double contraction ( $N = 1, 2, 3$ ), Dynkin normalisation  $T(\text{vector}) = 1$  — “level = flux quantum  $\times$  Dynkin index”, *linear* in the flux. (ii) *The pairing matrix is pinned, not postulated*: the  $\beta$ -stage  $\text{ch}_2$  ledger + integrality + unimodularity + effectivity force  $P_{mi} = c_1(T_m) \cdot [\Sigma_i] = \delta_{mi}$  by complete enumeration (row candidates 8/6/8; 48 unimodular solutions; with  $P \geq 0$  exactly 2: identity + the  $A_3$  diagram flip), so the glue fluxes are  $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$  (total  $188 = 248 - 60$ , flip-invariant) and every charged root current carries *exactly one* flux quantum through exactly one sphere. (iii) *The level formula, honest*: “level = total open-string flux” is *killed by the lockstep test itself* ( $k_i = (64, 60, 64)$  unequal —  $F_i$  is the flavour multiplicity, the brane matter content); what works is  $k = \text{flux per adjoint current} = F_i / \dim \mathfrak{g}_i = (1, 1, 1)$ , anchored non-circularly by the lattice current count  $(240, 0, 0, 0)$  for  $k = 1..4$  and the embedding index  $j(D_5 \subset E_8) = j(A_3 \subset E_8) = 1$  ( $60 = 2h^\vee$ ). The ord-4-vs-level-1 tension is *resolved*: the clock order counts *fractional flux sectors*, not the level ( $\mu = 16 = \text{ord}^2$ ; the  $\mu_4$  glue diagonal is isotropic and Lagrangian; KLM  $16 \rightarrow 16/4^2 = 1$ , zero residual fractional flux), and a *new dial* pins the value:  $\#\text{primaries}((E_8)_k) = (1, 3, 5, 10, 15, 27)$  for  $k = 1..6$  from the affine marks — exactly *one* sector iff  $k = 1$ , independent of the CPS transplantation. (iv) *The lockstep theorem* (the lifting of the  $\beta$ -stage “one scale  $\Rightarrow$  one level” to a *theorem*): clock invariance forces  $a_3 = a_2 = a_1 = 0$  sharply, the four branch points are one  $\mu_4$  orbit,  $|\Pi_j| = \sqrt{2}t$  is *equal* on all three spheres, the monodromy acts as the pure phase  $i$  and  $\det(A - 1) = -4 \neq 0$  (no invariant flux vector) — with the CPS flux-level dictionary,  $k_1 = k_2 = k_3$  is a *theorem* of clock invariance; the new falsifier: the clock-forbidden family  $Z^4 - Z$  (smooth, disc  $-27$ , branch points not a  $\mu_4$  orbit) gives *zero* lockstep chains over all 24 orderings. Honest limit: uniform flux doubling keeps the lockstep — the theorem proves *equality*, not the value; the value 1 comes from the current/sector counters and the CPS minimal quantum  $N = 1$ . Controls separate honestly:  $k = 2$  dies on the closure dials (count 0, residual 360, prefactor  $-31/48$ ,

#prim 3) while the lockstep dial honestly does *not* fire;  $SO(16)$  glue leaves spheres  $1/3$  flux-dark with condensation stuck at 4; the  $A_2$  form has  $\mu = 12$  (no Lagrangian subgroup exists at all) and Coxeter order  $3 \neq 4$ . Typed honestly: the CPS dictionary itself on  $\mathbb{P}T/\mathbb{Z}_4$  stays **[C]** (their geometry is the  $\mathbb{C}^2$  blow-up with one  $(-1)$ -sphere and  $\mathfrak{so}(8)$  at negative level — not the  $A_3$  ALE); the type-I B-model back-reaction on  $\mathbb{P}T/\mathbb{Z}_4$  stays **[O]** only in its  $\delta_{1d}$  derivation face (all three milestones M1–M3 have since been executed, **v515/v516/v517**; the  $\delta_1$  chain itself has since been *decided* by **v518** — kill under the derived measure — and the declared/derived measure question has since been decided at probe level in favour of the declared reading under the typed premises TP-1..TP-4, **v520**; the residual **[O]** is the constructive  $w_m$  derivation). **The  $\delta_2$  stage (*done, intermediate verdict, exact*) **[E]/[C] — the full-tensor ledger: the collapse is real, and one cubic door opens** —** is executed and machine-verified as CELEST.WP5E.DELTA2.01 [**verification/v511\_celestial\_wp5e\_delta2\_tensor\_ledger.py**] (exact Kostant/Weyl weight arithmetic + sympy/Fraction, 41 checks). The  $\gamma$ -stage collapse was a *Cartan-restricted* statement;  $\delta_2$  recomputes the ledger on the full adjoint tensor structure. (i) *Structure*:  $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$  is *semisimple* with no  $\mathfrak{u}(1)$  factor (class-0 roots split  $40+12$ , root span rank 8); the twisted sectors factorise multiplicity-free into minuscule Weyl orbits  $\mathfrak{g}_1 = (16_s, 4)$ ,  $\mathfrak{g}_2 = (10, 6)$ ,  $\mathfrak{g}_3 = (\mathfrak{g}_1)^*$ , each irreducible; and the *innerness theorem* decides the collapse wholesale: the grading element  $h$  lies in the Cartan of  $\mathfrak{g}_0$ , so every  $\mathfrak{g}_0$ -invariant tensor carries total sector charge  $0 \bmod 4$  — the  $\gamma$ -stage collapse is a *full-tensor* statement *and* holds across arities (all 15 non-neutral trilinear sector triples have  $\text{Hom} = 0$ : the sphere axions have no invariant partner operator at any arity  $\leq 3$ ). (ii) *The Hom tables*: bilinear invariants exist only at  $j' = -j$  (dims  $2/1/1/1$ , the Killing blocks); the neutral trilinear multisets carry dims  $(3, 2, 2, 1, 1)$ , all cross-sector trilinears are pure brackets (antisymmetric in their repeated legs) — the *only* totally symmetric trilinear on all of  $\mathfrak{e}_8$  is the  $\mathfrak{su}(4)$   $d$ -symbol ( $\mathfrak{so}(10)$  has no cubic Casimir:  $\text{Inv}(S^3 45) = 0$  protects  $T_5$ ). (iii) *The new channel*: the quartic projection of the  $d$ -symbol exchange is  $Q_{dd} = (0, 0, -\frac{1}{240}, 0, \frac{1}{60}) = \frac{1}{60}(T_3 - P_3/4)$  by two independent routes (Killing propagator  $2h^\vee = 60$  from the roots; hand route  $|x^2 - (\text{tr } x^2/4)\mathbf{1}|^2/60$ ), and  $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$  — the  $\gamma$ -stage master kill (the product theorem) covers *quadratic* vertices only and does not extend to cubic ones: the  $T_3$  direction *is* reachable through the  $d$ -symbol channel, while brackets, tadpoles and mixed  $d \times f$  die exactly and no  $T_5$  analogue exists. (iv) *The new certificate*:  $M = [E_{13}, E_{22}, E_{00}, Q_{dd}]$  has rank 3, augmented rank 4 — the pairing in the charge reading remains *obstructed*, but with the genuinely *weaker* certificate  $\{\Phi_{T_5}, \psi = \Phi_P - \Phi_{T_3}/4\}$ ,  $\psi(A_{\text{fix}}) = 72 - 8 = 64$ ; and the relaxed reading (charge bookkeeping dropped on the bilinear  $P$ -block) becomes *exactly solvable*:  $A_{\text{fix}} = -u + 8v + 2w + 1920 Q_{dd}$  (residual 0) — where the  $\gamma$  stage found  $T_5 = T_3 = 0$  persisting even relaxed, the  $d$ -channel closes the  $T_3$  gap with small integer coefficients. Number fences typed **[C]** only:  $64 = \dim \mathfrak{g}_1 = 2^6$ ;  $1920 = |W(D_5)| = 8 \cdot 240$  — and the 1920 fence has now been stress-tested and *fails as a derivation claim* (**[verification/v513\_celestial\_dterm\_nonderivation.py]**, CELEST.DTERM.NONDERIV.01): the  $1920 = |W(D_5)|$  reading is look-elsewhere-loaded (11/924 pre-declared catalog expressions hit 1920, vs. 8/924 for the control target 1800) and convention-contingent (only 1 of 5 normalisations produces 1920: Gell-Mann gives 120,  $\mathfrak{su}(4)$ -Killing 256, plain trace 32,  $\text{ch}_3$ -normalised 69120); the charge-legal flux pairings give 2048/1800 — they bracket and miss — while the only flux hit is the charge-forbidden  $(1, 2)$  pairing; quantisation delivers only lattices ( $60\mathbb{Z}$ , with 1920 the 32nd multiple; strict  $\text{ch}_3$  units exclude it); and the twisted cubic index  $32i$  clashes in class provenance with the true quartic 32 (classes  $1/3$  vs.  $0/2$ : same number, disjoint mechanism). The convention-stable **[E]** core is the factorisation  $c_d = \Phi_{T_3}(A_{\text{fix}}) \times 2h^\vee(E_8) = 32 \times 60$ ; the fence stays **[C]**, the physical generation of the 32 stays **[O]** — no marker moves. Controls: the Killing control replicates the  $\gamma$ -stage channels and  $A_{\text{fix}}$  exactly;  $SO(16)/D_8$  has *no*

symmetric cubic ( $\text{Inv}(S^3 \text{adj}_{D_8}) = 0$ ), no odd sectors and no  $A_3$  block — the  $\delta_2$  channel cannot even be posed ( $E_8/\mu_4$  doubly special); the false  $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_2 \oplus \mathfrak{u}(1)$  inflates every Hom dimension ( $(5, 2, 2) \neq (2, 1, 1)$ ,  $\text{Inv}(S^3 \mathfrak{g}_0) = 6 \neq 1$ ) and the false sector content ( $16_s, \bar{4}$ ) kills every pairing. What stays open after  $\delta_2$  [O]: the arity  $\geq 4$  ledger, the *physical* justification of the cubic GS term ( $c_d = 32 \times 60$ , the convention-stable core; the  $1920 = |W(D_5)|$  reading is look-elsewhere-loaded and convention-contingent, v513;  $\text{ch}_2$ -naturalness for cubic vertices *undecided* — no canonical geometric datum exists for a cubic vertex), and the burden on  $\delta_1$  is *sharpened*: it must supply either the  $\psi = 64$  slice or the selection-rule relaxation — a burden since *discharged* by the M2 milestone: the declared KS measure supplies the  $\psi = 64$  slice exactly (v516), so the cubic  $d$ -channel is not needed ( $c_d \text{ free} = 0$ ) and its physical-justification question stays dissolved — the  $\delta_1$  decision (v518) does *not* revive it (no branch fires under the derived measure). **The  $\delta_1$  stage (done, DECIDED — kill under the derived measure, the  $\mathbb{Z}_2$  anchor a method boundary) [E]/[C]** — is executed and machine-verified as CELEST.WP5E.DELTA1.01 [verification/v518\_celestial\_wp5e\_delta1\_derived\_kill.py] (exact  $\mathbb{Q}(\zeta_8)$ /phase arithmetic + 30-digit kernels, 30 checks; the  $\delta_{1b}/\delta_{1c}/\delta_{1d}$  probe chain consolidated). The exploration stage had left the verdict undecided with four exact sharpenings — (i) the strict holomorphic  $q^0$  reading *refuted at the  $\mathbb{Z}_2$ /Eguchi–Hanson anchor* (a method boundary, not a kill of the contract); (ii) the forced leading ( $T_5, T_3$ ) ratio 4:3; (iii) the contact term = the *modular completion* of the Atiyah–Bott data (a Harvey–Moore-type  $\tau$ -integral, orbit split  $15 = 12 + 3$ ); (iv) massive coset levels  $n \geq 2$  mandatory. The  $\tau$ -integral has now been *evaluated under a derived measure*, and the route is decided. Four exact results: (i) *the 16-component Weil completion closes*: the discriminant module of  $D_5 \oplus A_3$  is  $\mathbb{Z}_4 \times \mathbb{Z}_4$  with  $q = (5x^2 + 3y^2)/8$ , Gauss sums  $2\zeta_8^5 \times 2\zeta_8^3 = 4 = \sqrt{16}$  (signature 0 mod 8 = rank  $E_8$ ), the exact Weil relations hold in  $\mathbb{Q}(\zeta_8)$ , the diagonal and anti-diagonal Lagrangians are the two  $E_8$  gluings, and the 16-vector theta  $S$ -covariance drops the naive-4-rule residual from  $2.91 = O(1)$  to  $\sim 10^{-39}$ ; (ii) *the obstruction is a finite  $\mu_4$  character, identified*: the dressed gauge blocks  $M = G[a, b]\eta^{-8}$  carry a multiplier system whose  $SL(2, \mathbb{Z})$  relation defects are  $(1, 1, 1)$  exactly on all 15 pairs (a *character* of the orbit stabilisers  $\Gamma_1(4)/\Gamma_0(2)^\pm$ , not a genuine 2-cocycle), on  $\Gamma_1(4)$  explicitly  $\lambda(\gamma) = i^{2B+C/4}$ ; (iii) *the cancellation exists and is the twisted fibre block  $f_1 f_3$ , not the three sphere axioms*:  $G[a, b] = f_1 f_3$  is an exact identity, the T-fix mechanism is one line of exact phase arithmetic ( $(-1) \times e(-1/6) = e(1/3) = \chi_4(T)$ ), and the exact cancellation table reads none  $\rightarrow 4$ ,  $f_1 f_2 f_3 \rightarrow 4$ ,  $f_2 \rightarrow 6$ ,  $f_1 f_3 \rightarrow 1$  (both orientations), wrong weights  $\rightarrow 4$  — the strict solutions are exactly  $\chi_4$  (dims (3, 3)) and  $\chi_{10}$  (dims (1, 1)), certified on the dressed functions; (iv) *all three preregistered testers fail* under both derived solutions ( $T_5$  fractions 0.5/0.83 instead of 0;  $W_{13} = 0$  instead of  $W_2:W_{13} = 4:3$ ; spreads 1.05/1.47 against  $-A_{\text{fix}}$ ; no  $\psi = -64N$  slice), and the honest  $(N_1, N_2)$  orbit rebalance rescues nothing in the positive cone — a genuine KILL on the derived surface. Controls: the  $\mathbb{Z}_2$ /EH anchor (gauge-only residual order 2, cancelled to 1 by the same fibre dressing),  $SO(16)$  (order-4 supply structurally absent), wrong form/wrong signature (break exactly). Fences [C]: the  $f_1 f_3 = \text{KS-weight identification}$  (deck weights (1, 3) = the twistor fibre weights forced by the v514 incidence ledger — exact match, *not* a complete BCOV derivation), the kernel-family convention, the discrete  $\chi_k$  choice (both evaluated, both fail). TENSION, stated honestly (the mandatory fence): the DECLARED completion reading (v516/M2) delivers the  $\psi = 64$  slice and cancels  $32 T_3$ ; the DERIVED chiral measure (this module) fails all three testers — both are exact; the sharp open question was which of the two is the true BCOV measure (the declared reading is supported by the  $\delta_1$  modular-completion finding but not derived; the derived measure rests on the  $f_1 f_3$  identification). Neither result is hidden behind the other. That question is since *decided at probe level* by v520 in favour of the declared reading (single-valuedness derived from

F-independence + the Quillen pairing under the typed premises TP-1..TP-4; the kill of this module is *sharpened*: the column-canonicalised  $\chi_4$  member is boundary-consistent yet tester-dead); and the constructive BCOV derivation of the  $w_m$  normalisation itself is since *executed* by v523 (verdict ERFOLG; the residual [O] narrows to the global BCOV integral beyond the fibre zero-mode factor). **The  $\varepsilon_1$  stage (done) [E]/[C] — the  $O(-2)$  bulk-axion slot** — is executed and machine-verified as CELEST.WP5E.EPS1.01 [verification/v514\_celestial\_wp5e\_eps1\_axion\_slot.py] (exact sympy/Fraction arithmetic, 34 checks, verdict B). Three results, each exact. (i) *The slot is a construction*: on  $\mathbb{P}T' = \text{Tot}(O(1) \oplus O(1) \rightarrow \mathbb{P}^1)$  the pushforward ledger  $\pi_* O(-2)$  gives per spacetime degree  $d$  exactly  $(d+1)^2$  classes = the exact wave-operator nullspaces, and the *equivariant* bookkeeping under the v492 action  $(\mu_1, \mu_2) \mapsto (i\mu_1, i^{-1}\mu_2)$  (incidence-forced column weights  $(+1, +1, -1, -1)$ ) closes *block by block* for all  $d \leq 6$  and all four characters; the character series are  $P_0 = 1 + 3t^2 + 15t^4 + 21t^6 + 45t^8 + \dots$ ,  $P_1 = P_3 = 2t + 8t^3 + 18t^5 + \dots$ ,  $P_2 = 6t^2 + 10t^4 + 28t^6 + \dots$ , the  $d = 0$  slot has multiplicity  $(1, 0, 0, 0)$  — *the bulk axion survives the projection* — and the Molien invariant ring is  $(1 - t^8)/((1 - t^4)^2(1 - t^2))$ :  $Z \in O(2)$ ,  $X, Y \in O(4)$ , one relation in degree 8 = the  $a_0$  weight (the v492/v493 bridge); the E3 bijection is sharpened per character (twisted minimal content  $(2t, 6t^2, 2t)$ , no degree-0 mode, = the Coxeter eigenvalues,  $\det(A - 1) = -4$ ), and the graviton control separates  $a_0$  (the  $O(+2)$  slot starts invariant only at fibre degree 4 with multiplicity  $3 = \{X, Z^2, Y\}$ ). (ii)  $\tilde{\lambda} = 6$  *is triply pinned*: Okubo  $\text{Tr}_{\text{adj}} X^4 = 36\langle x, x \rangle^2 = (6\langle x, x \rangle)^2$  on the 240 glue roots ( $36 = h^\vee + 6$ ,  $\tilde{\lambda} = 6 = |\mathbb{Z}_2|N_{\text{fam}}$ ); the measure chain on  $\mathbb{P}T/\mathbb{Z}_4$  cancels the quotient factor  $\mu = 1/4$  *exactly* (vertex $^2$   $\mu^2$ , propagator  $1/\mu$ , anomaly  $\mu$ :  $\tilde{\lambda}_{\text{eff}}^2 = 36$  for arbitrary  $\mu$ ), with the wrong bookings quantified and *excluded* (anomaly-only  $\Rightarrow \tilde{\lambda} = 3$ , missing propagator renormalisation  $\Rightarrow \tilde{\lambda} = 12$ ); and the flux side (minimal CPS quantum  $N = 1$ , a single exchange channel exactly at  $k = 1$ ,  $(\kappa/c_3)^2 = 12$  exact). (iii) *The first honest back-reaction step (Gibbons–Hawking  $A_3$ )*: clock-invariant four-centre configurations form exactly two branches (axis $^4$  = pure resolution or *one* free  $\mu_4$  orbit = pure deformation); the free orbit gives  $\prod_p (Z - i^p z_0) = Z^4 - z_0^4$ , i.e.  $XY = Z^4 + a_0$  with  $a_0 = -z_0^4$  — the v493 family *re-derived from the geometry*, with the  $a_0$  monodromy = one clock step; the charge- $k$  GH point is exactly  $\mathbb{R}^4/\mathbb{Z}_k$  ( $k = 4$ : the seam boundary  $S^3/\mathbb{Z}_4$ ); the periods are  $\Pi_j = 4\pi t_0(i - 1)(1, i, i^2)$  with equal moduli (lockstep, the v509 counter-check from geometry) and the Coxeter clock re-derived ( $A^4 = 1$ ,  $\det(A - 1) = -4$ ,  $\Pi \cdot A = i\Pi$ ); the CPS “ $N \log$ ” analogue carries source charge  $4 = |\mu_4|$ , with the honest sharpening that the Ricci-flat ALE has asymptotic log coefficient exactly 0 (the CPS log is an exceptional-locus statement; Burns contrast  $\det g \neq 1$ ), and the multipole selection rule  $m \equiv 0 \pmod 4$  stores  $-a_0$  in the first symmetry-breaking  $(4, \pm 4)$  harmonic. Controls:  $\mathfrak{so}_8$  ( $\tilde{\lambda}^2 = 12$  irrational; perfect squares in the Deligne series only  $\{9, 36\}$ ),  $\text{diag}(i, i)$  (Veronese cone, no hypersurface),  $k = 2$  (three channels). [C]: conditional on Costello’s flat- $\mathbb{P}T$  matching  $\lambda_g^2 E = A$  (the measure chain is classical bookkeeping). [O] (the honest fence): the *quantised* BCOV one-loop coefficient on  $\mathbb{P}T/\mathbb{Z}_4$  and the twisted channels  $(32 T_3)$  — preregistered as M1–M3 below; all three have since been executed (v515/v516/v517), the  $\delta_1$  chain has since been *decided* by v518 (kill under the derived measure), the measure question has since been decided at probe level by v520 for the declared reading, and the  $w_m$  normalisation has since been *derived* at probe level by v523; what stays [O] is the global BCOV integral beyond the fibre zero-mode factor. **WP5e proper [O] (the global BCOV quantisation)**: the BCOV/Kodaira–Spencer theory on  $\mathbb{P}T/\mathbb{Z}_4$  itself (quantisation, non-fixed-point contributions, the twisted closed-string measure). The exchange sub-branch is *closed* by the  $\gamma$  stage, the full-tensor ledger ( $\delta_2$ ), the level-from-flux dial ( $\varepsilon_2$ ) and the bulk-axion slot ( $\varepsilon_1$ ) are *executed*, and  $\delta_1$  is *decided* by v518 (kill under the derived measure); the named remaining target, with preregistered success/kill criteria: **(the measure question — the  $\delta_1$  successor; decided at**

**probe level, ERFOLG-A)** the  $\delta_1$   $\tau$ -integral had been evaluated on *both* sides: the M2 milestone (v516) *delivers* the coefficient  $(9, -30, -15, 0, 32)$ -completion exactly on the *declared* measure (every sector the same Okubo square, the  $\psi = 64$  slice supplied), while the *derived* chiral measure (v518: blockwise  $SL(2, \mathbb{Z})$  covariance solved for, the  $\mu_4$  multiplier character cancelled by the twisted fibre block  $f_1 f_3$ ) fails all three preregistered testers — both are exact, and the stated tension was the sharp remaining question: *derive* from the BCOV integral on  $\mathbb{PT}/\mathbb{Z}_4$  which of the two measures is the true one. That decision is executed and machine-verified as CELEST.WP5E.MEASURE.01 [verification/v520\_celestial\_wp5e\_measure\_decision.py] (the  $\delta_{1e}/\delta_{1f}$  consolidation, 39 checks, ERFOLG-A on the preregistered decision layer): the exact invariant subspace of the dressed 16-component Weil system is  $\dim_{\mathbb{Q}}\{v : \rho(T)v = v, \rho(S)v = v\} = 8 = 4 \times 2$  with *every* kernel vector in the  $\mathbb{Q}(\zeta_8)$ -span of  $\{e_H, e_{H'}\}$  (both theta =  $E_4$  = the unphased trace), and single-valuedness of the physical one-loop lattice factor is *derived* from two independent constructive sources rather than postulated — (*source 1, the F-lemma*) every strict chiral closure at physical weight carries a nontrivial character ( $\chi_4(T) = e(1/3)$ ,  $\chi_{10}(T) = e(5/6)$ , exact; pointwise covariance of the canonical column-matched member certified at  $3.9 \times 10^{-16}$  with  $|G| \geq 59.6$ ), and an exact change of variables forces  $\int_{\gamma_F} G d\mu = \chi_4(\gamma) \int_F G d\mu$ , so fundamental-domain independence forces  $\int = 0$  for *every* chiral integrand: the v518 route was never a well-defined nonvanishing moduli integral (the kill is *sharpened*); (*source 2, the Quillen pairing*) the doubled hol $\times$ antihol transports satisfy  $|t\bar{t} - 1| < 8.3 \times 10^{-40}$  on all 15 pairs ( $|\chi_4|^2 = 1$  exactly vs one-sided  $\chi_4^2 = e(2/3) \neq 1$ ), the doubled 256-dim system closes *strictly* at trivial character and physical weight (nullspaces  $(28, 17) \geq 11$ ) containing the unphased diagonal *and* the physical columns  $w_b \otimes \bar{w}_b$ , while the hol $\otimes$ hol control is *empty*  $(0, 0)$ ; the forced unphased numerator then reproduces the v516 Okubo squares *levelwise* inside the v518 scaffold (exact 4-design on every level  $n \leq 8$ ,  $J$ -weighted spreads  $\sim 5 \times 10^{-41}$ ,  $\psi(\text{skeleton}) = +0.2302 = -\psi(\text{contact})$  — the  $J$ -weighted mirror of  $\pm 64$ ), the declared reading also wins the column-canonicalisation discriminator (the physical AB column is contained *exactly* in the  $\chi_4$  family, canonicalising  $(N_1, N_2)$ , yet the canonical member fails every tester) and the  $\mathbb{Z}_2/\text{EH}$  anchor (all three derived instantiations fail; only the declared reading hits  $9\langle x, x \rangle^2 = (1/4) \times 36$  with the scale tooth  $c = 1/2 = |\mathbb{Z}_2|h^{A_1}$ ). Typed premises [C] (the mandatory fence): TP-1 (F-independent moduli integral), TP-2 (nonvanishing — it must source  $\psi = 64$ ), TP-3 (the v518 kernel convention), TP-4 (the Quillen structure of BCOV  $F_1$ ). The residual [O] was the constructive BCOV derivation of the  $w_m$  normalisation itself — the success branch (“the BCOV derivation lands on the declared completion reading, carrying the forced 4:3 leading ratio”) has fired at probe level, the kill branch (“it lands on the derived chiral measure”) is dead — and that derivation is since *executed* (verdict ERFOLG) as CELEST.WP5E.WM.01 [verification/v523\_celestial\_wp5e\_wm\_derived.py]: the  $1/\det_j$  weight is COMPUTED from three independent sources — the equivariant mode ledger of the fibre  $\mathbb{C}^2$  (closed form  $1/((1 - q i^j)(1 - q i^{-j}))$  exact at every order  $n \leq 120$ , regular at  $q = 1$  with Abel value  $(1/2, 1/4, 1/2) = 1/\det_j$ , fixed-point splitting at every truncation level for the phased AND the v520-forced unphased trace, so  $\text{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$  is computed, not declared), the zeta/Quillen route ( $\det_{\zeta} = 4 \sin^2(\pi j/4) = \det_j$  exactly with the unique real positive holomorphic section via the  $SU(2)$  conjugation pairing, and the  $\delta_{1f}$  block constant term  $1/\det_b$ ), and the consistency chain (the v516 numbers reproduced one by one:  $w = 4h$ ,  $c = 4 = |\mu_4|$ , squares  $(18, 9, 18)$ , total  $45\langle x, x \rangle^2$ ,  $\psi \pm 64$  — with the negative controls behaving  $(1/\det^2$  and  $1/|1 - i^j|$  break the chain quantifiably, the  $\mathbb{Z}_2/\text{EH}$  anchor passes at  $c = 2 = |\mathbb{Z}_2|$ , the  $SO(16)/D_8$  kill fires,  $\text{diag}(i, i)$  breaks the zeta = AB identity itself, the  $k = 3, 5$  weights wander correctly) and the typed premises TP-REG/TP-Q/TP-NUM/TP-CH as its own mandatory [C] fence; the residual [O] narrows to the *global* BCOV integral on  $\mathbb{PT}/\mathbb{Z}_4$  beyond the fibre

zero-mode factor (the resolved-ALE route of **v514/v517** not built); **(the cubic GS term — dissolved)** the  $\psi = 64$  slice is *delivered* by the declared KS measure (**v516**); the cubic  $d$ -channel is *not needed* ( $c_d$  free = 0) — its physical-justification question (the honest **v513** target  $c_d = 32 \times 60$ ) stays dissolved: the  $\delta_1$  decision (**v518**) does not revive the channel (no branch fires under the derived measure); a  $\text{ch}_2$ -type naturalness datum for *cubic* vertices still does not exist; **(the back-reaction milestones, preregistered as M1–M3 in v514; all three executed — the v514 fence is fully worked off)** the type-I B-model *back-reaction* on  $\mathbb{PT}/\mathbb{Z}_4$ , which would upgrade the  $\varepsilon_2$  CPS dictionary from **[C]** to derived: **M1 (done, SUCCESS) [E]/[C]** the  $A_3$   $\Omega_N$  (the Beltrami/Kodaira–Spencer deformation sourced by the glue currents on the three lockstep spheres; preregistered success = a closed-form  $\Omega$ -deformation whose  $S^3/\mathbb{Z}_4$  periods are  $(2\pi i)^2 \times$  an integer flux vector with the lockstep phases  $(i-1)(1, i, i^2)$ ; kill = unequal sphere fluxes or a non-integer period — then the CPS transplant *and* the  $\varepsilon_2$  level dial die) — executed and machine-verified as CELEST.WP5E.M1.01 [verification/v515\_celestial\_wp5e\_m1\_omega\_n.py] (exact sympy, 30 checks): the residue form  $\omega_2 = dX \wedge dZ/X$  is *derived* (three ambient routes; orbifold-cover factor  $4 = |\mu_4|$ ; clock phase  $i$ ), the twistor family closes as  $XY = Z^4 + 4t_0^2 \lambda^2 Z^2 - t_0^4 (1 - \lambda^4)^2$  ( $e_1 = e_3 = 0$  identically; seam fibre  $\lambda = 0$  = the **v493** family) with the CY-compatible clock lift  $(Z, \lambda) \mapsto (iZ, -i\lambda)$ ,  $\Omega \rightarrow +\Omega$ , and  $\Omega_N = \Omega_0 + \sum_p N_p K_p$  is *closed form* (CPS/Bochner–Martinelli kernels on the four centre twistor lines) with every  $S^3/\mathbb{Z}_4$  period  $(2\pi i)^2$ -integral, the seam-fibre phase vector  $2\pi i t_0 (i-1)(1, i, i^2)$  with clock covariance  $\Pi \rightarrow i\Pi$ , the flux vector  $N(1, 1, 1, 1)$  forced uniform (clock orbit +  $K_4$  connectivity) and the lens geometry *forcing* the source charge  $4 = |\mu_4|$  ( $N \in 4\mathbb{Z}$ ); the 12 conifold nodes sit exactly on the 8 eighth roots of unity ( $8 = 2|\mu_4|$ ); honest fence: on the clock-forbidden  $Z^4 - Z$  family the  $(2\pi i)^2$  quantisation *also* holds — integrality alone is *not* the discriminator, the discriminator is the lockstep phase structure ( $0/24$  vs  $8/24$ ) plus the clock forcing of equal fluxes; neither kill fired; **M2 (done, SUCCESS on the declared completion measure — verdict B) [E]/[C]** the twisted KS measure (quantise the  $O(-2)$  tower on  $\mathbb{PT}/\mathbb{Z}_4$  with the  $\varepsilon_1$  mode ledger and compute the one-loop box coefficient per sector; preregistered success = the bulk sector reproduces  $\tilde{\lambda}^2 = 36$  *and* the twisted sectors pair with the three sphere axions cancelling the rigid  $32 T_3$  residual; kill = a leftover independent  $T_3$ -type quartic in any sector) — executed and machine-verified as CELEST.WP5E.M2.01 [verification/v516\_celestial\_wp5e\_m2\_twisted\_ks\_measure.py] (exact sympy/Fraction, 23 checks): the *declared* completion contact term  $\text{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$  (the twisted-sector loop contributes the *unphased* sector trace with the  $g^j$  zero-mode normalisation) carries the exact completion-weight identity  $w_m = \sum_j (1 - i^{jm})/\det_j = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = |\mu_4|h_m = -4\text{ch}_2(T_m)$  — the three sphere axions pair through their *own* McKay  $\text{ch}_2$  charges, *no free scale, no fit*; the locks are parameter-free ( $T_5 = 0$  for *any* scale, ratio  $h_2:h_1 = 4:3$  = the  $\delta_1$ -forced leading ratio *reproduced*, the  $T_3$  budget forcing  $c = 4 = |\mu_4|$  uniquely); every twisted channel becomes the *same* perfect Okubo square  $36\langle x, x \rangle^2/\det_j$  ( $T_5 = T_3 = 0$  in every sector — the KILL does not fire), the total is  $45\langle x, x \rangle^2 = \frac{5}{4} \times 36\langle x, x \rangle^2 = \text{Dedekind} \times \text{Okubo}$  = the unique quartic-free weighting of the  $\beta$ -stage rigidity theorem; both  $\gamma$ -stage certificates are killed ( $\Phi_{T_3} : 32 \rightarrow 0$ ,  $\Phi_P : 72 \rightarrow 0$ ) and the  $\delta_2$  slice  $\psi = 64$  is *supplied exactly* — no cubic  $d$ -channel needed ( $c_d$  free = 0); controls: wrong scale leaves  $T_3 = 32 - 8c$ , the shuffle breaks  $T_5$  and  $T_3$ ,  $SO(16)$  keeps  $T_5 = 20/T_3 = -40$  (the KILL fires there:  $E_8$  doubly special),  $\text{diag}(i, i)$  degenerates, the  $\mathbb{Z}_2/\text{EH}$  anchor passes at scale  $2 = |\mathbb{Z}_2|$ . Mandatory fence **[C]**: the completion reading (the loop of twisted states carries the *unphased* sector trace with the same zero-mode normalisation) is *declared*, supported by the  $\delta_1$  modular-completion finding, *not* derived from the BCOV integral within that module — the  $\delta_{1b}/\delta_{1c}/\delta_{1d}$  chain has since been executed and decided by **v518** (the *derived* chiral measure fails all three testers), the declared-vs-derived measure question

has since been decided at probe level by **v520** for the declared reading (typed premises TP-1..TP-4), and the  $w_m$  normalisation has since been derived at probe level by **v523** (the residual **[O]** narrowing to the global BCOV integral beyond the fibre zero-mode factor); **M3 (done, SUCCESS)** **[E]/[C]** the  $a_0$  uplift (lift the  $(4, \pm 4)$  GH multipole to the Beltrami differential and compute the induced Kähler-potential correction; preregistered success = a log-type correction with coefficient tied to the source charge  $4 = |\mu_4|$ ; kill = decoupling from the centre count) — executed and machine-verified as **CELEST.WP5E.M3.01** **[verification/v517\_celestial\_wp5e\_m3\_a0\_uplift.py]** (exact sympy, 23 checks): the uplift object is the generalized-Legendre-transform kernel  $\chi = \log P_4$  — the *log of the M1 family polynomial* — well-typed (the  $O(2)$  section is a null coordinate: *any* kernel is harmonic) and matching the  $V$ -ledger through the exact residue identity  $\partial_x[\text{transform}(\log \eta_p)] = 1/r_p$  (flux  $-4\pi$  per centre, source charge  $4 = |\mu_4|$ ); the log-type correction arrives on *four coupled centre-count scales*: (i) the asymptotic kernel  $\log \chi = 4 \log \eta + a_2/\eta^2 + (a_0 - a_2^2/2)/\eta^4 + \dots$  with coefficient  $4 = |\mu_4|$ , the seam-fibre first correction *exactly*  $a_0/\eta^4$  = the  $(4, \pm 4)$  multipole (exact  $m$ -grading, no  $\log \times$  power terms — the  $\varepsilon_1$  honest zero preserved); (ii) the GLT tower  $p_{4k} = 4(-a_0)^k$  with the  $n \equiv 0 \pmod 4$  selection rule; (iii) the exceptional-locus  $\log \chi(0) = \log a_0 = 4 \log t_0 + i(4\varphi_0 + \pi)$  (response 1 at the locus vs power-law at infinity); (iv) the period response  $d \log \Pi_j / d \log a_0 = 1/4 = 1/|\mu_4|$  uniformly, integrating to the monodromy  $i$  = one Coxeter clock step (the WP2 monodromy reproduced from perturbation theory), with  $a_0$ -rigid  $\mathbb{Z}_8$  node support and topological  $(2\pi i)^2$  fluxes; controls: the  $(4, 0)$  multipole is clock-invariant, the  $\mathbb{Z}_2/\text{EH}$  analogue reads  $2 = |\mathbb{Z}_2|$  on *every* dial,  $k = 3, 5$  orbits move the coefficient with the centre count ( $O(6)/O(10)$  slots), the forbidden family fails both dials ( $p_3 = 3, e_4 = 0$ ) — the KILL (decoupling) does not fire; fence **[C]**: the GLT dictionary ( $\eta_p \log \eta_p$  = the GH Kähler-potential kernel, Lindström–Roček / Ivanov–Roček); **[O]**: the full nonlinear Kähler potential of the resolved  $A_3$  ALE. Success (unchanged): partition function =  $E_4/\eta^8$  *including* the  $q^{-1/3}$  prefactor *derived from the twistor side*; kill: an anomaly mismatch at the Costello–Li grade — neither the CFT-side dials of **v502**, nor the equivariant skeleton of **v505**, nor the exchange no-go of **v508**, nor the flux/sector dials of **v509**, nor the full-tensor ledger of **v511**, nor the axion-slot construction of **v514**, nor the M1–M3 back-reaction milestones of **v515/v516/v517**, nor the  $\delta_1$  decision of **v518**, nor the measure decision of **v520** trigger it (all computable dials say  $k = 1$ , “one level” is now a theorem, the bulk axion survives the projection, the back-reacted  $\Omega_N$  carries the forced integral lockstep flux, the twisted KS measure lands on the unique quartic-free weighting, the  $a_0$  uplift couples to the centre count, the  $\delta_1$  kill is a statement about the derived measure, not about the algebra, and the measure decision selects the declared reading at probe level), but the global quantisation is untested.

Honest status line: WP5a–WP5d (both WP5d stages), the WP5e  $\alpha, \beta, \gamma, \delta_1, \delta_2, \varepsilon_2$  and  $\varepsilon_1$  stages and the M1–M3 back-reaction milestones are exact-arithmetic **[E]** constructions (plus ED-validated lattice witnesses in WP5d) with **[C]** named identifications (“radial decoupling = collar thickness  $\rightarrow 0$ ”; Weyl complete reducibility beyond BFS depth 4; the Kac mechanics beyond level 2; the quasi-free extension; the KLM/Longo–Xu/CHMP index dictionary; Longo split heredity; Kac generation of the maximal submodule for  $k \geq 3$ ; the WP5e- $\beta$  dictionaries —  $\text{CS}(\rho) \pmod 1$  = discriminant weight,  $32 = (h, h) + (h', h')$ ,  $-30 = -h^\vee$ , the  $\Omega$ -contraction; the WP5e- $\gamma$  number fences  $32 = 20 + 12$ ,  $72 = 2\lambda_{e_8}^2$ ; the WP5e- $\delta_2$  number fences  $64 = \dim \mathfrak{g}_1$ ,  $1920 = |W(D_5)|$  — the latter typed look-elsewhere-loaded and convention-contingent by **v513**; the WP5e- $\varepsilon_2$  CPS dictionary on  $\mathbb{P}^T/\mathbb{Z}_4$ ; the WP5e- $\varepsilon_1$  conditionality on Costello’s flat- $\mathbb{P}^T$  matching; the M1 global kernel patching, Hitchin small resolution and CPS brane dictionary, **v515**; the M2 *declared* completion reading of the twisted KS measure, **v516**; the M3 GLT dictionary, **v517**); *WP5a–WP5d plus WP5e- $\alpha/\beta/\gamma/\delta_1/\delta_2/\varepsilon_2/\varepsilon_1$  plus M1–M3 are landed — all three KLM ingredients of complete rationality carry lattice witnesses, both faces of the inflow*

are anchored, the exchange sub-branch is closed with a certificate, the collapse is confirmed full-tensorially with one cubic door open, “one level” is now a theorem of clock invariance with the sector-counter dial pinning  $k = 1$ , the bulk-axion slot is a construction with  $\bar{\lambda} = 6$  triply pinned, the back-reacted  $\Omega_N$  is closed-form with  $(2\pi i)^2$ -integral lockstep periods and the source charge  $4 = |\mu_4|$  forced, the twisted KS measure (on the declared completion reading) cancels the  $32T_3$  residual and supplies the  $\psi = 64$  slice without the cubic  $d$ -channel, the  $a_0$  uplift couples to the centre count on four scales, and the  $\delta_1$  chain is decided — kill under the derived measure, tension with the declared reading stated — WP5e proper (the global BCOV quantisation) is the single remaining open piece, with the continuum uplift of the WP5d lattice witnesses honestly fenced as Xu’s theorem (cited, not claimed); the exchange sub-branch is closed by v508, the full-tensor ledger executed by v511, the level-from-flux dial executed by v509, the axion slot executed by v514, the M1–M3 back-reaction milestones executed by v515/v516/v517 (the **v514** fence fully worked off), the  $\delta_1$  chain decided by v518, the measure question decided at probe level by v520 (the declared completion reading wins: single-valuedness derived from F-independence + the Quillen pairing under TP-1..TP-4; the derived chiral measure was never a well-defined nonvanishing moduli integral), the  $w_m$  normalisation itself derived constructively at probe level by v523 (Atiyah–Bott mode ledger + zeta/Quillen determinant + consistency chain, typed premises TP-REG/TP-Q/TP-NUM/TP-CH), and the named remaining target is the global BCOV integral beyond the fibre zero-mode factor (the cubic-GS-term question stays dissolved: **v518** does not revive the  $d$ -channel) — SEAM.EQUIV.01 stays **[O]** and nothing in WP5a–WP5e-M3, the  $\delta_1$  decision or the measure decision moves it.

**Kill tests (pre-registered).** **(K1)** the equivariant sector could have *contradicted* the  $52+64+60+64$  split (uniformity of  $D[d]$  needs exactly  $\dim \mathfrak{g}_j = (60, 64, 60, 64)$ ) — *survived by WP1 and WP2*: the false-glue control shows the test has teeth (3112/6720 violations), and WP2 adds the deformation side (a shape modulus surviving clock-invariance, or  $a_0$ -corrections leaking between  $\mu_4$  sectors, would have fired — neither occurred; the clock-variant controls fail in two independent ways each). **(K2)** if one-loop associativity of the celestial  $E_8$  theory demanded extra fields that TFPT does not possess, the contract dies (Costello’s axionic mechanism is the first pass:  $E_8$  is admissible; the TFPT-specific field content is the open half). **(K3)** — *evaluated by WP3* ([`verification/v495_celestial_wp3_gs_alignment.py`]): the trigger (“no exact alignment between the GS coefficient and the  $c_3$  normalisation”) does *not* fire — exact alignment exists ( $(\kappa/c_3)^2 = 12$  resp.  $1/300$ ). But the look-elsewhere battery *demotes the scope of K3 itself*: the alignment format passes 8/8 across Costello’s whole list, so surviving K3 certifies compatibility plus  $\lambda$ -arithmetic, *not*  $\mathfrak{e}_8$ -selective evidence — alignment survives, selectivity does not. **(K4)** — *evaluated by WP4* ([`verification/v496_celestial_wp4_character_shadow.py`]): K4 fires only against the *exact*-block reading (the character is provably not a conformal block in the jet grading); it does *not* degrade the contract to “ $E_8$  is an admissible gauge algebra for celestial SDYM”, because the sector arithmetic holds exactly — WP4 lands at the boundary-limit reading (verdict B(ii), the SEAM.EQUIV.01 scaling-limit shape), with the constructive limit handed to WP5 — and made precise there by WP5a ([`verification/v497_celestial_wp5a_null_ideal.py`]: the coefficientwise  $\chi_w$  limit with threshold  $w = n+1$  and the derived ideal), with the deleting operator itself constructed by WP5b ([`verification/v498_celestial_wp5b_singular_vector.py`]:  $|s\rangle$  explicit, level dial  $2(1-k)$ ), the limit state by WP5c ([`verification/v500_celestial_wp5c_gns_limit_state.py`]: the ideal in the GNS kernel, level-2 rank 4124 exactly), the two-interval index chain by WP5d- $\alpha$  ([`verification/v501_celestial_wp5d_two_interval_index.py`]:  $\mu$ -chain  $16 \rightarrow 4 \rightarrow 1$  anchored at both measured ends), the split + strong-additivity legs by WP5d- $\beta$  ([`verification/v504_celestial_wp5d_klm_completeness.py`]: all three KLM ingredients of complete rationality witnessed on the lattice; the continuum uplift is Xu’s theorem, cited not claimed), the CFT-side anchors of the uplift itself by WP5e- $\alpha$  ([`verification/v502_celestial_wp5e_prefactor_level.py`]: the  $q^{-1/3}$  prefactor as exact  $\mu_4$  vacuum-energy bookkeeping,  $k = 1$  forced three ways), the equivariant twistor skeleton by WP5e- $\beta$  ([`verification/v505_celestial_wp5e_beta_equivariant_ledger.py`]:

the index bridge  $f(m) = \text{ch}_2(T_m)$  exact, glue defect  $-78$  two routes, geometric  $k = 1$  dials, the rigid  $32T_3$  residual, and the honest  $a_0$  refutation — the kill branch does not fire on the skeleton), the exchange no-go by WP5e- $\gamma$  ([[verification/v508\\_celestial\\_wp5e\\_gamma\\_pairing.py](#)]: the sphere-axion pairing by quadratic-vertex exchange refuted with the three-functional certificate  $(0, 32, 72)$  — the  $32T_3$  burden moves to the twisted contact terms,  $\delta_1$  binary), the full-tensor ledger by WP5e- $\delta_2$  ([[verification/v511\\_celestial\\_wp5e\\_delta2\\_tensor\\_ledger.py](#)]: the collapse confirmed full-tensorially by the innerness theorem, the cubic  $d$ -symbol channel opening  $T_3$  with the weaker certificate  $\psi(A_{\text{fix}}) = 64$  and the relaxed solution  $c_d = 1920Q_{dd}$ -coefficient — typed by the v513 negative certificate as  $c_d = 32 \times 60$  with the  $|W(D_5)|$  reading look-elsewhere-loaded), and the level-from-flux dial by WP5e- $\varepsilon_2$  ([[verification/v509\\_celestial\\_wp5e\\_eps2\\_level\\_from\\_flux.py](#)]: the lockstep theorem — “one level” as a theorem of clock invariance, falsifier  $Z^4 - Z$  at  $0/24$  — plus the sector-counter dial  $\#\text{prim}((E_8)_k) = 1 \Leftrightarrow k = 1$ ), the bulk-axion slot by WP5e- $\varepsilon_1$  (v514: the equivariant  $O(-2)$  slot construction,  $\tilde{\lambda} = 6$  triply pinned, the first  $\text{GH}/A_3$  back-reaction step), the back-reacted  $\Omega_N$  by the M1 milestone (v515: closed form,  $(2\pi i)^2$ -integral lockstep periods, forced source charge  $4 = |\mu_4|$ ), the twisted KS measure by the M2 milestone ([[verification/v516\\_celestial\\_wp5e\\_m2\\_twisted\\_ks\\_measure.py](#)]: the completion-weight identity  $w = 4h = -4\text{ch}_2$ , per-sector Okubo squares, the  $32T_3$  cancelled and the  $\psi = 64$  slice supplied on the declared completion reading — the M2 kill did not fire) and the  $a_0$  uplift by the M3 milestone ([[verification/v517\\_celestial\\_wp5e\\_m3\\_a0\\_uplift.py](#)]: the log-type correction coupled to the centre count on four scales, monodromy  $i$  — the M3 kill did not fire), and the  $\delta_1$  chain decided by the consolidated  $\delta_1$  module ([[verification/v518\\_celestial\\_wp5e\\_delta1\\_derived\\_kill.py](#)]: the derived chiral measure fails all three preregistered testers — kill under the derived measure, the declared/derived tension stated), and the measure question decided at probe level by the consolidated  $\delta_{1e}/\delta_{1f}$  module (v520: single-valuedness derived from two constructive sources, the declared reading wins under TP-1..TP-4), and the  $w_m$  normalisation derived constructively by the WM module ([[verification/v523\\_celestial\\_wp5e\\_wm\\_derived.py](#)]: the  $1/\det$ ; fixed-point weight computed from the Atiyah–Bott mode ledger, the zeta/Quillen determinant and the consistency chain — the WM kill did not fire), leaving the global BCOV quantisation (WP5e proper, narrowed to the global BCOV integral beyond the fibre zero-mode factor) as the single remaining open piece.

**Honest success estimate.** The probability of *full* success (WP5e: a second continuum route to SEAM.EQUIV.01) remains low — it is constructive holography at the Costello–Li grade. The *partial*-findings expectation is now realised and *exceeded*: WP1–WP4, all five WP5 constructive milestones (both WP5d stages), the WP5e  $\alpha$ ,  $\beta$ ,  $\gamma$ ,  $\delta_1$ ,  $\delta_2$ ,  $\varepsilon_2$  and  $\varepsilon_1$  stages and the M1–M3 back-reaction milestones are executed with sharp structural results (verdicts B, B, B, B(ii); WP5a–WP5d passed; WP5e- $\alpha$  pins the CFT side, WP5e- $\beta$  the equivariant twistor skeleton, WP5e- $\gamma$  closes the exchange sub-branch with a certificate, WP5e- $\delta_2$  confirms the collapse full-tensorially and opens the one cubic door, WP5e- $\varepsilon_2$  makes “one level” a theorem and pins its value by the sector counter, WP5e- $\varepsilon_1$  builds the bulk-axion slot with  $\tilde{\lambda} = 6$  triply pinned, M1 delivers the closed-form back-reacted  $\Omega_N$  with forced charge  $4 = |\mu_4|$ , M2 delivers the twisted KS measure on the declared completion reading (the  $32T_3$  cancelled, the  $\psi = 64$  slice supplied, no cubic  $d$ -channel), M3 delivers the  $a_0$  uplift coupled to the centre count on four scales, and the  $\delta_1$  decision (v518) closes the chain with an honest kill under the derived measure, the measure decision (v520) selects the declared reading at probe level, and the  $w_m$  derivation (v523) computes the fixed-point weight from three independent sources; *WP5e proper* — the global BCOV quantisation, narrowed to the global BCOV integral beyond the fibre zero-mode factor — is the single remaining open piece) — the WP1 critical correction (clock  $\neq$  deck;  $\text{clock}^2 = \text{deck}$ ), the WP2 sharpenings (the clock *is* the Picard–Lefschetz/Coxeter monodromy of the seam deformation;  $a_0$  pure scale with the  $\tau=i$  shape frozen), the WP3 non-selective alignment (exact  $\lambda$ -arithmetic, honestly typed by the 8/8 look-elsewhere table), the WP4 boundary-limit localisation of the character mismatch (three exactly-located obstructions, one surviving limit structure), the WP5a null-ideal derivation (stabilisation threshold  $w = n+1$ ;  $27000 = (h^\vee)^3$  derived; quotient = sector sum = 4124;  $SO(16)_1$  contrast), the WP5b singular-vector construction (the deletion operator explicit, case tally  $190/57/1$ , level dial  $2(1-k)$ , one-block closure typed  $E_8/\mu_4$ -specific), the WP5c

limit state (the ideal in the GNS kernel, the full 9361-block Gram at rank 4124, the family proved *necessary* by a CCR obstruction), the WP5d- $\alpha$  two-interval index (the  $\ln 2$  plateau at  $10^{-15}$ , the  $\mu$ -chain  $16 \rightarrow 4 \rightarrow 1$  measured at both ends), the WP5d- $\beta$  KLM completion (strong additivity exact on the lattice with the shared boundary point; the bounded-vs-divergent entropic discriminator with the Ising  $\frac{1}{4}$ -exponent approach against the diverging  $U(1)$  control; the elliptic-nome split ladder with exact orbifold inheritance; Pimsner–Popa  $\lambda = \frac{1}{2}$  and  $\lambda_{E_4} = \frac{1}{4} = 1/\mu$  with integer attainment — all three KLM ingredients witnessed, Xu’s continuum theorem cited not claimed), the WP5e- $\alpha$  prefactor/level pinning (the  $q^{-1/3}$  prefactor as exact  $\mu_4$  vacuum bookkeeping;  $k = 1$  forced three ways; the glue-integrality route honestly retired), the WP5e- $\beta$  equivariant ledger (the index bridge  $f(m) = \text{ch}_2(T_m)$  exact with the  $-78$  defect; the rigid  $32 T_3$  twisted residual; the honest  $a_0$  refutation with the three sphere classes filling the three twisted axion slots), the WP5e- $\gamma$  exchange no-go (the invariant vertex spaces exactly  $\dim 2/5$ ; the product theorem; the certificate  $(0, 32, 72)$ ; naturalness dissolved scale-independently — an honest negative result that converts the pairing question into the binary  $\delta_1$  contact-term test), the WP5e- $\delta_2$  full-tensor ledger (the innerness theorem confirming the collapse across arities; the unique symmetric survivor = the  $\mathfrak{su}(4)$   $d$ -symbol with  $Q_{dd} = \frac{1}{60}(T_3 - P_3/4)$  opening the  $T_3$  channel; the weaker certificate  $\psi(A_{\text{fix}}) = 64$  and the exactly solvable relaxed reading with  $c_d = 32 \times 60$ , the  $1920 = |W(D_5)|$  fingerprint honestly retired as look-elsewhere-loaded by the v513 negative certificate), and the WP5e- $\varepsilon_2$  level-from-flux dial (the CPS skeleton exact; the pairing matrix pinned by complete enumeration with fluxes  $(64, 60, 64)$  and one quantum per current; the naive “level = total flux” killed by the lockstep test itself; the lockstep theorem with its  $Z^4 - Z$  falsifier; the sector counter  $\#\text{prim}((E_8)_k) = 1 \Leftrightarrow k = 1$ ), the WP5e- $\varepsilon_1$  axion slot (the equivariant Penrose ledger closing block by block; the  $d = 0$  slot surviving the projection;  $\tilde{\lambda} = 6$  triply pinned with the wrong bookings 3 and 12 excluded; the GH/ $A_3$  step re-deriving the v493 family and the Coxeter clock from geometry), the M1 back-reacted  $\Omega_N$  (the residue form derived with cover factor  $4 = |\mu_4|$ ; the closed twistor family with its CY-compatible clock lift; every  $S^3/\mathbb{Z}_4$  period  $(2\pi i)^2$ -integral with the forced uniform flux vector and the lens-forced charge  $4 = |\mu_4|$ ; the honest integrality-is-not-the-discriminator fence), the M2 twisted KS measure (the completion-weight identity  $w = 4h = -4\text{ch}_2$  with no free scale; every twisted channel the same perfect Okubo square; both  $\gamma$ -stage certificates and the  $\delta_2$   $\psi = 64$  slice supplied exactly; the declared completion reading honestly fenced [C], v516) and the M3  $a_0$  uplift (the GLT kernel  $\chi = \log P_4$  with the log coefficient  $4 = |\mu_4|$  on four coupled centre-count scales, the period response  $1/4$  integrating to the Coxeter monodromy  $i$ , v517) and the  $\delta_1$  decision (the 16-component Weil completion closing exactly with the E1.5 residual  $2.91 \rightarrow \sim 10^{-39}$ ; the  $\mu_4$  multiplier obstruction identified as a *character* of the orbit stabilisers by the exact koboundary test; the cancellation achieved by the twisted fibre block  $f_1 f_3 = G$  — not the three sphere axions; all three preregistered testers failing under both derived solutions with the honest  $(N_1, N_2)$  rebalance rescuing nothing; the declared/derived measure tension stated as the mandatory fence, v518) and the measure decision (the exact invariant subspace =  $\text{span}\{e_H, e_{H'}\}$  forcing the unphased  $E_4$  numerator once single-valuedness is derived from F-independence + the Quillen pairing; the consistency circle to v516 closed levelwise; the typed premises TP-1..TP-4 named, v520) — exactly the kind of sharp, falsifiable refinement the contract format is for. Status: CELEST.SEAM.01 [O] (research contract, WP5e proper the single remaining open milestone); CELEST.WP1.01 [E]/[C] (executed); CELEST.WP2.01 [E]/[C] (executed); CELEST.WP3.01 [E]/[C] (executed); CELEST.WP4.01 [E]/[C] (executed); CELEST.WP5A.01 [E]/[C] (executed); CELEST.WP5B.01 [E]/[C] (executed); CELEST.WP5C.01 [E]/[C] (executed); CELEST.WP5D.01 [E]/[C] (executed,  $\alpha$  stage); CELEST.WP5DB.01 [E]/[C] (executed,  $\beta$  stage — the continuum uplift stays [O], Xu’s theorem cited); CELEST.WP5E.ALPHA.01 [E]/[C] (executed,  $\alpha$  stage — the CFT-side prefactor + level pinning); CELEST.WP5E.BETA.01 [E]/[C] (executed,  $\beta$  stage — the equivariant anomaly ledger; the global BCOV quantisation itself stays [O]); CELEST.WP5E.GAMMA.01 [E]/[C] (executed,  $\gamma$  stage — the exchange no-go with certificate; the  $\delta_2$  ledger since executed by v511, the twisted contact terms  $\delta_1$  since decided by v518 — kill under the derived measure, tension with v516 stated); CELEST.WP5E.DELTA2.01 [E]/[C] (executed,  $\delta_2$  stage — the full-tensor ledger, intermediate verdict: collapse confirmed by innerness, the cubic  $d$ -symbol channel opens  $T_3$  with

the weaker certificate  $\psi = 64$ ; the arity  $\geq 4$  ledger stays **[O]**; the  $\psi = 64$  burden since discharged by M2/v516 — the cubic-GS question stays dissolved, the  $\delta_1$  decision (v518) does not revive it); CELEST.DTERM.NONDERIV.01 **[E]** (executed, the  $c_d$  negative certificate — the  $1920 = |W(D_5)|$  fence typed look-elsewhere-loaded and convention-contingent, the convention-stable core  $c_d = 32 \times 60$ ; the fence stays **[C]**, the generation of the 32 stays **[O]**); CELEST.WP5E.EPS2.01 **[E]/[C]** (executed,  $\varepsilon_2$  stage — the level-from-flux dial, verdict B: “one level” a theorem, the sector counter pins  $k = 1$ ; the CPS dictionary on  $\mathbb{PT}/\mathbb{Z}_4$  typed **[C]**, the type-I B-model back-reaction milestones M1–M3 all since executed via v515/v516/v517); CELEST.WP5E.EPS1.01 **[E]/[C]** (executed,  $\varepsilon_1$  stage — the  $O(-2)$  bulk-axion slot construction with  $\tilde{\lambda} = 6$  triply pinned and the first GH/ $A_3$  back-reaction step; the M1–M3 milestones it preregistered are all since executed; the BCOV-integral derivation stays **[O]**); CELEST.WP5E.M1.01 **[E]/[C]** (executed, back-reaction milestone M1, SUCCESS on the preregistered criterion — the closed-form  $\Omega_N$  with  $(2\pi i)^2$ -integral lockstep periods and the lens-forced source charge  $4 = |\mu_4|$ ; honest fence: integrality alone is not the discriminator; M2–M3 since executed via v516/v517); CELEST.WP5E.M2.01 **[E]/[C]** (executed, back-reaction milestone M2, SUCCESS on the preregistered criterion *on the declared completion measure* — the completion-weight identity  $w = 4h = -4\text{ch}_2$  with no free scale, every twisted channel the same perfect Okubo square  $36\langle x, x \rangle^2 / \det_j$ , the  $32 T_3$  cancelled, both  $\gamma$ -stage certificates and the  $\psi = 64$  slice supplied exactly, no cubic  $d$ -channel needed; the completion reading stays **[C]**-declared within that module — the  $\delta_1$  chain since decided by v518, and the completion reading since *derived at probe level* from two independent constructive sources (F-independence + Quillen pairing) under the typed premises TP-1..TP-4 by v520; the residual **[O]** is the constructive  $w_m$  derivation); CELEST.WP5E.M3.01 **[E]/[C]** (executed, back-reaction milestone M3, SUCCESS on the preregistered criterion — the  $a_0$  uplift on the GLT kernel  $\chi = \log P_4$  with the log coefficient  $4 = |\mu_4|$  coupled to the centre count on four scales, the period response  $1/4$  integrating to the Coxeter monodromy  $i$ ; the GLT dictionary stays **[C]**, the full nonlinear Kähler potential stays **[O]**); CELEST.WP5E.DELTA1.01 **[E]/[C]** (executed,  $\delta_1$  stage — DECIDED: kill under the derived measure; the 16-component Weil completion closes exactly, the  $\mu_4$  multiplier obstruction is a character (koboundary defects  $(1, 1, 1)$ ,  $\lambda(\gamma) = i^{2B+C/4}$  on  $\Gamma_1(4)$ ), the cancellation is the twisted fibre block  $f_1 f_3 = G$ , and all three preregistered testers fail under both derived solutions; the  $f_1 f_3 = \text{KS-weight}$  identification and the kernel-family convention stay **[C]**; the declared-vs-derived BCOV measure question — the stated v516 tension — was the named **[O]** and is since decided at probe level by v520 for the declared reading, sharpening this kill; the residual **[O]** is since narrowed by v523 to the global BCOV integral beyond the fibre zero-mode factor); CELEST.WP5E.MEASURE.01 **[E]/[C]** (executed, the measure decision — ERFOLG-A on the preregistered decision layer: single-valuedness derived from two independent constructive sources (F-independence + the Quillen pairing), the invariant subspace exactly  $\text{span}\{e_H, e_{H'}\} = E_4$ , the consistency circle to v516 closed levelwise, the  $\mathbb{Z}_2/\text{EH}$  anchor and the column canonicalisation both selecting the declared reading; typed premises TP-1..TP-4 stay **[C]**; the constructive BCOV derivation of the  $w_m$  normalisation itself — the named residual **[O]** — is since executed by v523); CELEST.WP5E.WM.01 **[E]/[C]** (executed, the constructive  $w_m$  derivation — verdict ERFOLG per the frozen preregistration: the  $1/\det_j$  fixed-point weight COMPUTED from three independent sources (the equivariant mode ledger with Abel value  $(1/2, 1/4, 1/2)$ ; the zeta/Quillen determinant  $4 \sin^2(\pi j/4)$  with the unique real positive section; the  $\delta_{1f}$  block constant term  $1/\det_b$ ), the v516 chain reproduced number by number ( $w = 4h$ ,  $c = 4 = |\mu_4|$ , squares  $(18, 9, 18)$ , total  $45\langle x, x \rangle^2$ ,  $\psi \pm 64$ ), all negative controls behaving (the  $\mathbb{Z}_2/\text{EH}$  anchor at  $c = 2 = |\mathbb{Z}_2|$ , the  $SO(16)$  kill, the  $\text{diag}(i, i)$  break, the  $k = 3, 5$  wandering); typed premises TP-REG/TP-Q/TP-NUM/TP-CH stay **[C]**; the global BCOV integral beyond the fibre zero-mode factor stays **[O]**); SEAM.EQUIV.01 untouched, stays **[O]**.

## Research Contract — WOIT.OS.TWISTOR.01 (the Osterwalder–Schrader twistor bridge)

*Typing.* This is the actual bridge from compiler to physics — everything in CELEST.SEAM.01 is preparation for it. It is a research contract [O] (ledger row WOIT.OS.TWISTOR.01: Open, research contract), *not* a claim; nothing below asserts that the construction exists. The external programme it engages — Woit’s Euclidean Twistor Unification (arXiv:2104.05099) — is a *named reference frame* for the shape of the target, never a confirmation in either direction (the Woit-bridge paragraph in tfpt\_3 states exactly what is missing).

**Input.**  $X_+ = (\mathbb{PT}/\Gamma)_+$  (the positive-time half of the quotiented projective twistor space;  $\Gamma = \Gamma_{\text{ALE}} \cong \mathbb{Z}_4$ , normalised by the clock — the definition block of CELEST.SEAM.01);  $\mathcal{A}_{\text{hol}}$ , the *interacting* open+closed twistorial algebra (SDYM( $E_8$ ) open strings + BCOV closed strings — the WP5e-proper object, whose free/equivariant skeleton is pinned by v492–v517);  $\rho$ , the order-4 clock diag( $i, 1$ );  $\Theta$ , the anti-linear real structure induced by the seam reflection; and  $\mu_{\text{BCOV}+\text{SDYM}}$ , the interacting functional (the global BCOV+SDYM quantisation on  $\mathbb{PT}/\Gamma$ ). *Precision (i), from the  $\alpha$  stage:* “induced by the seam reflection” means the seam-circle *reflection*, not the deck/covering involution — the deck (whose free-ness is topology, v510) furnishes the  $(-1)^F$  Kramers class  $\Theta_t^2 = (-1)^F$ , not the OS  $\Theta$ ; the seam-circle reflection furnishes  $\Theta_{\text{Fock}}^2 = +1$  ([verification/v519\_woit\_theta\_rp\_free.py]). *Precision (ii):* on the RP side the  $\mu_4$  marks sit at the *bond midpoints* of the 16-Majorana seam circle (the half-offset/NS-natural placement, 4 sites per mark quadrant) — the cut through the sites fails reflection positivity exactly ([verification/v519\_woit\_theta\_rp\_free.py]). *Precision (iii), from the  $\beta_1$  stage:* “gauge-invariant subalgebra” at the free level means the GSO/fermion-parity  $\mathbb{Z}_2$  — the only  $\theta$ -compatible part of the  $\mu_4$  tower (the free-fermion shadow of the  $E_8$  glue:  $(E_8)_1 = \text{even NS of } SO(16)_1 + \text{R spinor}$ ); the clock is the euclidean rotation itself (*timelike*: all 16 seam mirror axes invert it, none commutes), and its equivariant statement moves to  $\beta_2$  — OS quotient first, then the clock as reconstructed transfer operator ([verification/v522\_woit\_beta1\_gso\_gauge.py]).

**Target theorem.**  $\Theta^2 = 1$  and  $\Theta\rho\Theta = \rho^{-1}$ ; the gauge-invariant algebra is reflection-positive; and the Osterwalder–Schrader quotient produces  $(\mathcal{H}, \Omega, U(\mathcal{P}_+^\uparrow), \mathcal{A}_{\text{Mink}})$  with: a positive Hilbert metric, positive energy, local causality, *one chiral fermion generation without mirror doubling*, the TFPT charge lattice, an internal  $SU(3) \times SU(2) \times U(1)$  action, and the  $(E_8)_1$  seam net as an *actual* boundary net (not only as a character).

**Kill tests (pre-registered, all seven live).** (1)  $\Theta$  is incompatible with the clock (no anti-linear  $\Theta$  with  $\Theta^2 = 1$  and  $\Theta\rho\Theta = \rho^{-1}$  exists on  $\mathcal{A}_{\text{hol}}$ ). (2) Reflection positivity fails after gauge fixing (the gauge-invariant subalgebra of  $\mu_{\text{BCOV}+\text{SDYM}}$  is not RP). *Named toy threat (no status change):* on the first genuinely interacting 16-Majorana Fidkowski–Kitaev seam toy (SEAM.INT.FKTOY.01, [verification/v529\_seam\_interacting\_toy\_fk.py]) Kill-Test 1 does *not* fire ( $\Theta$  exists with  $\Theta^2 = +1$  and clock-tower inversion), but Kill-Test 2 *fires at toy level* — RP breaks for every  $g > 0$  by an interference mechanism, with the *straddle law* (RP fails exactly on quartet-straddled cuts, stays PD on quartet-avoiding ones; 24/24) as a concrete hurdle for the interacting stages ( $\gamma$  and beyond; the free/quotient-level  $\beta_3$  is since executed, v565). Fence: one toy / one interaction class / [C] flat-band parent — this narrows the route (every  $\mathcal{A}_{\text{hol}}$  must protect RP against exactly this mechanism), it does not close it. *The hurdle, executed as a selector (SEAM.STRADDLE.CONE.01, [verification/v534\_seam\_straddle\_cone.py]; no status change):* on the equivariant interacting mark family (independent couplings on the four mark quartets, both signs, every admissible cut — including the  $\pi/2$  straddled clock axes the 24-entry law had not covered) reflection positivity survives for *exactly one* member  $\times$  sign:  $\delta = \pi/2$  with positive coupling, PD on all four cuts over  $g \in \{1/32..8\}$  with the minimum eigenvalue lifted *away* from the RP boundary; every asymmetric member dies. The straddle law refines (asymmetric straddling kills, symmetric straddling protects), the literal leading-order cone formalization is dead (the protection is nonperturbative), and the threat paragraph gains its complement: a passing interaction class *exists* and its unique survivor is the alignment bit — the same member that carries the v528 twist-class order parameter. Toy-level evidence for a dynamical origin of the bit, not a derivation; the kill tests stay formally live on  $\mathcal{A}_{\text{hol}}$ .

(3) The reconstruction produces a vector-like mirror generation (chirality is lost in the OS quotient). (4) The Penrose transform reaches only free / self-dual states (the interacting extension does not exist at the Costello–Li grade). (5) The reconstructed net has the right character but the wrong OPE / fails net equivalence (character-level agreement without operator-algebraic agreement). (6) The internal  $SU(2)$  remains a spacetime factor (no genuine internal weak factor separates from the Euclidean rotation group). (7) The four  $\mu_4$  marks are not incidence-compatibly extendable over spacetime (the mark data of the seam cannot be propagated through the incidence relation).

**The  $\alpha$  stage — executed (*done*; the free/equivariant milestone `WOIT.THETA.FREE.01`, [`verification/v519_woit_theta_rp_free.py`]).** The real structure *exists*, and free reflection positivity picks the *same* family. Exactly *two* families of anti-linear structures on  $\mathbb{C}^2$  normalise the clock: family D ( $z \mapsto \mu\bar{z}$ ,  $|\mu| = 1$ ) satisfies  $\Theta\rho\Theta = \rho^{-1}$  *exactly* with  $\Theta^2 = +1$  ( $-1$  impossible:  $MM^\dagger = \text{diag}(|\mu|^2, 1)$  — Kramers-free), inverts the deck and reflects the seam circle (2 cut points); family A ( $z \mapsto \mu/\bar{z}$ ) centralises the clock projectively and *never* inverts it — it *defines* the euclidean section (Woit’s  $\rho_{\text{tw}}$  side:  $\rho_{\text{tw}}^2 = -1$ , no real points, replicated exactly on  $\mathbb{C}^4$ ), while family D is the OS conjugation ( $\sigma_{\text{std}}$ , real points  $\mathbb{RP}^3$ ). Mark compatibility pins  $\mu \in \mu_4$  (a 4-element torsor, 2 clock orbits); the  $\mathbb{Z}_8$  spin plane has no phase leaks; in exact  $\text{Cl}(16)$ ,  $\Theta_{\text{Fock}} = U_r \circ K$  has  $\Theta_{\text{Fock}}^2 = 2^7 > 0$  (normalised  $+1$ ), inverts the Fock clock tower ( $V \mapsto 4096 V^{-1}$ ) and normalises the deck, while the deck-induced candidate has  $\Theta_t^2 = (-1)^F$  — the `v510` split/nonsplit dichotomy *is* the  $\Theta^2 = +1$  vs  $(-1)^F$  dichotomy. Free RP holds on the bond cut with *no* degree truncation (one-particle Gram  $(8, 0, 0)$  PD, even  $\deg \leq 2$  sector  $(29, 0, 0)$ , complete  $N = 8$  half-sided algebra PD; twist  $\eta = +i$  forced), the site cut and the clock-centralising family fail RP exactly, and the anti-chiral state flips the odd sector to negative definite (the free shadow of kill test 3). **Kill test (1) therefore does *not* fire at the free/equivariant level — and stays formally *live* on the interacting algebra  $\mathcal{A}_{\text{hol}}$** ; kill tests (2)–(7) are untouched. No marker moves: the contract stays **[O]**.

**The  $\beta_1$  stage — executed (*done*, verdict **UNDECIDED** per the frozen preregistration; the typed milestone `WOIT.BETA1.GSO.01`, [`verification/v522_woit_beta1_gso_gauge.py`]).** “The clock is time-like, GSO is the gauge datum.” The preregistered clock-tower reading of “gauge” (the  $\mu_4/\Gamma_{\text{ALE}}$  equivariance averaged *before* the OS quotient) is structurally the *wrong* operationalisation: the one-step clock insertion  $T_1 = \langle \theta(e_a), \alpha_S(e_b) \rangle$  violates Hermiticity *exactly* (witness entry  $-i/(8\sin(5\pi/16))$  where the conjugate transpose demands  $+i/(8\sin(5\pi/16))$ ); the pairing is complex-*symmetric*: 745 matching, 96 anti-matching, 0 violations — “OS-symmetric” is strictly weaker than Hermitian, so “positivity after gauge fixing” is not even well-posed for the clock reading). The mechanism is typed exactly: *all* 16 dihedral reflection axes of the NS seam circle invert the clock lift ( $RSR^{-1} = S^{-1}$ ), none commutes — the  $\mu_4$  clock *is* Woit’s euclidean rotation, and the  $\theta$ -compatible (gaugeable) part of its  $\mathbb{Z}_8$  Fock tower is exactly the 2-torsion  $\{1, (-1)^F\}$  = the GSO/fermion-parity  $\mathbb{Z}_2$  (Hermiticity census over all powers;  $\alpha^4$  = the  $2\pi$  rotation of the seam = fermion parity). Under that corrected typing **gauge-fixed RP holds exactly**: the parity-projected bond-cut Gram is  $(29, 0, 0)$  PD at  $N = 16$  ( $\deg \leq 2$ , min eigenvalue  $1.78 \times 10^{-6}$  at 40 digits) and  $(8, 0, 0)$  PD on the complete  $N = 8$  half algebra (no degree truncation); the site-cut defect *survives* gauge fixing  $((7, 9, 6)$  — the bond placement of precision (ii) is gauge-invariant information); family A stays indefinite  $((17, 12, 0)$  vs family D’s  $(29, 0, 0)$ : the  $\alpha$ -stage triangulation survives); the Ramond (wrap  $+1$ ) “true  $\mathbb{Z}_4$ ” lift fails state invariance (the NS/ $\mathbb{Z}_8$  tower is forced); and the Kramers class stays trivial on the invariant sector ( $\Theta_t^2 = \text{id}$  on  $\mathcal{A}^G$ ,  $\Theta$  restricts with  $\Theta^2 = +1$ ). **Kill test (2)’s free shadow does *not* fire** (no legitimate Hermitian gauge-fixed form has a negative direction), and kill test (1) stays discharged also gauge-invariantly; both stay formally live on  $\mathcal{A}_{\text{hol}}$ . Transparency (documented in the module): the first frozen run scored 6/13 — the clock-reading checks failed exactly as the mechanism predicts — and the preregistered  $\deg$ -2 clock-invariant dimension 30 was a combinatorial error (correct orbit census  $28 + 4 = 32$ ); the verdict follows the frozen logic (Hermiticity failure  $\Rightarrow$  structural-failure branch  $\Rightarrow$  UNDECIDED).  $\beta_1$  *proper* (the clock-equivariant statement) is re-routed through  $\beta_2$ . No marker moves: the contract stays **[O]**.

**The  $\beta_2$  stage — executed (*done*, verdict **SUCCESS** per the frozen preregis-**

tration, **[C]-typed per contract precision (iii)**; the milestone **WOIT.BETA2.OS.01**, [\[verification/v524\\_woit\\_beta2\\_os\\_quotient.py\]](#). “The OS quotient of the free system made explicit.”  $\mathcal{H}_{\text{phys}} = \mathcal{A}_+/\mathcal{N}$  is *explicit and nondegenerate* at both levels: at  $N = 16$  ( $\deg \leq 2$ ) the  $37 \times 37$  bond-cut OS Gram is exactly Hermitian and parity-block-diagonal with inertia  $(37, 0, 0)$  PD (min eigenvalue  $1.7801 \times 10^{-6}$  at 40 digits) and null space  $\{0\}$ ,  $\dim 37 = 29 \oplus 8$ ; at  $N = 8$  the *complete* half algebra gives  $(16, 0, 0)$  PD with  $\dim 16 = 8 \oplus 8 = 4^2$  — compact euclidean time reconstructs a *thermal* (KMS) representation, with the exact certificate  $\sin^2(3\pi/8) - \sin(\pi/8)\sin(5\pi/8) = 1/2$ . The euclidean rotation becomes a Klein–Landau *local symmetric semigroup*:  $\tau_k = \omega(\theta(a)\alpha_k(b))$  is exactly Hermitian on every shrinking domain  $D_k$  ( $k = 1..7$ , dims  $29, 22, 16, 11, 7, 4, 2$ ), the chain identities are exact and the vacuum is fixed; the positivity pattern *is* the  $\alpha$ -stage site/bond dichotomy (even steps PSD via the exact square identity  $T(2j) = A_j^* A_j$ , odd steps indefinite with exactly zero one-particle diagonal — the one-step transfer is *not* positive: the free chirality datum, kill-test-3 shadow sharpened). **The clock is implemented on  $\mathcal{H}_{\text{phys}}$  as the positive self-adjoint transfer step  $T^{N/4}$**  ( $(11, 0, 0)$  PD, min  $6.7 \times 10^{-5}$ ; compressed spectrum  $[0.0195, 1.6414]$  with vacuum eigenvalue exactly 1; spectral projections certified at  $\sim 10^{-40}$  — exactly the calculus the non-Hermitian pre-quotient average of v522 could not have), and at  $N = 8$  the compressed clock spectrum is *exactly*  $\{1, \sqrt{2} - 1\} = \{1, 1/\delta_S\}$  in both parity sectors (the silver axes of the v519  $\mu_4$  torsor returning as the clock eigenvalue); the reconstructed rotation group  $U(s) = e^{isH}$  is unitary with the group law — per precision (iii) the **[C]**-operationalisation of the contract phrase “the clock acting unitarily”. The v522 non-Hermiticity is *resolved*: the census  $(745, 96, 0)$  is reproduced, every anti-matching entry is a wrap overlap and every overlap-free entry matches — the pre-quotient failure was exactly the domain/wrap artifact. The pre-declared KMS deviation arrives with exactly the declared witnesses: *no contraction* on the compact circle (edge-mode expansion  $C(1)/C(3) = 1 + \sqrt{2} = \delta_S$  exactly;  $\det(G - \tau_4) < 0$ ) — a typed KMS finding, not a  $\beta_2$  failure (the contract demands no contraction). GSO/ $\Theta$ : the grading survives;  $\theta_{\text{cut}}$  and the deck candidate do *not* act on  $\mathcal{H}_{\text{phys}}$  (support censuses); the *perpendicular* torsor mirror descends anti-unitarily ( $\eta_{\perp} \in \{\pm i\}$ ) with  $\Theta_{\text{phys}}^2 = +1$  on *every* sector — the quotient is Kramers-free — and  $\theta_{\text{cut}} \circ \theta_{\perp} = \alpha_{N/2}$  exactly (the torsor closes onto the deck). Controls: the site cut is indefinite  $((3, 3, 1)/(2, 2, 4)$  — the contract kill branch fires for the site placement), family A admits no quotient  $((4, 4, 0))$ , the anti-chiral state flips to  $(8, 8, 0)$  (quotient existence itself selects the chiral orientation), the wrong semigroup direction is non-Hermitian exactly off-domain (all mismatches wrap-overlap). **Kill tests (1) and (2) are strengthened and the (3) shadow is sharpened; none fires** — all seven stay formally live on  $\mathcal{A}_{\text{hol}}$ . Transparency (documented in the module): run 1 scored 18/20 — a sympy simplify failure on a *true*  $\pi/16$  product formula ( $\sin(7\pi/16) - \sin(5\pi/16) = 2 \cos(3\pi/8) \sin(\pi/16)$ ), fixed by a Laurent-polynomial certificate; no criterion, tolerance or expectation changed. No marker moves: the contract stays **[O]**;  $\beta_3$  is executed next (below).

**The  $\beta_3$  stage, executed (WOIT.BETA3.DUALITY.01, [\[verification/v565\\_woit\\_beta3\\_pt\\_duality.py\]](#); verdict **SUCCESS** per the frozen preregistration; no status change).** The  $\mathbb{P}\mathbb{T} \leftrightarrow \mathbb{P}\mathbb{T}^*$  duality is *constructed as a compatibility class* and typed against  $\sigma_{\text{std}}$ , with the contract kill branch decided *empty by algebra*. **(i) The class:** a duality is a nondegenerate Hermitian form  $\Phi$  on  $\mathbb{C}^4$  ( $\sigma(Z) = Z^\dagger \Phi$ , mapping  $\mathbb{P}\mathbb{T} \rightarrow \mathbb{P}\mathbb{T}^*$ ); clock compatibility has the entrywise residual  $(\bar{\lambda}_j \lambda_k - 1) \Phi_{jk}$  and forces the four cross-eigenspace entries to zero exactly: two  $2 \times 2$  Hermitian blocks,  $16 \rightarrow 8$  real parameters (the deck adds nothing; the wrong clock lift  $\text{diag}(i, i, 1, 1)$  breaks the split). **(ii) The dichotomy and the forced  $(2, 2)$ :** pairing against Woit’s Euclidean structure  $\rho_{\text{tw}}$  via  $M_{\text{tw}}^\dagger \Phi M_{\text{tw}} = \varepsilon \bar{\Phi}$  splits the class exactly in two. The *Euclidean* branch ( $\varepsilon = +1$ ;  $\sigma_{\text{std}} = \mathbf{1}$  lives here) has seam-line norm  $a(|z|^2 + 1)$ : no null point, no OS cut, ever — exactly the section-defining role Woit assigns it. The *Minkowski* branch ( $\varepsilon = -1$ ) has seam-line norm  $a(|z|^2 - 1)$ : **the null locus meets the seam line exactly in the seam circle** — the OS cut is *forced*, not imposed — and the second block is exactly minus the conjugate of the first (charpoly mirrored), so the **total signature is  $(2, 2)$  for every nondegenerate member**: Woit’s Minkowski signature is a theorem of clock +  $\rho_{\text{tw}}$  pairing, not an input. **(iii) The kill is empty:** every compatible duality inverts

clock *and* deck exactly (family D); a clock-centralising (family A) duality would need  $\rho_4^2 = \text{scalar}$ , but  $\rho_4^2 = \text{diag}(-1, 1, -1, 1)$  — no solution; and family A independently has no quotient ((4, 4, 0), the v524 control). **(iv) The identification:** every seam-preserving Minkowski member induces *one* projective conjugation  $z \rightarrow -\bar{z}$  ( $\mu = -1$ , fixed points  $\pm i$ ), perpendicular to  $\sigma_{\text{std}}$ 's  $z \rightarrow +\bar{z}$ ; under the declared RP-side placement (v519 precision (ii); exact rational- $\pi$  arithmetic, offsets  $\mp\pi/2$ ) that member **is**  $\theta_\perp$  — the mirror whose quotient descent v524 certified:  $\Theta_{\text{phys}}$  is the shadow of the  $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$  duality, anti-unitary, Kramers-free ( $\Theta^2 = +1$  per sector), time-reversing the semigroup, closing with the  $\sigma_{\text{std}}$ -type cut onto the deck ( $\theta_{\text{cut}} \circ \theta_\perp = \alpha_{N/2}$ ; upstairs: the clock conjugation exchanges  $\mu = +1 \leftrightarrow \mu = -1$ ). *Role separation complete:*  $\sigma_{\text{std}} \sim$  the OS cut (the metric), the  $\Phi$ -twisted Minkowski duality  $\sim$  the descending mirror. Free/quotient level only;  $\mathcal{A}_{\text{hol}}$  untouched; all seven kill tests stay live there; the declared conventions (seam line, seam preservation, placement, torsor relabeling) are cited, not invented. No marker moves: the contract stays **[O]**;  $\gamma$  is the next milestone.

**Thermal follow-on (no status change).** The free OS quotient of  $\beta_2$  is KMS, and its temperature normalisation closes onto the Nariai/BH chain as the *third leg* of  $c_3$  (SEAM.THERMAL.KMS.01, [verification/v526\_seam\_thermal\_kms\_nariai\_bridge.py]:  $\beta$  measured =  $N$  at both levels;  $\beta_{\text{angle}} = 2\pi = 1/(4c_3)$ ,  $T_{\text{seam}} = 4c_3$ ; modular Hamiltonian closed; class-1 anti-numerology selection; non-compact  $\Rightarrow T = 0$ ). The silver clock eigenvalue is explained and demystified in parallel (SEAM.CLOCK.SILVER.01, [verification/v527\_seam\_clock\_silver\_spectrum.py]: structure SUCCESS, connection KILL). No marker moves.

**The  $\beta/\gamma$  roadmap (named milestones, each with success and kill criteria).** ( $\beta_1$ )  $\Theta$  on the gauge-invariant subalgebra + gauge-fixed RP on the equivariant SDYM( $E_8$ ) sector — success: the v519 OS form extends to the gauge-invariant subalgebra with positivity after gauge fixing; kill: kill test (2) fires (RP fails after gauge fixing). *Executed via v522 (above): UNDECIDED under the frozen preregistration — the clock reading is time-like (wrong operationalisation), the gauge datum is the GSO  $\mathbb{Z}_2$ , gauge-fixed RP holds under that typing, neither kill fires; the clock-equivariant statement moves to  $\beta_2$ .* ( $\beta_2$ ) The OS quotient of the free system made explicit — success:  $(\mathcal{H}, \Omega)$  constructed from the v519 bond-cut form with the clock acting unitarily; kill: the quotient degenerates (zero or indefinite completion). *Executed via v524 (above): SUCCESS, [C]-typed per precision (iii) —  $\mathcal{H}_{\text{phys}}$  explicit and PD at both levels, the clock a positive transfer step with spectral calculus (exact  $N = 8$  spectrum  $\{1, \sqrt{2} - 1\}$ ) and a reconstructed unitary rotation group; the pre-declared KMS deviation carries exactly the silver witnesses; neither kill fires;  $\beta_3$  is next.* ( $\beta_3$ ) The  $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$  duality typed against  $\sigma_{\text{std}}$  — success: Woit's Minkowski conjugation ( $\mathbb{PT} \rightarrow \mathbb{PT}^*$ ) identified with the family-D conjugation on the quotient; kill: the duality forces a clock-centralising (family A) structure instead. *Executed via v565 (above): SUCCESS — the kill branch is empty by algebra (every clock-compatible duality is family D; family A would need  $\rho_4^2 = \text{scalar}$ ), the OS cut and the (2, 2) signature are forced on the Minkowski branch, and the unique induced seam member is  $\theta_\perp$ , the v524  $\Theta_{\text{phys}}$  carrier.* ( $\gamma$ ) The chirality theorem + the mark incidence — success: the odd-sector definiteness flip of v519 upgraded to “one chiral generation without mirrors” on the reconstructed net, and the four  $\mu_4$  marks extended incidence-compatibly; kill: kill tests (3)/(6)/(7) fire. *The FREE slice is executed via [verification/v608\_gamma\_free\_slice.py] (9/9, GAMMA-FREE-SLICE-LANDED): on the explicit v524 net the mirror state's odd sector is strictly negative definite (sector-exact flip — every mirror fermion vector has negative OS norm, so no mirror mode survives reconstruction: kill test (3) cannot fire at the free level), the chiral generation is multiplicity-free (clock spectrum exactly  $\{1, \sqrt{2} - 1\}$ ), and the mark transport exists ( $\tau_2$  is the mark-A $\rightarrow$ B transport, Hermitian PD: kill test (7) does not fire at the free level).  $\gamma$  itself stays open on  $\mathcal{A}_{\text{hol}}$  (kill tests (3)/(6)/(7) formally live there); kill test (6) is untouched by the free slice. The INTERACTING toy slice is since executed via [verification/v615\_gamma\_toy\_interacting.py] (11/11, GAMMA-TOY-LANDED): on the RP-surviving interaction class (the alignment bit  $\delta = \pi/2$ ,  $s = +$ , v534) the mirror state's odd sector is strictly negative definite on every admissible cut and every coupling  $g \in \{1/32..8\}$  (bounded away from zero — kill test (3) cannot fire on the surviving class), the mark transport  $\alpha_4$  is Hermitian PD at every  $g$  (kill test (7) does not fire at the interacting toy level), the odd transfer spectrum stays*

*multiplicity-free (honest finding: strong-coupling near-doubling at  $g = 8$  without closing), and the RP-dead member has no gamma home — the dynamical bit selection and the chirality data co-locate on the same member. Fence: one toy, one interaction class;  $\gamma$  proper stays open on  $\mathcal{A}_{\text{hol}}$ . Kill test (6) has since received its first (kinematic) slice via `[verification/v616_su2_internal_kinematic.py]` (17/17, SU2-KINEMATIC-SEPARATED): on  $\mathbb{C}^4 = \mathbb{H}^2$  the internal  $\text{su}(2)$  (the right quaternion action, with  $\rho_{\text{tw}} = \text{right-j verbatim}$ ) preserves every twistor fiber, commutes exactly with the spacetime quaternion action and intersects it in  $\{0\}$  — the internal factor separates kinematically, the clock factorizes exactly as  $\text{left-}e^{i\pi/4} \circ \text{right-}e^{i\pi/4}$ , the ALE deck is purely spacetime-side, and the complex choice breaks the internal  $SU(2)$  to  $U(1)$  with the v519 mark torsor as its  $\mu_4$ ; the DYNAMICAL half of kill test (6) (a genuine gauge action on the reconstructed net) stays open and formally live on  $\mathcal{A}_{\text{hol}}$ .*

**Scope fence (explicit non-claims of the celestial/twistor branch).** Self-dual is *not* full 4D physics. Until and unless this contract closes, the celestial/twistor branch claims *none* of the following: both helicities (the anti-self-dual completion), generic scattering amplitudes, local matter fields, full Einstein dynamics (beyond the self-dual sector), electroweak symmetry breaking, confinement. The classical field-equation route (entanglement equilibrium, v358/v359) is a separate, non-competing statement about the full equation; SEAM.EQUIV.01 and its route split (SEAM.EQUIV.MMST.01/SEAM.EQUIV.TWISTOR.01) are stated in their own rows and are not moved by anything here. Status: WOIT.OS.TWISTOR.01 [O] (research contract; the  $\alpha$  stage is executed — WOIT.THETA.FREE.01 [E], v519 — with kill test (1) discharged at the free level only; the  $\beta_1$  stage is executed — WOIT.BETA1.GSO.01, v522, verdict UNDECIDED per the frozen preregistration, with kill test (2)'s free shadow not firing, the gauge datum typed as the GSO  $\mathbb{Z}_2$  and contract precision (iii) added; the  $\beta_2$  stage is executed — WOIT.BETA2.OS.01, v524, verdict SUCCESS, [C]-typed per precision (iii), with the OS quotient explicit and PD at both levels, the clock a positive transfer step with spectral calculus and a reconstructed unitary rotation group, and the pre-declared KMS deviation carrying exactly the silver witnesses; the  $\beta_3$  stage is executed — WOIT.BETA3.DUALITY.01, v565, verdict SUCCESS, with the kill branch empty by algebra, the OS cut and the (2, 2) signature forced on the Minkowski branch of the duality class, and the unique induced seam member identified with the v524  $\Theta_{\text{phys}}$  carrier; no marker move).

## Certifiability and recommended order

| Building block                                | Machine                 | Hardness             |
|-----------------------------------------------|-------------------------|----------------------|
| $W_{\text{wall}}$ enumeration ( $C_U^{(1)}$ ) | Lean / Python / Wolfram | easy ( <i>done</i> ) |
| $D_4$ fixed-locus polynomial ( $C_U^{(2)}$ )  | Sage / Mathematica      | medium               |
| selector uniqueness ( $C_U^{(3)}$ )           | interval arithmetic     | medium               |
| SNF, determinants, spectra                    | Lean                    | easy–medium          |
| principal-symbol ellipticity (G1)             | xAct / Cadabra / Lean   | medium               |
| gap inequality (G5)                           | Lean                    | easy ( <i>done</i> ) |
| operator-norm bound (G5)                      | analysis + numerics     | hard                 |
| projective limit (G6)                         | not primarily machine   | very hard            |
| Ward identities (G7)                          | BV/BRST formal + checks | hard                 |

### Recommended order.

Selector-triangle pairings  $\rightarrow v_{\text{geo}}$  scale anchor  $\rightarrow G_{\text{net}}$  inclusion theorem

The flavor interface (historically  $U_{\text{wall}}$ ) is now reduced to the *selector-triangle pairings* (the dual normal pair  $(d, n)$  forcing  $R$  columnwise, v136/v139); it is finite, falsifiable and the cheapest visible upgrade. The  $v_{\text{geo}}$  scale anchor is the one dimensional input (same nature as  $1/G$ ).  $G_{\text{net}}$  — the index-4 seam-net inclusion certifying the metric sector — is more fundamental but a *programme*,

not a single proof: heavy machinery, with the projective limit the genuine summit.  $F_{\text{transfer}}$  (source→pole / relic / cosmology) is the remaining downstream interface.

$F_{\text{transfer}}$  is a typed functor, not a bag of open topics. It is standard physics fed TFPT *source* data,  $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ , with exactly four interfaces:

| Interface              | Map                                                                         | Status    | Source data → observable                                                                                                                                                                                                                                                 |
|------------------------|-----------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $F_{\text{pole}}$      | source→pole masses                                                          | [E] / [C] | Koide 53/54 = $a^T(R+Q)\mathbf{1}/(2\mathbf{1}^T R a)$ is an exact operator factor [E] (the missing Sheet-Diamond corner); the pole-transfer interpretation is [C] ([verification/v183_koide_f_corner_transfer.py])                                                      |
| $F_{\text{Boltzmann}}$ | CP source→ $\eta_B$                                                         | [C]       | $\tilde{m}_1=m_3/A_\Lambda$ anchors the washout, but $M_1=M_R\varphi_0^4$ only relocates the free input ( $M_R$ is the seesaw scale, not a compiler power), so $\eta_B$ stays a <i>sharper scenario</i> , not a closure ([verification/v184_etaB_anchored_boltzmann.py]) |
| $F_{\text{relic}}$     | $(f_a, m_a, \theta_i, N_{\text{DW}}) \rightarrow \Omega_a$                  | [C]       | the determinant-line axion can be all of DM, but only on the $\theta_i \approx 170^\circ$ hilltop where the relic is exponentially sensitive — a viable scenario ([verification/v185_axion_relic_solver.py])                                                             |
| $F_{\text{QCD}}$       | $(\alpha_s, b_3, \Lambda_{\text{QCD}}, \text{lattice}) \rightarrow m_p/m_e$ | [O]       | a QCD matching contract ( $b_3=-7$ a carrier output), <i>not</i> a compiler number — the rejected 1836 near-formula is the discipline working                                                                                                                            |

A machine guard enforces the typing: the four families must always carry a [C]/[O] marker and must never be promoted to a primitive [E] compiler prediction ([verification/v187\_fttransfer\_laws.py]). Exact *algebraic* sub-parts (the Koide factor,  $b_3=-7$ ) may be [E]; the *physical* prediction never is.

**The functor contract CONTRACT.F.01, four axioms** ([verification/v213\_fttransfer\_functor.py]). Beyond the typing guard, the functor itself satisfies four checkable structural axioms, each instance a *discrete* compiler kernel [E] composed with an *external* continuous solver [C]: (1)  **$\mu_4$ -deck equivariance** — the discrete kernels are carrier/deck invariants  $\{|\mu_4|, N_{\text{fam}}, g_{\text{car}}, A_\Lambda\}$ , and the Koide multiplier  $\lambda_2=(2/3)^6=64/729$  is the  $\mu_4$ -deck transfer eigenvalue (v82/v54); (2) **Plücker preservation** — the dimensionless readouts are Plücker readings ( $53=a^T(R+Q)\mathbf{1}$ ,  $\text{Pl}(F)=(1, 22, 30)$ ;  $\|\text{Pl}(K)\|_1=11$ ); (3) **positivity/stochasticity** —  $\text{spec } T=\{1, (2/3)^6, (1/3)^6\}$  is positive (RP/OS),  $\kappa_f \in (0, 1)$ ; (4) **external modules explicit** — the continuous content (pole transfer, the flavored Boltzmann network, the finite- $T$   $\chi(T)$  ODE, the lattice  $C_N$ ) is named and external, never a compiler number. So CONTRACT.F.01 is the third research contract alongside ( $U_{\text{wall}}$ ) and ( $G_{\text{metric}}$ ): a typed functor, *not* a bag of open topics — but a consolidation, *not* a closure.

**A preregistered cross-ratio check for the four solvers — designed, and since executed on its discrete half.** The design is machine-checked in [verification/v566\_parabolic\_anchor\_selfcode.py], the discrete-half execution in [verification/v571\_cr\_discrete\_corners.py]. The disc  $-7$  norm line  $\alpha_{t+1} = \alpha_t + 2$  (v533) equips the transfer path with a projective invariant:  $\text{CR}(\alpha_{-2}, \alpha_{-1}; \alpha_0, \alpha_1) = \frac{4}{3}$  *exactly* — and, honesty first,  $\frac{4}{3}$  is the *universal* cross-ratio of any arithmetic progression, so the test content is not the number but the *discipline*: if the four solvers ( $J, K, C, F$ ) export canonical one-dimensional projective states  $y_J, y_K, y_C, y_F$  — declared *before* any evaluation — then Möbius transport along the path forces  $\text{CR}(y_J, y_K; y_C, y_F) = \frac{4}{3}$ , and the fourth state is determined fit-free by the first three ( $y_F$  reconstruction identity). Negative control with teeth: the raw determinant path (2, 4, 14, 32) gives  $\text{CR} = \frac{28}{25} \neq \frac{4}{3}$  — the already-known norm sequence cannot pass. *The discrete half is now executed*: the four corners are the in-repo kernels  $J, K, C, F = M(1, t) = R + Q \text{diag}(1, t, t)$ ,  $t = -2.1$  (v224), and the corpus's one canonical declared solver reading — the Koide anchor response  $\ell(M) = a^T M \mathbf{1}$  (v183) — reads (26, 35, 44, 53) on them:  $\text{CR} = \frac{4}{3}$  exactly, and the fit-free reconstruction *forces*  $y_F = 53$  from (26, 35, 44) alone — the Koide numerator is Möbius-determined by the three upstream corners (a new exact cross-corner identity); honesty theorem stated out loud:  $\ell$  is affine on the path (step 9 =  $N_{\text{fam}}^2$ ), and *every* linear functional of the kernel passes — the check separates linear from nonlinear exports (the

dominant-eigenvalue export fails at 1.3222, the determinants at  $\frac{28}{25}$ , both as they must) [E]. Status: the *continuous* half (the external solvers exporting their own states) stays open exactly as this contract fences it; contract markers unchanged [C]. *The frozen export protocol for the continuous half* (design [X], preregistered here): each instance must export, *before* evaluation, the canonical one-dimensional datum of its own native flow — the fixed-point multiplier of its contraction (cf. the native rates of v425:  $F\_pole$   $(2/3)^6$  exact in-corpus,  $F\_Boltzmann/F\_relic/F\_QCD$  external) — as a projective state on the path; only then is  $CR = \frac{4}{3}$  tested, with no renormalisation after the fact. *Executed* ([verification/v575\_cr\_continuous\_uniflow.py]): the in-repo native exports form the quadruple  $\{(2/3)^6, (2/3)^6, 1, -7\}$  — the repetition is *forced* by the single-flow theorem (v425:  $F\_pole$  and  $F\_Boltzmann$  are literally the same seam contraction) — and a repeated quadruple gives  $CR \in \{0, 1, \infty\}$  in every corner assignment (theorem, plus all 24 permutations): the continuous half can never pass *natively*; it closes onto the single-flow reduction, verdict UNIFLOW-DEGENERATE. The four independent external data are the dimensionful anchors ( $v_{geo}$ ,  $M_R$ ,  $C_p$ ,  $\theta_i$ ) — [O] walls, not projective states — so a future continuous test needs genuinely *external* anchor-level exports; the meaningful cross-ratio content was and remains the discrete half. Both halves of the check are now executed [E]/[C]. *The jet-level successor is executed natively too* ([verification/v578\_native\_jets.py]): when point values degenerate, the Möbius-invariant *Schwarzian* of the exported trajectories is the right jet test — and natively, the exact v99 Koide trajectory has  $\{q, t\} = -\Delta^2/2$  in wall time (constant, trajectory-independent;  $F\_Boltzmann$  identical by the single-flow theorem) while being Möbius in its *own* exponential clock  $u = e^{\Delta t}$  (Schwarzian 0), and the 1-loop QCD flow is Möbius in its own RG time already (Schwarzian 0): the native quadruple  $\{-\Delta^2/2, -\Delta^2/2, 0, \text{degenerate}\}$  shares *no* wall-time jet invariant — each instance is *projective in its own clock*, the obstruction is the clock, and the clocks are anchored by the external data. Verdict MOEBIUS-IN-OWN-CLOCK; a future joint jet test needs external anchor-level jets; markers unchanged [E]/[C].

**The discrete→dynamic lens:  $F\_transfer$  is the readout end of the one principle** ([verification/v303\_fttransfer\_dynamics.py], FR.DYNAMICS.01). The typing above is sharpened by a *dynamical* reading. The main-branch mechanism is a gapped, positivity-preserving relaxation to a *unique* attractor (the local-averaging update  $m \leftarrow (A+2I)/4m \rightarrow$  the  $E_8$  marks, gap  $6 \ln(3/2)$ ). All four  $F\_transfer$  instances share that *same shape*: [E]  $F\_pole$  (Koide) *is* the seam dynamics — the source→pole/branch relaxation is the Möbius map with fixed-point multiplier  $F'=(2/3)^6=\lambda_2$ , the seam-clock subleading eigenvalue, so  $|F'| < 1 \Rightarrow$  a unique attractor at the *seam* rate (v82); [C]  $F\_Boltzmann$  ( $\eta_B$ ) is a gapped positive contraction (washout  $\kappa_f \in (0, 1)$  toward the unique balance, an H-theorem); [C]  $F\_relic$  (axion) freezes at an adiabatic fixed point (the comoving number relaxes to a constant); [E]  $F\_QCD$  ( $m_p/m_e$ ) is the 1-loop RG flow with the carrier  $b_3 = -7$ , whose Gaussian UV fixed point is the attractor (asymptotic freedom,  $\Lambda_{QCD}$  generated). *Honest boundary*: only  $F\_pole$  has the seam rate  $(2/3)^6$  exactly; the other three share the *shape* with external rates (thermal, cosmological, RG). So  $F\_transfer$  is not a bolt-on but the downstream manifestation of the one discrete→dynamic principle — the compiler fixes the discrete seed + integer kernel, a gapped flow carries it to the observable — and the theory’s simplicity is preserved precisely because the external rates are honestly fenced (v187), not because they reduce to the seam. (Even  $F\_pole$ ’s flow time is non-integer, v93: the relaxation is the *continuous* semigroup  $e^{t \log F}$  generated by the discrete operator — exactly the diffusion-semigroup reading of the main branch.)

**The single-flow reduction** (DYN.TRANSFER.UNIVERSAL.01). The dynamic lens is sharpened to a precise reduction ([verification/v425\_dyn\_transfer\_universal.py]): the shared “shape” is literally the *one* native flow of the theory — the seam modular/recovery semigroup (v238/v240, gap  $6 \ln(3/2)$ ) — restricted to each sector, extending the static universal-gap principle (v383) to a dynamic one. [E] the v99 Koide generator  $\dot{q}=(\Delta/N_{fam})(q-2)(q-5)$  linearised at the attractor  $q=2$  has rate  $-\Delta$ , so the time-1 contraction is  $(2/3)^6$  (the recovery eigenvalue,  $q=5$  unstable); [E] the Boltzmann washout is that same contraction, with the CP phase  $4\pi/3$  native (v320); [E] the QCD rate is  $b_3 = -7$ . So the transfer *dynamics* is native; only the *anchors* stay external — the IR pole scale

and  $M_R$  are  $v_{\text{geo}}$ -class,  $C_p$  is the lattice  $O(1)$ , and  $F_{\text{relic}}$ 's cosmological initial condition  $\theta_i$  is the one genuine wall. A reduction, *not* internalisation: no new premise, and the firewall (v187) holds — the observable stays [C]. So the visible residual is one geometric postulate (QGeo.SYM.01), one unit gauge ( $v_{\text{geo}}$ ), and this one typed transfer functor — everything else is out-of-sample data.

## External works built upon

The TFPT discrete core (the two axioms, the  $E_8$  glue, the integer skeleton) is self-contained and machine-checked. Its *bridges* to continuum/AQFT physics, to gravity and to the lattice/singularity picture build on established mathematics, listed here by topic; the body refers to these by author name.

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- Y. Kawahigashi, R. Longo, M. Müger, *Multi-interval subfactors and modularity of representations in conformal field theory*, Comm. Math. Phys. **219** (2001) 631–669.

### Conformal nets, vertex operator algebras and lattice constructions.

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- T. J. Osborne, A. Stottmeister, *Conformal field theory from lattice fermions*, Comm. Math. Phys. **398** (2023) 219–289, [arXiv:2107.13834](#), [doi:10.1007/s00220-022-04521-8](#). (This is the seam scaling-limit theorem the TFPT continuum leg cites — the Koo–Saleur  $\rightarrow$  Virasoro convergence,  $c = \frac{1}{2}$ , and the smeared WZW-current Theorem 5.1. The internal TFPT code-token “MMST” and the legacy claim-ids SEAM.MMST.\* / filenames v\*\_mmst\_\* denote *this* Osborne–Stottmeister theorem.)
- V. Morinelli, G. Morsella, A. Stottmeister, Y. Tanimoto, *Scaling limits of lattice quantum fields by wavelets*, Comm. Math. Phys. **387** (2021) 299–360, [arXiv:2010.11121](#), [doi:10.1007/s00220-021-04152-5](#); and *Operator-algebraic renormalization and wavelets*, Phys. Rev. Lett. **127** (2021) 230601. (The operator-algebraic (wavelet/Wilson–Kadanoff) renormalization *framework* the Osborne–Stottmeister theorem above builds on.)
- M. S. Adamo, Y. Moriwaki, Y. Tanimoto, *Osterwalder–Schrader axioms for unitary full vertex operator algebras*, [arXiv:2407.18222](#) (2024).
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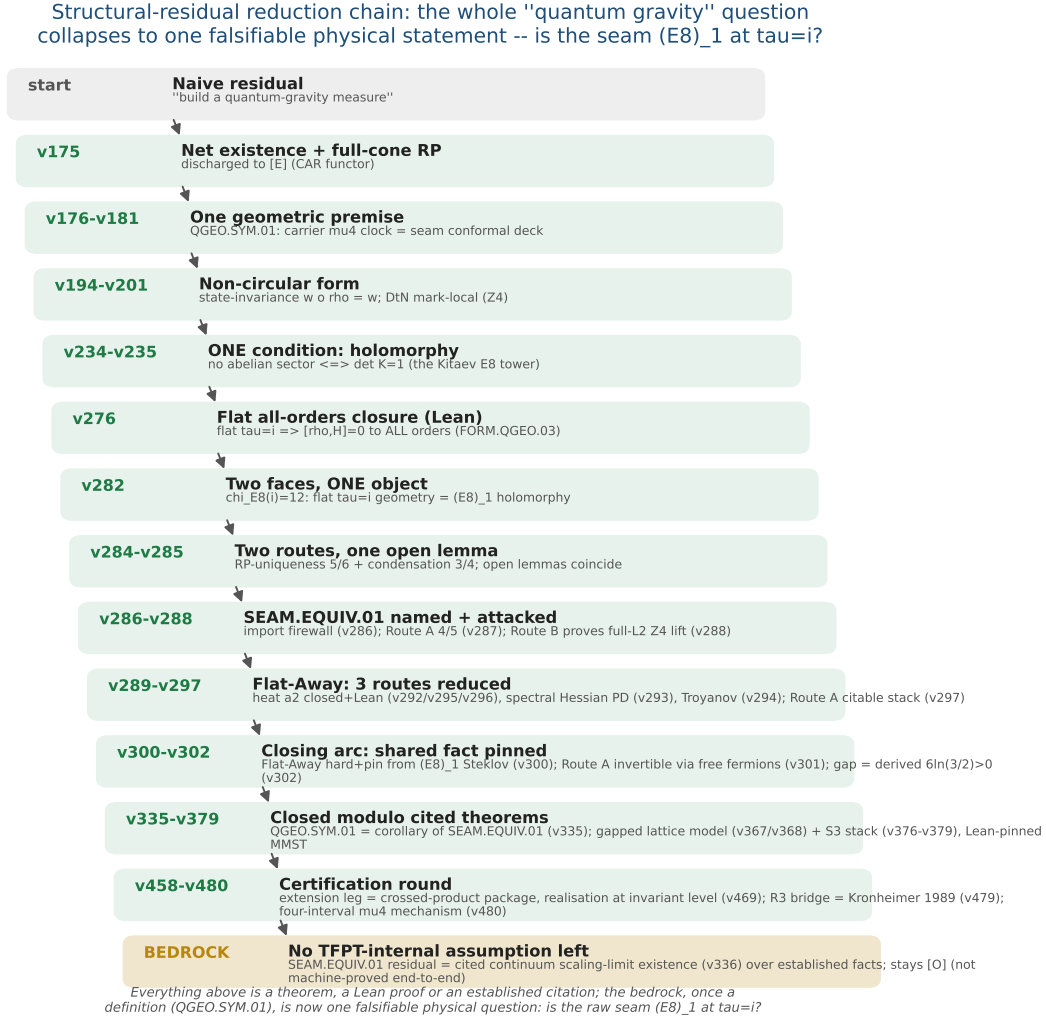
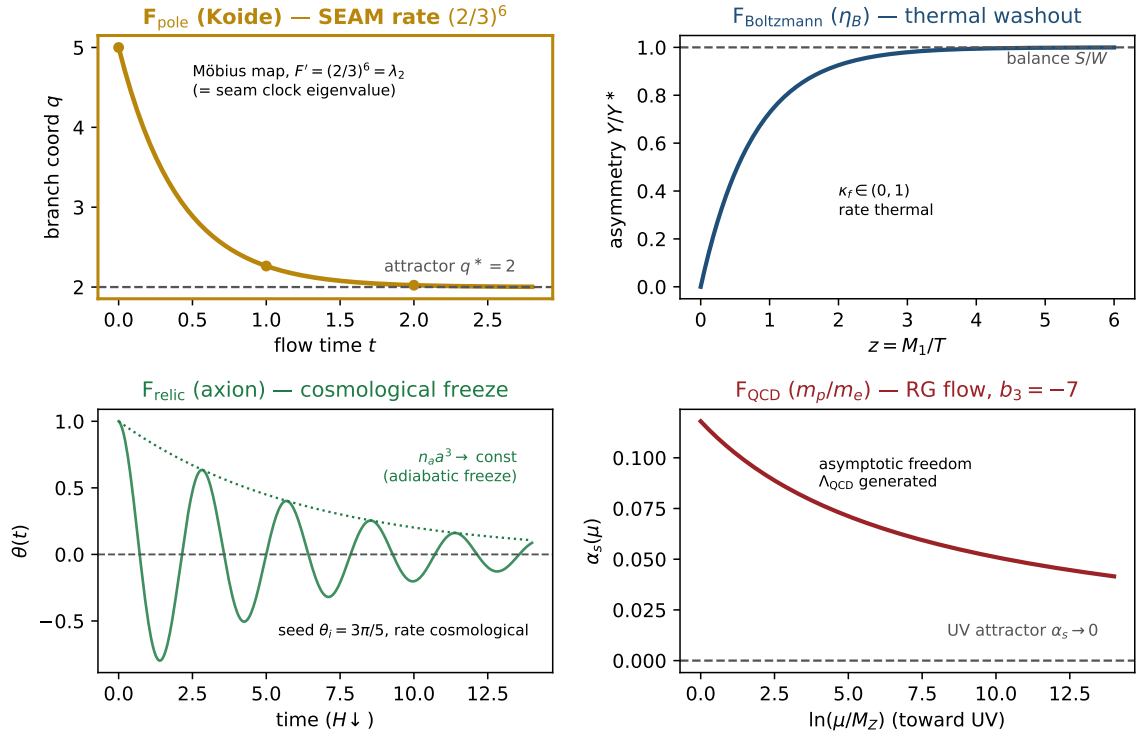


Figure 3: The structural-residual reduction chain ( $v175 \rightarrow$  the certification round). The naive “build a quantum-gravity measure” shrinks, one machine-checked step at a time, from a vague measure to one falsifiable physical statement — *is the raw seam  $(E_8)_1$  at  $\tau = i$ ?* (SEAM.EQUIV.01; its conformal-deck face QGEO.SYM.01 is a corollary, v335). The closing arc ( $v276$ – $v302$ ) pins the shared fact and derives the gap  $6\ln(3/2) > 0$ ; the certification round discharges the extension leg to the crossed-product package (v469), the R3 graph→geometry bridge to Kronheimer 1989 (v479) and exhibits the four-interval  $\mu_4$  mechanism (v480). Every step above the bedrock is a theorem, a Lean proof or an established citation; the residual — the cited continuum scaling-limit existence (v336) — stays honestly [O].

$F_{\text{transfer}}$ : one shape, four rates — a gapped relaxation to a **UNIQUE** attractor



Only  $F_{\text{pole}}$ 's rate is the seam eigenvalue  $(2/3)^6$ ; the other three share the shape with external rates (v303).

Figure 4: **The discrete→dynamic lens on  $F_{\text{transfer}}$**  ([verification/v303\_ftransfer\_dynamics.py]). All four transfer instances share *one* shape — a gapped, positivity-preserving relaxation to a *unique* attractor, the same Perron-Frobenius / H-theorem / RG-fixed-point structure as the main-branch update  $m \leftarrow (A+2I)/4m \rightarrow$  the  $E_8$  marks. *Only*  $F_{\text{pole}}$  (Koide, gold) runs it at the *seam* rate: its Möbius multiplier is  $F'=(2/3)^6=\lambda_2$ , the seam-clock subleading eigenvalue.  $F_{\text{Boltzmann}}$  ( $\eta_B$ , thermal washout),  $F_{\text{relic}}$  (axion, cosmological freeze) and  $F_{\text{QCD}}$  ( $m_p/m_e$ , the RG flow to the Gaussian UV fixed point with the carrier  $b_3 = -7$ ) share the *shape* with *external* rates. So  **$F_{\text{transfer}}$**  is the readout end of the one principle, not a bolt-on; the theory's simplicity is preserved because the external rates are honestly fenced (v187), not reduced to the seam.